

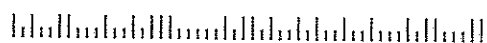
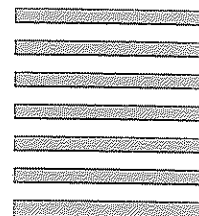
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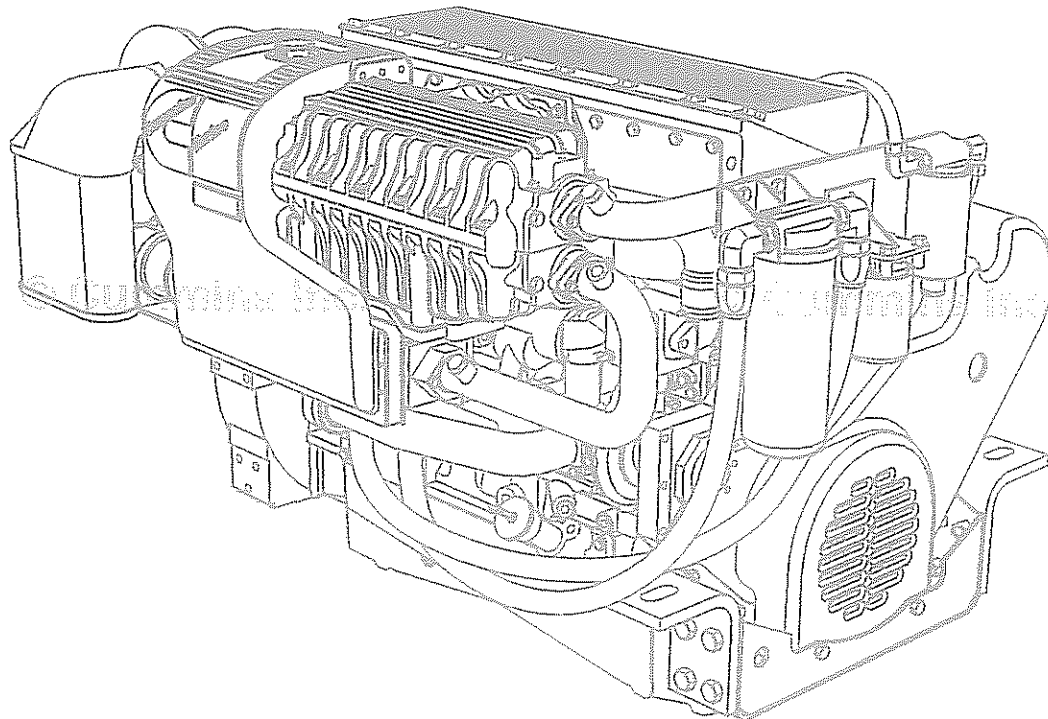
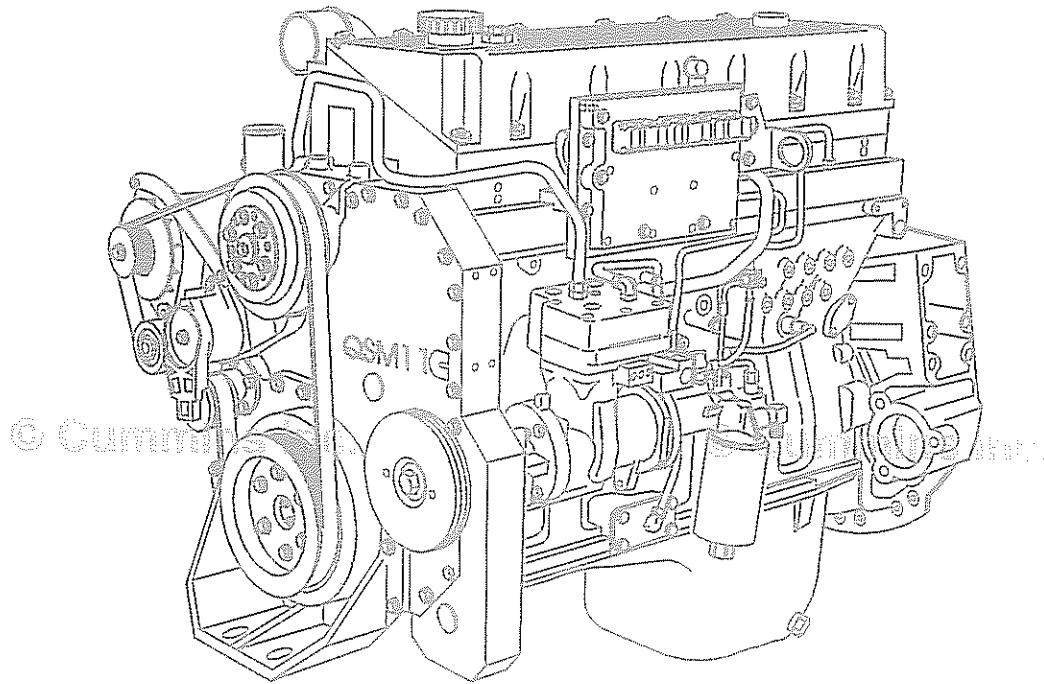
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Operation and Maintenance Manual Marine and Industrial QSM11 Engine



Foreword

This manual contains information for the correct operation and maintenance of your Cummins engine. It also includes important safety information, engine and systems specifications, troubleshooting guidelines, and listings of Cummins Authorized Repair Locations and component manufacturers.

Read and follow all safety instructions. Refer to the **WARNING** in the General Safety Instructions in Section I - Introduction.

Keep this manual with the equipment. If the equipment is traded or sold, give the manual to the new owner.

The information, specifications, and recommended maintenance guidelines in this manual are based on information in effect at the time of printing. Cummins Inc. reserves the right to make changes at any time without obligation. If you find differences between your engine and the information in this manual, contact your local Cummins Authorized Repair Location or call 1-800-DIESELS (1-800-343-7357) toll free in the U.S. and Canada.

The latest technology and the highest quality components were used to produce this engine. When replacement parts are needed, we recommend using only genuine Cummins or ReCon® exchange parts.

NOTE: Note: Warranty information is located in Section W. Make sure you are familiar with the warranty or warranties applicable to your engine.

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Important Reference Numbers

Fill in the part name and number in the blank spaces provided below. This will give you a reference whenever service or maintenance is required.

Name	Number	Number
Engine Model		
Engine Serial Number (ESN)		
Control Parts List (CPL)		
Fuel Pump Part Number		
Electronic Control Module (ECM)		
Electronic Control Module Serial Numbers (ECM)		
Filter Part Numbers:		
◊ Air Cleaner Element		
◊ Lubricating Oil		
◊ Fuel		
◊ Fuel-Water Separator		
◊ Coolant		
◊ Crankcase Ventilation		
◊ Cummins Particulate Filter		
Governor Control Module (GCM) (if applicable)		
Belt Part Numbers:		
◊		
◊		
◊		
Clutch or Marine Gear (if applicable):		
◊ Model		
◊ Serial Number		
◊ Part Number		
◊ Oil Type		
◊ Sea Water Pump		
- Model		
- Part Number		

Section i - Introduction

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To the Owner and Operator

General Information

Preventive maintenance is the easiest and least expensive type of maintenance. Follow the maintenance schedule recommendations outlined in Maintenance Guidelines (Section 2).

Keep records of regularly scheduled maintenance.

Use the correct fuel, lubricating oil, and coolant in your engine as specified in Maintenance Specifications (Section V). Blending engine oil with fuel is prohibited for engines with an aftertreatment system.

Cummins Inc. uses the latest technology and the highest quality components to produce its engines. Cummins Inc. recommends using genuine Cummins new parts and ReCon® exchange parts.

Personnel at Cummins Authorized Repair Locations have been trained to provide expert service and parts support. If you have a problem that can not be resolved by a Cummins Authorized Repair Location, follow the steps outlined in the Service Assistance (Section S).

Product coverage, warranty limitations and owner responsibilities are available in Warranty (Section W).

CAUTION

Disconnect both the positive (+) and negative (-) battery cables from the battery before welding on the vehicle. Attach the welder ground cable no more than 0.61 meters [2 feet] from the part being welded. Do not connect the ground cable of the welder to the ECM cooling plate or ECM. Welding on the engine or engine mounted components is not recommended.

About the Manual

General Information

This manual contains information needed to correctly operate and maintain your engine as recommended by Cummins Inc. For additional service literature and ordering locations, refer to Service Literature (Section L).

This manual does not cover vehicle, vessel, or equipment maintenance procedures. Consult the original vehicle, vessel, or equipment manufacturer for specific maintenance recommendations.

Both metric and U.S. customary values are listed in this manual. The metric value is listed first, followed by the U.S. customary in brackets.

Numerous illustrations and symbols are used to aid in understanding the meaning of the text. Refer to Symbols in this section for a complete listing of symbols and their definitions.

Each section of the manual is preceded by a Section Contents to aid in locating information.

How to Use the Manual

General Information

This manual is organized according to intervals at which maintenance on your engine is to be performed. A maintenance schedule, that states the required intervals and maintenance checks, is located in Maintenance Guidelines (Section 2). Locate the interval at which you are performing maintenance; then follow the steps given in that section for all the procedures to be performed.

Keep a record of all the checks and inspections made. A maintenance record form is located in Maintenance Guidelines (Section 2).

Engine troubleshooting procedures for your engine are located in Troubleshooting Symptoms (Section TS).

Specifications for your engine are located in Maintenance Specifications (Section V).

Symbols

General Information

The following symbols have been used in this manual to help communicate the intent of the instructions. When one of the symbols appears, it conveys the meaning defined below:

	WARNING - Serious personal injury or extensive property damage can result if the warning instructions are not followed.		PERFORM a mechanical or time MEASUREMENT .
	CAUTION - Minor personal injury can result or a part, an assembly, or the engine can be damaged if the caution instructions are not followed.		LUBRICATE the part or assembly.
	Indicates a REMOVAL or DISASSEMBLY step.		Indicates that a WRENCH or TOOL SIZE will be given.
	Indicates an INSTALLATION or ASSEMBLY step.		TIGHTEN to a specific torque.
	INSPECTION is required.		PERFORM an electrical MEASUREMENT .
	CLEAN the part or assembly.		Refer to another location in this manual or another publication for additional information.
			The component weighs 23 kg [50 lb] or more. To avoid personal injury, use a hoist or get assistance to lift the component.

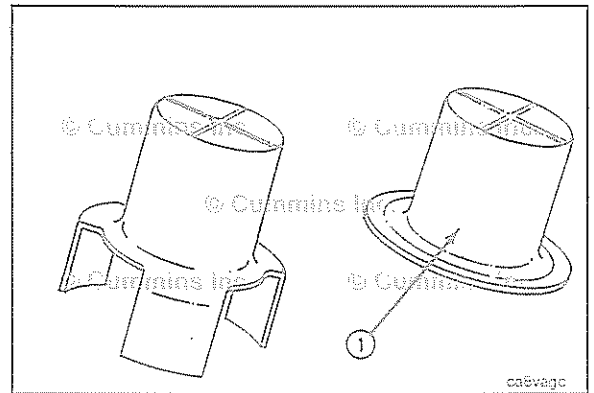
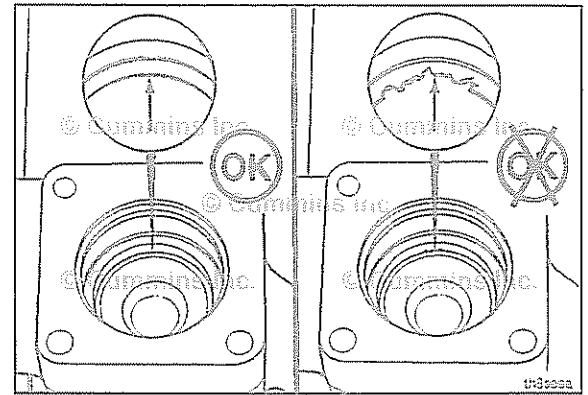
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Illustrations

General Information

Some of the illustrations throughout this manual are generic and will not look exactly like the engine or parts used in your application. The illustrations can contain symbols to indicate an action required and an acceptable or not acceptable condition.

The illustrations are intended to show repair or replacement procedures. The procedure will be the same for all applications, although the illustration can differ.



General Safety Instructions

Important Safety Notice



Improper practices, carelessness, or ignoring the warnings can cause burns, cuts, mutilation, asphyxiation or other personal injury or death.

Read and understand all of the safety precautions and warnings before performing any repair. This list contains the general safety precautions that must be followed to provide personal safety. Special safety precautions are included in the procedures when they apply.

- Work in an area surrounding the product that is dry, well lit, ventilated, free from clutter, loose tools, parts, ignition sources and hazardous substances. Be aware of hazardous conditions that can exist.
- Always wear protective glasses and protective shoes when working.
- Rotating parts can cause cuts, mutilation or strangulation.
- Do not wear loose-fitting or torn clothing. Remove all jewelry when working.
- Disconnect the battery (negative [-] cable first) and discharge any capacitors before beginning any repair work. Disconnect the air starting motor if equipped to prevent accidental engine starting. Put a "Do Not Operate" tag in the operator's compartment or on the controls.
- Use ONLY the proper engine barring techniques for manually rotating the engine. Do not attempt to rotate the crankshaft by pulling or prying on the fan. This practice can cause serious personal injury, property damage, or damage to the fan blade(s) causing premature fan failure.
- If an engine has been operating and the coolant is hot, allow the engine to cool before slowly loosening the filler cap to relieve the pressure from the cooling system.
- Always use blocks or proper stands to support the product before performing any service work. Do not work on anything that is supported ONLY by lifting jacks or a hoist.
- Relieve all pressure in the air, oil, fuel, and cooling systems before any lines, fittings, or related items are removed or disconnected. Be alert for possible pressure when disconnecting any device from a system that utilizes pressure. Do not check for pressure leaks with your hand. High pressure oil or fuel can cause personal injury.
- To reduce the possibility of suffocation and frostbite, wear protective clothing and ONLY disconnect liquid refrigerant (Freon) lines in a well ventilated area. To protect the environment, liquid refrigerant systems must be properly emptied and filled using equipment that prevents the release of refrigerant gas (fluorocarbons) into the atmosphere. Federal law requires capturing and recycling refrigerant.
- To reduce the possibility of personal injury, use a hoist or get assistance when lifting components that weigh 23 kg [50 lb] or more. Make sure all lifting devices such as chains, hooks, or slings are in good condition and are of the correct capacity. Make sure hooks are positioned correctly. Always use a spreader bar when necessary. The lifting hooks must not be side-loaded.
- Corrosion inhibitor, a component of SCA and lubricating oil, contains alkali. Do not get the substance in eyes. Avoid prolonged or repeated contact with skin. Do not swallow internally. In case of contact, immediately wash skin with soap and water. In case of contact, immediately flood eyes with large amounts of water for a minimum of 15 minutes. IMMEDIATELY CALL A PHYSICIAN. KEEP OUT OF REACH OF CHILDREN.
- Naptha and Methyl Ethyl Ketone (MEK) are flammable materials and must be used with caution. Follow the manufacturer's instructions to provide complete safety when using these materials. KEEP OUT OF REACH OF CHILDREN.
- To reduce the possibility of burns, be alert for hot parts on products that have just been turned off, exhaust gas flow, and hot fluids in lines, tubes, and compartments.
- Always use tools that are in good condition. Make sure you understand how to use the tools before performing any service work. Use ONLY genuine Cummins or Cummins ReCon® replacement parts.
- Always use the same fastener part number (or equivalent) when replacing fasteners. Do not use a fastener of lesser quality if replacements are necessary.
- Do not perform any repair when fatigued or after consuming alcohol or drugs that can impair your functioning.
- Some state and federal agencies in the United States of America have determined that used engine oil can be carcinogenic and can cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil.
- Liquified petroleum gas is heavier than air and can accumulate near the floor, in sumps, and low-lying areas.
- Natural gas is lighter than air and can accumulate under hood and awnings.
- To reduce the possibility of suffocation and frostbite, wear protective clothing and ONLY disconnect natural gas and liquified petroleum gas lines in a well ventilated area.
- Coolant is toxic. If not reused, dispose of in accordance with local environmental regulations.
- The catalyst reagent contains urea. Do not get the substance in your eyes. In Case of contact, immediately flood eyes with large amounts of water for a minimum of 15 minutes. Avoid prolonged contact with skin. In case of contact, immediately wash skin with soap and water. Do not swallow internally. In the event the catalyst reagent is ingested, contact a physician immediately.

- ◊ The catalyst substrate contains Vanadium Pentoxide. Vanadium Pentoxide has been determined by the State of California to cause cancer. Always wear protective gloves and eye protection when handling the catalyst assembly. Do not get the catalyst material in your eyes. In Case of contact, immediately flood eyes with large amounts of water for a minimum of 15 minutes. Avoid prolonged contact with skin. In case of contact, immediately wash skin with soap and water.
- ◊ The Catalyst substrate contains Vanadium Pentoxide. Vanadium Pentoxide has been determined by the State of California to cause cancer. In the event the catalyst is being replaced, dispose of in accordance with local regulations.
- ◊ California Proposition 65 Warning - Diesel engine exhaust and some of its constituents are known to the State of California to cause cancer, birth defects, and other reproductive harm.

General Repair Instructions

General Information

This engine incorporates the latest technology at the time it was manufactured; yet, it is designed to be repaired using normal repair practices performed to quality standards.

- Cummins Inc. does not recommend or authorize any modifications or repairs to engines or components except for those detailed in Cummins Service Information. In particular, unauthorized repair to safety-related components can cause personal injury or death. Below is a partial listing of components classified as safety-related:

- | | |
|---------------------------------|---|
| 1. Air Compressor | 12. Flywheel Mounting Capscrews |
| 2. Air Controls | 13. Fuel Shutoff Assemblies |
| 3. Air Shutoff Assemblies | 14. Fuel Supply Tubes |
| 4. Balance Weights | 15. Lifting Brackets |
| 5. Cooling Fan | 16. Throttle Controls |
| 6. Fan Hub Assembly | 17. Turbocharger Compressor Casing |
| 7. Fan Mounting Bracket(s) | 18. Turbocharger Oil Drain Line(s) |
| 8. Fan Mounting Capscrews | 19. Turbocharger Oil Supply Line(s) |
| 9. Fan Hub Spindle | 20. Turbocharger Turbine Casing |
| 10. Flywheel | 21. Vibration Damper Mounting Capscrews |
| 11. Flywheel Crankshaft Adapter | |

- Follow all safety instructions noted in the procedures
- Follow the manufacturer's recommendations for cleaning solvents and other substances used during the repair of the engine. Some solvents and used engine oil have been identified by government agencies as toxic or carcinogenic. Avoid excessive breathing, ingestion and contact with such substances. Always use good safety practices with tools and equipment.
- Provide a clean environment and follow the cleaning instructions specified in the procedures
- The engine and its components must be kept clean during any repair. Contamination of the engine or components will cause premature wear.
- Perform the inspections specified in the procedures
- Replace all components or assemblies which are damaged or worn beyond the specifications
- Use genuine Cummins new or ReCon® service parts and assemblies
- The assembly instructions have been written to use again as many components and assemblies as possible. When it is necessary to replace a component or assembly, the procedure is based on the use of new Cummins or Cummins ReCon® components. All of the repair services described in this manual are available from all Cummins Distributors and most Dealer locations.
- Follow the specified disassembly and assembly procedures to reduce the possibility of damage to the components

Complete rebuild instructions are available in the shop manual which can be ordered or purchased from a Cummins Authorized Repair Location. Refer to Section L — Service Literature for ordering instructions.

Welding on a Vehicle with an Electronic Controlled Fuel System



Disconnect both the positive (+) and negative (-) battery cables from the battery before welding on the vehicle. Attach the welder ground cable no more than 0.61 meters [2 feet] from the part being welded. Do not connect the ground cable of the welder to the ECM cooling plate or ECM. Welding on the engine or engine mounted components is not recommended or damage to the engine or components can result.

General Cleaning Instructions

Definition of Clean

Parts must be free of debris that can contaminate any engine system. This does not necessarily mean they have to appear as new.

Sanding gasket surfaces until the factory machining marks are disturbed adds no value and is often harmful to forming a seal. It is important to maintain surface finish and flatness tolerances to form a quality sealing surface. Gaskets are designed to fill small voids in the specified surface finish.

Sanding gasket surfaces where edge-molded gaskets are used is most often unnecessary. Edge-molded gaskets are those metal carriers with sealing material bonded to the edges of the gasket to seal while the metal portion forms a metal to metal joint for stability. Any of the small amounts of sealing material that can stick to the parts are better removed with a blunt-edged scraper on the spots rather than spending time polishing the whole surface with an air sander or disc.

For those gaskets that do not have the edge molding, nearly all have a material that contains release agents to prevent sticking. Certainly this is not to say that some gaskets are not difficult to remove because the gasket has been in place a long time, has been overheated or the purpose of the release agent has been defeated by the application of some sealant. The object however is just to remove the gasket without damaging the surfaces of the mating parts without contaminating the engine (don't let the little bits fall where they can not be removed).

Bead blasting piston crowns until the dark stain is removed is unnecessary. All that is required is to remove the carbon build-up above the top ring and in the ring grooves. There is more information on bead blasting and piston cleaning later in this document.

Cummins Inc. does not recommend sanding or grinding the carbon ring at the top of cylinder liners until clean metal is visible. The liner will be ruined and any signs of a problem at the top ring reversal point (like a dust-out) will be destroyed. It is necessary to remove the carbon ring to provide for easier removal of the piston assembly. A medium bristle, high quality, steel wire wheel that is rated above the rpm of the power tool being used will be just as quick and there will be less damage. Yes, one must look carefully for broken wires after the piston is removed but the wires are more visible and can be attracted by a magnet.

Oil on parts that have been removed from the engine will attract dirt in the air. The dirt will adhere to the oil. If possible, leave the old oil on the part until it is ready to be cleaned, inspected and installed, and then clean it off along with any attracted dirt. If the part is cleaned then left exposed it can have to be cleaned again before installation. Make sure parts are lubricated with clean oil before installation. They do not need to be oiled all over but do need oil between moving parts (or a good lube system priming process conducted before cranking the engine).

Bead blasting parts to remove exterior paint is also usually unnecessary. The part will most likely be painted again so all that needs happen is remove any loose paint.

Abrasive Pads and Abrasive Paper

The keyword here is "abrasive". There is no part of an engine designed to withstand abrasion. That is they are all supposed to lock together or slide across each other. Abrasives and dirt particles will degrade both functions.



Abrasive material must be kept out of or removed from oil passages and parts wear points. Abrasive material in oil passages can cause bearing and bushing failures that can progress to major component damage beyond reuse. This is particularly true of main and rod bearings.

Cummins Inc. does not recommend the use of emery cloth or sand paper on any part of an assembled engine or component including but not limited to removing the carbon ridge from cylinder liners or to clean block decks or counterbores.

Great care must be taken when using abrasive products to clean engine parts, particularly on partially assembled engines. Abrasive cleaning products come in many forms and sizes. All of them contain aluminum oxide particles, silicon carbide, or sand or some other similar hard material. These particles are harder than most of the parts in the engine. Since they are harder, if they are pressed against softer material they will either damage the material or become embedded in it. These materials fall off the holding media as the product is used. If the products are used with power equipment the particles are thrown about the engine. If the particles fall between two moving parts, damage to the moving parts is likely.

If particles that are smaller than the clearance between the parts while they are at rest (engine stopped), but larger than the running clearance then damage will occur when the parts move relative to each other (engine started). While the engine is running and there is oil pressure, particles that are smaller than the bearing clearance are likely to pass between the parts without damage and be trapped in the oil filter. However, particles larger than the bearing clearance will remove material from one part and can become embedded in one of the parts. Once embedded in one part it will

abrade the other part until contact is no longer being made between the two parts. If the damage sufficiently degrades the oil film, the two parts will come into contact resulting in early wear-out or failure from lack of effective lubrication.

Abrasive particles can fly about during cleaning it is very important to block these particles from entering the engine as much as possible. This is particularly true of lubricating oil ports and oil drilling holes, especially those located downstream of the lubricating oil filters. Plug the holes instead of trying to blow the abrasive particles and debris with compressed air because the debris is often simply blown further into the oil drilling.

All old gasket material must be removed from the parts gasket surfaces. However, it is not necessary to clean and polish the gasket surface until the machining marks are erased. Excessive sanding or buffing can damage the gasket surface. Many newer gaskets are of the edge molded type (a steel carrier with a sealing member bonded to the steel). What little sealing material that can adhere is best removed with a blunt-edged scraper or putty knife. Cleaning gasket surfaces where an edge-molded gasket is used with abrasive pads or paper is usually a waste of time.

WARNING

Excessive sanding or grinding the carbon ring from the top of the cylinder liners can damage the liner beyond reuse. The surface finish will be damaged and abrasive particles can be forced into the liner material which can cause early cylinder wear-out or piston ring failures.

Tape off or plug all openings to any component interior before using abrasive pads or wire brushes. If really necessary because of time to use a power tool with abrasive pads, tape the oil drillings closed or use plug and clean as much of the surface as possible with the tool but clean around the oil hole/opening by hand so as to prevent contamination of the drilling. Then remove the tape or plug and clean the remaining area carefully and without the tool. DO NOT use compressed air to blow the debris out of oil drilling on an assembled engine! More likely than not, the debris can be blown further into the drilling. Using compressed air is fine if both ends of the drilling are open but that is rarely the case when dealing with an assembled engine.

Gasket Surfaces

The object of cleaning gasket surfaces is to remove any gasket material, not refinish the gasket surface of the part.

Cummins Inc. does not recommend any specific brand of liquid gasket remover. If a liquid gasket remover is used, check the directions to make sure the material being cleaned will not be harmed.

Air powered gasket scrapers can save time but care must be taken to not damage the surface. The angled part of the scraper must be against the gasket surface to prevent the blade from digging into the surface. Using air powered gasket scrapers on parts made of soft materials takes skill and care to prevent damage.

Do not scrape or brush across the gasket surface if at all possible.

Solvent and Acid Cleaning

Several solvent and acid-type cleaners can be used to clean the disassembled engine parts (other than pistons. See Below). Experience has shown that the best results can be obtained using a cleaner that can be heated to 90° to 95° Celsius (180° to 200° Fahrenheit). Kerosene emulsion based cleaners have different temperature specifications, see below. A cleaning tank that provides a constant mixing and filtering of the cleaning solution will give the best results. Cummins Inc. does not recommend any specific cleaners. Always follow the cleaner manufacturer's instructions. Remove all the gasket material, o-rings, and the deposits of sludge, carbon, etc., with a wire brush or scraper before putting the parts in a cleaning tank. Be careful not to damage any gasket surfaces. When possible, steam clean the parts before putting them in the cleaning tank.

WARNING

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturers recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

Experience has shown that kerosene emulsion based cleaners perform the best to clean pistons. These cleaners should not be heated to temperature in excess of 77°C (170°F). The solution begins to break down at temperatures in excess of 82°C (180°F) and will be less effective.

Do not use solutions composed mainly of chlorinated hydrocarbons with cresols, phenols and/or cresylic components. They often do not do a good job of removing deposits from the ring groove and are costly to dispose of properly.

Solutions with a pH above approximately 9.5 will cause aluminum to turn black; therefore do not use high alkaline solutions.

Chemicals with a pH above 7.0 are considered alkaline and those below 7.0 are acidic. As you move further away from the neutral 7.0, the chemicals become highly alkaline or highly acidic.

Remove all the gasket material, o-rings, and the deposits of sludge, carbon, etc., with a wire brush or scraper before putting the parts in a cleaning tank. Be careful to not damage any gasket surfaces. When possible use hot high

pressure water or steam clean the parts before putting them in the cleaning tank. Removing the heaviest dirt before placing in the tank will allow the cleaner to work more effectively and the cleaning agent will last longer.

Rinse all the parts in hot water after cleaning. Dry completely with compressed air. Blow the rinse water from all the capscrew holes and the oil drillings.

If the parts are not to be used immediately after cleaning, dip them in a suitable rust proofing compound. The rust proofing compound must be removed from the parts before assembly or installation on the engine.

Steam Cleaning

Steam cleaning can be used to remove all types of dirt that can contaminate the cleaning tank. It is a good method for cleaning the oil drillings and coolant passages



When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

Do not steam clean the following components:

- Electrical Components
- Wiring Harnesses
- Injectors
- Fuel Pump
- Belts and Hoses
- Bearings (ball or taper roller)
- Electronic Control Module (ECM)
- ECM Connectors
- Dosing Control Unit

Plastic Bead Cleaning

Cummins Inc. does not recommend the use of glass bead blast or walnut shell media on any engine part. Cummins Inc. recommends using only plastic bead media, Part Number 3822735 or equivalent on any engine part. Never use sand as a blast media to clean engine parts. Glass and walnut shell media when not used to the media manufacturer's recommendations can cause excess dust and can embed in engine parts that can result in premature failure of components through abrasive wear.

Plastic bead cleaning can be used on many engine components to remove carbon deposits. The cleaning process is controlled by the use of plastic beads, the operating pressure and cleaning time.



Do not use bead blasting cleaning methods on aluminum pistons skirts or the pin bores in any piston, piston skirt or piston crown. Small particles of the media will embed in the aluminum or other soft metal and result in premature wear of the cylinder liner, piston rings, pins and pin bores. Valves, turbocharger shafts, etc., can also be damaged. Follow the cleaning directions listed in the procedures.



Do not contaminate wash tanks and tank type solvent cleaners with the foreign material and plastic beads. Remove the foreign material and plastic beads with compressed air, hot high pressure water or steam before placing them in tanks or cleaners. The foreign material and plastic beads can contaminate the tank and any other engine parts cleaned in the tank. Contaminated parts may cause failures from abrasive wear.

Plastic bead blasting media, Part Number 3822735, can be used to clean all piston ring grooves. Do not use any bead blasting media on piston pin bores or aluminum skirts.

Follow the equipment manufacturer's cleaning instructions. Make sure to adjust the air pressure in the blasting machine to the bead manufacturer's recommendations. Turning up the pressure can move material on the part and cause the plastic bead media to wear out more quickly. The following guidelines can be used to adapt to manufacturer's instructions:

1. Bead size: U.S. size Number 16 — 20 for piston cleaning with plastic bead media, Part Number 3822735
2. Operating Pressure — 270 kPa (40 psd) for piston cleaning. Pressure should not cause beads to break.
3. Steam clean or wash the parts with solvent to remove all of the foreign material and plastic beads after cleaning. Rinse with hot water. Dry with compressed air.



The bead blasting operation must not disturb the metal surface. If the metal surface is disturbed the engine can be damaged due to increased parts clearance or inadequate surface finish on parts that move against other parts.

When cleaning pistons, it is not necessary to remove all the dark stain from the piston. All that is necessary is to remove the carbon on the rim and in the ring grooves. This is best done by directing the blast across the part as opposed to straight at the part. If the machining marks are disturbed by the blasting process, then the pressure is too high or the blast is being held on one spot too long. The blast operation must not disturb the metal surface.

Walnut shell bead blast material is sometimes used to clean ferrous metals (iron and steel). Walnut shell blasting produces a great amount of dust particularly when the pressure of the air pressure on the blasting machine is increased above media manufacturer's recommendation. Cummins Inc. recommends not using walnut shell media to clean engine parts due to the risk media embedment and subsequent contamination of the engine.

Cummins Inc. now recommends glass bead media NOT used to clean any engine parts. Glass media is too easily embedded into the material particularly in soft materials and when air pressures greater than media manufacturer's recommend are used. The glass is an abrasive so when it is in a moving part, that part is abrading all the parts in contact with it. When higher pressures are used the media is broken and forms a dust of a very small size that floats easily in the air. This dust is very hard to control in the shop, particularly if only compressed air (and not hot water) is used to blow the media after it is removed from the blasting cabinet (blowing the part off inside the cabinet may remove large accumulations but never removes all the media).

Bead blasting is best used on stubborn dirt/carbon build-up that has not been removed by first steam/higher pressure washing then washing in a heated wash tank. This is particularly true of pistons. Steam and soak the pistons first then use the plastic bead method to safely remove the carbon remaining in the grooves (instead of running the risk of damaging the surface finish of the groove with a wire wheel or end of a broken piston ring. Make sure the parts are dry and oil free before bead blasting to prevent clogging the return on the blasting machine.

Always direct the bead blaster nozzle "across" rather than directly at the part. This allows the bead to get under the unwanted material. Keep the nozzle moving rather than hold on one place. Keeping the nozzle directed at one-place too long causes the metal to heat up and be moved around. Remember that the spray is not just hitting the dirt or carbon. If the machining marks on the piston groove or rim have been disturbed then there has not been enough movement of the nozzle and/or the air pressure is too high.

Never bead blast valve stems. Tape or use a sleeve to protect the stems during bead blasting. Direct the nozzle across the seat surface and radius rather than straight at them. The object is to remove any carbon build up and continuing to blast to remove the stain is a waste of time.

Acronyms and Abbreviations

General Information

The following list contains some of the acronyms and abbreviations used in this manual.

API	American Petroleum Institute
ASTM	American Society of Testing and Materials
BTU	British Thermal Unit
°C	Celsius
CARB	California Air Resources Board
C.I.D.	Cubic Inch Displacement
CNG	Compressed Natural Gas
CPL	Control Parts List
cSt	Centistokes
ECM	Electronic Control Module
EGR	Exhaust Gas Recirculation
EPA	Environmental Protection Agency
°F	Fahrenheit
FMI	Failure Mode Identifier
GVW	Gross Vehicle Weight
LPG	Liquified Petroleum Gas
Hg	Mercury
hp	Horsepower
H ₂ O	Water
ICM	Ignition Control Module
km/l	Kilometers per Liter
kPa	Kilopascal
LNG	Liquid Natural Gas
LTA	Low Temperature Aftercooling
MPa	Megapascal
mph	Miles Per Hour
mpq	Miles Per Quart
N•m	Newton-meter
NG	Natural Gas
OEM	Original Equipment Manufacturer
PID	Parameter Identification Descriptions
ppm	Parts Per Million
psi	Pounds Per Square Inch
PTO	Power Takeoff
rpm	Revolutions Per Minute
SAE	Society of Automotive Engineers
SCA	Supplemental Coolant Additive
STC	Step Timing Control
SID	Subsystem Identification Descriptions
VS	Variable Speed
VSS	Vehicle Speed Sensor

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Section E - Engine Identification

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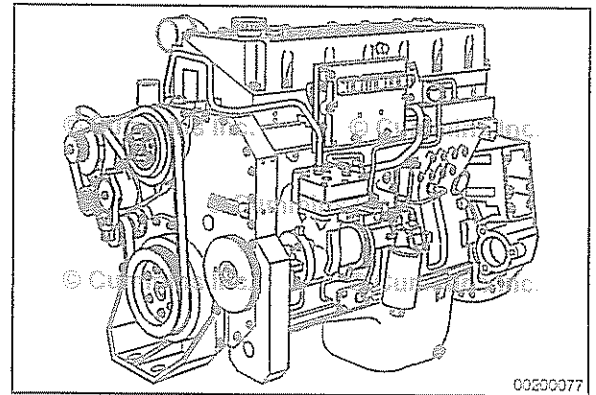
Engine Identification

Engine Dataplate

Marine Applications

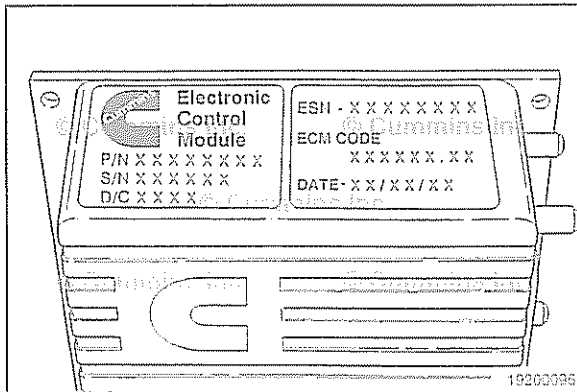
The engine dataplate shows important facts about your engine. The engine serial number and CPL provide data for ordering parts and service. The engine dataplate must not be changed unless approved by Cummins Inc.

The dataplate is located on the fuel pump side of the engine, on the rocker housing. Have the following engine data available when communicating with a Cummins Authorized Repair Location. The following information on the dataplate is mandatory when sourcing service parts:



The image shows a Cummins engine identification plate. It contains various fields for engine specifications. Numbered callouts point to specific areas: 1 points to the 'Engine Serial Number' field, 2 points to the 'Model' field, 3 points to the 'Horsepower and rpm rating' field, 4 points to the 'CPL' (Cummins Part Label) field, and 5 points to the 'Emission statement (if applicable)' field.

1. Engine serial number
2. Model
3. Horsepower and rpm rating
4. CPL
5. Emission statement (if applicable).



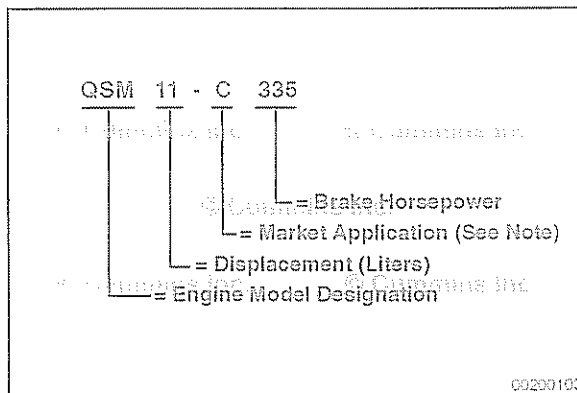
ECM Dataplate

Marine Applications

There are two dataplates on top of the ECM.

The dataplate on the left contains the part number (P/N), serial number (S/N), and data code (D/C) of the ECM.

The dataplate on the right contains the engine serial number (ESN), ECM code, and date.



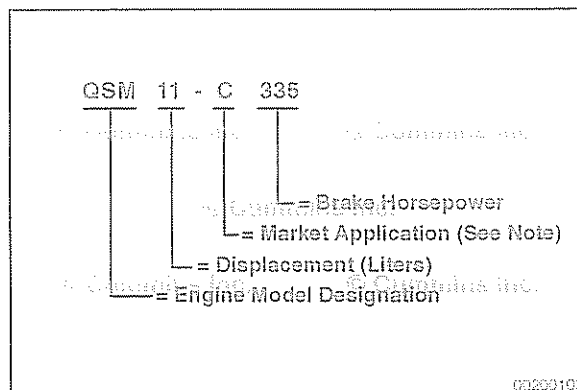
Cummins® Engine Nomenclature

Industrial and Power Generation

The Cummins® engine nomenclature provides the data as illustrated in the graphics.

NOTE: The following letters designate some of the different market applications for a Cummins® engine.

- A = agriculture
- C = construction
- M = marine
- G = generator drive



Marine Applications

The Cummins® engine nomenclature provides the data as illustrated in the graphics.

NOTE: The following letters designate some of the different market applications for a Cummins® engine.

- A = agriculture
- C = construction
- M = marine
- G = generator drive

Fuel Injection Pump Dataplate

The fuel pump dataplate is located on top of the fuel pump. It provides the data needed to calibrate the fuel pump.

FUEL PUMP DATAPLATE		
Fuel Code		Revision
0749 4219-A		840508
3043327		3052282
Service Part No.	Pump Production Part No.	

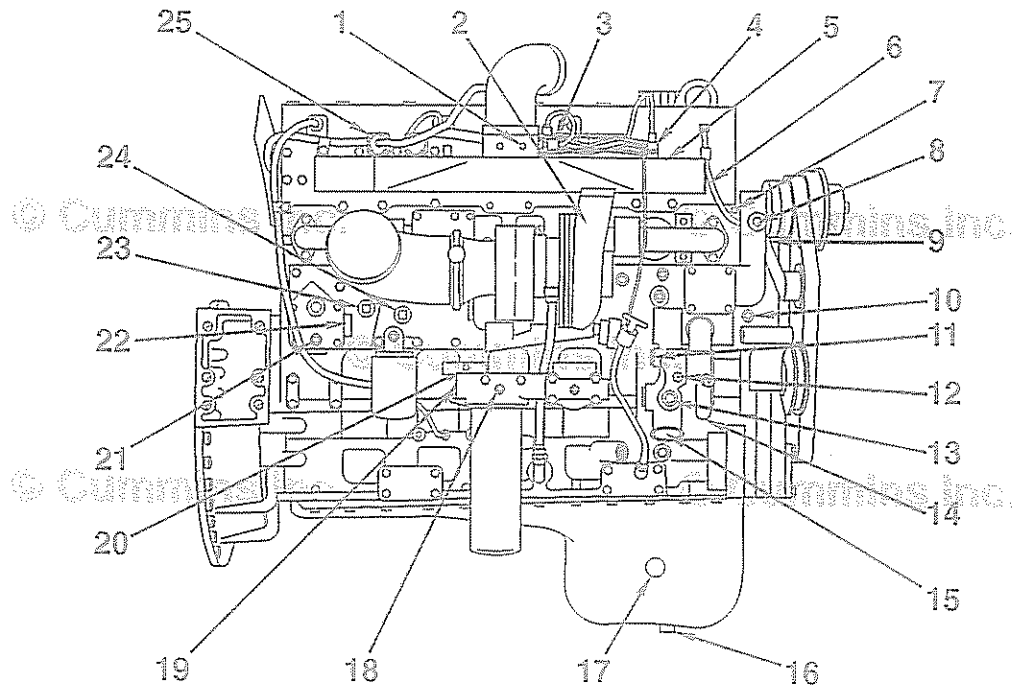
fp5p1ga

Engine Diagrams

Engine Views

The following illustrations contain information about engine components, filter locations, drain points, and access locations for instrumentation and engine controls. The information and configuration of components shown in these drawings are of a general nature. Some component locations will vary depending on applications and installations.

Industrial and Generator Drive

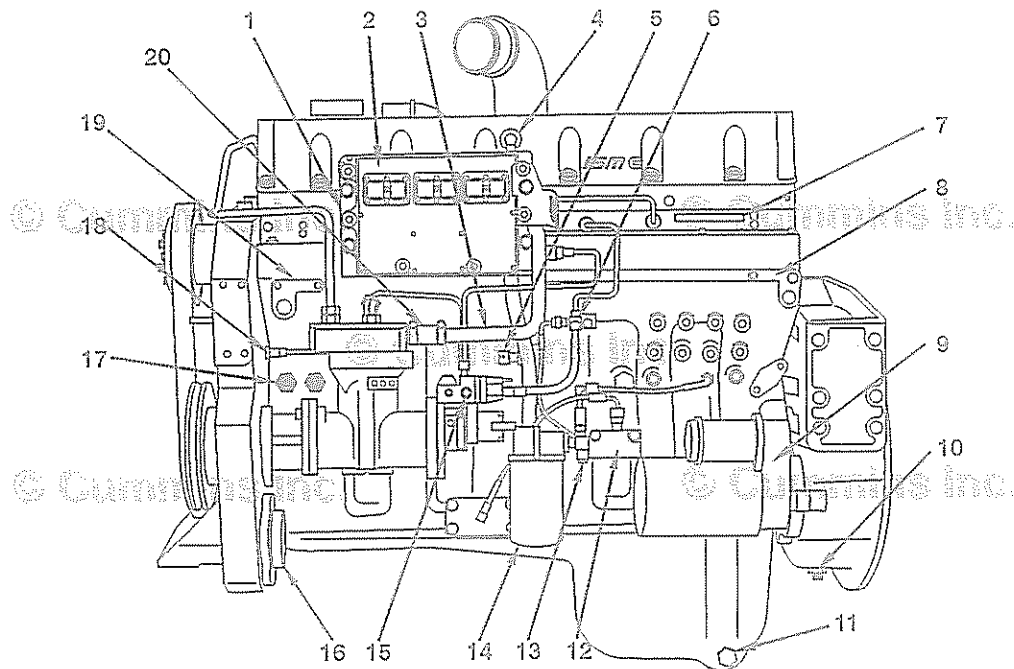


Industrial and Generator Drive - Exhaust Side

- | | |
|--|--------------------------------|
| 1. Ether hookup | 14. Water pump pressure |
| 2. Before charge air cooler | 15. Coolant inlet |
| 3. Boost temperature sensor | 16. Oil drain |
| 4. Intake manifold pressure sensor | 17. Oil pan pump heater |
| 5. After charge air cooler (intake manifold) | 18. Filter outlet pressure |
| 6. Shutter stat | 19. Engine brake oil supply |
| 7. Engine vent coolant | 20. Filter inlet pressure |
| 8. Shutter stat | 21. Block coolant pressure |
| 9. Fan sensor | 22. Coolant heater |
| 10. Coolant temperature sensor | 23. Heater supply |
| 11. Heater return | 24. Coolant temperature pickup |
| 12. Coolant inlet pressure | 25. Wastegate controller. |
| 13. Makeup line from radiator | |

01200109

Industrial and Generator Drive

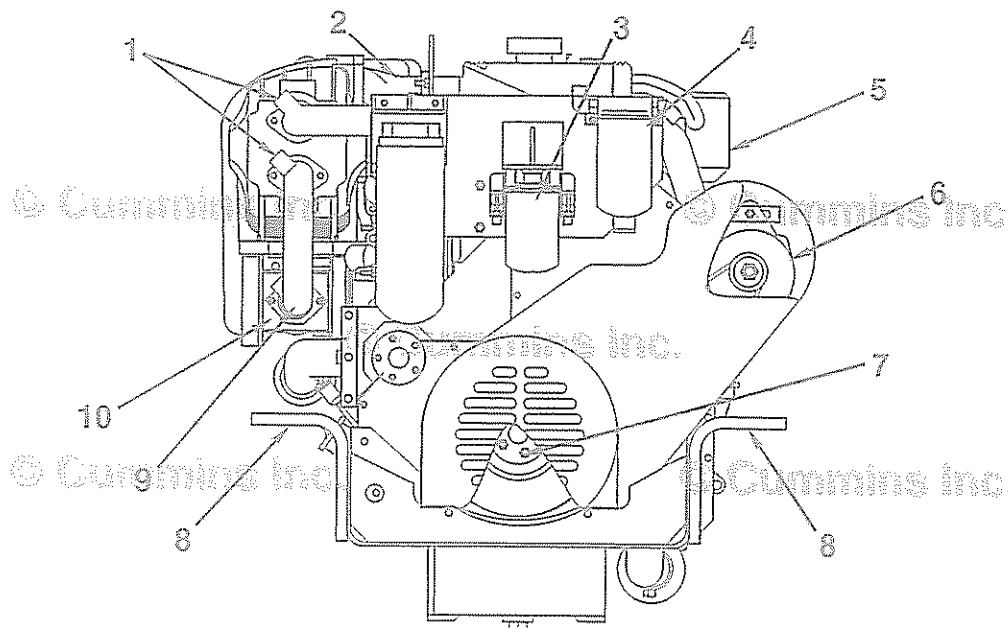


Q1200141

Industrial and Generator Drive - Fuel Pump Side

- | | |
|-------------------------------------|--|
| 1. ECM cooling plate | 11. Side oil drain |
| 2. ECM | 12. Centinel™ (optional) |
| 3. Compressor air inlet | 13. Fuel inlet to pump |
| 4. Blowby measurement | 14. Fuel filter |
| 5. Ambient air pressure sensor | 15. Rail pressure |
| 6. Fuel return to tank | 16. Power steering pump mounting location |
| 7. Engine data tag | 17. Oil pressure and temperature sensor |
| 8. Engine serial number | 18. Engine position sensor |
| 9. Starter | 19. Freon compressor mounting location. |
| 10. Flywheel ring gear speed sensor | 20. Compressor air discharge (not visible) |

Marine Applications

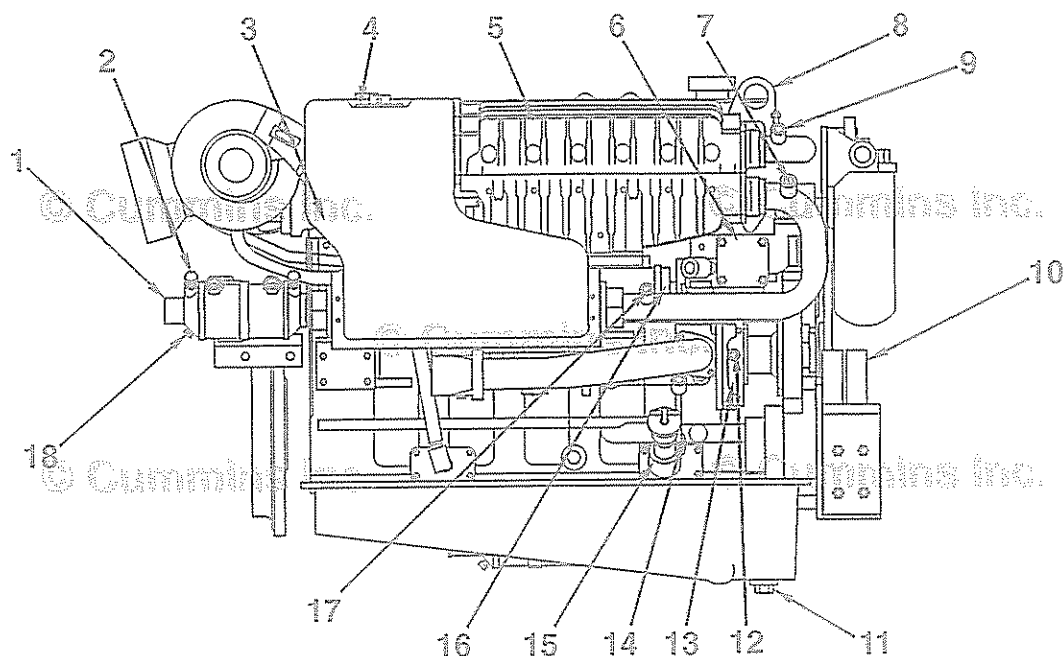


Marine - Front View

- | | |
|-----------------------|---------------------------------------|
| 1. Zinc plugs | 6. Alternator |
| 2. Air crossover | 7. Vibration damper |
| 3. Coolant filter | 8. Front engine mounts |
| 4. Fuel Filter | 9. Aftercooler-to-heat-exchanger pipe |
| 5. Crankcase breather | 10. Heat exchanger. |

17200015

Marine Applications

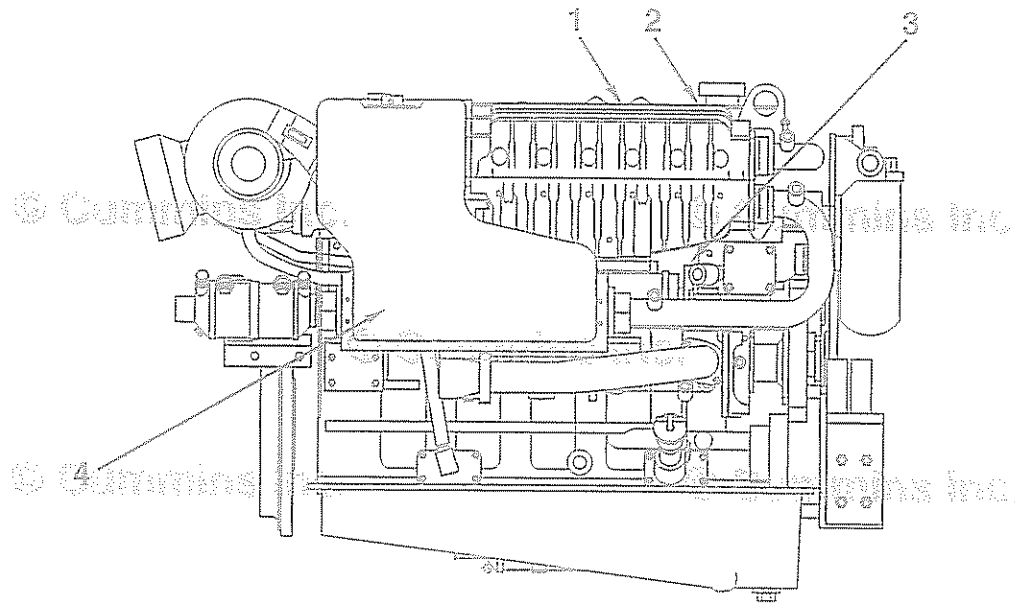


17200014

Marine - Starboard View

- | | |
|-----------------------------|---------------------------------|
| 1. Gear oil cooler | 10. Vibration damper |
| 2. Zinc anode (gear cooler) | 11. Oil pan drain plug |
| 3. Heat exchanger | 12. Coolant drain |
| 4. Expansion tank | 13. Water pump |
| 5. Aftercooler | 14. Oil dipstick |
| 6. Thermostat housing | 15. Oil fill |
| 7. Zinc anode | 16. Engine oil cooler |
| 8. Engine lifting bracket | 17. Zinc anode |
| 9. Zinc anode | 18. Gear oil cooler drain plug. |

Marine Applications

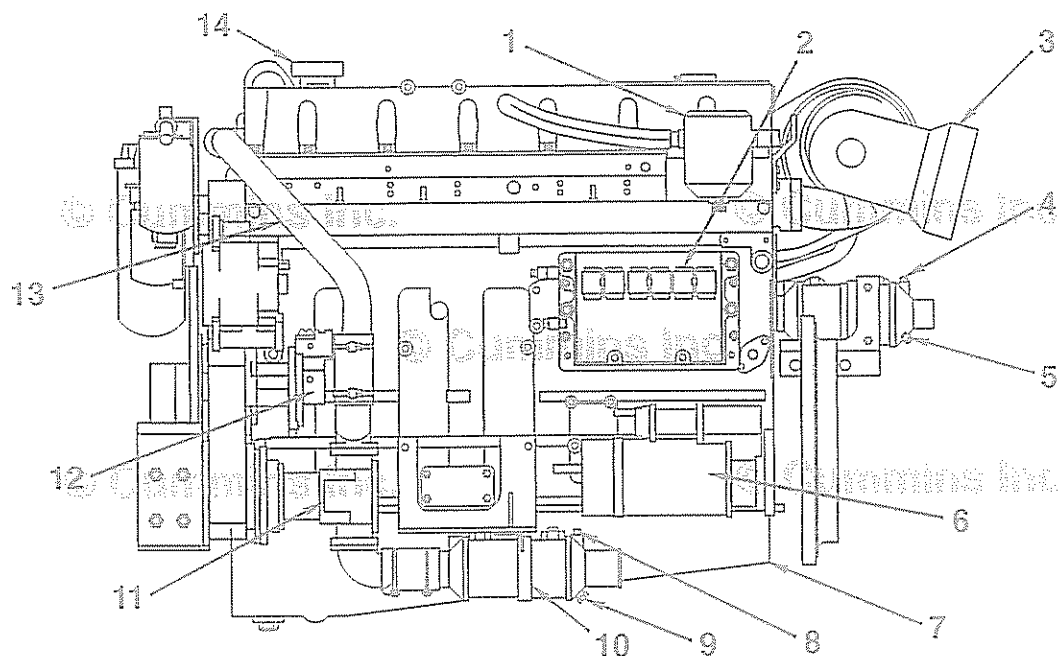


QSM11 Starboard View

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1. Intake manifold temperature sensor
2. Intake manifold pressure sensor
3. Coolant temperature sensor
4. Coolant level sensor

Marine Applications

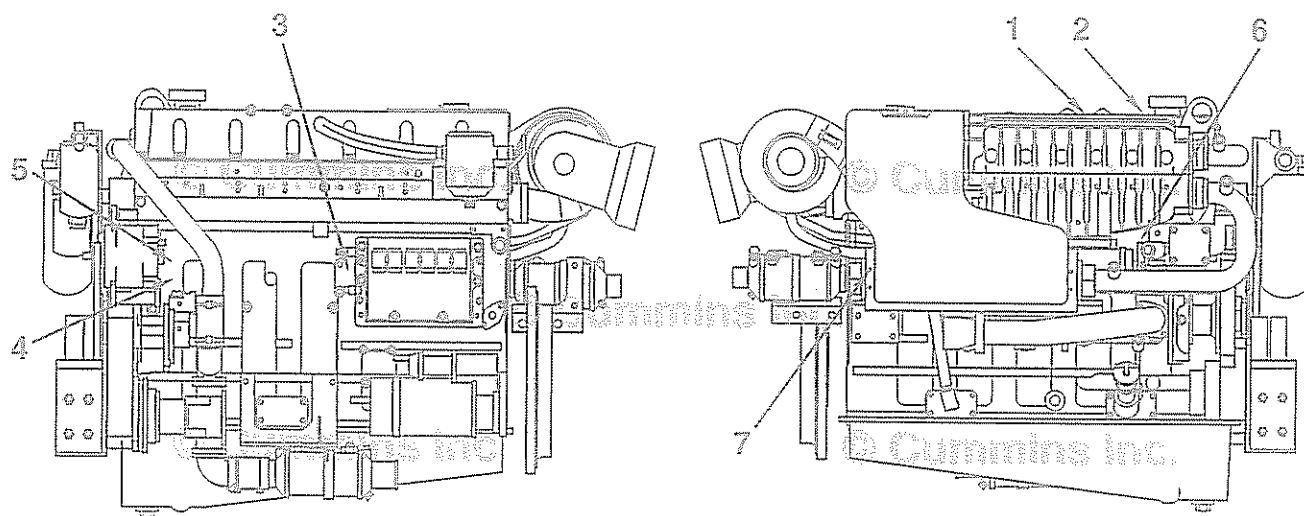


17200015

Marine - Port View

- | | |
|---------------------------|--|
| 1. Crankcase breather | 8. Zinc anode (fuel cooler) |
| 2. ECM | 9. Fuel cooler drain plug |
| 3. Exhaust elbow | 10. Fuel cooler |
| 4. Zinc anode | 11. Sea water pump |
| 5. Gear cooler drain plug | 12. Fuel pump |
| 6. Starter | 13. Sea-water-pump-to-aftercooler pipe |
| 7. Oil pan | 14. Oil fill. |

Marine Applications

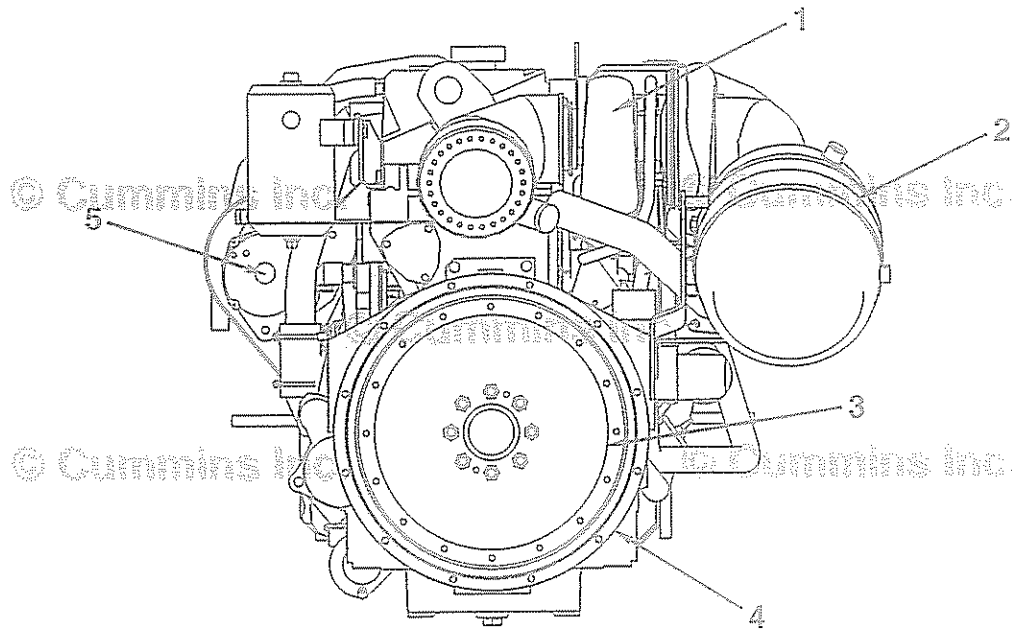


QSM 11 Port View

15d00831

1. Ambient air pressure sensor
2. Oil pressure/temperature sensor (behind fuel pump)
3. Engine position sensor (behind fuel pump)

Marine Applications



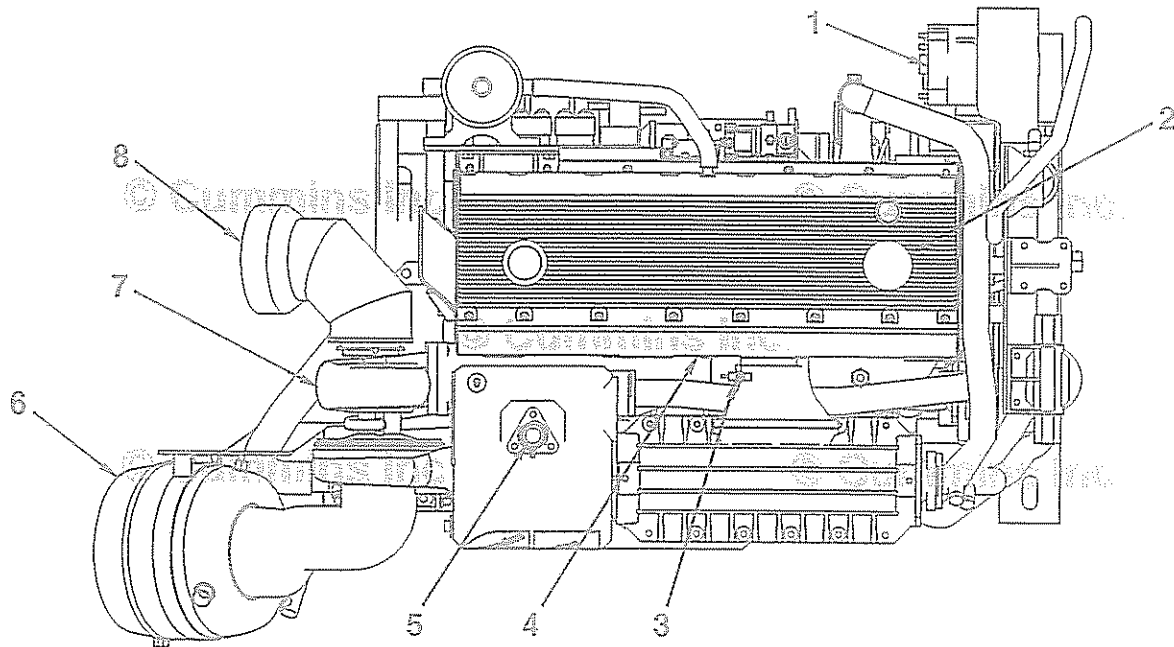
Marine - Rear View

- 1. Turbocharger
- 2. Air cleaner
- 3. Flywheel

- 4. Flywheel housing
- 5. Alternator.

17200017

Marine Applications

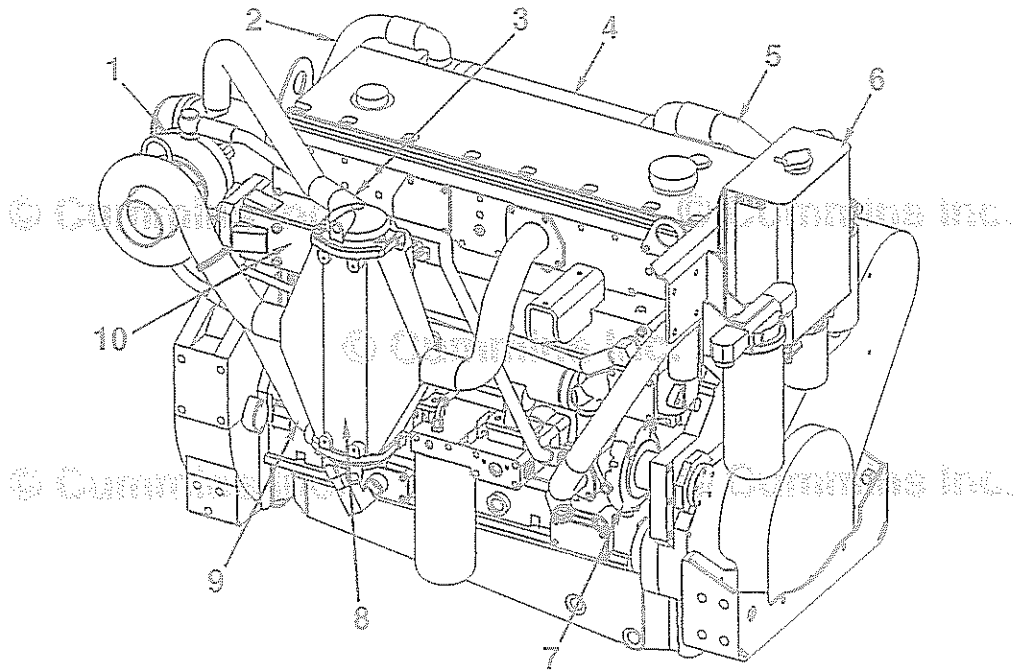


17200016

Marine - Top View

- | | |
|---------------------------------------|-------------------|
| 1. Alternator | 5. Coolant fill |
| 2. Oil fill (alternate) | 6. Air cleaner |
| 3. Intake manifold temperature sensor | 7. Turbocharger |
| 4. Intake manifold pressure sensor | 8. Exhaust elbow. |

Marine Applications

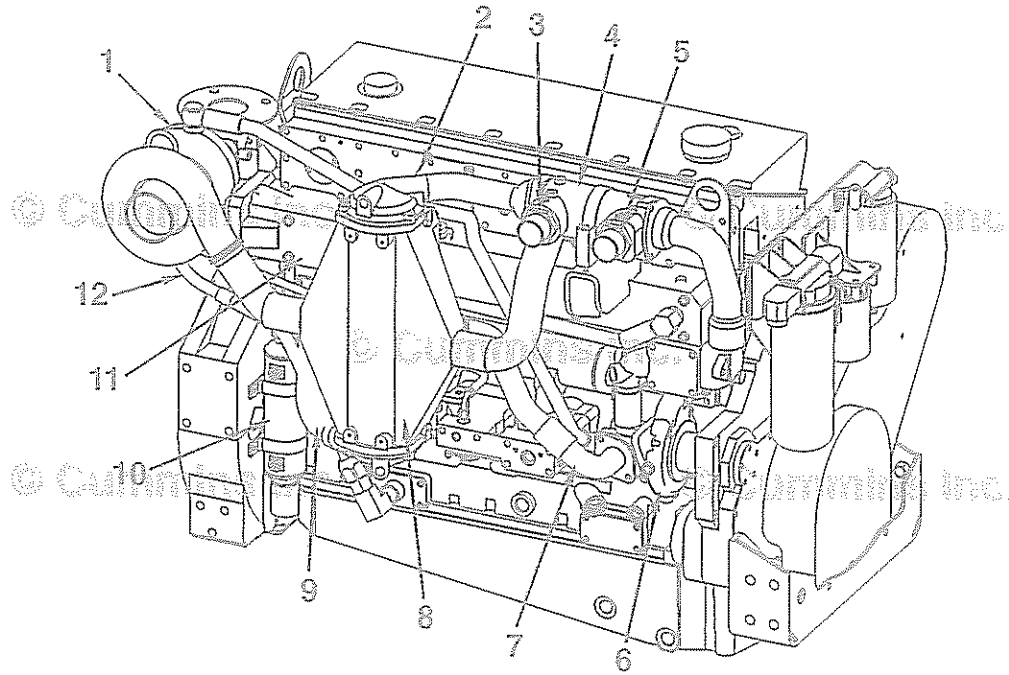


Heat Exchanger

1. Turbocharger with water cooled turbine housing
2. Coolant flow from heat exchanger
3. Aftercooler sea water outlet
4. Heat exchanger
5. Coolant flow to heat exchanger
6. Expansion tank
7. Water pump
8. Aftercooler
9. Aftercooler sea water inlet
10. Water cooled exhaust manifold

08206244

Marine Applications



08200245

1. Turbocharger with water cooled turbine housing
2. Coolant inlet (aftercooler)
3. Coolant in from Keel Cooler
4. LTA thermostat housing
5. Coolant out to Keel Cooler
6. Water pump
7. Coolant to water pump
8. Aftercooler
9. Coolant outlet (aftercooler) to water pump
10. Gear oil cooler (optional)
11. Water cooled exhaust manifold
12. Coolant out from turbocharger

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Section 1 - Operating Instructions

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Operating Instructions - Overview



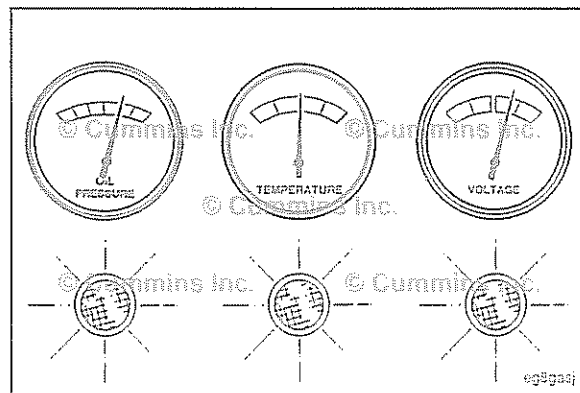
General Information

Correct care of your engine will result in longer life, better performance, and more economical operation.

Follow the daily maintenance checks listed in Maintenance Guidelines (Section 2).

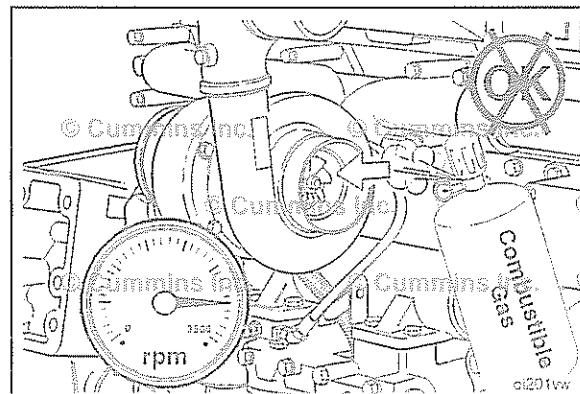
The new Cummins engine associated with this manual does not require a "break-in" procedure. This section of the manual provides all of the necessary information required for proper engine operation.

Check the oil pressure indicators, temperature indicators, warning lights, and other gauges daily to make sure they are operational.

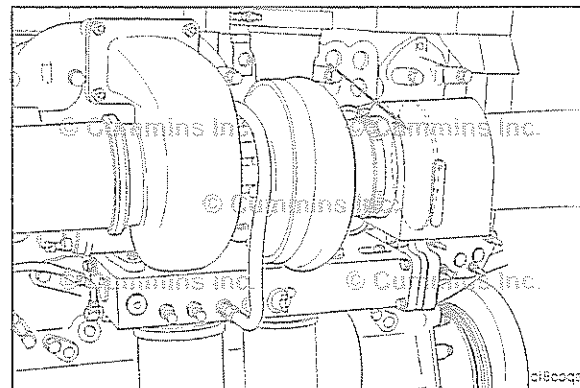


▲ WARNING ▲

DO NOT OPERATE A DIESEL ENGINE WHERE THERE ARE OR CAN BE COMBUSTIBLE VAPORS. The vapors can be sucked through the air intake system and cause engine acceleration and overspeeding that can result in a fire, an explosion, and extensive property damage. Numerous safety devices are available, such as air intake shutoff devices, to minimize the risk of overspeeding where an engine, due to its application, due to a fuel spill or gas leak. Remember, Cummins has no way of knowing the use you have for your engine. **THE EQUIPMENT OWNER AND OPERATOR ARE RESPONSIBLE FOR SAFE OPERATION IN A HOSTILE ENVIRONMENT. CONSULT YOUR CUMMINS AUTHORIZED REPAIR LOCATION FOR FURTHER INFORMATION.**

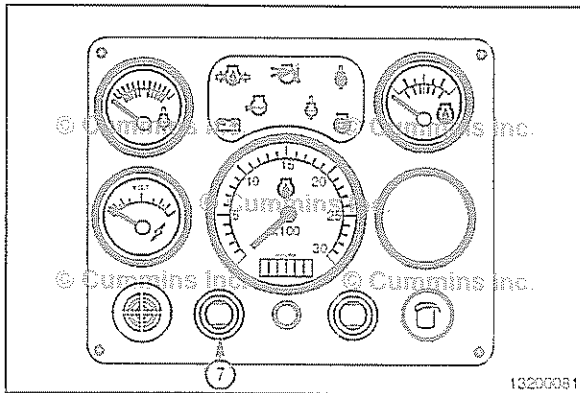


Cummins recommends the installation of an air intake shutoff device or a similar safety device to minimize the risk of overspeeding where an engine, due to the vehicle, vessel or equipment being operated in a combustible environment, such as due to a fuel spill or gas leak.



CAUTION

Do not expose the engine to corrosive chemicals. Corrosive chemicals can damage the engine.

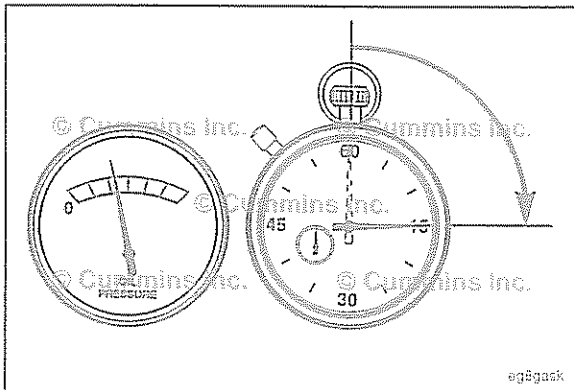


Normal Starting Procedure

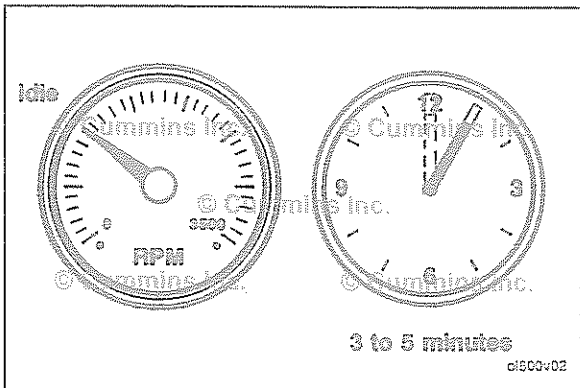
General Information

NOTE: There is a separate keyswitch wired to the primary panel. It will be installed in the helm at the boat manufacturer's or installer's desired location. The keyswitch must be in the ON position to crank-start the engine.

The engine start button (7) is used to engage the starter motor.

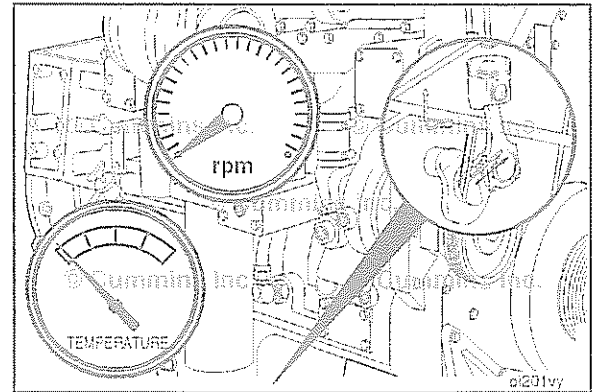


The engine must have adequate oil pressure within 15 seconds after starting. If the warning light indicating low oil pressure has not gone out or there is no oil pressure indicated on a gauge within 15 seconds, shut off the engine immediately to avoid engine damage. Confirm the correct oil level in the oil pan.

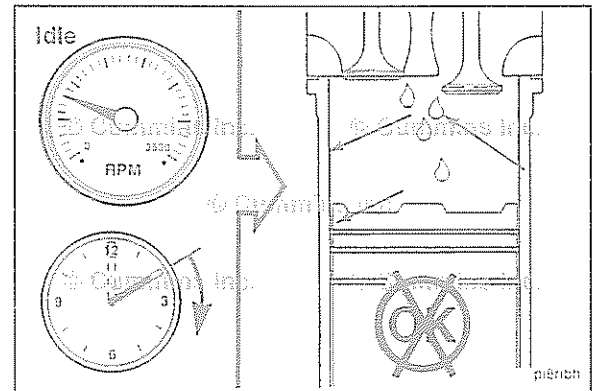


Idle the engine 3 to 5 minutes before operating with a load.

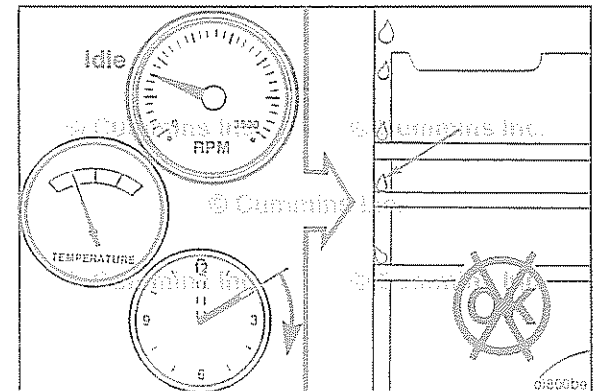
Increase the engine speed (rpm) slowly to provide adequate lubrication to the bearings and to allow the oil pressure to stabilize.

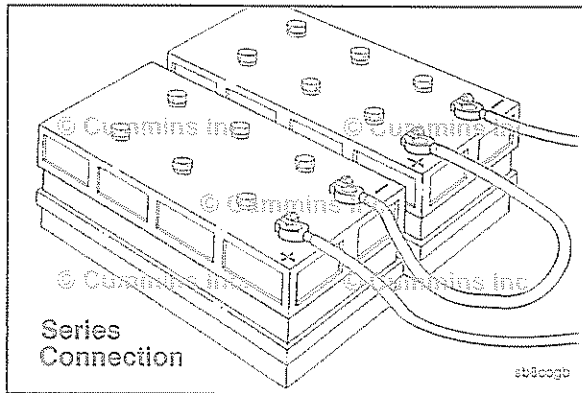
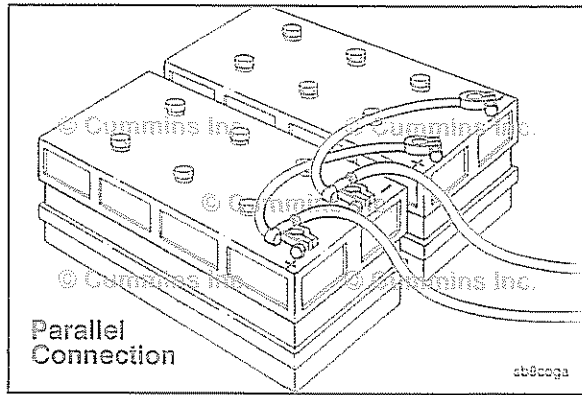


Do not keep the engine at low idle for long periods. Idling for periods longer than 10 minutes can damage an engine, causing combustion chamber temperatures to drop so low the fuel will not burn completely. This will cause carbon to build up around the injector spray holes and piston rings, and can cause the valves to stick.



If the engine coolant temperature becomes too low, below 60° C [140° F], raw fuel will wash the lubrication oil off the cylinder walls and dilute the crankcase oil. Fuel dilution adversely affects lubricating oil properties and can shorten engine life. Utilize the fast idle to prevent these conditions.





⚠ WARNING ⚠

Batteries can emit explosive gases. To avoid personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ CAUTION ⚠

When using jumper cables to start the engine, make sure to connect the cables in parallel: Positive (+) to positive (+) and negative (-) to negative (-). When using an external electrical source to start the engine, turn the disconnect switch to the OFF position. Remove the key before attaching the jumper cables.

The accompanying illustration shows a typical parallel battery connection. This arrangement doubles the cranking amperage.



This illustration shows a typical series battery connection. This arrangement, positive (+) to negative (-), doubles the voltage.

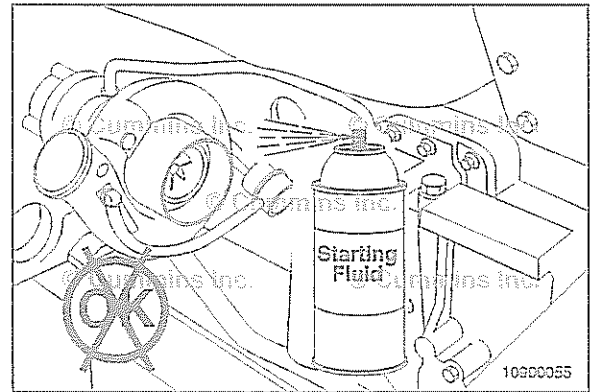
Cold Weather Starting

General Information

Follow the Normal Starting Procedures in this section. In cold weather, the engine can run at idle only longer.

Using Starting Aids

Cold weather starting aids are available for your engine. Contact the local Cummins Authorized Repair Location for more information.



Starting Procedure After Extended Shutdown or Oil Change

General Information

Follow the Normal Starting Procedure in this section. The engine will run at idle only until the minimum oil pressure is detected by the ECM.

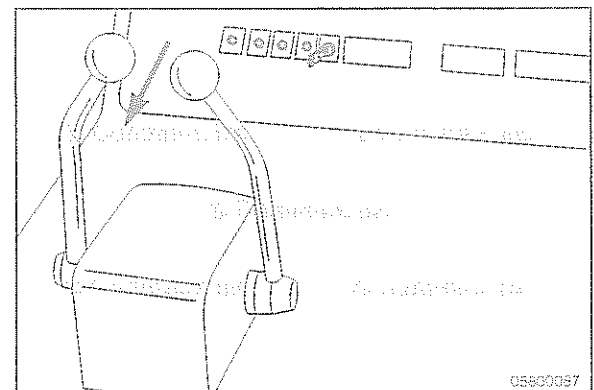
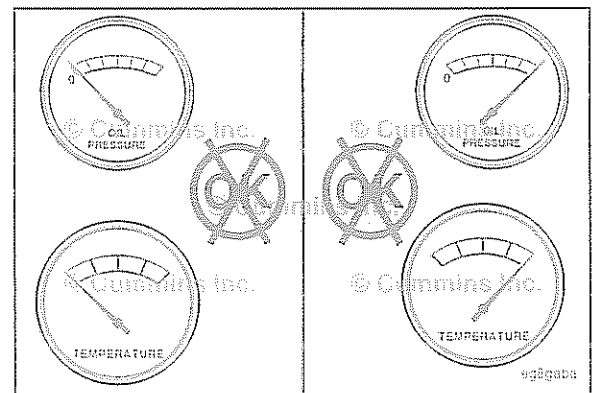
Operating the Engine

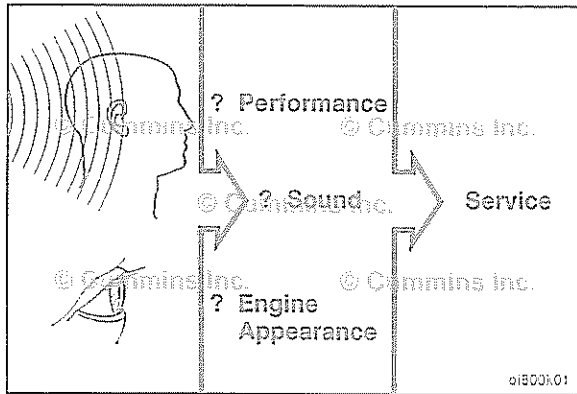
Normal

If equipped, monitor the oil pressure and coolant temperature gauges frequently. Refer to Lubricating Oil System specifications and Cooling System specifications, in Maintenance Specifications (Section V) for recommended operating pressures and temperatures. Shut off the engine if any pressure or temperature does not meet the specifications.

Continuous operation with engine coolant temperature above or below the engine coolant temperature specifications listed in Maintenance Specifications (Section V) can damage the engine.

If an overheating condition starts to occur, reduce the power output of the engine by reducing the throttle setting until the temperature returns to normal operating range. If the engine temperature does not return to normal, further reduce the engine speed and contact a Cummins Authorized Repair Location.





Most failures give an early warning. Look and listen for changes in performance, sound, or engine appearance that can indicate service or engine repair is needed. Some changes to look for are:

- Engine misfires
- Vibration
- Unusual engine noises
- Sudden changes in engine operating temperatures or pressures
- Excessive smoke
- Loss of power
- An increase in oil consumption
- An increase in fuel consumption
- Fuel, oil, or coolant leaks.

Cold Weather

It is possible to operate engines in extremely cold environments if they are properly prepared and maintained. Satisfactory performance of an engine in low ambient temperature conditions requires modification of the engine, surrounding equipment, operating practices and maintenance procedures.

The correct engine coolant lubricating oil and fuels must be used for the cold weather range in which the engine is being operated. Below are the recommendations for these critical engine fluids:

Ambient Temperature

0 to -32°C [32 to -25°F]

Use 50-percent glycol antifreeze and 50-percent water for the engine coolant mixture.

Refer to Lubricating Oil recommendations and Specifications and Coolant Recommendations and Specifications in Section V for the correct specifications.

The Diesel fuel must have maximum cloud and pour points 6°C [10°F] lower than the ambient temperature in which the engine operates.

-32 to -54°C [-25 to -65°F]

Use 60-percent glycol antifreeze and 40-percent water for the engine coolant mixture.

Refer to Lubricating Oil recommendations and Specifications and Coolant Recommendations and Specifications in Section V for the correct specifications.

The Diesel fuel must have maximum cloud and pour points 6°C [10°F] lower than the ambient temperature in which the engine operates.

Engine Operating Range General Information



Do not operate the engine at full throttle operation below peak torque rpm (refer to engine dataplate for peak torque rpm) for more than 30 seconds. Operating the engine at full throttle below peak torque will shorten engine life to overhaul, can cause serious engine damage, and is considered engine abuse.

Cummins® engines are designed to operate successfully at full throttle under transient conditions down to peak torque engine speed. This is consistent with recommended operating practices.



Do not operate the engine beyond the maximum engine speed. Operating the engine beyond the maximum engine speed can cause severe engine damage. Use proper operating techniques for the vehicle, vessel, or equipment to prevent engine overspeed. The maximum engine speed specification is listed in Maintenance Specifications (Section V).

Engine Shutdown General Information

For engines with shielded exhaust manifold and turbochargers, allow the engine(s) to idle for 10 to 12 minutes before shutting off after full load operation.

Otherwise, allow the engine(s) to idle 3 to 5 minutes before shutting off after full load operation. This allows the cool down of pistons, cylinders, bearings and turbocharger components.

If the engine does not shut down, refer to Troubleshooting Symptom (Section TS).

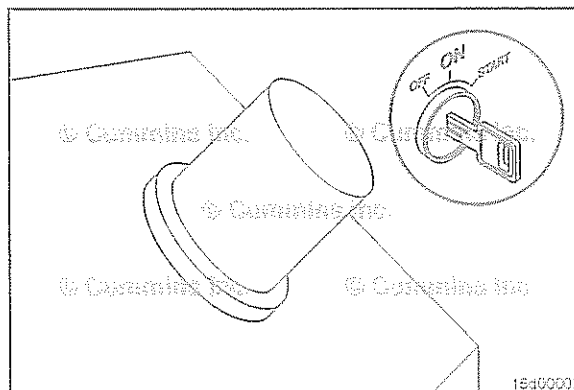
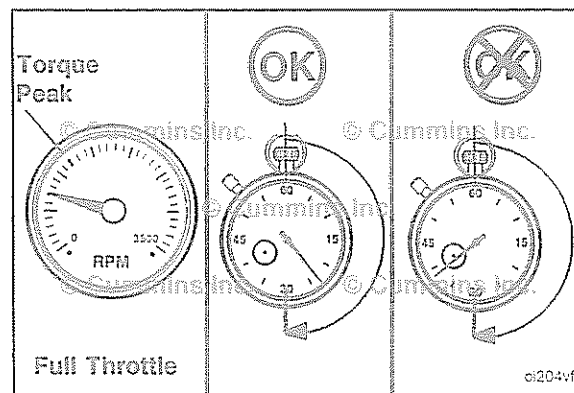
Turn the keyswitch to the OFF position.

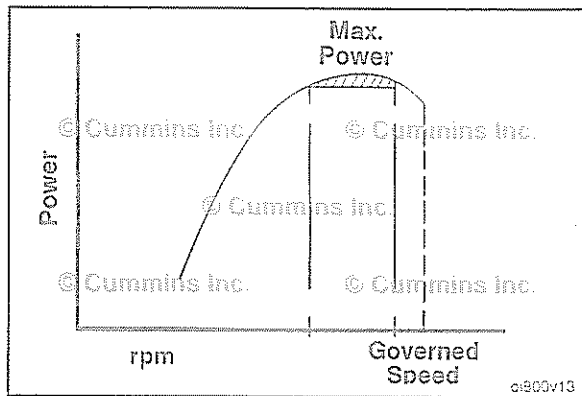
Push the STOP button to stop the engine(s), or if equipped with a keyswitch only, turn the keyswitch to the OFF position.

NOTE: Damage to the engine resulting from freezing is not covered under the Cummins warranty.



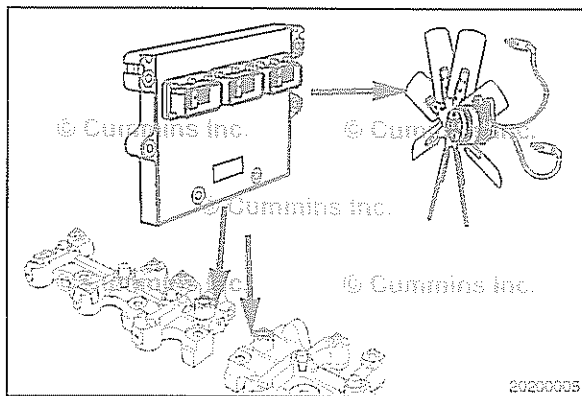
Procedures must be followed to protect the engine from freeze damage during a cold weather storage period. The procedure applies anytime freezing conditions are present or could occur. Refer to Procedure 000-014 (Seasonal Storage) - in Section SS of this manual for steps to protect the engine during these periods.





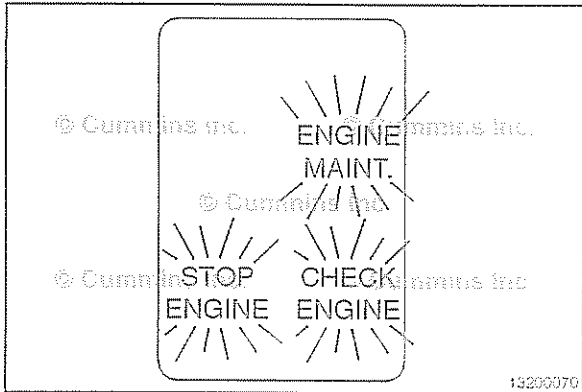
Electronic Controlled Fuel System General Information

The system is an electronically controlled fuel injection system that optimizes fuel economy and reduces exhaust emissions. It does this by controlling the torque and horsepower curve, AFC function, engine high speed, low idle, and road speed.



The engine has the capability of controlling the fan clutch actuator if an electronically controlled fan clutch is used.

The engine also allows the engine brakes to be activated by controlling the engine brake solenoids.



Diagnostic Fault Codes

Engine Fault and Maintenance Lamps

The ENGINE FAULT and MAINTENANCE lamps are illuminated for 2 seconds when the keyswitch is turned to the ON position.

After 2 seconds, the red STOP ENGINE lamp will turn off. After an additional 1/2 second, the amber CHECK ENGINE lamp will turn off. After an additional 1/2 second, the amber ENGINE MAINT lamp will turn off.

The lamps will remain off until a fault is detected.

NOTE: This is a self-test feature of the lamp wiring and lamps.

NOTE: The names and colors of the lamps can vary with vessel manufacturer if non-Cummins panels are used.

The following chart summarizes the different lamps and their operation.

Feature	Operator Message	Lamp Operation		
		Check Engine	Stop Engine	Engine Maint.
Lamp Display	Power-up lamp test	On then off	On then off	On then off
Diagnostics	Fault code flashout	Flash once/code	Flash code Number	
Engine Protection	System problem		Slow flash	
Maintenance Monitor	Interval expired			3x5 fast flash
Maintenance Monitor	Interval rest			3x5 fast flash
Diagnostics	Nonfatal system error	On steady		
Diagnostics	Fatal system error		On steady	
Diagnostics	Maintenance required			On steady

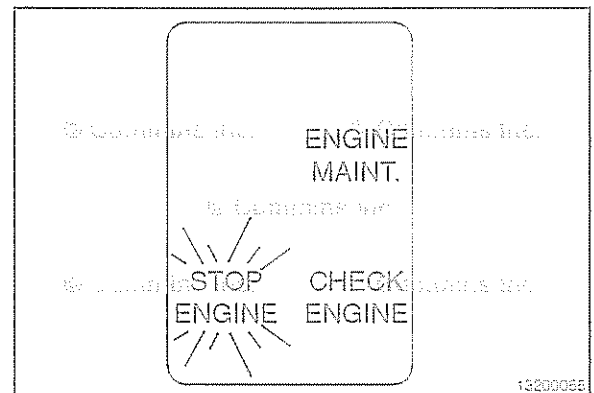
If the STOP or CHECK ENG lamp comes on when the engine is running, it means a fault code has been recorded. The lamp will remain on as long as the fault exists. The severity of the fault will determine the lamp that will come on.

Stop Engine Lamp

The STOP ENGINE lamp is a red lamp. This lamp indicates that the engine needs to be shut down before permanent damage occurs to the engine.

NOTE: The engine should be shut off as soon as it can be shut off safely. The engine should not be run until the fault is corrected.

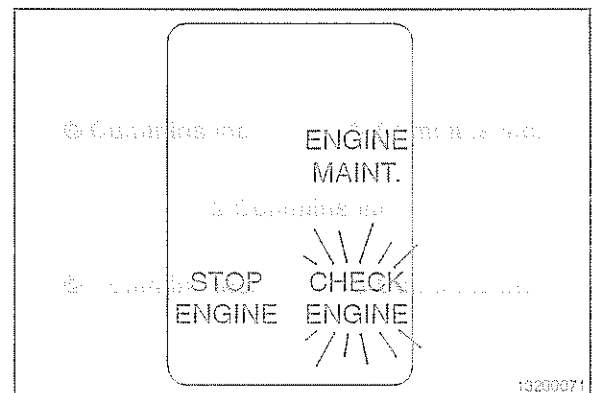
This lamp is also used to flash out the fault code number in the diagnostic mode.



Check Engine Lamp

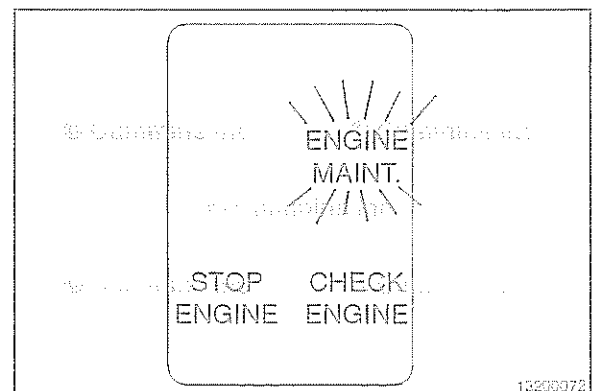
The CHECK ENGINE lamp comes on during a nonfatal system error. The engine can still be run, but the fault should be corrected as soon as possible.

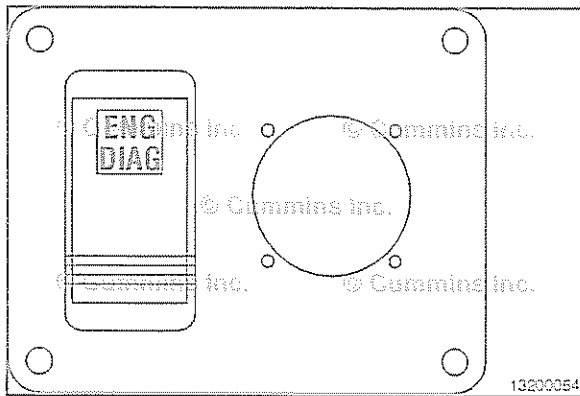
NOTE: In the diagnostic mode, the check engine lamp will flash after the stop engine lamp completes the three-digit fault code.



Engine Maintenance Lamp

The ENGINE MAINT lamp comes on when engine maintenance is required.

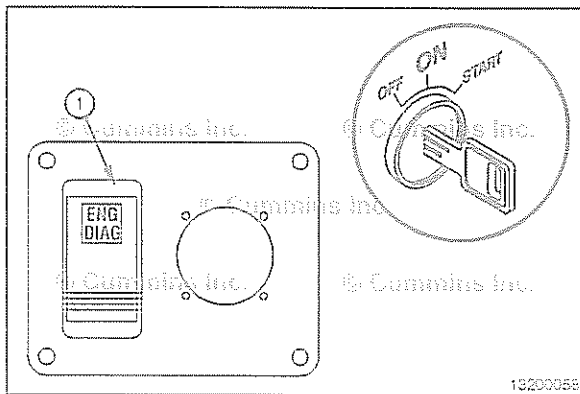




Engine Diagnostics

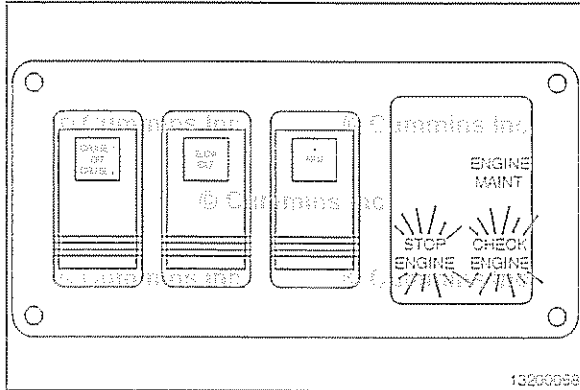
When a fault or maintenance lamp is lit, the engine diagnostics switch allows the operator to view the fault codes. The receptacle to the right of the switch is for the technician's computer connection using INSITE™ or Echet™ service tool.

Active fault codes can be viewed using the stop engine warning lamp as described below.



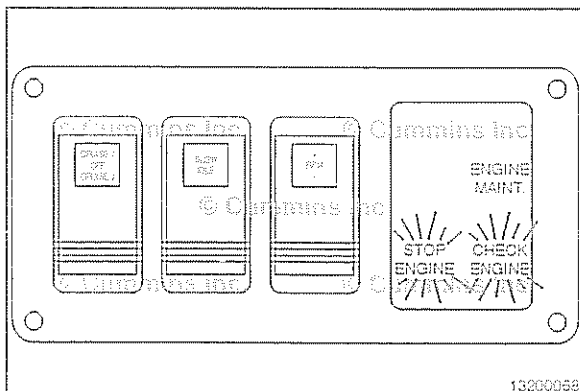
To view the fault codes:

1. The engine must be shut off (not running)
2. The keyswitch must be in the ON position
3. The ENG DIAG switch (1) must be in the ON position.



The check engine and stop engine lamps flash if there are any fault codes to display.

If there are no fault codes to display, the check engine and stop engine lamps will remain lit.



If there are fault codes to be displayed, the check engine lamp will flash momentarily. Then the stop engine lamp will flash the first, second, and third digits of the fault code.

Example:

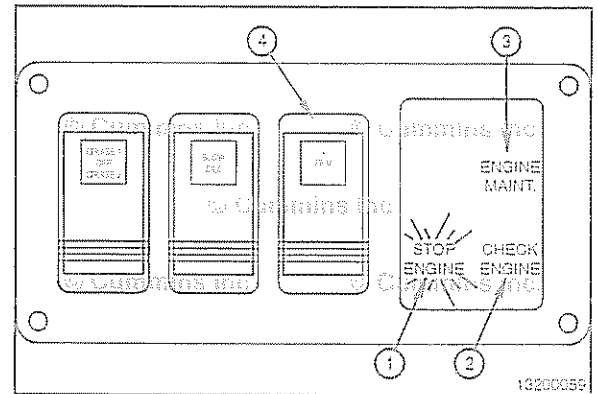
Fault Code 432
4 flashes, pause
3 flashes, pause
2 flashes.

NOTE: The check engine lamp will flash between each fault code.

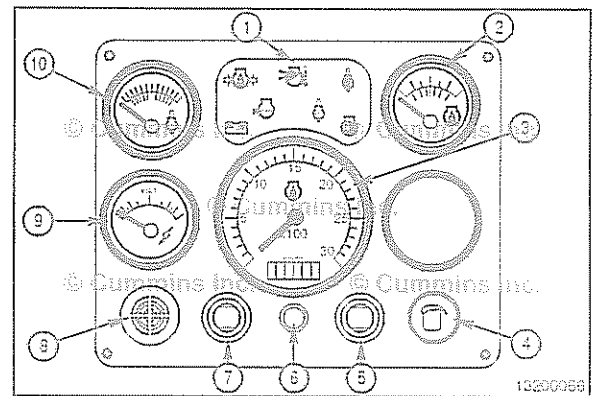
The pattern repeats itself until the fault is cleared or the switch is turned off.

To view the next fault code, press the rpm $\pm$ switch (4) in the + direction.

To view the previous fault code, press the rpm $\pm$ switch (4) in the - direction.

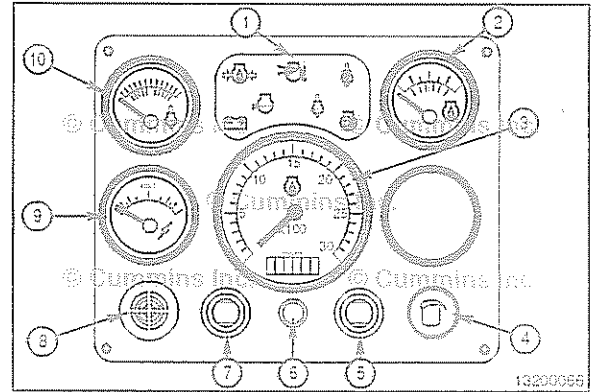


The audible alarm (8) comes on anytime the warning or caution symbols are illuminated.



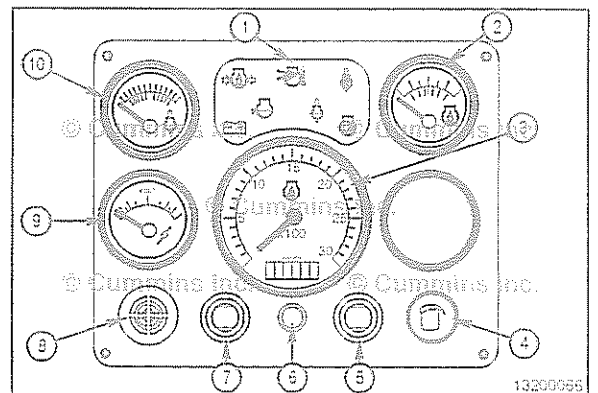
The alarm silence button (6) will temporarily silence the audible alarm.

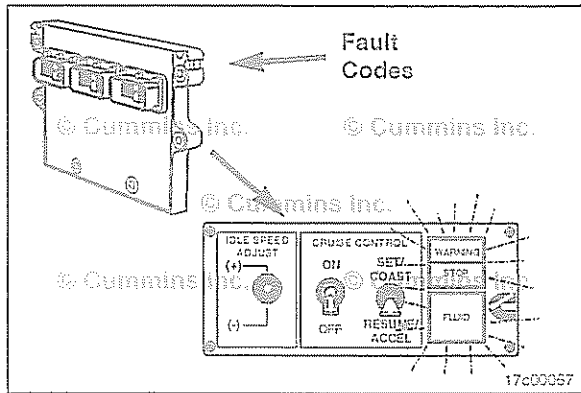
NOTE: The alarm will be silenced for up to 2 minutes. As long as the fault condition exists, the alarm will "chirp" every 2 minutes to remind the operator that a fault exists.



The alarm silence button (6) is also used to test the warning and caution symbol lamps (1) and the gauges.

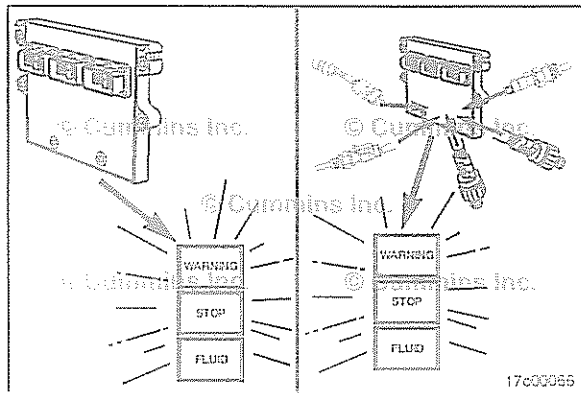
NOTE: To test the gauges and symbol lamps, press the alarm silence button (6) while turning on the keyswitch. The alarm will come on for 5 seconds and for 25 seconds all symbols will illuminate and the gauge needles will move from the lowest position to the highest position and back to the lowest position.





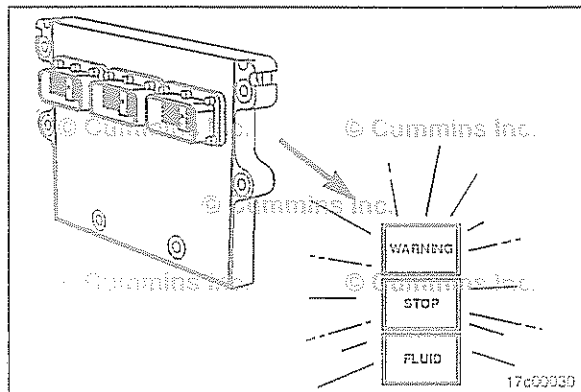
The system can show and record operation irregularities that present themselves as fault codes. These codes will make troubleshooting easier. The fault codes are recorded in the ECM. They can be read using the two fault lamps in the cab panel or with INSITE™.

NOTE: Not all engine irregularities are shown as fault codes.



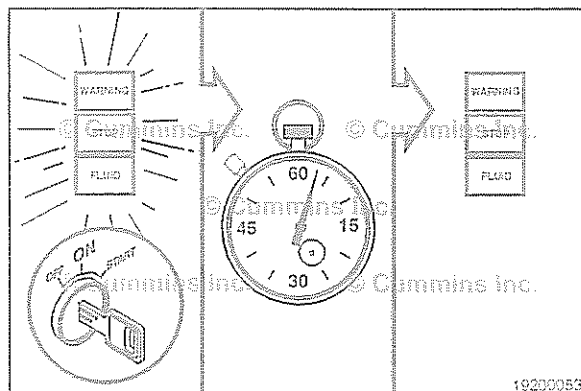
There are two types of fault codes:

- Engine electronic fuel system codes
- Engine protection system codes.



The engine electronic fuel system fault codes can be seen on the WARNING and STOP lights in the cab panel.

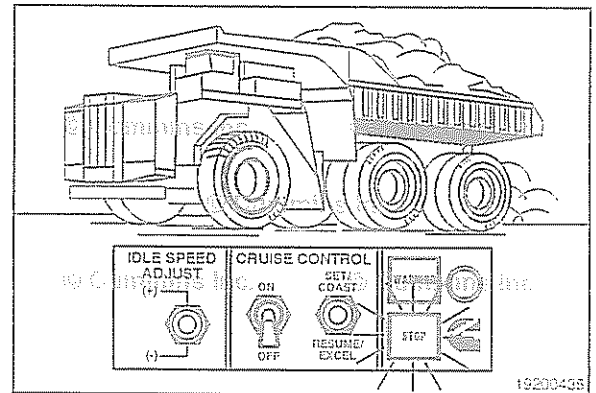
NOTE: Inactive fault codes can not be blinked out on the two lights in the cab panel. An INSITE™ service tool must be used to read inactive faults in the ECM. Refer to your Cummins Authorized Repair Location.



The STOP fault light will be red. The WARNING light will be yellow or red, depending on the OEM's preference. When the vehicle keyswitch is turned on and the diagnostic switch is off, all three lights will illuminate to check their operation. The lights will go off in sequence after about 2 seconds.

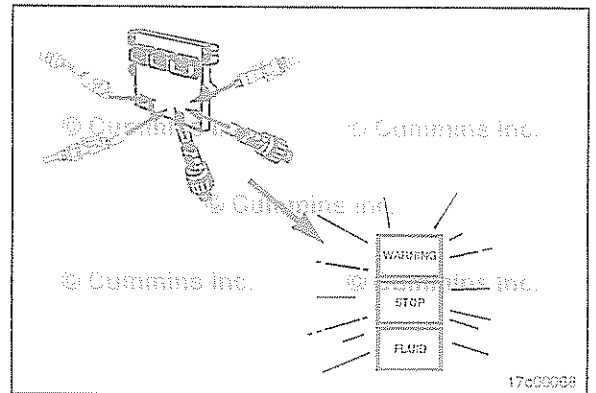
The lights will remain off until a fault code is recorded. If a light remains on, an active fault exists.

If the STOP light (red) is illuminated while operating, the fault can be engine disabling. The equipment **must** be shut off as soon as it can be done in a safe manner. The equipment **must** remain parked as long as this fault exists. If the WARNING light (yellow or red) is illuminated, the equipment can be safely operated, but the fault **must** be corrected as soon as possible.

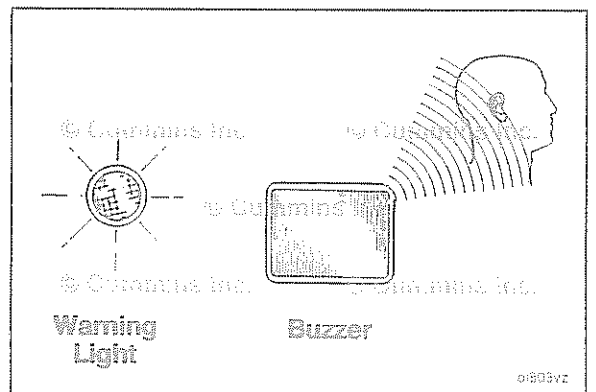


The engine protection system logs separate fault codes for out-of-range conditions associated with any of the following sensors:

- Coolant temperature
- Coolant level
- Oil temperature
- Oil pressure
- Intake manifold temperature.

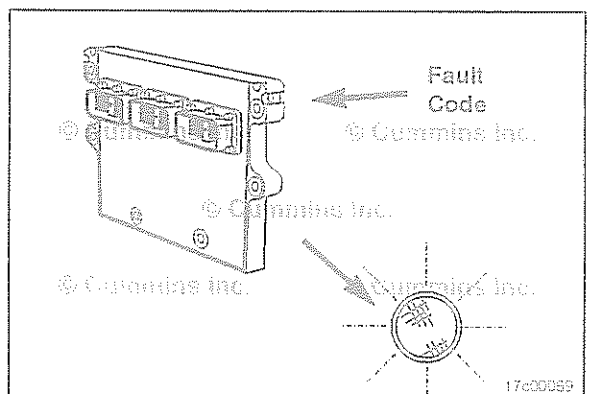


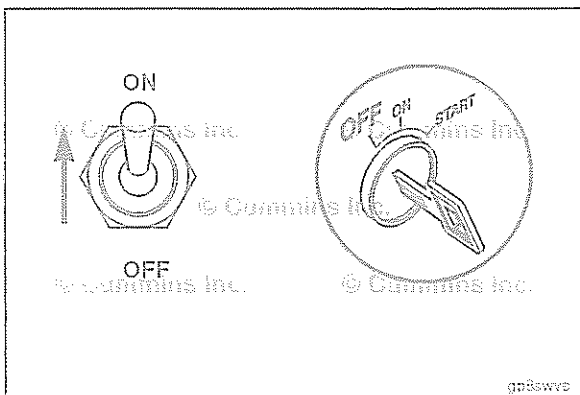
This system will activate an in-cab warning device when an out-of-range condition occurs. The warning device is a light, a buzzer, or both. This system will also activate the yellow fluid lamp, if equipped.



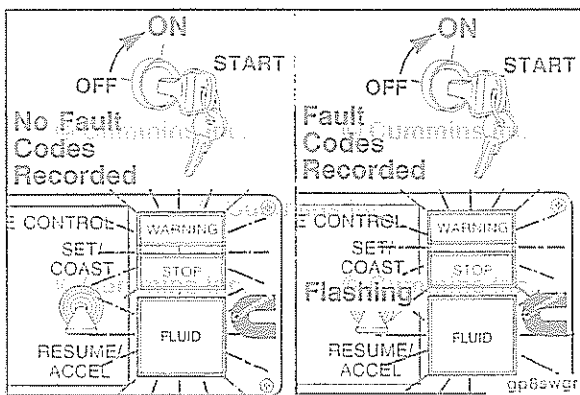
If the engine protection system light or buzzer comes on while driving, it means a fault code has been recorded. The lamp will remain on as long as the fault exists, and engine power and speed will gradually be reduced. If the out-of-range conditions continue, the light will start to flash or blink. If the engine protection shutdown feature is enabled, the engine will be shut down to help prevent engine damage.

The fault must be corrected as soon as possible.





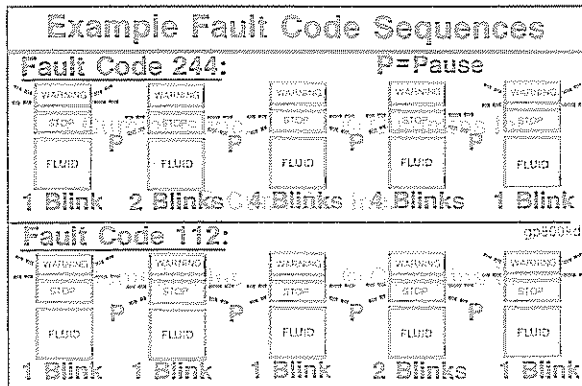
Turn off the engine. To check for engine electronic fuel system and engine protection system fault codes, move the diagnostic switch to the ON position, or connect the shorting plug into the diagnostic connector.



Turn the keyswitch to the ON position.

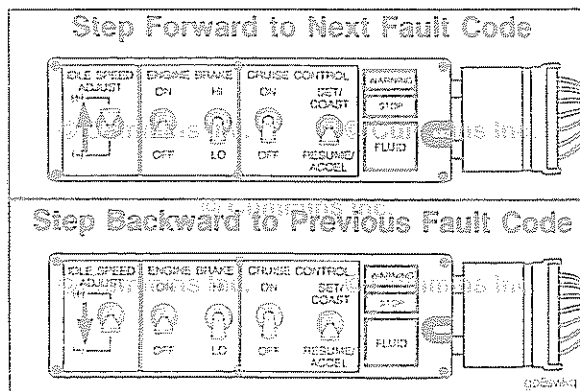
If no active fault codes are recorded, both lights will come on and stay on.

If active fault codes are recorded, both lights will come on momentarily, then begin to flash the code of the recorded fault.



The fault code will flash in the following sequence:

First, a WARNING (yellow) light will flash. Then there will be a short 1- or 2-second pause after which the number of the recorded fault code will flash in STOP (red). There will be a 1- or 2-second pause between each number. When the number has finished flashing in red, a yellow light will appear again. The three-digit code will repeat in the same sequence.



The lights will continue to flash the same code until the system is told to do something else. To go to the next fault code, move the CRUISE CONTROL/PTO switch momentarily to the RESUME/ACCEL position. You can go back to the previous fault code by momentarily moving the CRUISE CONTROL/PTO switch to the SET/COAST position. If only one active fault is recorded, the system will continuously display the same fault code when either RESUME/ACCEL or SET/COAST switch is depressed.

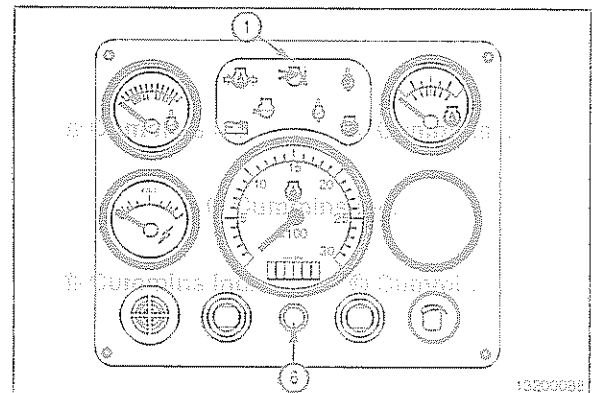
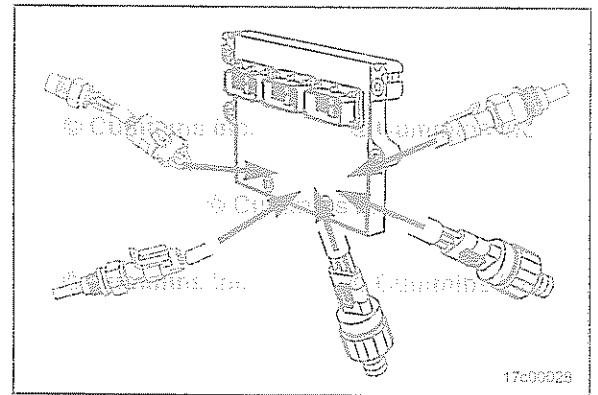
Engine Protection System

The engines are equipped with an engine protection system. The system monitors critical engine temperatures and pressures, and will log diagnostic faults when an abnormal operating condition occurs. If an out-of-range condition exists and engine derate action is to be initiated, the operator will be alerted by an in-cab warning light. The warning light will blink or flash when out-of-range conditions continue to get worse. The driver must pull to the side of the road, when it is safe to do so, to reduce the possibility of engine damage.

NOTE: Engine power and speed will be gradually reduced, depending on the level of severity of the observed condition. The engine protection system will not shut down the engine unless the engine protection shutdown feature has been selected. If the feature has been selected and the engine does shut down, the engine can be restarted by turning OFF the keyswitch, then turning it back ON.

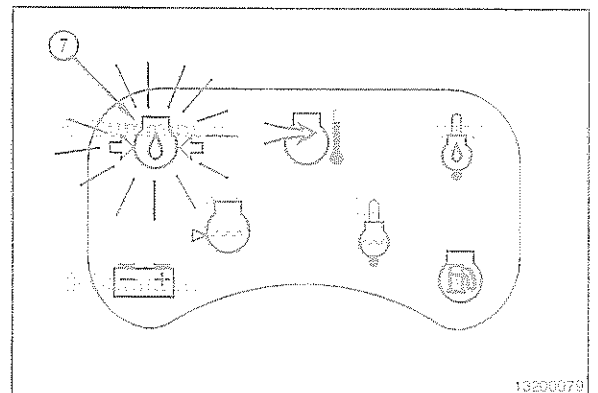
The indicator symbols (1) provide additional information on the type of fault that the ECM has detected. The individual symbols will flash during a fault condition.

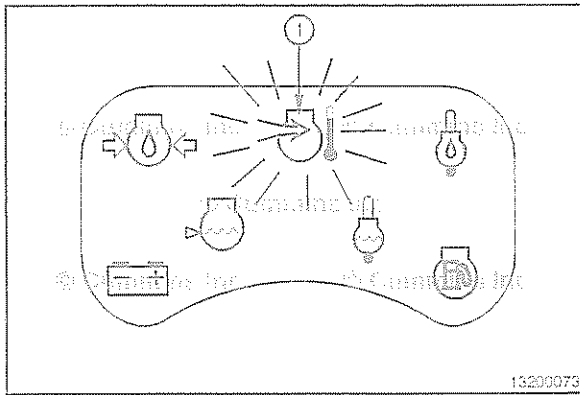
NOTE: Pressing the alarm cancel button (6) when the keyswitch is turned on will illuminate the symbols for a self-test.



Low Engine Oil Pressure

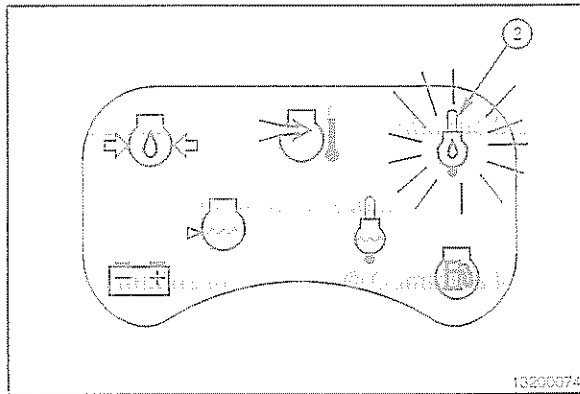
The low engine oil pressure lamp (7) comes on when the engine oil pressure is below specification. Refer to 018-017 (Lubricating Oil System) in Section V.





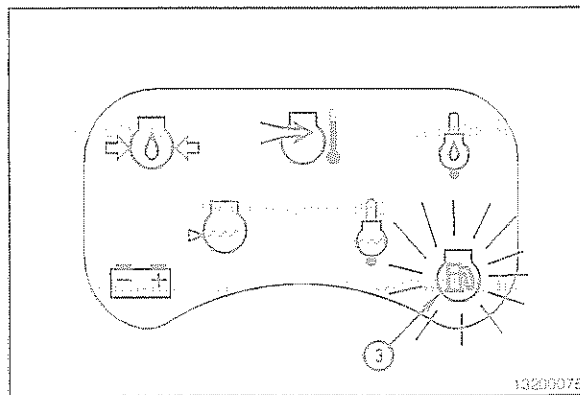
High Intake Manifold Temperature

The high intake manifold temperature lamp (1) comes on when the intake manifold temperature is above specification.



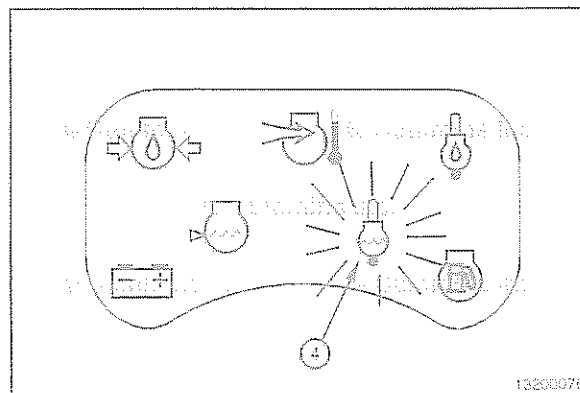
High Engine Oil Temperature

The high engine oil temperature lamp (2) comes on when the engine oil temperature is above specification.



Water in Fuel

The water-in-fuel lamp (3) interfaces with the optional water-in-fuel sensor in the primary fuel filter. It comes on when there is water in the fuel filter.

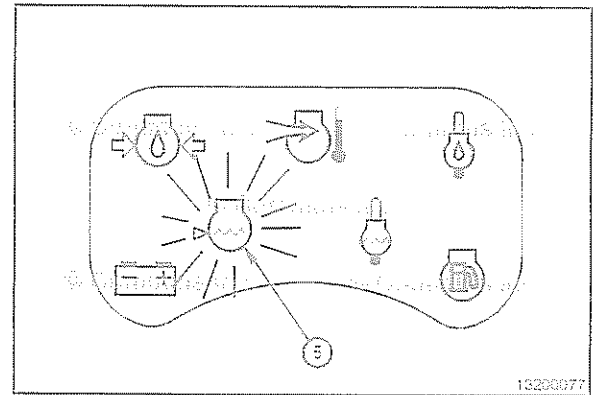


High Coolant Temperature

The high coolant temperature lamp (4) comes on when the engine coolant temperature is above specification.

Low Coolant Level

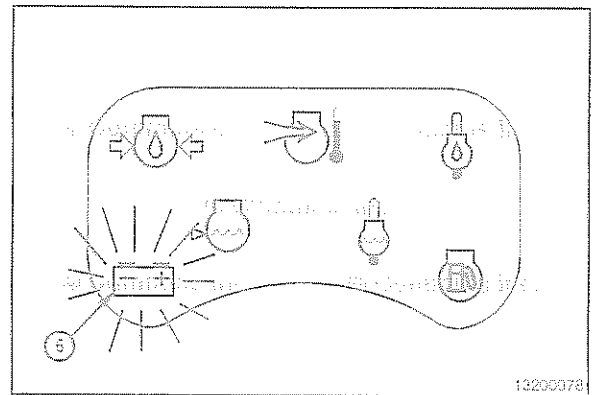
The low coolant level lamp (5) comes on when the coolant level is below specification. Refer to 018-018 (Cooling System) in Section V.



Low Battery Voltage

NOTE: This voltage lamp only applies to marine applications.

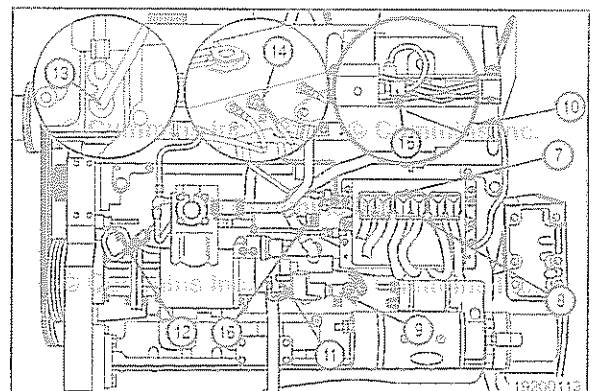
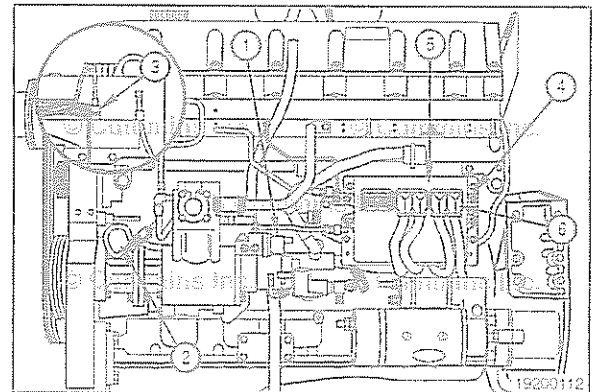
The low battery voltage lamp (6) comes on when the battery voltage is below specification.

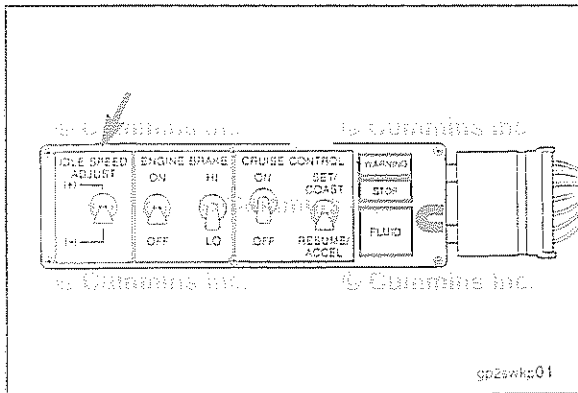


Fuel System Description

The ISM fuel system on an engine consists of:

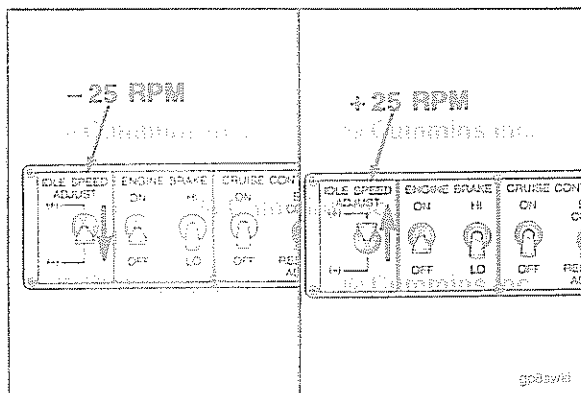
1. Fuel Shutoff Valve
2. Oil Pressure Sensor and Temperature Sensor
3. Intake Manifold Boost Sensor
4. Cooling Plate
5. ECM
6. Engine Wiring Harness Deutsch Connector.
7. OEM Wiring Harness
8. Engine Wiring Harness Deutsch Connector
9. Fuel In
10. Fuel Out
11. Fuel Gear Pump
12. Engine Position Sensor (EPS)
13. Coolant Temperature Sensor (in thermostat support)
14. Coolant Level Sensor (in overflow tank) - Optional
15. Intake Manifold Temperature Sensor
16. Ambient Air Pressure Sensor.



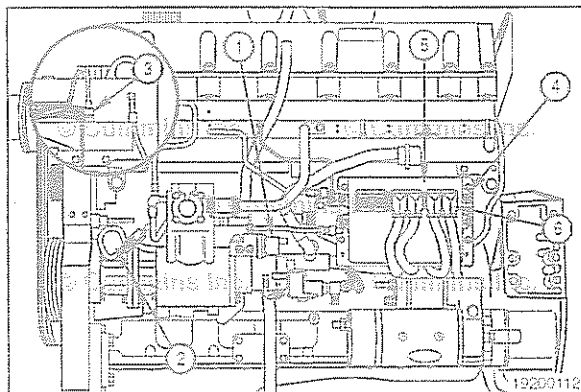


The idle adjustment is in the cab panel. Use this switch to adjust the engine idle speed in increments 25 rpm.

- Automotive - 600 to 800 rpm
- Transit Bus - 650 to 800 rpm.

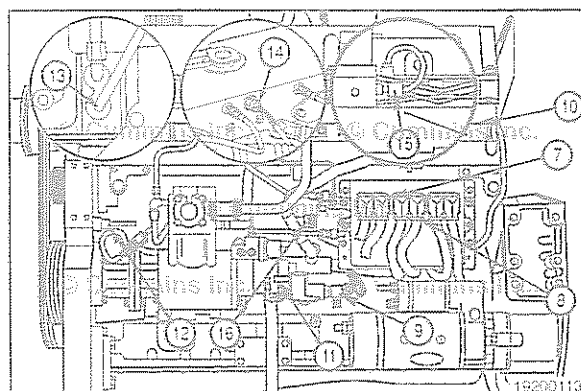


Each time the switch is briefly moved to the minus (-) position, the idle speed is decreased by 25 rpm. When the switch is briefly moved to the plus (+) position, the idle speed is increased by 25 rpm.



The fuel system on a QSM11 engine consists of:

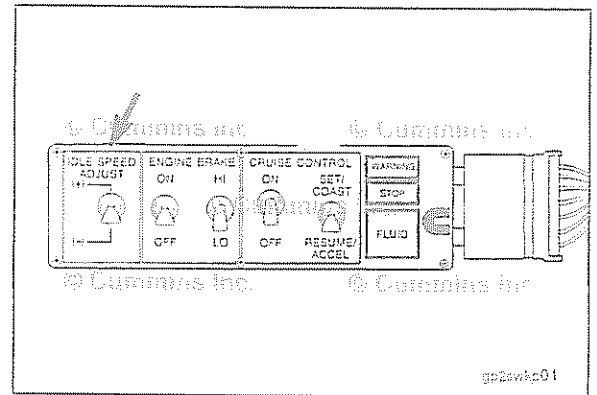
1. Fuel shutoff valve
2. Oil pressure sensor and temperature sensor
3. Intake manifold boost sensor
4. Cooling plate
5. ECM
6. Engine wiring harness Deutsch connector.



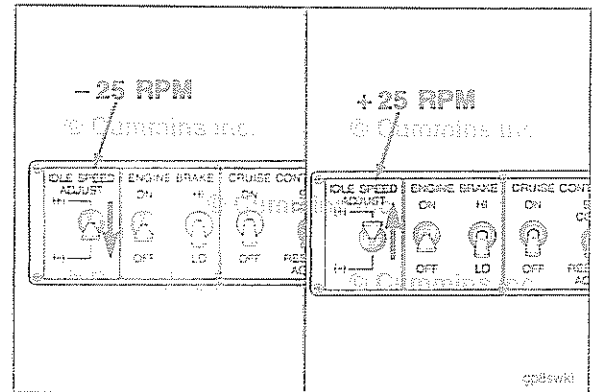
7. OEM wiring harness
8. Engine wiring harness Deutsch connector
9. Fuel in
10. Fuel out
11. Fuel gear pump
12. Engine position sensor (EPS)
13. Coolant temperature sensor (in thermostat support)
14. Coolant level sensor (in overflow tank) - optional
15. Intake manifold temperature sensor
16. Ambient air pressure sensor.

The idle adjustment is in the cab panel. Use this switch to adjust the engine idle speed in increments of 25 rpm.

- Industrial - 650 to 800 rpm.



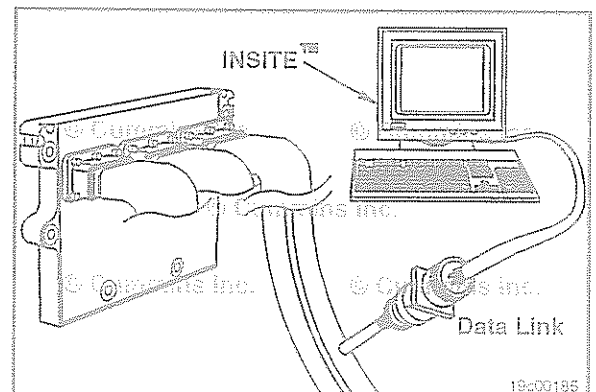
Each time the switch is briefly moved to the minus (-) position, the idle speed is decreased by 25 rpm. When the switch is briefly moved to the plus (+) position, the idle speed is increased by 25 rpm.



Programmable Features

General Information

The system provides additional electronic features that enhance engine and vehicle performance and control. These are programmable features in the ECM. For more information on additional features, refer to the Troubleshooting and Repair Manual, Electronic Control System, ISM and QSM11 Engines, Bulletin 3666266; or contact a Cummins Authorized Repair Location.



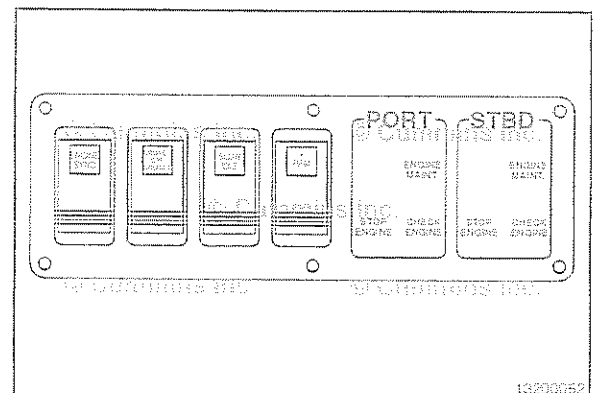
Marine Applications

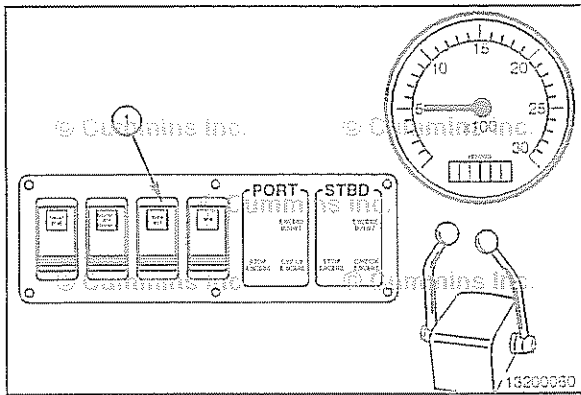
Slow Idle



Deactivate the SLOW-IDLE feature during docking maneuvers. In slow-idle mode, propeller inertia can cause the engine to stall when shifting into forward or reverse, which can cause the user to collide with the dock. Failure to turn off the SLOW-IDLE feature can cause personal injury, damage to the vessel, and/or the dock.

The SLOW-IDLE feature allows for lower idle speed for operation in "No Wake Zones." When activated, the slow-idle feature reduces the engine speed to 550 rpm.



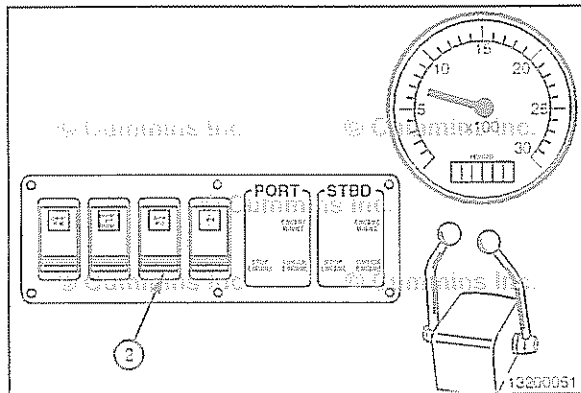


NOTE: A single switch controls both engines on twin-engine vessels.

To use the SLOW-IDLE feature:

1. The vessel must be in gear
2. The throttle must be in the idle position
3. Put the SLOW-IDLE switch in the ON position (1) by depressing the top of the rocker switch.

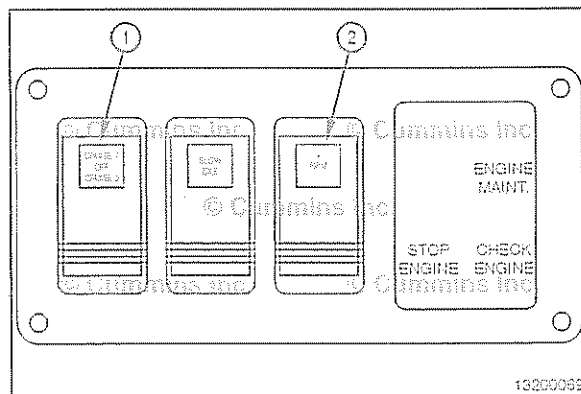
The engine(s) idle will decrease to 550 rpm.



To turn the slow-idle feature off, put the SLOW-IDLE switch in the OFF position (2) by depressing the bottom of the rocker switch.

The engine(s) will adjust to the idle set speed.

NOTE: When the slow-idle switch is on, increasing the throttle will temporarily turn off the slow-idle feature. When the throttle is moved back into the idle position, the slow-idle feature will automatically turn itself on again.

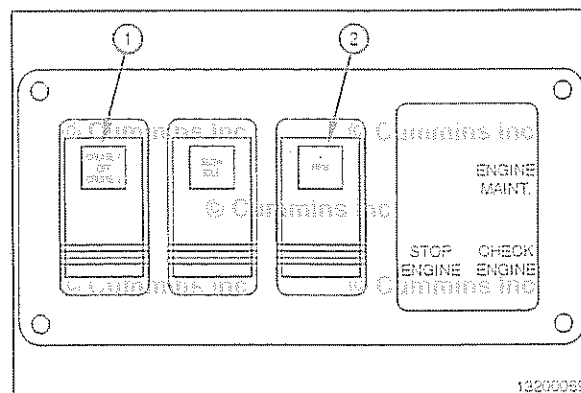


Engine Speed Control

The engine speed control (rpm $\pm$) switch (2) allows the idle speed to be adjusted in 25-rpm increments by pressing the rocker switch.

Pressing the top of the switch increases (+) engine rpm.

Pressing the bottom of the switch decreases (-) engine rpm.



When the rpm $\pm$ switch (2) is used with the engine at idle, the switch will only work from 600 to 1000 rpm.

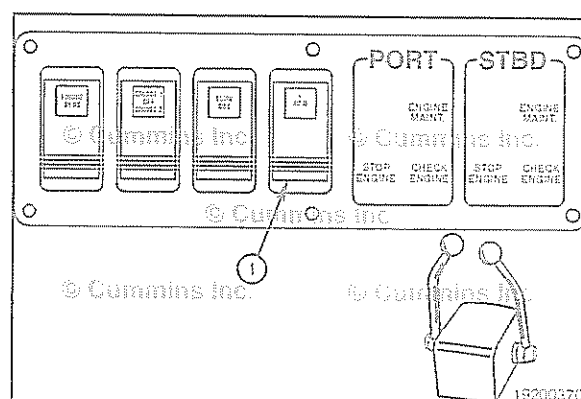
NOTE: The rpm $\pm$ switch (2) will not change the idle speed when the engine is in SLOW-IDLE mode.

Use of the slow-idle feature will deactivate the rpm $\pm$ feature.

Twin-Engine Vessels

If the engine speed between the two engines is not the same, adjust both engines to the minimum rpm setting, 600 rpm, by depressing the bottom (-) of the switch until both engines are at 600 rpm.

Then using the rpm $\pm$ switch (1), adjust the idle to the desired speed, rpm.



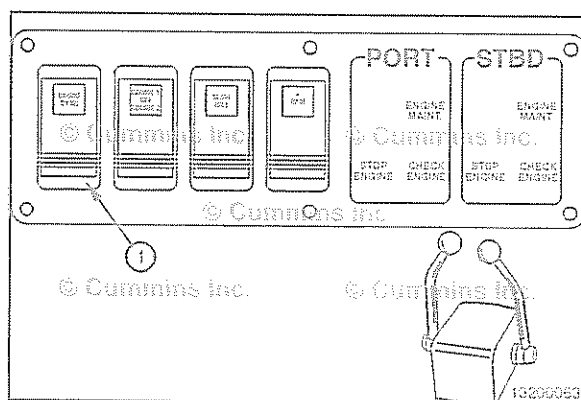
Engine Synchronization Switch



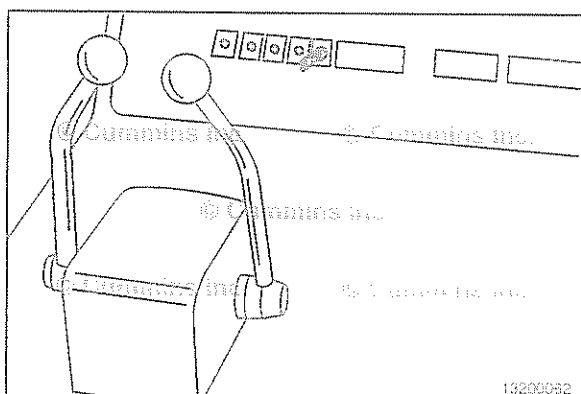
The engine synchronization feature must be turned off before docking or low-speed maneuvering; this allows full and separate control of each engine. Failure to turn off the engine synchronization feature can cause personal injury, damage to the vessel, and/or the dock.

NOTE: The engine synchronization feature is an option available only on twin-engine vessels.

The engine synchronization feature is used to control the speed of both engines electronically, using one throttle lever.



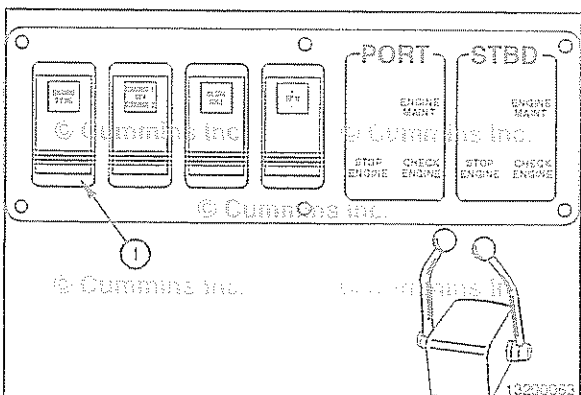
Normally the starboard throttle is used as the master throttle. The distributor or dealer can change it to the port throttle, if desired, by changing the master and slave plugs in the engine wiring. Refer to the OEM service manual.

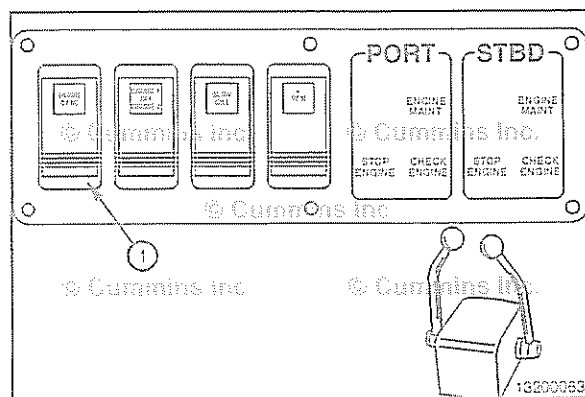


To use the engine synchronization feature:

1. Both engines must be in the same throttle position, running at the same rpm.
2. Put the ENG SYNC switch (1) in the ON position by depressing the top of the rocker switch.

Both engines will adjust to the same rpm. The master throttle will control both engines.





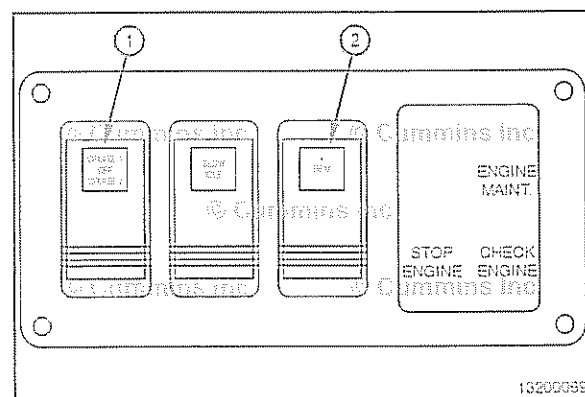
⚠CAUTION⚠

Adjust both throttles to the same position before turning the engine synchronization feature off. Failure to do so can cause a sudden port or starboard turn.

To turn the engine synchronization feature off:

1. Both throttles must be in the same position.
2. Put the ENG SYNC switch (1) in the OFF position by depressing the bottom of the rocker switch.

The engines speed will slowly ramp to their physical throttle settings.



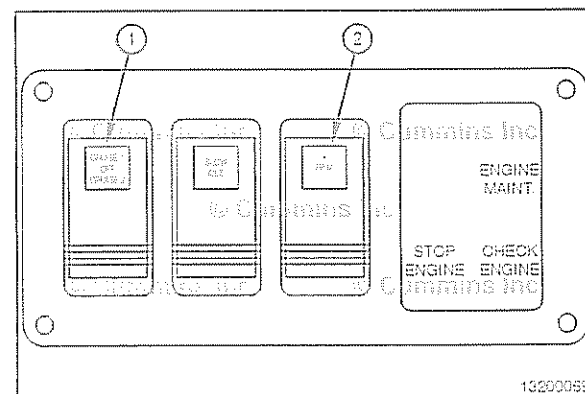
Marine Cruise Control

The marine cruise control feature provides two adjustable engine speeds. The cruise control can be used to run the engine(s) at the optimal cruise rpm or trolling rpm.

CRUISE 1 has a default setting of 2100 rpm.

CRUISE 2 has a default setting of 1200 rpm.

NOTE: The marine cruise control feature can be used in conjunction with the engine synchronization feature.

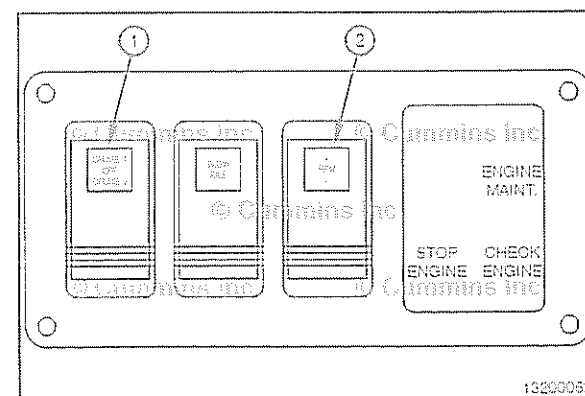


To activate the marine cruise control feature:

1. Put the cruise switch (1) in the desired position, CRUISE 1 or CRUISE 2.
2. Move the throttle(s) past the desired cruise speed.

The engine(s) speed will increase to the cruise setting.

NOTE: When using the engine synchronization feature, the master throttle is the only throttle that must be moved past the cruise point.



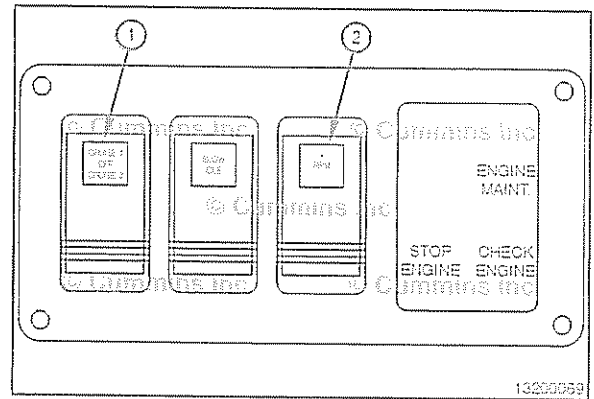
To turn the marine cruise control off, put the cruise switch (1) in the OFF position (rocker switch in the center position).

The engine(s) rpm will slowly ramp to the throttle lever setting.

To change the CRUISE 1 or CRUISE 2 default setting rpm:

1. With the engine running, select the cruise setting that you want by setting the cruise control switch (1) to CRUISE 1 or CRUISE 2.
2. Move the throttle to the full-throttle position while underway, or at the dock with the gear in neutral. The engine will not exceed the existing cruise setting. For example, if CRUISE 2 is at the factory setting of 1200 rpm, the engine will only reach 1200 rpm at full throttle.
3. Using the rpm $\pm$ switch (2), adjust the cruise rpm to the new desired engine speed.

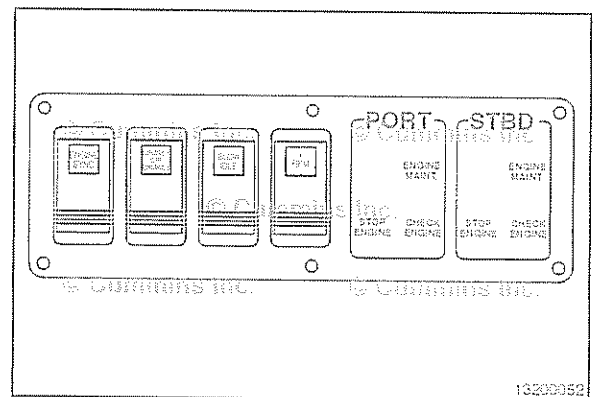
NOTE: The cruise speeds can be set between 600 and 2100 rpm. This new setting will be saved until the cruise is readjusted.



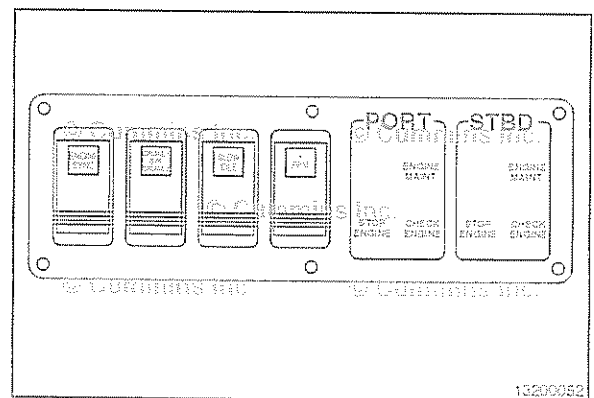
Twin-Engine Applications

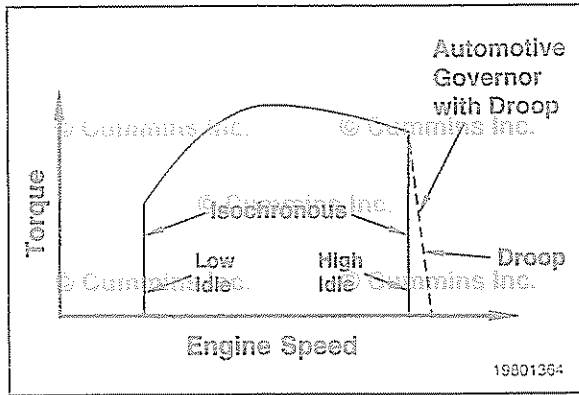
If the engines are running at different speeds:

- Adjust the cruise to the minimum or maximum setting, using the rpm $\pm$ switch.
- Adjust the cruise to the desired setting, using the rpm $\pm$ switch.



NOTE: If an engine will not reach rated rpm, make sure the marine cruise control feature is in the OFF position. If the marine cruise control feature is left in the CRUISE 1 or CRUISE 2 position, maximum engine speed will be limited to the set point of the position that the cruise switch is in.



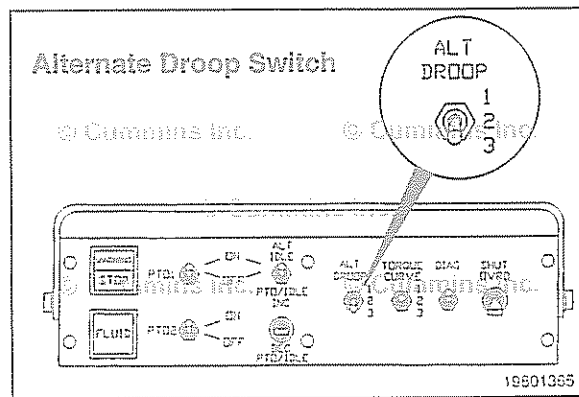


Industrial Applications

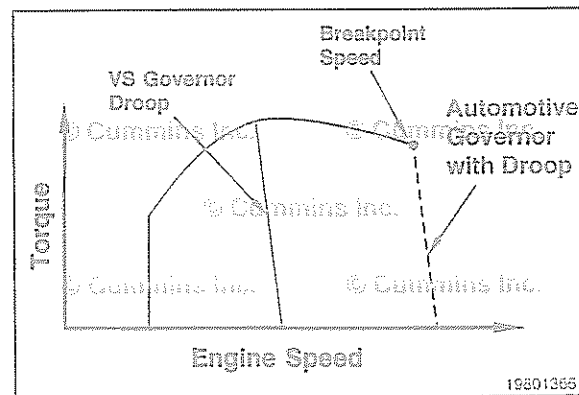
Alternate Droop

The alternate droop feature allows characteristics to be changed for the automotive governor and for the variable speed governor. Droop is usually expressed as a percentage. The graph to the right illustrates the isochronous (0-percent droop) and droop (more than 0-percent droop) governor characteristics. Less governor droop provides a more responsive governor for more precise engine control. More governor droop provides smoother shifting and smoother mechanical clutch engagement.

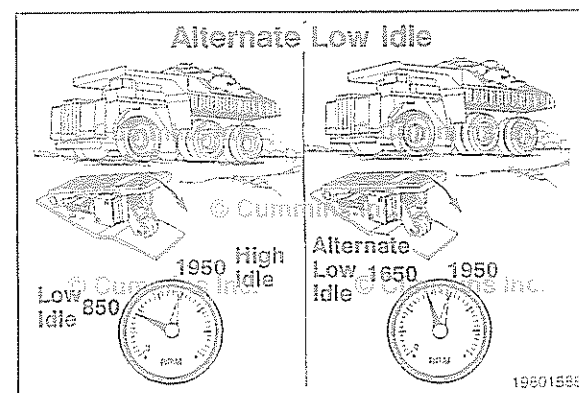
Alternate droop is available for industrial calibrations only and is specific to each calibration and application.



The alternate droop feature provides the ability to select different droop settings as follows. An OEM-provided switch can be used to select up to three droop settings (base, alternate 1, and alternate 2) and/or vehicle speed can be used to select up to five droop settings (base, alternate 1, alternate 2, Jcomm, and no selection). The final droop is based on a user-defined priority assigned to either the switch or the vehicle speed. The availability of these droop settings and abilities is dependent on the individual OEM setup and application.



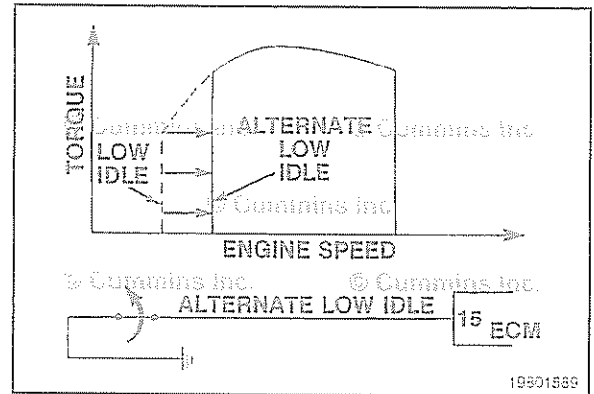
Each alternate droop setting provides the ability to select the breakpoint droop percent for the automotive governor and droop percent for VS governor. The breakpoint speed determines what position on the engine torque curve the automotive governor will start to limit engine torque output.



Alternate Low Idle Speed

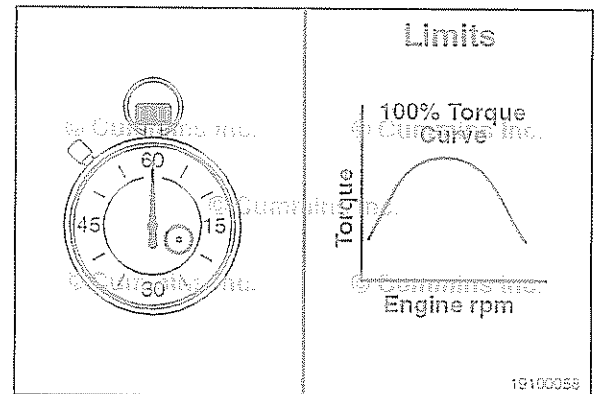
This feature allows for two different low-idle speed settings with normal throttle control above the low-idle speed setting. This feature is often used on electric drive haul trucks when they are traveling above 5 kph [3 mph].

The alternate low-idle feature is activated whenever the normally closed alternate low-idle switch is opened and 5 VDC are detected on the alternate low-idle signal line. The following art illustrates an alternate low-idle speed that is above the normal low-idle speed.



Automatic Boost Power

The automatic boost power feature will limit the time that the engine can operate on the 100-percent torque curve. Engine load intake manifold temperature and coolant temperature will be compared to thresholds to determine if boost power is available. Once boost power has been selected, the amount of the time spent on the 100-percent torque curve is limited. Boost power will be active and not time-limited when the engine speed is below a calibration threshold. A boost power lamp will be illuminated while active, and will flash as boost power is about to time out. The availability of this feature is dependent on the individual OEM application.



Sensed Parameters

This feature allows an OEM-provided pressure sensor and/or an OEM-provided temperature sensor to be read by the ECM. It also determines the out-of-range limits for each sensor. If the sensor value is outside the limits, then a fault will be logged corresponding to the out-of-range sensor and a default value will be used for the out-of-range sensor. This feature can be enabled or disabled by INSITE. The Number of Inputs, Upper and Lower Pressure Thresholds, and Upper and Lower Temperature Thresholds can be adjusted by INSITE.

The temperature and pressure sensors can be used by the ECM to control other features, such as the Electronic Fan Clutch or Switched Outputs.

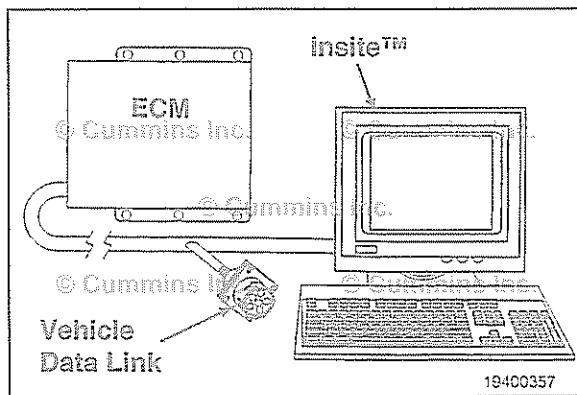
Low Idle Shutdown

The low-idle shutdown feature automatically shuts off an engine after a period of idling when there is no activity from the driver such as clutch, brake, or throttle actuation. The time interval can be changed using an electronic service tool.

The idle shutdown system will not be active at coolant temperatures below 43°C [109°F].

After an engine has been automatically shut off, the key must be turned off for 5 seconds before attempting a restart.

NOTE: This feature will shut off the engine only. It will not remove power from other accessories powered by the keyswitch.



Hot Shutdown Monitor

The hot shutdown monitor is a selectable feature within the electronic service tool INSITE™. If this feature is enabled, the ECM will log an inactive fault when the engine is turned off while still "hot" by the operator or by the engine protection feature.

An engine is considered "hot" when the hot shutdown load percent of the engine is above the threshold set by the INSITE™ tool. The hot shutdown load percent is based on the duty cycle load factor, which is determined from engine fueling levels.

Engine Protection Shutdown Manual Override

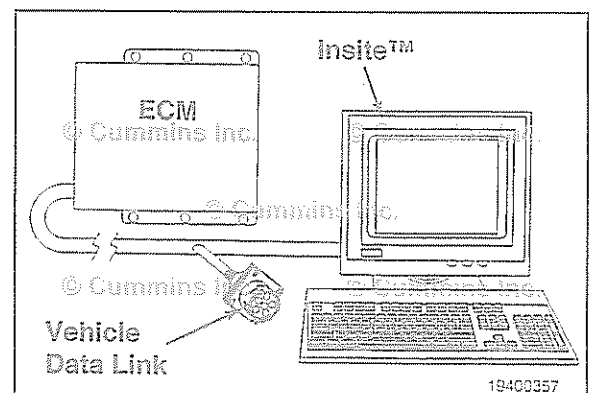
The engine protection shutdown manual override feature can override the engine shutdown process when it is necessary. The engine protection shutdown manual override feature detects the operator request for engine protection shutdown manual override, records the times of occurrences, and sends a flag to inform the engine protection shutdown feature and lamp control to restart the shutdown warning period. This feature can be enabled by INSITE.

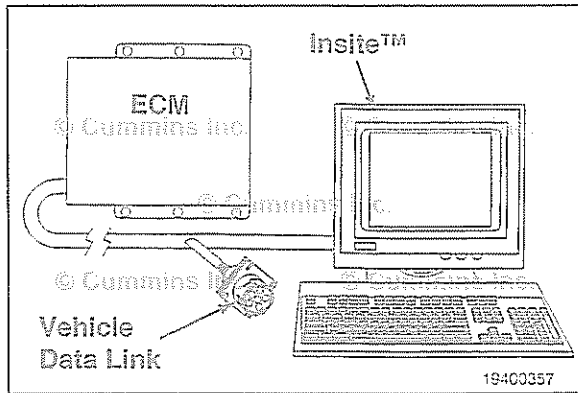
During an engine protection shutdown sequence, the warning lamp will flash. With the engine protection shutdown manual override feature enabled, the shutdown will be overridden if the dash-mounted override switch is turned on as the warning lamp is flashing. The feature will also record a snapshot that can be viewed under the Engine Protection Data in INSITE.

Electronic Fan Clutch

The electronic fan clutch feature provides a pulse-width modulated signal to control a variable-speed fan clutch based on the need provided by eight possible sensor inputs, or an input from an electronic service tool. This feature will reduce fuel consumption by minimizing fan on-time and lengthen belt life by eliminating belt hop and slippage.

The electronic fan clutch can be enabled or disabled with the INSITE™ service tool. The fan percent ON (0 to 100 percent) can be monitored and the fan clutch ramp rate adjusted in units of percent/second. Adjusting the ramp rate allows different applications to be tuned for different fan characteristics.

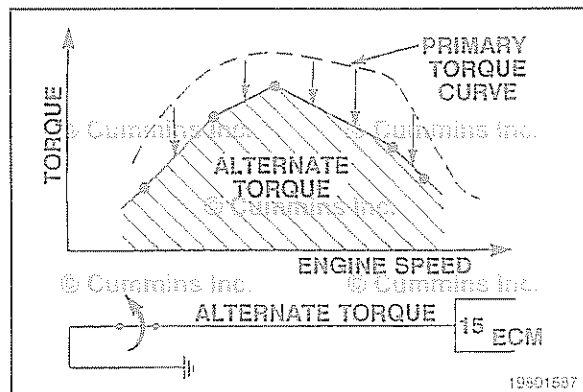




The INSITE™ service tool also has the capability to control the fan clutch percent ON with the fan clutch control dialog. While the engine is running and the service tool is connected to the datalink, the user can enter the fan percent ON request between 0 and 100 percent. Controlling the fan clutch with the service tool will aid in troubleshooting and diagnostics of the fan clutch and engine cooling system. Refer to INSITE™ Electronic Service Tool Manual, Bulletin 3666148.

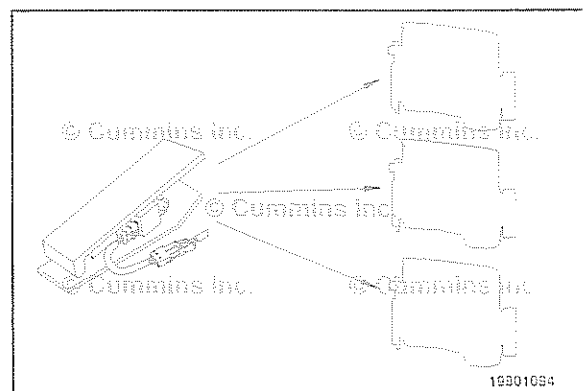
Switchable (Alternate) Torque

This feature enables an alternative electronically controlled maximum engine torque curve for optimum operating efficiency in loaded versus unloaded conditions.



The alternate torque feature is activated whenever the normally closed alternate torque switch is opened and 5 VDC are detected on the alternate torque signal line. Five torque versus engine speed points define the alternate torque curve.

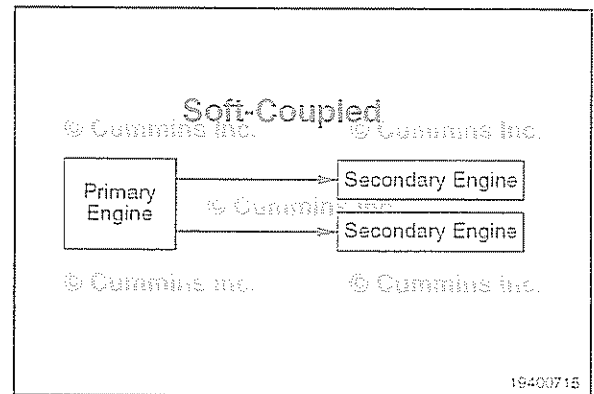
Shown is a graph illustrating an alternate torque curve that is below the normal torque curve.



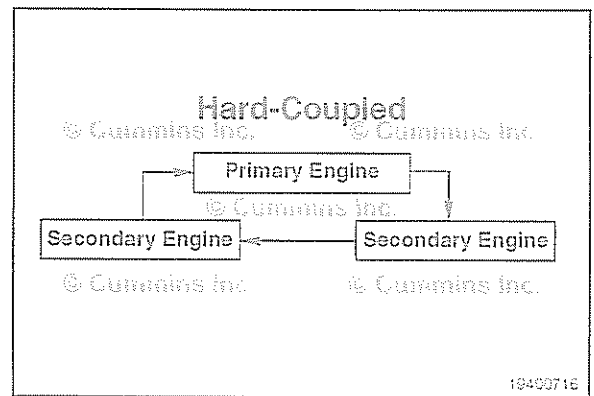
Multiple Unit Synchronization

The multiple unit synchronization feature allows two or more engines (up to a maximum of 11) to be controlled by a single throttle signal. There are three engine configurations available with this feature. They are soft-coupled, hard-coupled, and soft-coupled marine.

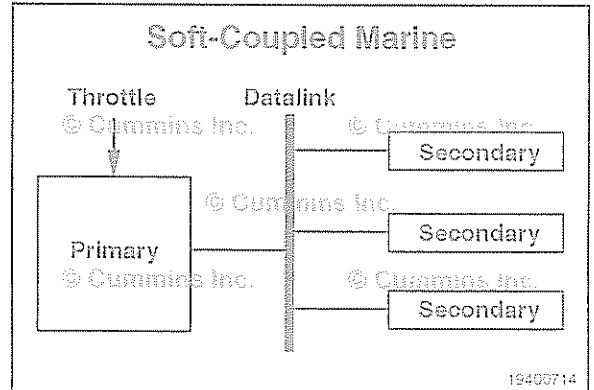
The soft-coupled configuration has all secondary engines in parallel.



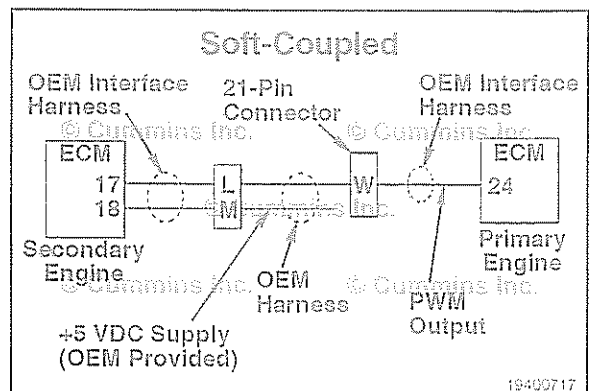
The hard-coupled configuration has all secondary engines in series.

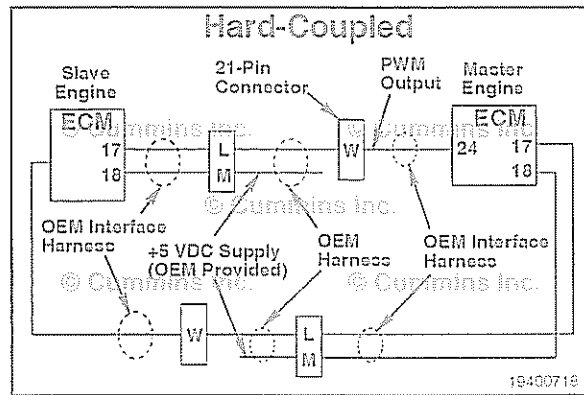


The soft-coupled marine configuration has all engines connected to a J1939 datalink.

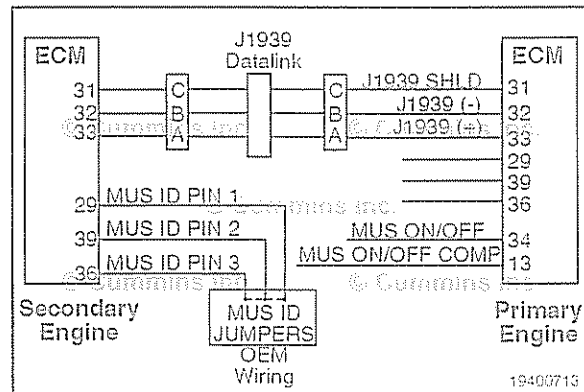


In the soft-coupled configuration, the primary engine outputs a throttle signal on ECM pin 24. This signal is received by the secondary engine(s) on ECM pin 17. The secondary engine also receives a +5-VDC supply (OEM provided) to ECM pin 18 to complete the circuit. This setup allows all engines to remain running if a secondary engine stops running.

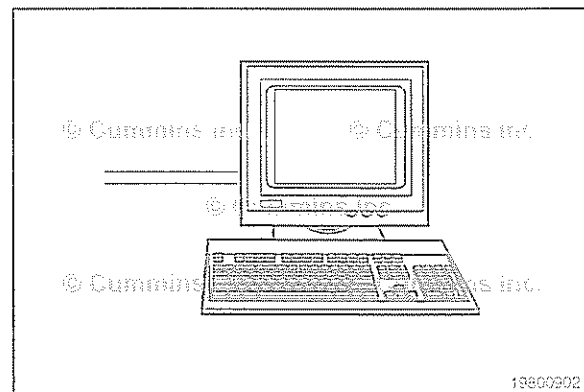




In the hard-coupled configuration, the primary engine outputs a throttle signal on ECM pin 24. This signal is received by the secondary engine(s) on ECM pin 17. The secondary engine also receives a +5-VDC supply (OEM provided) to ECM pin 18. The secondary engine then outputs the throttle signal on ECM pin 24 to the next secondary engine in the series. This process is repeated by all secondary engines until the throttle signal and +5-VDC supply are received by the primary engine.



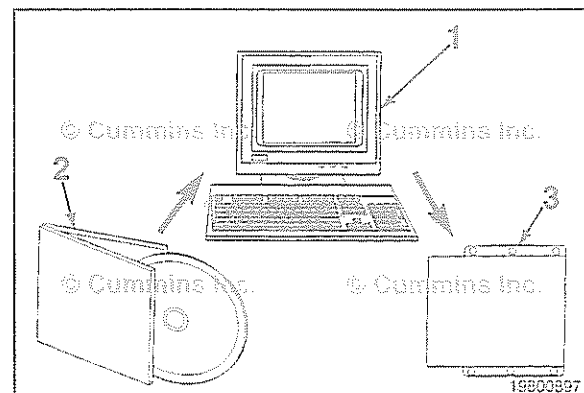
In the soft-coupled marine configuration, the primary engine outputs a throttle signal to the J1939 datalink pins 31, 32, and 33. Secondary engines that are connected to the datalink then read the throttle value. The soft-coupled marine application also allows for primary/secondary configuration via wiring harness jumpers (pins 29, 39, and 36) instead of specific calibrations. The soft-coupled marine configuration uses a multiple unit synchronization on/off switch (pins 13 and 34) to control the enabling or disabling of the engine synchronization.



The INSITE™ service tool can be used to monitor the following parameters:

- Multiple unit synchronization couple type (soft-coupled, hard-coupled, or soft-coupled marine)
- Multiple unit synchronization engine type (primary or secondary)
- Multiple unit synchronization on/off switch
- Percent pulse-width modulate output (throttle signal).

The multiple unit synchronization couple type is not adjustable by the INSITE™ service tool, but the type can be changed by downloading a calibration to the ECM.



This feature can only be enabled or disabled by calibration. An electronic service tool (1) will be required to download a calibration from the ESDN (2) to the electronic control module (3) if this feature needs to be enabled or disabled.

Intermediate-Speed Control

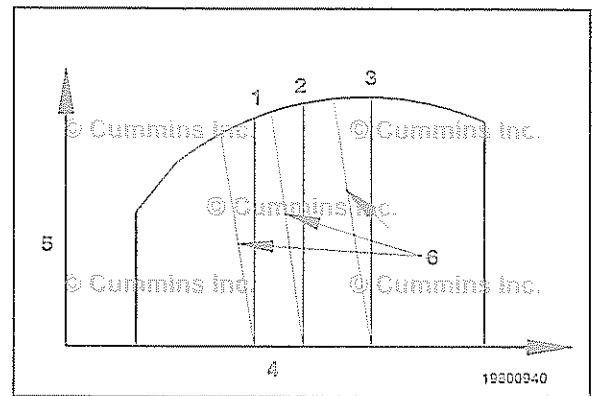
The intermediate-speed control feature can control the engine to three programmable engine set speeds (1, 2, 3) in three different modes of operation. The three set speeds are determined by a three-position switch and a two-position switch. The three modes of operation are a manual normal set speed operation, a low speed limit operation, and a high speed limit operation.

NOTE: These three modes of operation are set by the calibration and are not customer-adjustable.

An intermediate-speed droop is also programmable.

NOTE: Depending on the electronic calibration, the third intermediate speed switch can act as a validation switch. The purpose of the validation switch is to eliminate accidentally switching to intermediate speed 1 or 2. To switch to intermediate speed 1 or 2, the validation switch must be engaged first; then move the intermediate-speed switch to 1 or 2.

This feature also uses a momentary increment and decrement switch to raise the engine set speed while in operation.



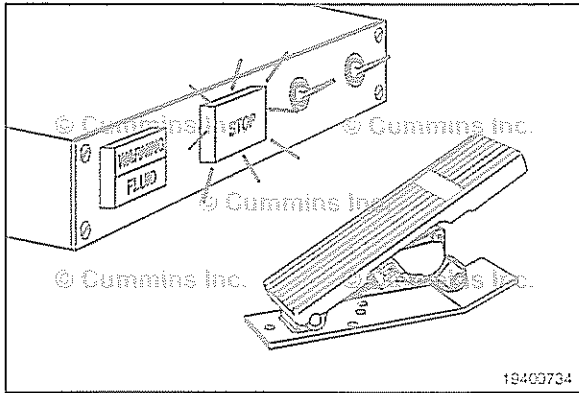
Switched Outputs

This feature provides up to two fully independent switched outputs for OEM use. The state of each switched output is determined by up to 12 different inputs to the ECM. The ECM can provide different outputs if any of the inputs are above or below calibrated thresholds. Each switched output is independent of the other in controlling parameter input and threshold settings. This feature can be enabled or disabled with INSITE. The thresholds can not be adjusted with INSITE.

The ECM can determine the state of the switched outputs based on the following possible inputs (either or both switched outputs can use the same inputs):

- Engine speed
- Fueling command (cubic mm per second)
- Boost pressure (in Hg)
- Auxiliary speed input (rpm)
- Oil pressure (psi)
- Coolant temperature (degrees Fahrenheit)
- Commanded throttle (percent)
- ISC 2 status (off or active)
- OEM temperature (degrees Fahrenheit)
- OEM pressure (psi)
- OEM supplied sensor (volts)
- OEM supplied switch (open or ground)

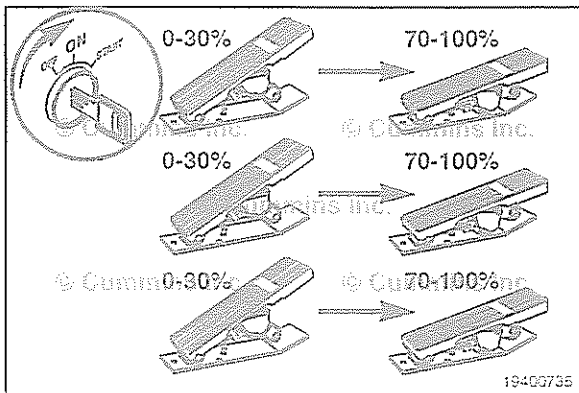
Each of these inputs can have a specified threshold and threshold type (over or under). Each of the switched outputs can be calibrated to either "on" or "off" after a threshold is crossed. Also, each output can be calibrated to change states if either any threshold is crossed or all thresholds are crossed.



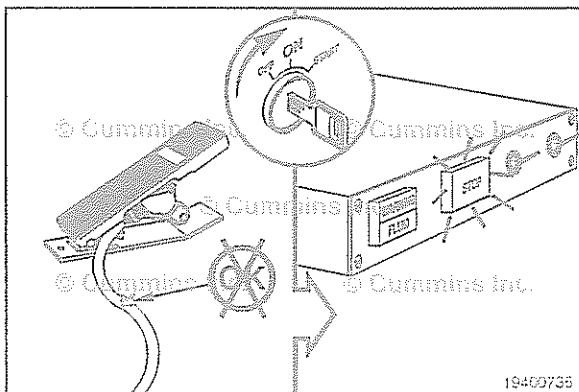
Throttle Activated Diagnostic Switch

Throttle-activated diagnostic switch is intended to eliminate the need for a dash-mounted diagnostic switch, which is used to activate the diagnostic mode to display active fault codes in a sequence of flashing lamps. The throttle-activated diagnostic switch feature eliminates the need for a dash-mounted diagnostic switch by providing a simple sequence of throttle movements that activate the diagnostic mode.

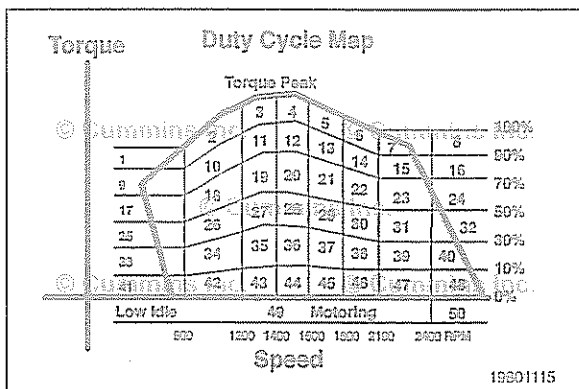
NOTE: This feature will work with all throttle types.



When the engine is not running, the keyswitch is turned on, and the feature flag is enabled, a sequence of three throttle cycles from 30 to 70 percent within a specific time will activate the diagnostic mode. The increment/decrement switch can be used to navigate to the next or previous fault code. In case these switches are not available, a throttle cycle from 30 to 70 percent will also increment to the next fault.



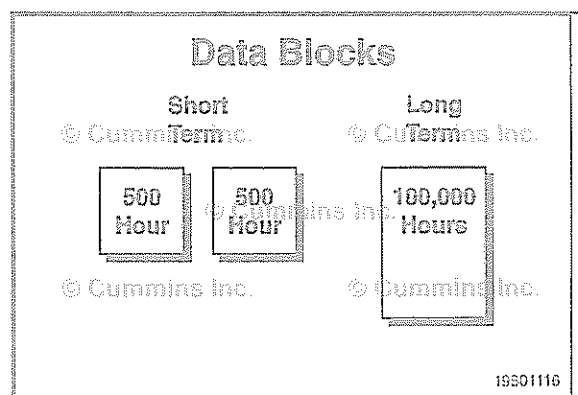
A sensitive mode is enabled when this feature is enabled. The sensitive mode is an algorithm that will automatically activate the diagnostic mode when the engine is not running, the keyswitch is turned on, and any throttle-related errors are detected. This allows the user to activate the diagnostic mode when throttle errors prevent them from doing so with throttle movements.



Duty Cycle Monitor

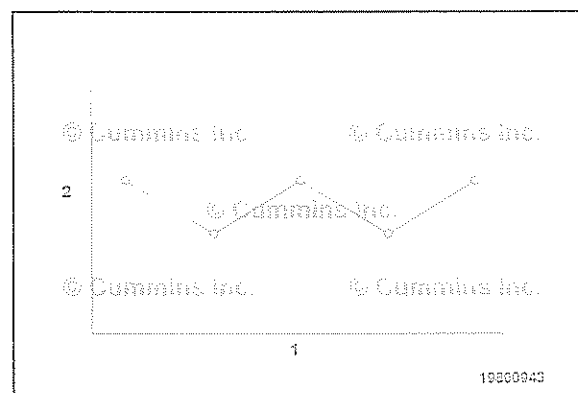
The duty cycle monitor tracks the time the engine spends in 50 different operating regions. These operating regions are based on engine speed and engine torque.

This feature provides two short-term 500-hour resettable data blocks and one long-term 100,000-hour nonresettable data block.

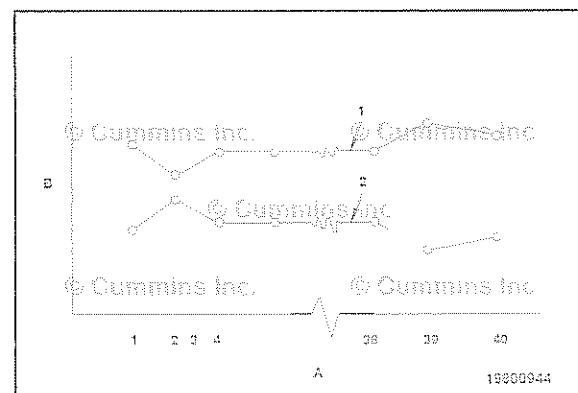


Fuel Consumption Rate Logger

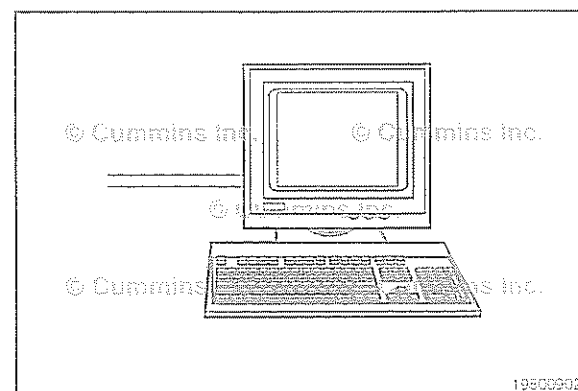
The fuel consumption rate feature allows an electronic service tool to access fuel consumption data (1 = time, 2 = gallons/hour).

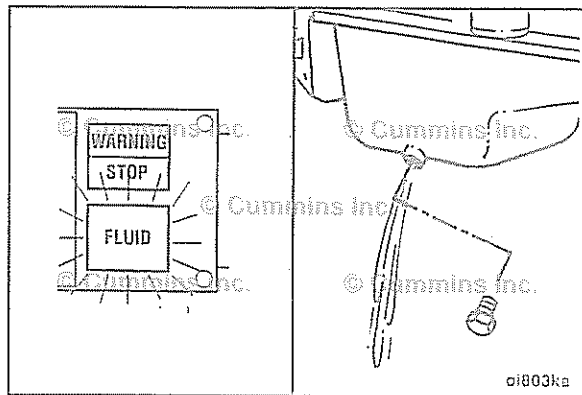


This feature provides two resettable 40-hour fuel consumption periods (1 and 2). Each period records fuel consumption data in forty 1-hour segments. These 40 data segments can be graphed to show fuel consumption over both 40-hour periods (A = hours, B = gallons/hour).



An instantaneous fuel consumption rate and a nonresettable lifetime or running average fuel consumption rate are available on the monitor screen of an electronic service tool. In addition, the two 40-hour fuel consumption periods are individually resettable using an electronic service tool.





Maintenance Monitor

NOTE: The maintenance monitor is designed to alert the operator of the need for a routine maintenance stop. Maintenance records must still be maintained for historical purposes.

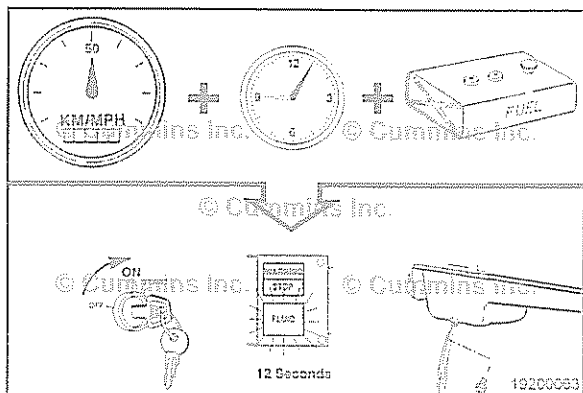
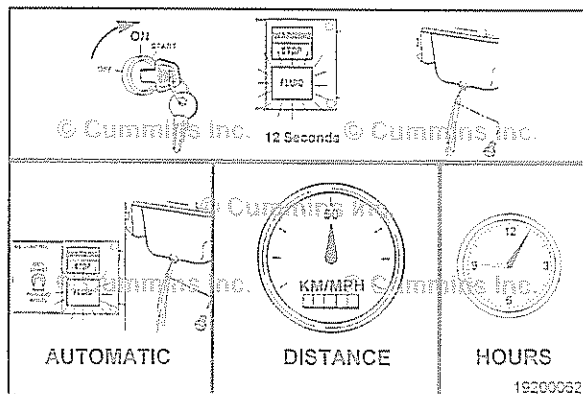
NOTE: The maintenance monitor uses data received from the ECM to determine the amount of fuel burned. Whenever a VSS, injector circuit, or battery voltage fault has occurred, the maintenance monitor data can be inaccurate.

The maintenance monitor is an optional feature that will alert the operator when it is time to change oil and perform any other simultaneous maintenance tasks. The maintenance monitor continuously monitors the time the engine has been operating and the amount of fuel burned to determine when it is time to change oil.

The operator must still be alert for any indications that the engine needs other service.

The maintenance monitor has three modes of operation:

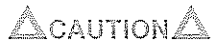
- Automatic mode
- Time mode.



The automatic mode alerts the operator when it is time to change oil based on Cummins recommended interval. It determines the maintenance interval based on engine operating time and fuel burned.

When the automatic mode is selected, the severe oil drain interval duty cycle is the default.

The original factory programmed value is AUTOMATIC.



Select the correct oil change interval for your application. Cummins Engine Company, does not recommend exceeding these published intervals and is not responsible for damage sustained due to overextended drain intervals.

The maintenance monitor automatic mode is easily adjusted to accommodate severe-, normal-, or light-duty applications and Valvoline® Premium Blue 2000™ engine oil. This is done by using a Compulink™ or EcheK™ service tool to enter an interval factor that corresponds to the appropriate duty cycle and type of product used.

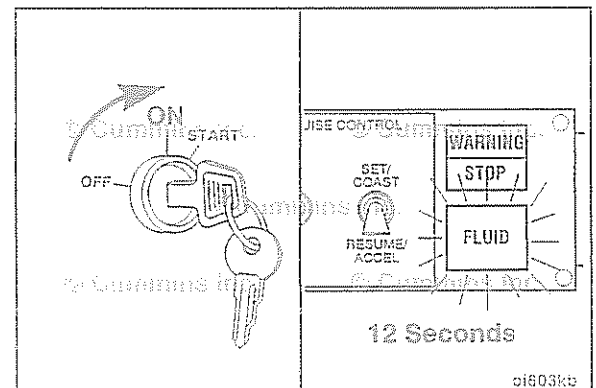
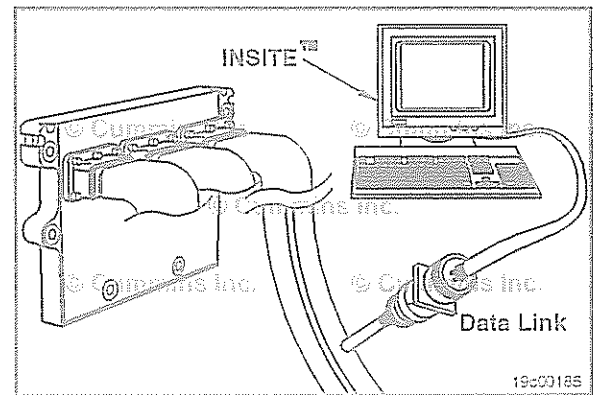
When selecting the correct interval factor for your application, refer to 102-002 (Maintenance Schedule) in Section 2, to determine which oil drain interval duty cycle fits your application: Severe duty, normal duty, or light duty. Once the duty cycle is chosen, use an interval factor of 1.00 for severe duty, 1.50 for normal duty, and 2.00 for light duty.

NOTE: These interval factors must be adjusted accordingly if Valvoline® Premium Blue® 2000 oil is used to extend the oil drain interval. See the Valvoline® Premium Blue® 2000 product literature to determine how much your drain interval can be extended.

The time mode allows the customer to enter a desired time interval. The maintenance monitor will then monitor the time the engine has been operating and alert the operator when the interval has been consumed.

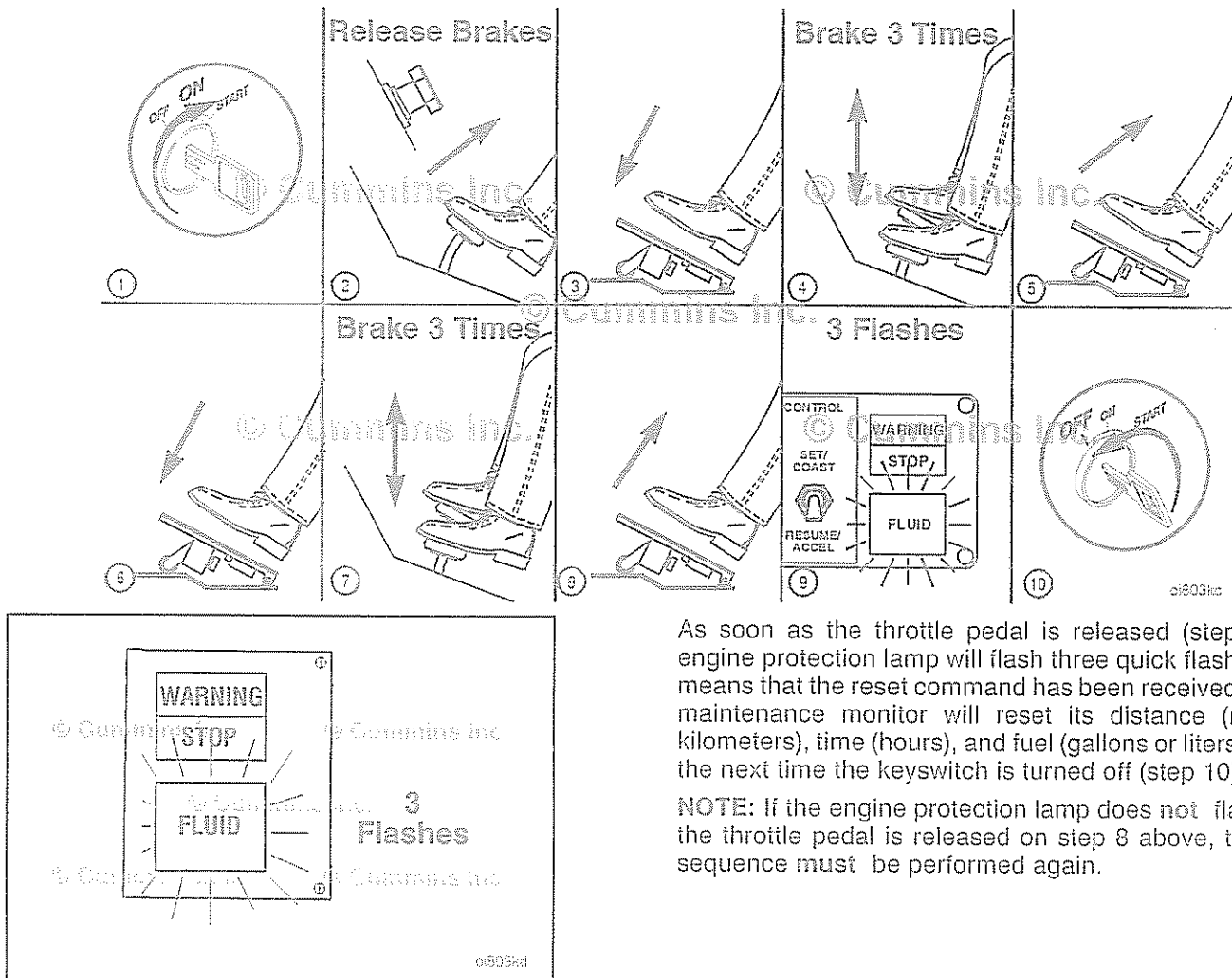
Alerting the operator: The maintenance monitor will alert the operator of the need to change oil by flashing the engine protection lamp (fluids lamp) for approximately 12 seconds after key-on. The flashing sequence will be three quick flashes, followed by a pause. This flash sequence will go through five cycles in the 12-second period. This sequence will occur at every key-on until the maintenance monitor has been reset.

NOTE: The diagnostic switch must be in the OFF position for the flashing sequence to occur.



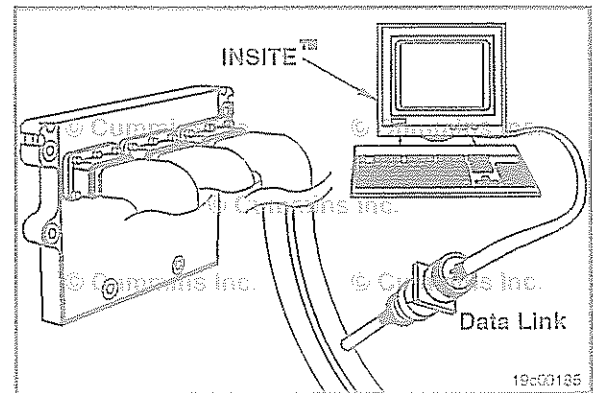
Resetting the Maintenance Monitor. The maintenance monitor can be reset using an INSITE™ service tool or by following steps 1 through 10 below. Steps 3 through 8 must be done within 12 seconds in order for the maintenance monitor to be reset. The diagnostic switch must be in the OFF position and the vehicle air system must be fully charged.

1. Turn on the keyswitch (engine must not be operating)
2. Make sure brakes are released (service and trailer brakes)
3. Hold throttle pedal at 100-percent throttle
4. Press and release service brake three times
5. Release throttle pedal
6. Depress and hold throttle pedal at 100-percent throttle again
7. Press and release service brake three more times
8. Release throttle pedal
9. The light will flash three times
10. Turn off the keyswitch.



When viewing maintenance monitor data with an INSITE™ service tool, the following maintenance data can be viewed or printed from the ECM:

- Percent of present interval consumed (by either distance, time, or fuel burned)
- Distance since last reset
- Time since last reset
- Reset log (distance and time at the last three resets).



The interval factor is only used in the maintenance monitor Auto mode to adjust the following applications:

- Severe duty
- Normal duty
- Light duty
- Extended interval when using Valvoline® Premium Blue® 2000
- Other products that extend maintenance intervals.

The interval factor can be adjusted between 0 and 5.0.

The original factory programmed value is 1.

The maintenance monitor interval alert percentage allows the user to enter the percentage of the present interval at which the light should come on indicating the need for an oil change. The parameter allows the user to obtain an early warning of the need for a maintenance stop.

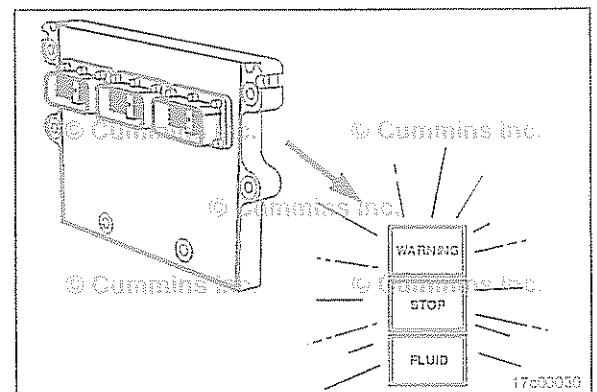
For example, if the distance mode is set to 24,194 km [15,000 mi] and the interval alert percentage is set to 90 percent, the lamp will illuminate at 21,774 km [13,500 mi], which is 90 percent of 24,174 km [15,000 miles].

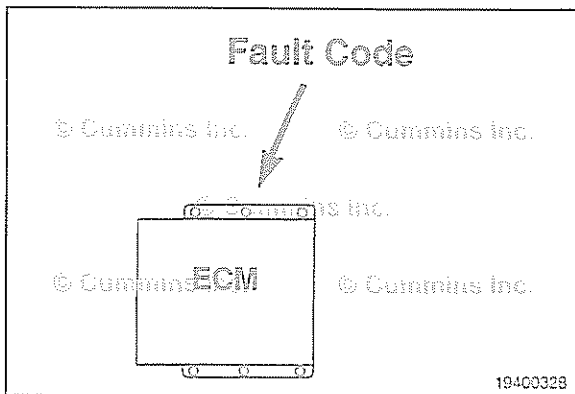
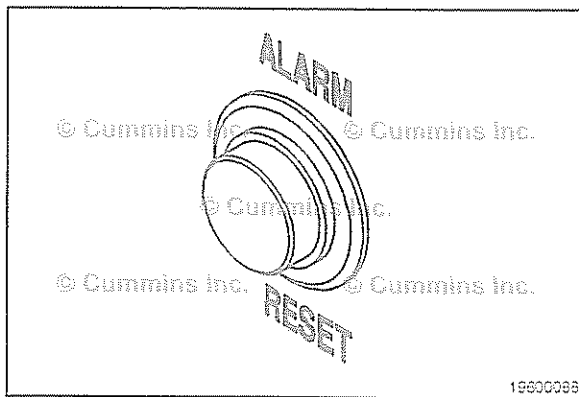
The interval alert percentage can be set between 0 and 100 percent.

The original factory programmed value is 100 percent.

Fault Lamp Sequencing

The fault lamp sequencing feature is used to determine whether the diagnostic dashboard lamps have been installed and wired correctly. After key-on, all dashboard diagnostic lamps will turn on at once and then turn off in sequence.





Diagnostic Fault Codes

Power Generation

Fault Acknowledge

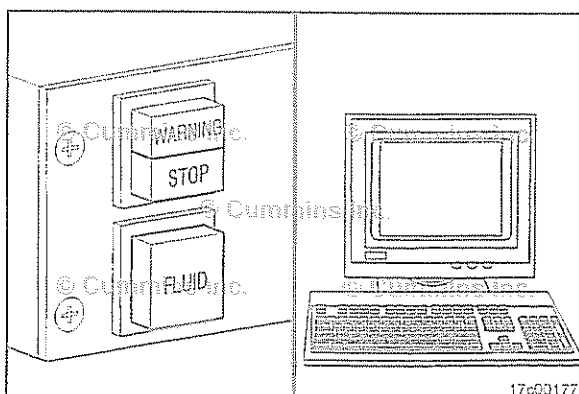
A customer-supplied switch acknowledges all shutdown/warning functions and inactive fault codes. Warning relay drivers and relay contacts can be acknowledged while the engine is running or shut down. Shutdown relay drivers and contacts, and inactive fault codes can be acknowledged only when the engine is shut down. Before restarting the engine after a fault induced shutdown, check the ECM for fault codes via the electronic service tool.

The fuel system can display and record certain detectable fault conditions. These failures are displayed as fault codes, which make troubleshooting easier. The fault codes are retained in the electronic control module (ECM).

There are two types of fault codes. There are engine electronic fuel system fault codes and engine protection system fault codes.

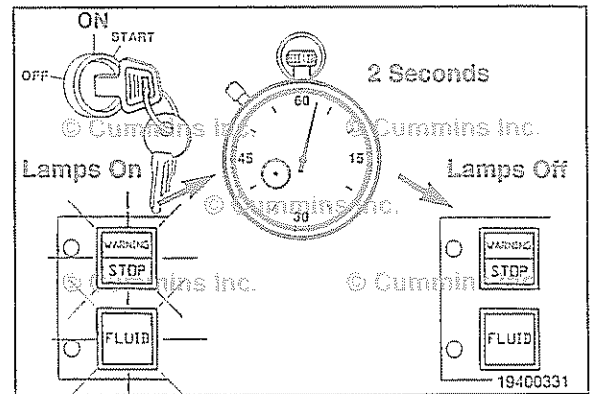
All fault codes recorded will either be active (fault code is presently active on the engine) or inactive (fault code was active at some time, but is not presently active).

Diagnostic Fault Codes
• Engine Electronic Fuel System Fault Codes
• Engine Protection System Fault Codes.



Active fault codes can be read using the warning (amber) and stop lamps (red) typically located on the generator set control panel or electronic service tool. Inactive fault codes can only be viewed with an electronic service tool.

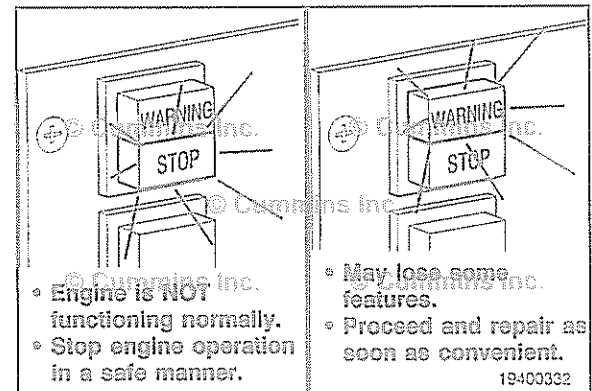
When the vehicle keyswitch is turned on and the diagnostic switch is off, the fault code lamps (red, yellow, and maintenance) will illuminate for approximately 2 seconds, one after the other, to check their operation.



The lights will remain off until a fault code is recorded. If a stop (red) light comes on while the engine is in operation, the fault can be engine-disabling. Stop the engine in a safe manner as soon as possible.



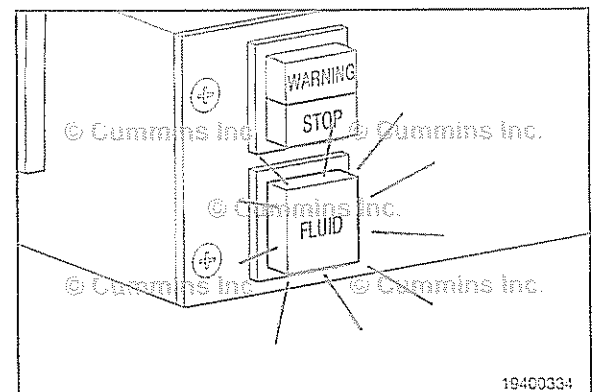
If the warning (amber) light illuminates, the engine can still be operated, but it can lose some system features that can sometimes result in a power loss. The failure must be repaired as soon as is convenient.

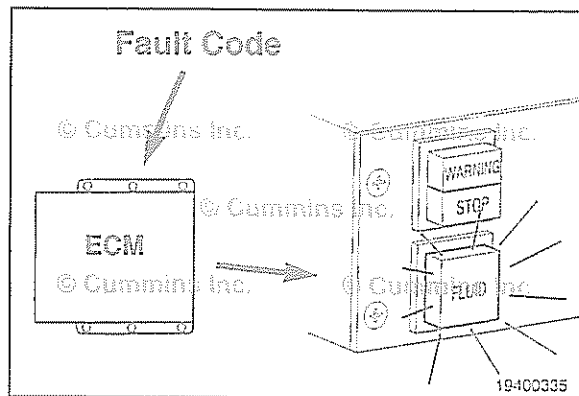


The engine protection system records separate fault codes when an out-of-range condition is found for any of the sensors in the engine protection system. Engine protection is only available when the engine protection feature is enabled.

- Coolant Temperature
- Coolant Level
- Intake Manifold Temperature
- Oil Pressure
- Water-in-Fuel (optional protection feature).

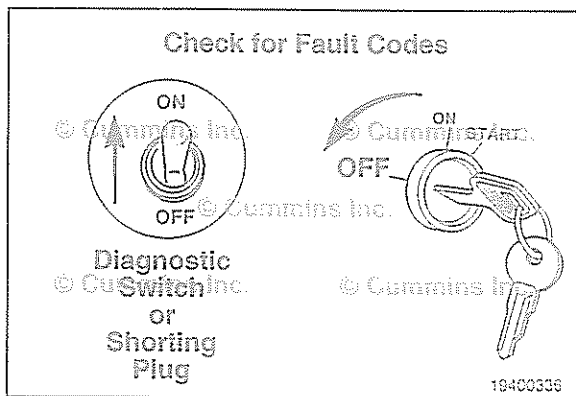
The engine protection system will light the maintenance amber (warning) lamp when an out-of-range condition occurs.





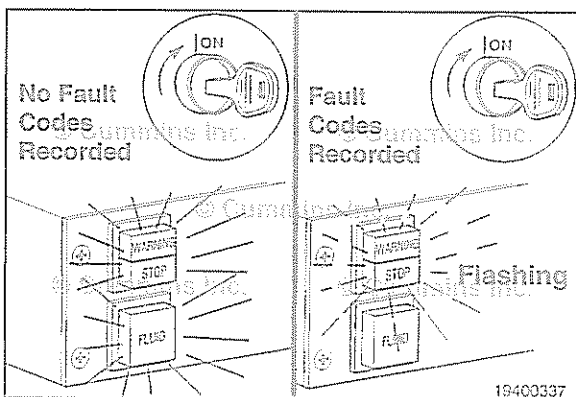
If the engine protection maintenance lamp comes on, it means that a fault code has been recorded. The light will remain on as long as the fault is occurring.

If the engine protection shutdown feature is enabled, the engine will shut down to prevent engine damage.

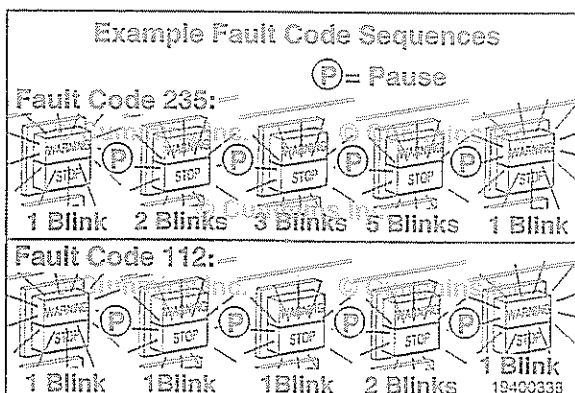


To check for active fault codes, first turn the vehicle keyswitch to the OFF position. Move the diagnostic switch to the ON position.

NOTE: Some OEMs use a shorting plug.



Turn the vehicle keyswitch to the ON position. If no active fault codes are recorded, all lights will come on and stay on. If active fault codes are recorded, all three lights will come on momentarily. The amber (warning) and red (stop) lights will begin to flash the code of the recorded fault.



The fault code will flash in the following sequence. First, the amber (warning) lamp will flash. Then there will be a short 1-second pause when both the amber and red lights are off. Then the numbers of the recorded fault code will flash in red. There will be a 1-second pause between each number. When the number has stopped flashing, an amber light will appear again. The number will repeat in the same sequence.

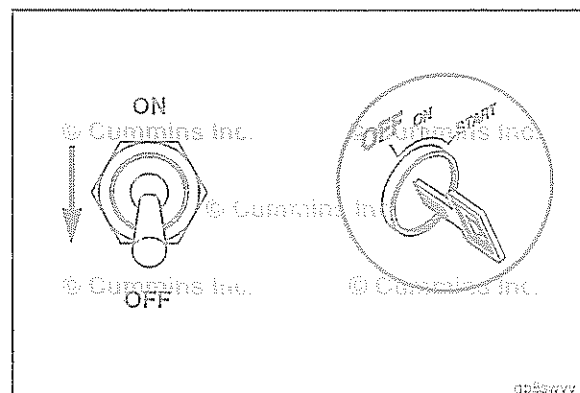
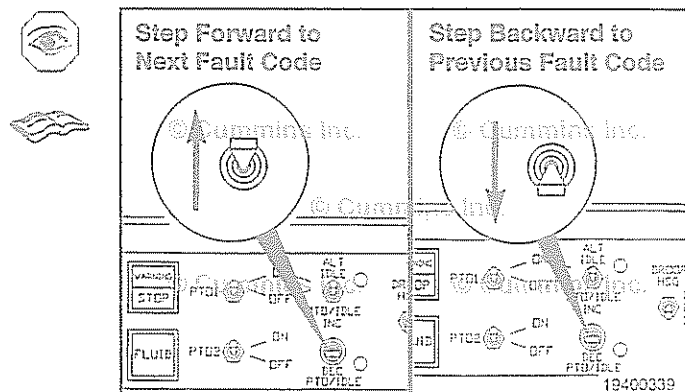
The lights will continue to flash the same fault code until the system has advanced to the next active fault code. To go to the second fault code, move the idle-speed adjust switch to "+", then release it. You can also go back to the previous fault code by moving the switch to "-", then releasing it. To check the third or fourth fault code, move the switch to "+", then release it when all active fault codes have been viewed. Moving the switch to "+" will go back to the first fault code.

The explanation and correction of all fault codes is in the troubleshooting charts of the fuel manual. Refer to Troubleshooting and Repair Manual, Electronic Control System, ISM/QSM11 Electronic Control System, Bulletin Number 3666266.

Electronic fault code troubleshooting trees are in ascending numerical order. An index is located at the beginning of the section.

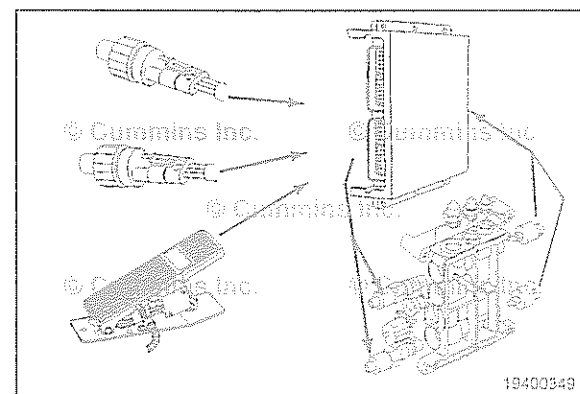
To stop the diagnostic system, move the diagnostic switch to the OFF position, or remove the shorting plug. Turn the vehicle keyswitch to the OFF position.

NOTE: Some OEMs use a shorting plug.



Fault Code Snapshot Data

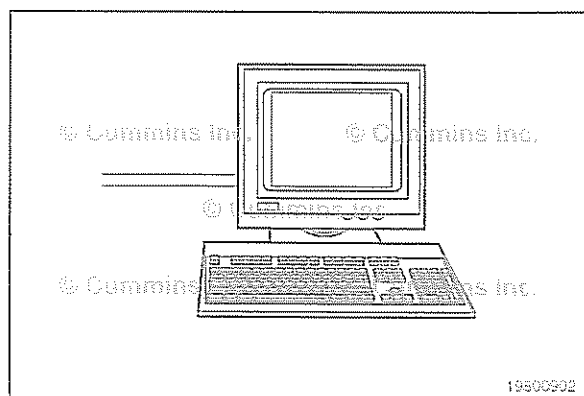
When a diagnostic fault code is recorded in the ECM, ECM input and output data are recorded from all sensors and switches. Snapshot data allow the relationships between ECM inputs and outputs to be viewed and used during troubleshooting.



Control System Relay Drivers

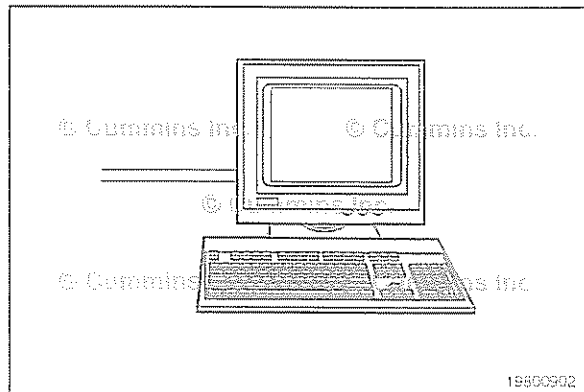
The control system has five relay drivers for customer-supplied relays.

- Overspeed
- Low Oil Pressure
- High Engine Temperature
- Engine Protection Shutdown Has Occurred
- Engine Protection Warning Condition Exists.



Fault Codes — Electronic Service Tool

The electronic service tool can be used to read the fault codes. Connect a personal computer, with the electronic service tool installed, to the engine using the service harness. Refer to the electronic service tool manual for specifics about how to use the tool to read the fault codes.



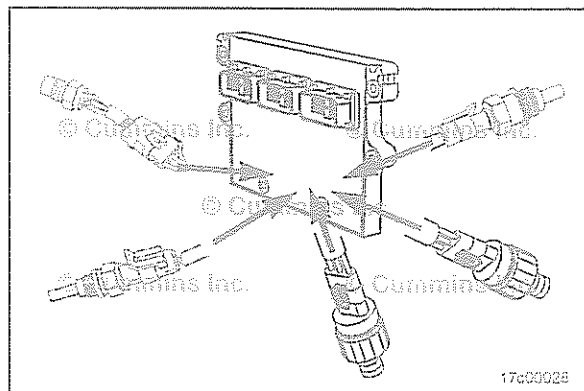
To Clear a Fault Code

Only inactive fault codes can be cleared. There are two ways to clear an inactive fault code:

- The fault acknowledge switch on the operator interface panel
- The electronic service tool.

NOTE: The engine must be shut down to clear inactive shutdown faults.

All fault codes recorded will either be active (fault code is presently active on the engine) or inactive (fault code was active at some time, but is not presently active).



Engine Protection System

QSM11-G engine are equipped with an engine protection system. The system monitors critical engine temperatures and pressures and will log diagnostic faults when an abnormal operating condition occurs. If an out-of-range condition exists and engine derate action is to be initiated, the operator will be alerted by the warning light.

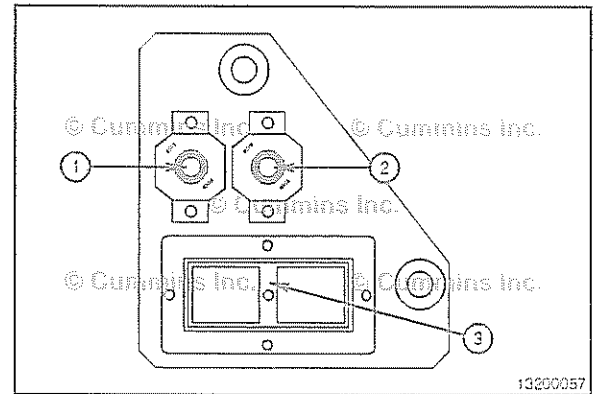
NOTE: The engine protection system will not shut down the engine unless the engine protection shutdown override feature has been selected. If the feature has been selected and the engine does shut down, the engine can be started again by turning off the keyswitch and then turning it back on.

Circuit Breakers

Marine Applications

The QSM11 marine engine is equipped with two circuit breakers, located on the ECM side of the engine.

A 5-amp circuit breaker (1) is used for keyswitched power and a 10-amp circuit breaker (2) is used for nonkeyswitched power. The circuit breaker panel also houses a 40-pin OEM connector (3).



Electromagnetic Interference (EMI)

General Information

Some engine applications utilize accessories (CB radios, mobile transmitters, etc.) that generate and use radio frequency energy that, if not installed and used properly, can cause electromagnetic interference (EMI) conditions to exist between the accessory and Cummins electronic controlled fuel system. Cummins is not liable for any performance problems with either the fuel system or the accessory due to EMI. EMI is not considered by Cummins to be an engine failure and therefore is not warrantable.

System EMI Susceptibility

Your Cummins product has been designed and tested for minimum sensitivity to incoming electromagnetic energy. Testing has shown that there is no engine performance degradation at relatively high energy levels; however, if very high energy levels are encountered, then some noncritical diagnostic fault code logging can occur. The fuel system EMI susceptibility level will protect your engine from most, if not all, electromagnetic energy-emitting devices that meet the Federal Communications Commission legal requirements.

System EMI Radiation Levels

Your Cummins product has been designed to emit minimum electromagnetic energy. Electronic components are required to pass various Cummins and industry EMI specifications. Testing has shown that when the engine is properly installed, it will not interfere with onboard communication equipment or with the vehicle's, equipment's, or vessel's ability to meet any applicable EMI standards and regulated specifications.

If an interference condition is observed, follow the suggestions below to reduce the amount of interference:

1. Locate the receiving antenna as far away from the engine and as high as possible.
2. Locate the receiving antenna as far away as possible from all metal obstructions (e.g., exhaust stacks)
3. Consult a representative of the accessory supplier in your area to:
 - Calibrate accurately the device for proper frequency, power output, and sensitivity (both base and remote site devices must be properly calibrated)
 - Obtain antenna reflective energy data measurements to determine the optimum antenna location
 - Obtain optimum antenna type and mounting arrangement for your application
 - Make sure your accessory equipment model is built for maximum filtering to reject incoming electromagnetic noise.

Notes

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Section 2 - Maintenance Guidelines

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Maintenance Guidelines - Overview

General Information

Cummins Inc. recommends that the engine be maintained according to the Maintenance Schedule in this section.

If the engine is operating in ambient temperatures below -18°C [0°F] or above 38°C [100°F], perform maintenance at shorter intervals. Shorter maintenance intervals are also required if the engine is operated in a dusty environment or if frequent stops are made. Contact your local Cummins Authorized Repair Location for recommended maintenance intervals.

Some of these maintenance procedures require special tools or must be completed by qualified personnel. Contact your local Cummins Authorized Repair Location for detailed information.

If your engine is equipped with a component or accessory not manufactured by Cummins Inc., refer to the component manufacturer's maintenance recommendations.

Use the chart provided in this section as a convenient way to record maintenance performed.

Tool Requirements

General Information

Most of the maintenance operations described in this manual can be performed with common hand tools, metric and SAE wrenches, sockets, and screwdrivers.

The following is a list of special service tools required for some of the maintenance operations:

Tool Part Number:	Description:
3400157	Oil filter wrench
3376592	Torque wrench (valve and injector adjustment)
3163196	Torque wrench adapter (used with 3376592 torque wrench)
3400158	Engine coolant and fuel filter wrench
3822524	Belt tension gauge, click type (v-belts and v-ribbed with 4 or 5 ribs)
3822525	Belt tension gauge, click type (v-ribbed with 6 to 12 ribs)
ST-1138	Belt tension gauge (v-belts)
ST-537	Dial depth gauge
3164019	Chip removing tool
ST-1293	Belt tension gauge (v-ribbed belts)
CC-2800	Refractometer
CC-2602	Coolant test kit

Contact your nearest Cummins Authorized Repair Location for the required service tools.

Maintenance Schedule

General Information

Marine and Industrial Applications

Perform maintenance at whichever interval occurs first. At each scheduled maintenance interval, perform all previous maintenance checks that are due for scheduled maintenance.

Daily or Refueling - Maintenance CheckSection 3

- Fuel-Water Separator - drain
- Lubricating Oil Level - check
- Coolant Level - check
- Drive Belt - inspect
- Air Cleaner Restriction - check
- Seawater Strainer - disassemble, clean, prime, and assemble
- Marine Gear ² - check
- Air Tanks and Reservoirs - drain
- Crankcase Breather Tube - check
- Cooling Fan - inspect
- Charge Air Piping - check

Maintenance Procedures at 250 Hours or 6 Months ¹Section 4

- Coolant Filter ⁴ - change
- Lubricating Oil and Filters ³ - change
- Fuel Filter - replace
- SCA and Antifreeze Concentration - check
- Engine Wiring - check
- Sacrificial Plug ⁵ - check

Maintenance Procedures at 600 Hours or 1 Year ¹Section 5

- Overhead Set - adjust

Maintenance Procedures at 1500 Hours or 1 Year ¹Section 6

- Water Pump - check
- Turbocharger - check
- Engine Mounting Bolts - check
- Air Leaks, Intake and Exhaust Systems - check
- Air Cleaner Assembly (Engine-Mounted) - check
- Cooling System Hoses - check
- Batteries - check
- Air Cleaner Restriction - check
- Engine - clean

Maintenance Procedures at 6000 Hours or 2 Years ¹Section 7

- Cooling System - clean, flush, and fill
- Vibration Damper - check
- Fan Drive Idler Pulley - check
- Fan Hub, Belt Driven - check
- Turbocharger - check
- Air Compressor - check

Reference the appropriate sections for complete inspection and maintenance procedures.

NOTE: These maintenance intervals must be performed at hours or months, whichever interval occurs first. At each scheduled maintenance interval, perform all previous maintenance procedures that are due for scheduled maintenance.

NOTE: Follow the gear manufacturer's recommended maintenance procedures for checking, filling, and changing oil. Some gear manufacturers require an initial oil change at low hours, in addition to their regular maintenance.

NOTE: Fuel burn is the most accurate method for determining oil change intervals. Oil change intervals can be required more frequently, depending on the quality of the oil available. Reference the Oil Drain Interval table for marine and industrial engines in this procedure for additional oil change intervals.

NOTE: Do not change the coolant filter if the SCA concentration level is over three units.

NOTE: Depending upon the quality of electrical bonding and water conditions, increased maintenance of the zinc anode plugs could possibly be necessary.

Oil Drain Intervals

Marine and Industrial Applications

QSM11 Marine Engines - Oil Drain Intervals				
Oil Type	Fuel Burned	Hours	Months	Acceptable Filter
CF-4 (1)	6625 liters [1750 gal]	125	6	LF9001 (4)
CG-4	13,249 liters [3500 gal]	250	6	LF9001 (4)
CH-4 (CES 20071 (2)	15,142 liters [4000 gal]	300	12	LF9001 (4)
CES 20076 (3)	15,142 liters [4000 gal]	300	12	LF9001 (4)
Notes:				
1. CF-4 oil is not recommended and can only be used if better quality oil is not available.				
2. Valvoline® Premium Blue® meets and exceeds API CH-4 and CES20071 specifications.				
3. Valvoline® Premium Blue® 2000 meets and exceeds CES 20076 specifications.				
4. Oil filters must meet CES 10065 specifications. LF9001 meets and exceeds CES 10065 specifications.				

QSM11 Industrial Engines - Oil Drain Intervals by Duty Cycle (hours)				
	Severe	Heavy	Medium	Light
Fuel Consumption	Greater than 3 gallon/hours	10 to 13 gallon/hours	8.5 to 10 gallon/hours	Less than 8.5 gallon/hours
API CF-4	150	200	250	300
API CG-4	250	300	400	500
API CH-4 or ACEA E5	300	350	500	600
CES20076 or CES20077	350	400	550	700
Notes:				
1. Valvoline® Premium Blue® and Premium Blue® 2000 meet 20076 standards.				
2. Limit oil change interval to 250-hour maximum for all oil grades if 40 percent of the operating time is in ambient temperatures above 38°C [100°F] to prevent engine damage due to oil degradation resulting from high oil temperatures.				
3. Cummins does not recommend the use of CE or CD grade oils. If CD or CE oils must be used, make the oil drain intervals half of the API CF-4 values.				
4. Engine oil drain intervals need to be reduced by 50 percent when fuel sulfur content exceeds 0.5 percent.				
5. Oil filters must meet CES 10065 specifications. LF9001 meets and exceeds CES 10065 specifications.				

Maintenance Data

Maintenance Record	
Engine Serial No.:	Engine Model:
Owner's Name:	Equipment Name/Number:

Key to table headings:

- A = Date
- B = km [Miles], Hours or Time Interval
- C = Actual km [Miles] or Hours
- D = Maintenance Check Performed
- E = Check Performed By
- F = Comments

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Section 3 - Maintenance Procedures at Daily Interval

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Daily Maintenance Procedures - Overview

General Information

Preventative maintenance begins with day-to-day awareness of the engine and its system. Before starting the engine, check the oil and coolant levels. Look for:

- Leaks
- Loose or damaged parts
- Worn or damaged belts
- Any change in engine appearance.
- Odor of fuel

Engine Operation Report

The engine must be maintained in top mechanical condition if the operator is to get optimum satisfaction from its use. The maintenance department needs daily running reports from the operator to make necessary adjustments in the time allocated. The daily running report also helps to make provisions for more extensive maintenance work as the reports indicate the necessity.

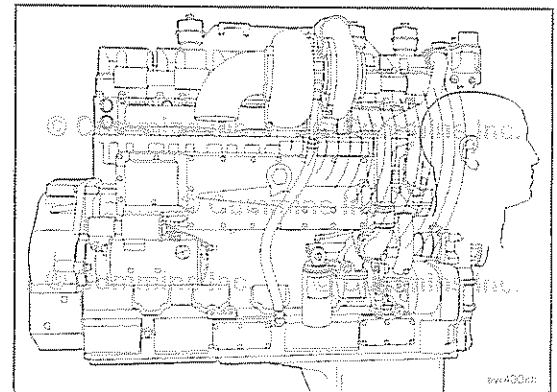
Comparison and intelligent interpretation of the daily report, along with a practical follow-up action, will eliminate most failures and emergency repairs.

Report to the maintenance department any of the following conditions:

- Low lubricating oil pressure
- Low power
- Power increases or engine surge
- Erratic or no accelerator control or response
- Any warning lights flashing or staying on
- Abnormal water or oil temperature
- Unusual engine noise
- Excessive smoke
- Excessive use of coolant, fuel, or lubricating oil
- Any fuel, coolant, or lubricating oil leaks
- Loose or damaged parts
- Worn or damaged belts

Unusual Engine Noise

During daily maintenance checks, listen for any unusual engine noise that can indicate that service is required.



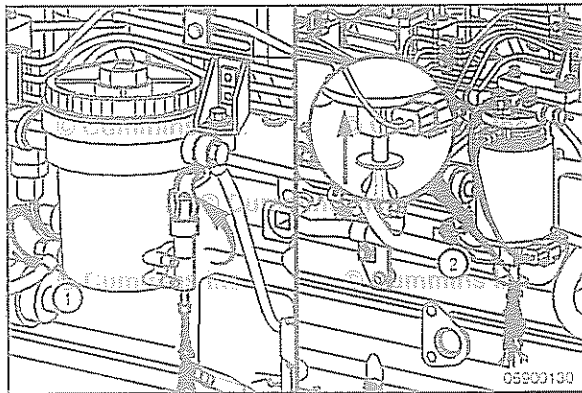
Fuel-Water Separator Drain



Drain the water-fuel separator into a container and dispose of in accordance with local environmental regulations.

Cummins Inc. requires a fuel-water separator or fuel filter be installed in the fuel supply system.

Drain the water and sediment from the separator daily.

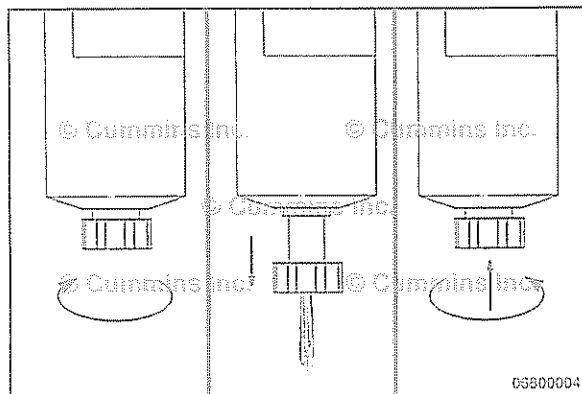


Canister Type

Shut off the engine.

Pull up on the drain valve lever until fluid drains out of the drain tube. Drain the filter sump until clear fuel is visible.

Push up on the drain valve until fluid drains out of the drain tube.



Spin-on Type

Shut off the engine.

Use your hand to open the drain valve. Turn the valve counterclockwise approximately $3\frac{1}{2}$ turns until the valve drops down 25.4mm [1 in] and draining occurs.

Drain the filter sump until clear fuel is visible.



When closing the drain valve, do not overtighten the valve. Overtightening can damage the threads.

To close the valve, lift the valve and turn clockwise until it is hand-tight.

Lubricating Oil Level Maintenance Check



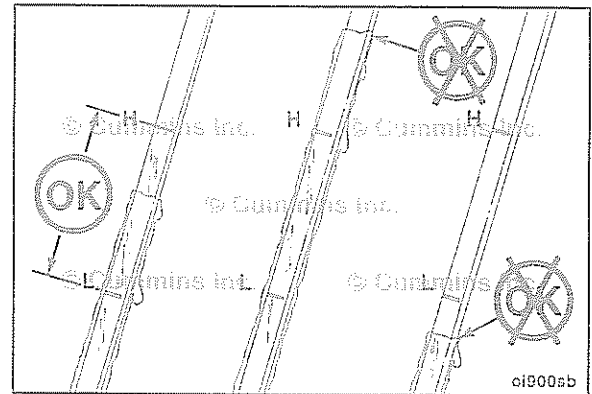
Never operate the engine with oil level below the L (low) mark or above the H (high) mark. Poor engine performance or engine damage can occur.

The engine must be level when checking the oil level to make sure the measurement is correct.

Shut off the engine for an accurate reading.

Wait at least 15 minutes after shutting off the engine to check the oil level. This allows time for the oil to drain into the oil pan.

For additional lubricating oil recommendations and oil pan capacity information, refer to Maintenance Specifications (Section V).



Coolant Level Maintenance Check

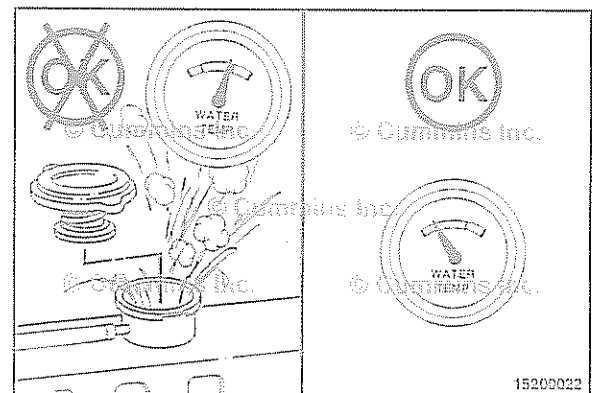


Do not remove a pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



Never use a sealing additive to stop leaks in the cooling system. This can result in cooling system plugging and inadequate coolant flow, causing the engine to overheat.

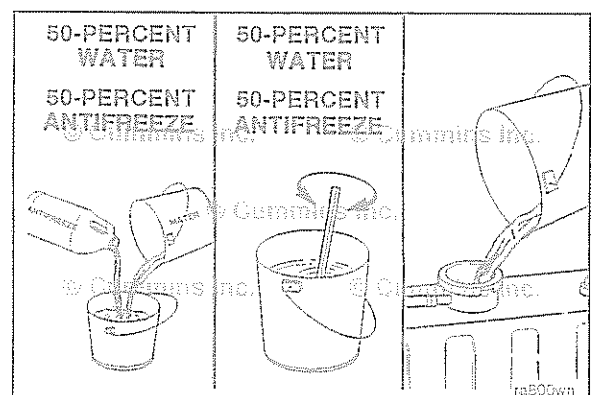
The coolant level must be checked daily.

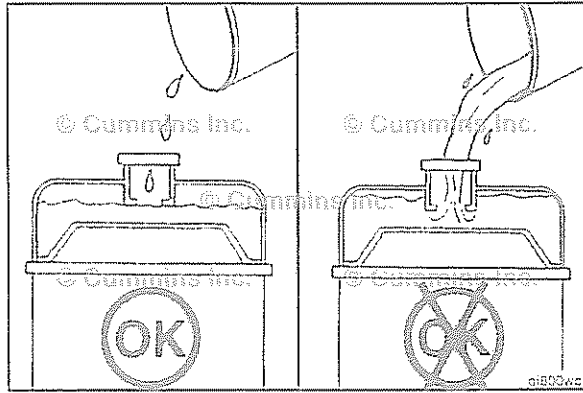


Do not add cold coolant to a hot engine. Engine castings can be damaged. Allow the engine to cool to below 50°C [120°F] before adding coolant.

Make up coolant added to the engine must be mixed with the correct proportions of antifreeze, supplemental coolant additive, and water to avoid engine damage.

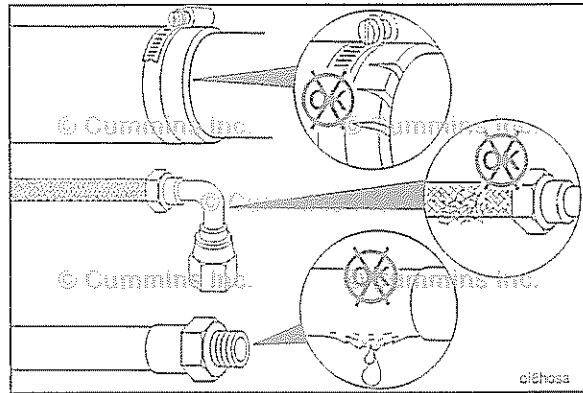
Coolant recommendations and specification details on correct mixing of coolant can be found in Maintenance Specifications (Section V).





Fill the cooling system with coolant to the bottom of the fill neck in the radiator fill or expansion tank.

NOTE: Some radiators have two fill necks, both of which must be filled when the cooling system is drained.

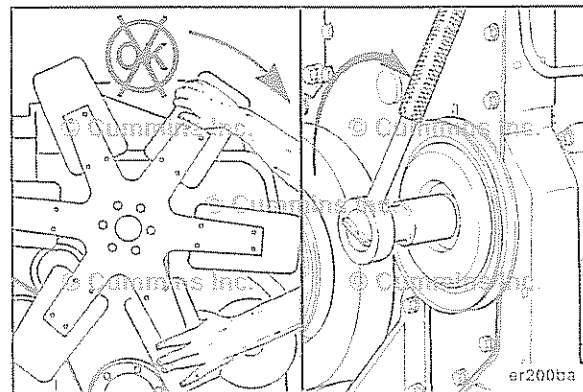


Cooling System Hoses

Maintenance Check

Marine and Industrial Applications

Inspect the cooling system hoses and hose connections for leaks or deterioration. Particles of deteriorated hose can be carried through the cooling system and restrict or clog small passages, especially the heat exchanger and lubricating oil cooler, and partially stop circulation. Replace as necessary.

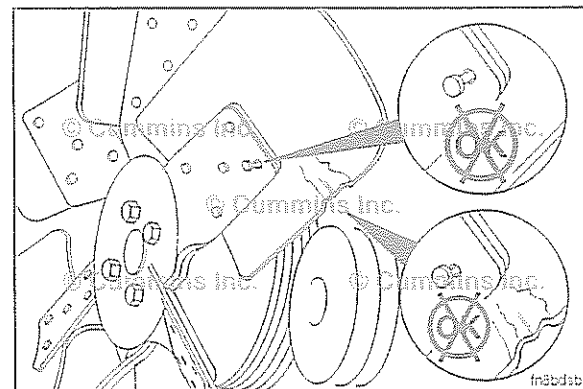


Fan, Cooling

Inspect for Reuse



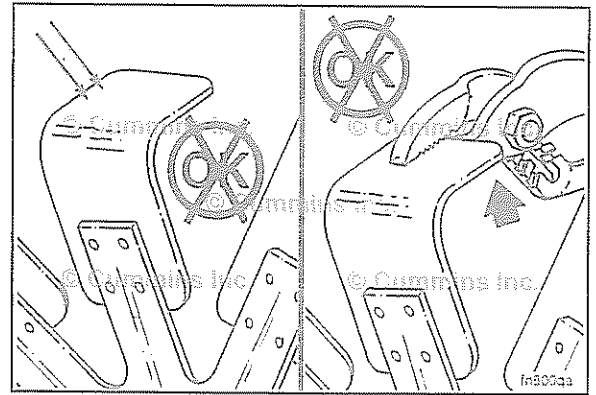
Do not rotate the engine by pulling or prying on the fan. The fan blade(s) can be damaged and cause the fan to fall and cause personal injury or property damage. Use the accessory drive shaft or the crankshaft barring tool to rotate the crankshaft.



A visual inspection of the cooling fan is required daily. Check for cracks, loose rivets, and bent or loose blades. Check the fan to make sure it is securely mounted. Tighten the capscrews, if necessary.

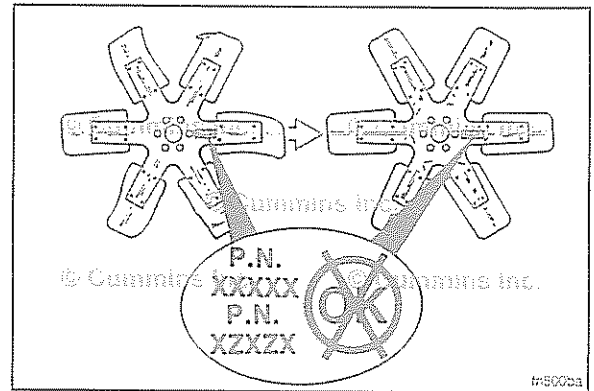
▲WARNING▲

Do not straighten a bent fan blade or continue to use a damaged fan. A bent or damaged fan blade can fail during operation and cause personal injury or property damage.



Replace original equipment fan that is damaged with a fan of the identical part number. Cummins Inc. must approve any other fan changes to be covered under warranty.

Refer to the vehicle or equipment manufacturer's specifications for capscrew torque.



Drive Belts

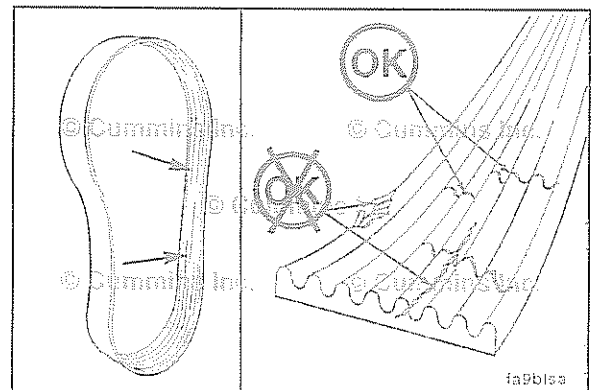
Maintenance Check

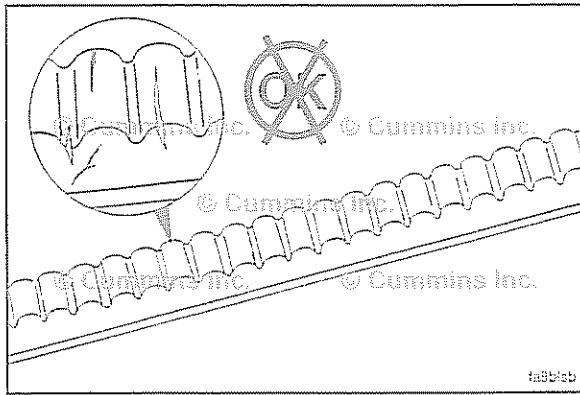
Poly-Vee Belt

Inspect the belts daily. Check the belt for intersecting cracks. Traverse (across the belt width) cracks are acceptable. Longitudinal (direction of belt length) cracks that intersect with transverse cracks are not acceptable. Replace the belt if it is frayed or has pieces of material missing. Refer to Section A for belt adjustment and replacement procedures.

Belt damage can be caused by:

- Incorrect tension
- Incorrect size or length
- Pulley misalignment
- Incorrect installation
- Severe operating environment
- Oil or grease on the side of belts.





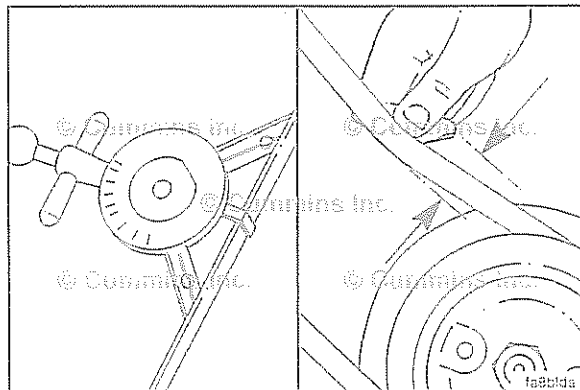
Cogged Belt

Inspect the belts daily. Replace the belts if they are cracked, frayed, or have chunks of material missing. Small cracks are acceptable.

Adjust the belts that have a glazed or shiny surface, which indicates belt slippage. Correctly installed and tensioned belts will show even pulley and belt wear. Refer to Section A for belt adjustment and replacement procedures.

Belt damage can be caused by:

- Incorrect tension
- Incorrect size or length
- Pulley misalignment
- Incorrect installation
- Severe operating environment
- Oil or grease on the belts

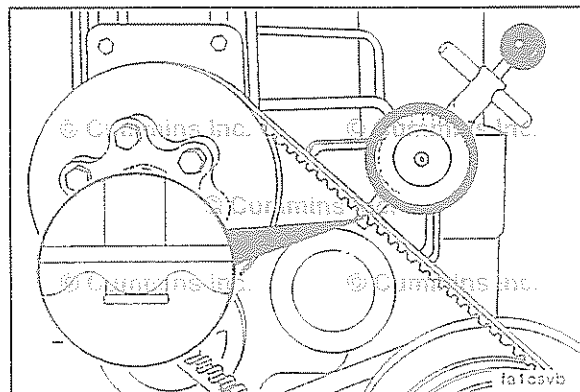


Measure the belt tension in the center span of the pulleys. Refer to the Belt Tension Chart in Section V for the correct gauge and tension value for the belt width used.

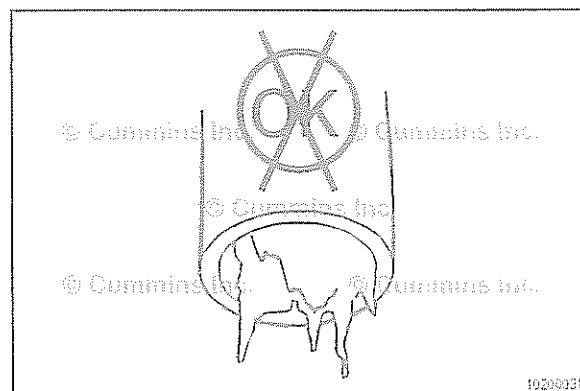


An alternate method (deflection method) can be used to check belt tension by applying 110 N [25 lbf] force between the pulleys on v-belts. If the deflection is more than one belt thickness per foot of pulley center distance, the belt tension must be adjusted.

Refer to Section A for adjustment procedures.



For cogged belts, make sure that the belt tension gauge is positioned so that the center tensioning leg is placed directly over the high point (hump) of a cog. Other positioning will result in incorrect measurement.



Crankcase Breather Tube Maintenance Check

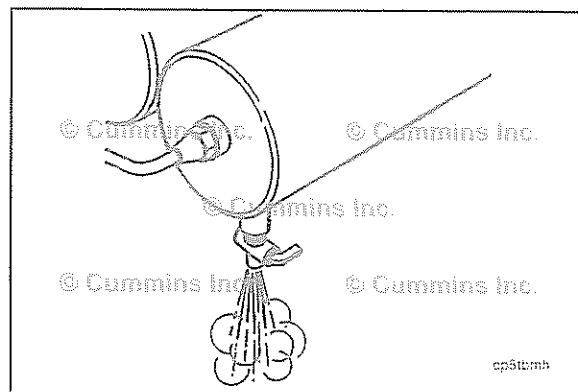
Inspect the breather tube for sludge, debris, or ice in the tube.

Inspect the tube more frequently in icy conditions.

Air Tanks and Reservoirs

Drain

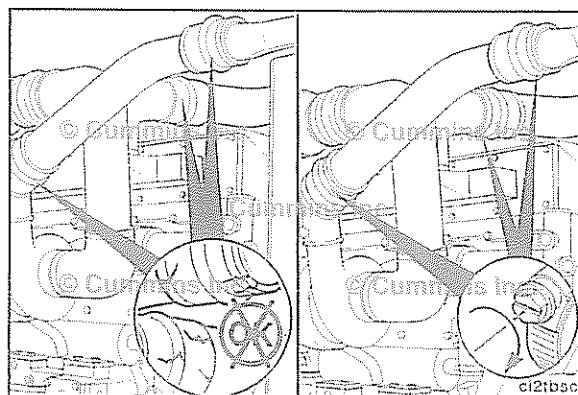
If automatic purging or spitter valves are used, confirm the valves are operating correctly. If a manual drain valve is used on the wet tank, open the draincock on the wet tank to drain any moisture accumulated in the air system. If oil is present, the air compressor system **must** be checked. Contact your Cummins Authorized Repair Location.



Charge-Air Piping

Maintenance Check

Inspect the charge-air piping and hoses for leaks, holes, cracks, or loose connections. Tighten the hose clamps if necessary. Refer to the vehicle or equipment manufacturer's specifications for the correct torque value.



Air Cleaner Restriction

Maintenance Check

Mechanical Indicator

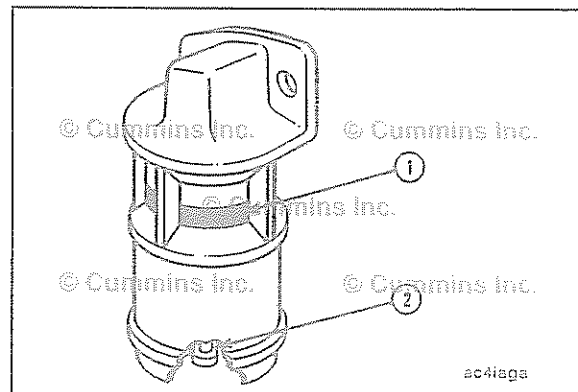
NOTE: Do not remove the felt washer from the indicator. The felt washer absorbs moisture.

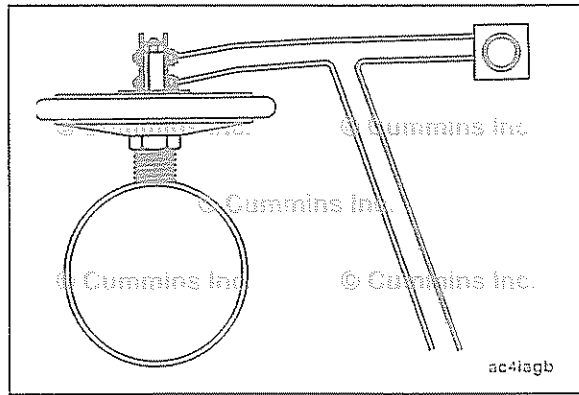
A mechanical restriction indicator is available to indicate excessive air restriction through a dry-type air cleaner. This instrument can be mounted in the air cleaner outlet or on the instrument panel. The red flag (1) in the window gradually rises as the cartridge loads with dirt. After changing or replacing the cartridge, reset the indicator by pushing the reset button (2).

Restriction or vacuum indicators need to be installed as close as possible to the turbocharger air inlet in order to obtain a true indication of restrictions.



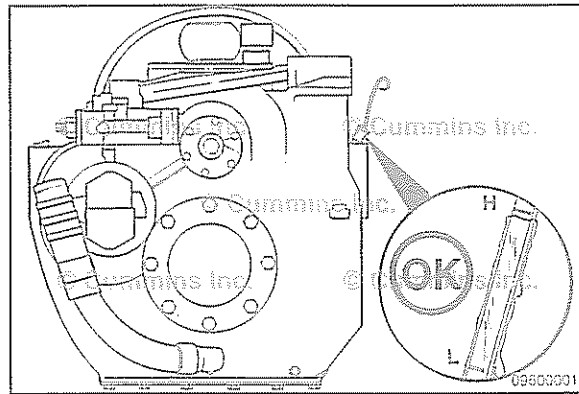
Never operate the engine without an air cleaner. Intake air must be filtered to prevent dirt and debris from entering the engine and causing premature wear.





Vacuum Indicator

Vacuum switches actuate a warning light on the instrument panel when the air restriction becomes excessive.



Marine Gear

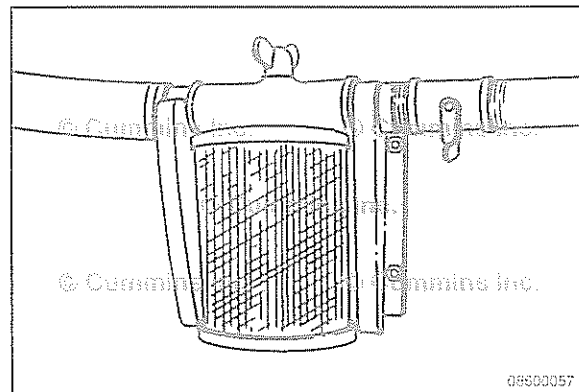
Maintenance Check

Marine Applications

Check the marine gear oil level daily.

Refer to the OEM's recommendations for the gear oil requirements.

NOTE: Different models of marine gears will have the marine gear oil dipstick and fill cap in different locations. The location shown is typical.

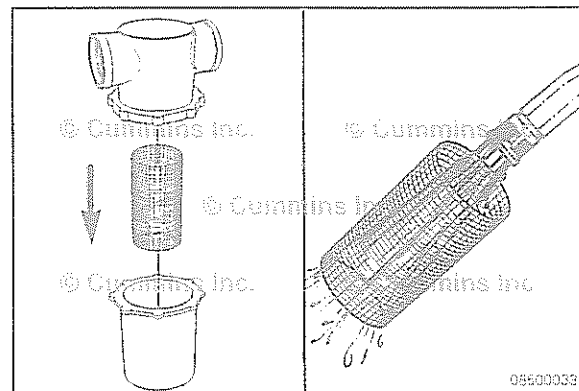


Sea Water Strainer

General Information

Marine Applications

NOTE: Most sea water systems for heat exchanger cooled engines use a sea water strainer. The strainer removes debris from the sea water before it enters the sea water pump.



NOTE: Sea water strainer arrangements may differ.

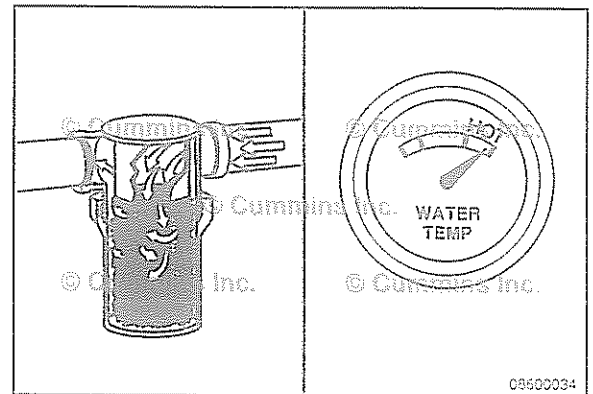
Inspect the sea water strainer daily for any foreign objects that could restrict water flow.



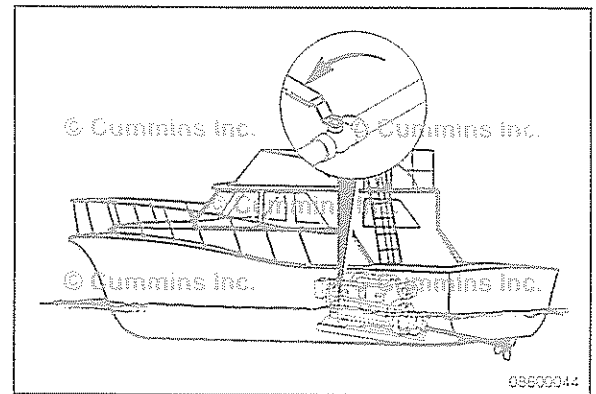
CAUTION

A restricted or clogged sea water strainer can result in hotter than normal, or overheated, engine coolant and marine gear oil temperatures.

For more detailed information, refer to Sea Water Strainer Cleaning in this section.



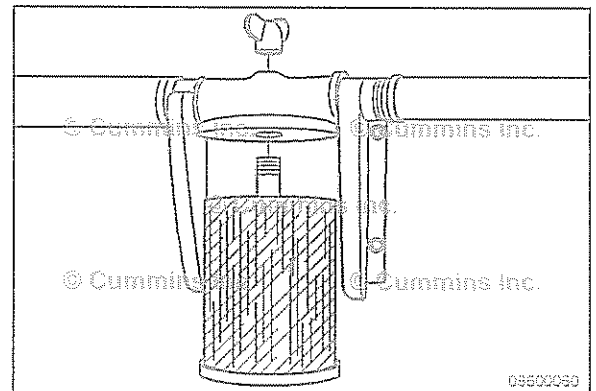
NOTE: If the sea water strainer is below the water level, close the sea water inlet valves before servicing the sea water strainer.



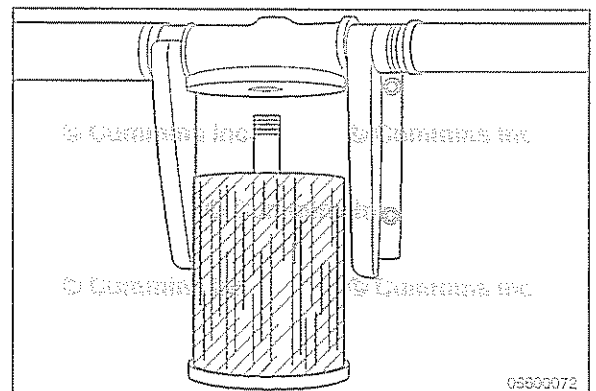
Disassemble

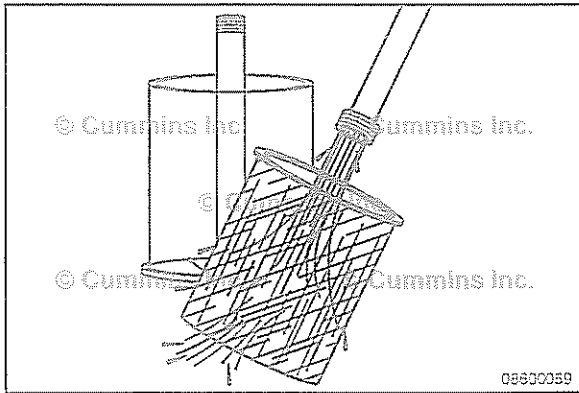
Marine Applications

Loosen and remove the sea water strainer wing nut(s) as required.

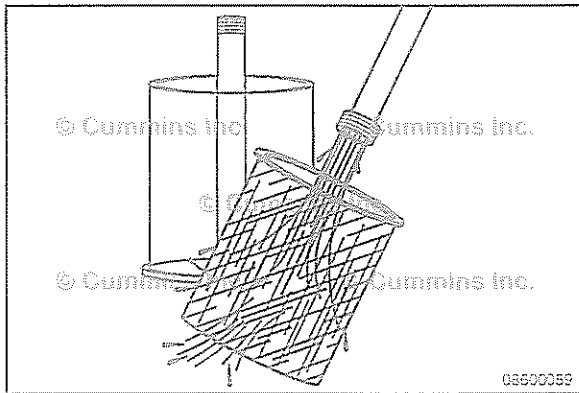


Remove the sea water strainer assembly. Be careful not to damage o-ring. Retain and inspect for damage.





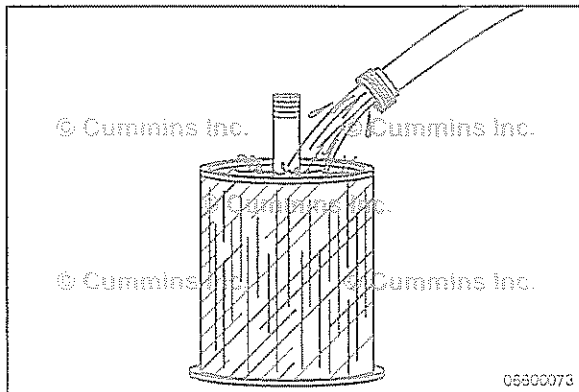
Remove the sea water strainer basket.



Clean

Marine Applications

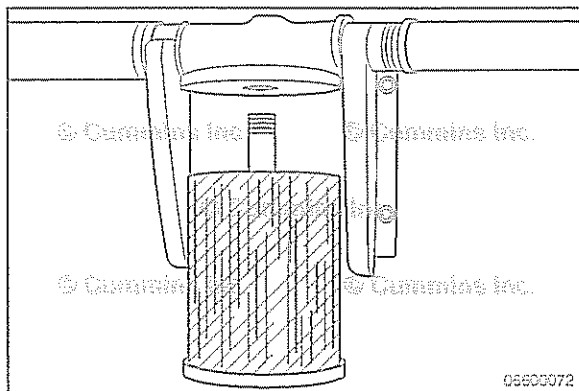
Empty all debris from the sea water strainer basket.
Clean the sea water strainer.



Prime

Marine Applications

Prime the sea water strainer with clean water.



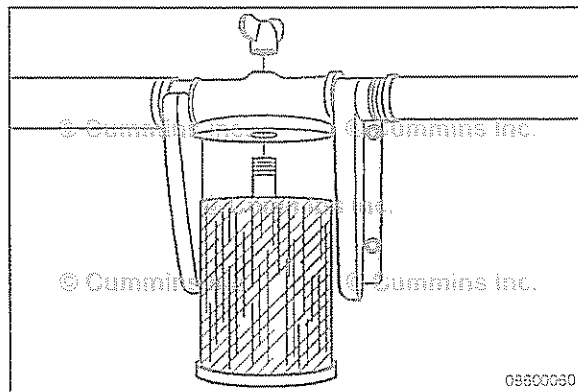
Assemble

Marine Applications

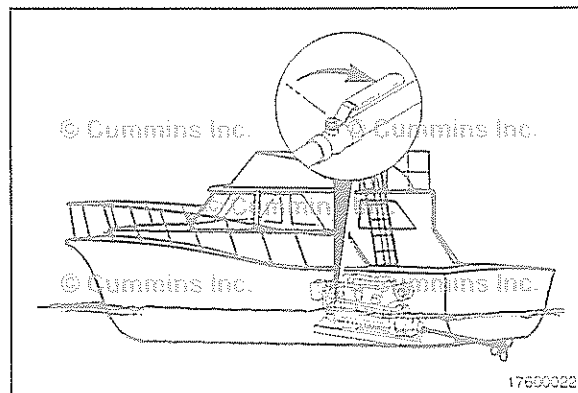
Install the sea water strainer.

Inspect o-ring seal for tears or damage. Replace if necessary. Be sure o-ring is seated properly.

Install and tighten the wing nut(s).



Open the sea water inlet valves.



This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper appears to be a standard notebook page or a sheet of stationery. There is no handwriting or other markings on the page.

Section 4 - Maintenance Procedures at 250 Hours or 6 Months

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Marine Applications.....	4-9
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Maintenance Check.....	4-9
Supplemental Coolant Additive (SCA).....	4-9
Antifreeze.....	4-9

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Maintenance Procedures - Overview

General Information

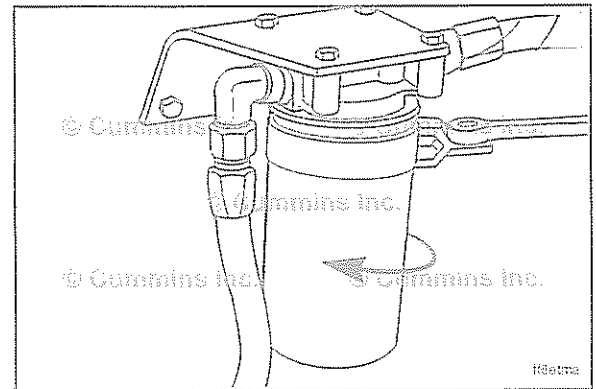
All maintenance checks and inspections listed in previous maintenance intervals must also be performed at this time, in addition to those listed under this maintenance interval.

Fuel Filter (Spin-On Type)

Remove

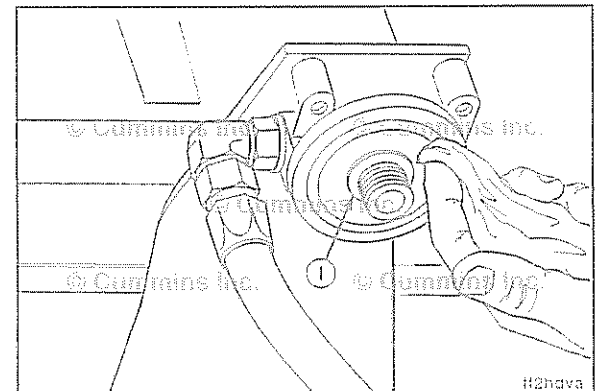
Clean the area around the fuel filter and fuel filter head.

Remove the fuel filter with filter wrench, Part Number 3376807, or equivalent.



Remove the thread adapter sealing ring (1).

Use a clean, lint-free towel to clean the gasket surface on the fuel filter head.



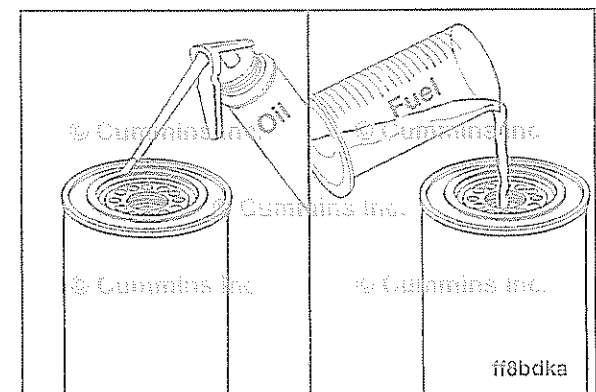
Install

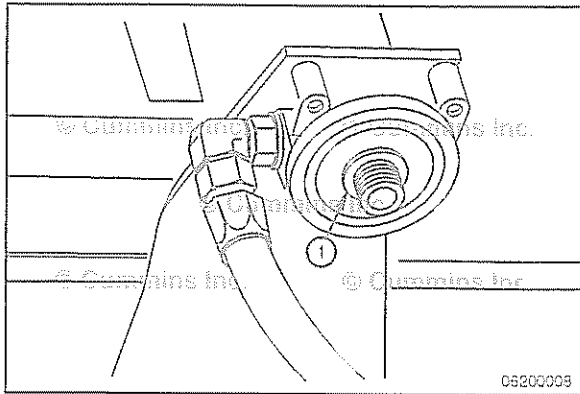
Use the correct filter(s) for your engine. Refer to Procedure 018-024 (Cummins/Fleetguard® Filter Specifications) in Section V for the fuel filter specifications.

Clean the area around the fuel filter and fuel filter head.

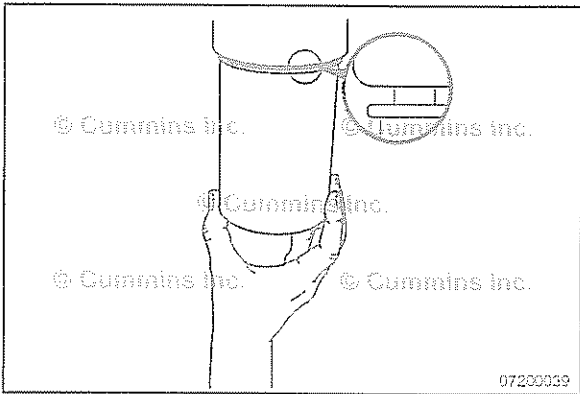
Lubricate the fuel filter gasket with clean engine oil.

Fill the new fuel filter with clean diesel fuel.





Install the new thread adapter sealing ring (1) supplied with the new filter.



CAUTION

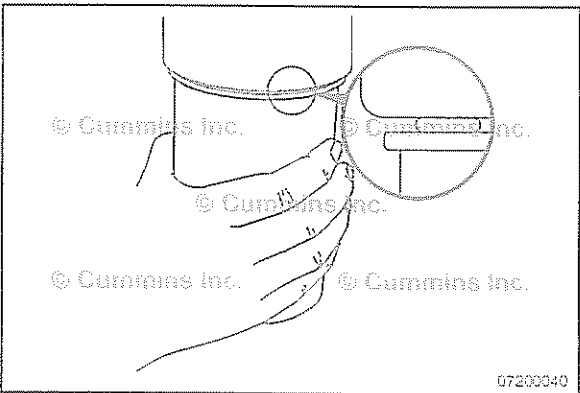
If the filter canister is damaged in any way, do not use it. Dents or scrapes can lead to a rupture or premature failure of the filter.

CAUTION

Mechanical overtightening of the filter can distort the threads and damage the fuel filter seal.

Install the filter on the filter head.

Turn the filter until the gasket contacts the filter head surface.



Tighten the filter, by hand, an additional $\frac{1}{2}$ to $\frac{3}{4}$ of a turn after the gasket contacts the filter head surface, or as specified by the filter manufacturer.

Lubricating Oil and Filters

Drain

▲WARNING▲

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused dispose of in accordance with local environmental regulations.

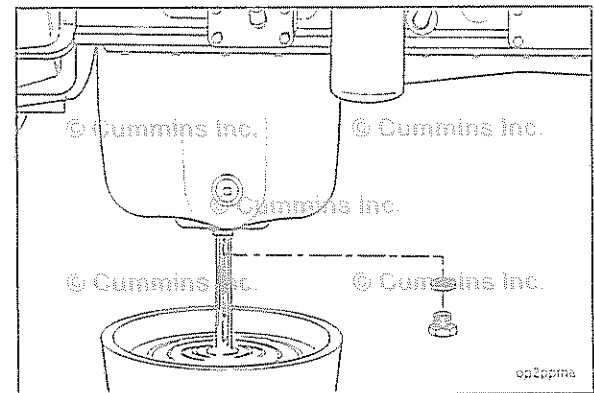
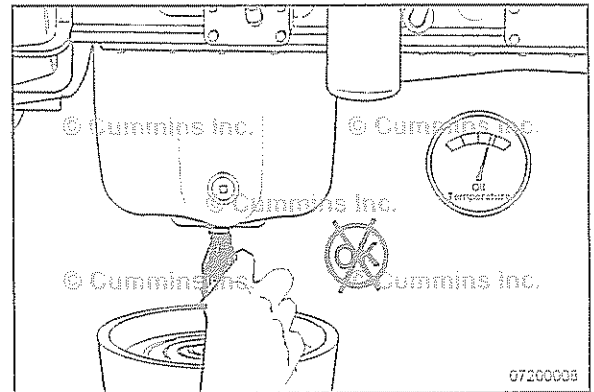
▲WARNING▲

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

Change the lubricating oil and filter at the specified oil change interval. Refer to the Procedure 102-002 (Maintenance Schedule) in Section 2 to find the correct change interval for your application.

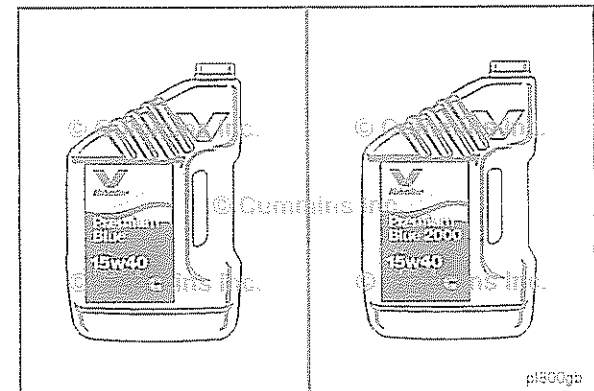
Operate the engine until the water temperature reaches 60°C [140°F]. Shut off the engine. Remove the oil drain plug from the bottom of the lubricating oil pan. Do not remove the plugs on either side of the oil pan to drain the oil. They will not allow the oil to drain completely.

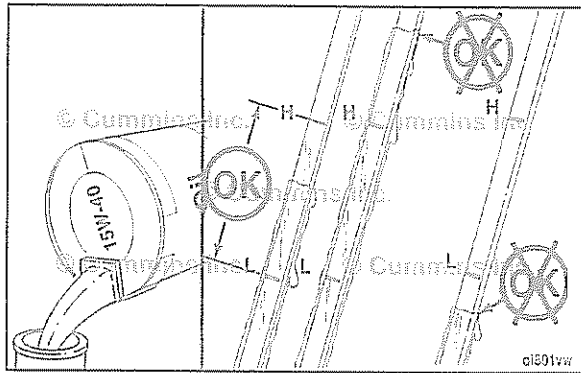
NOTE: Fittings used in the bottom drain opening of the oil pan other than Cummins-specified parts must not exceed the size or weight limits. Refer to Procedure 018-017 (Lubricating Oil System Specifications) in Section V.



Fill

Use a high-quality multiviscosity oil such as Valvoline® Premium Blue™, Valvoline® Premium Blue™ 2000, or equivalent. Choose the correct oil for your operating climate as outlined in Procedure 018-003 (Cummins Recommendations and Specifications) in Section V.

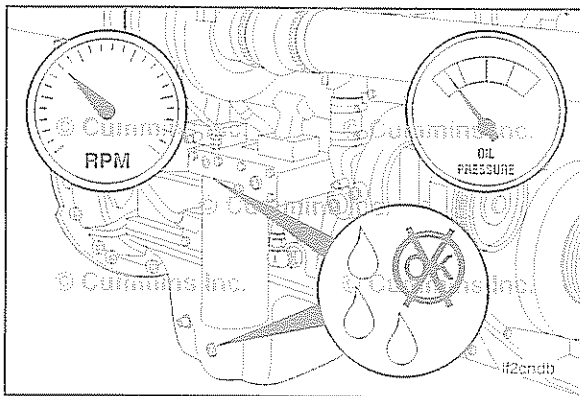




Fill the engine with clean oil to the H (high) mark on the dipstick. For total system capacity including filter, refer to Procedure 018-017 (Lubricating Oil System Specifications) in Section V.



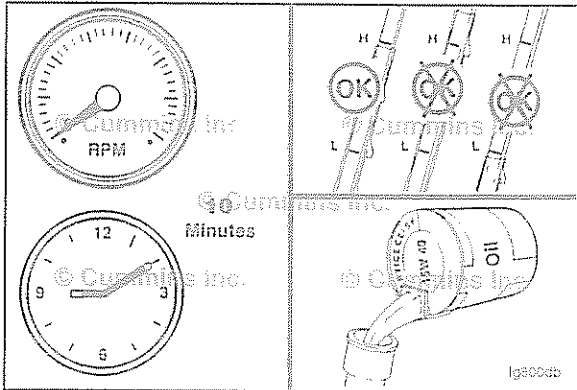
NOTE: Check the Centinel™ makeup tank oil level, if equipped.



⚠CAUTION⚠

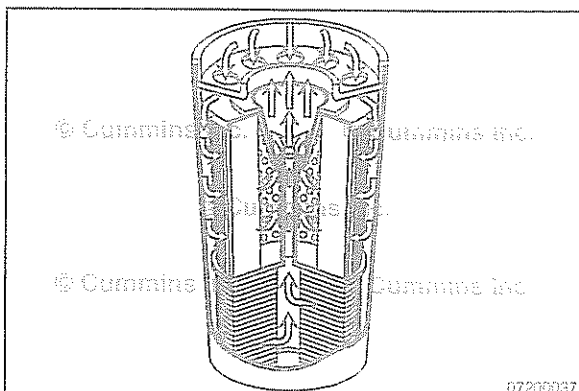
Engine oil pressure must be indicated on the gauge within 15 seconds after starting. If oil pressure is not registered within 15 seconds, shut off the engine immediately to avoid engine damage. Confirm the correct oil level is in the oil pan.

Operate the engine at idle speed to inspect for leaks at the filter and the drain plug.



Shut off the engine. Wait approximately 10 minutes for the oil to drain back from the upper parts of the engine to the oil pan.

Check the oil level again. Add oil as necessary to bring the level up to the "H" (high) mark on the dipstick.



Lubricating Oil Filter (Spin-On) General Information

A combination oil filter is present on the engine. The upper portion of the combination filter contains the full-flow filter element while the lower portion contains the bypass filter element.

The venturi design has been optimized to distribute oil flow during normal engine operation; however during extreme engine oil conditions (additive dropout, contamination, soot etc.) the improvements increase the overall filtration effectiveness.

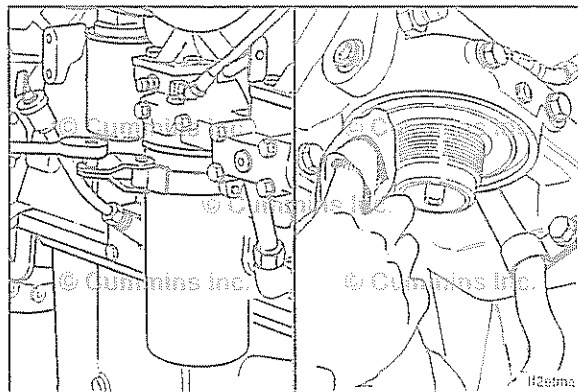
Oil circulates through the full flow and bypass portions of the filter, into the filter head, and is directed to the engine's main oil rifle.

Remove

Clean the area around the lubricating oil filter head.

Use oil filter wrench, Part Number 3375049, or equivalent, to remove the filter.

Clean the gasket surface of the filter head. The o-ring can stick on the filter head. Make sure it is removed.



Install

CAUTION

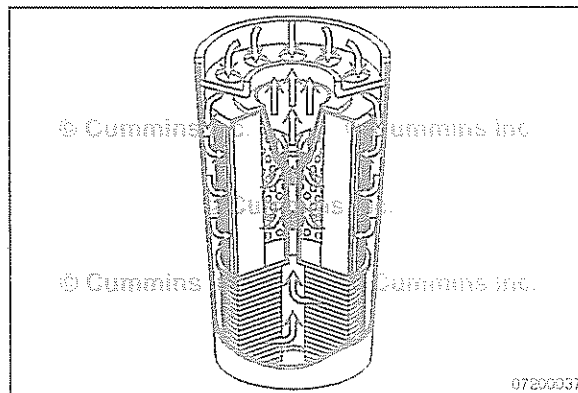
Do not replace LF9001 lubricating filter with extended length LF9000 lubricating filter without verifying adequate clearance with suspension and/or chassis at extremes of travel. The engine can be damaged.

Use the correct oil filter for your engine. Cummins Inc. requires a lubricating oil filter(s) be used that meets the specifications given in Procedure 018-024 (Cummins/Fleetguard Filter Specifications) in Section V.

Combination lubricating filter

Cummins Part Number 3101868 or Fleetguard® Part Number LF9000.

Cummins Part Number 3101869 or Fleetguard® Part Number LF9001.



CAUTION

Fill the lubricating oil filter with clean lubricating oil. The lack of lubrication during the delay until the filter is pumped full of oil at start-up is harmful to the engine.



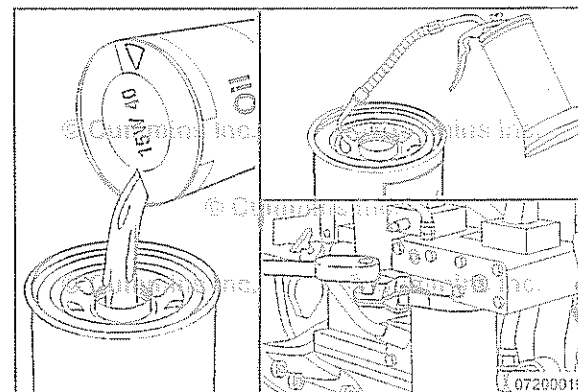
CAUTION

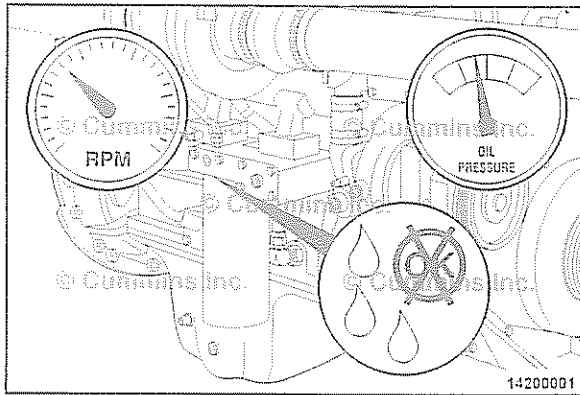
Mechanical overtightening of the filter can distort the threads or damage the filter element seal.

Fill the filter with clean oil. Choose the correct oil for your operating climate as outlined in Procedure 018-003 (Cummins Recommendations and Specifications) in Section V. The LF9001 filter capacity is 2.6 liters [0.7 gal]. The LF9000 filter capacity is 3.0 liters [0.8 gal]. Apply a light film of oil to the gasket sealing surface before installing the new filter.

Install the filter on the filter head. Tighten the filter until the gasket contacts the filter head surface.

Use oil filter wrench, Part Number 3375049, or equivalent, to tighten the filter to the specifications supplied with the filter.

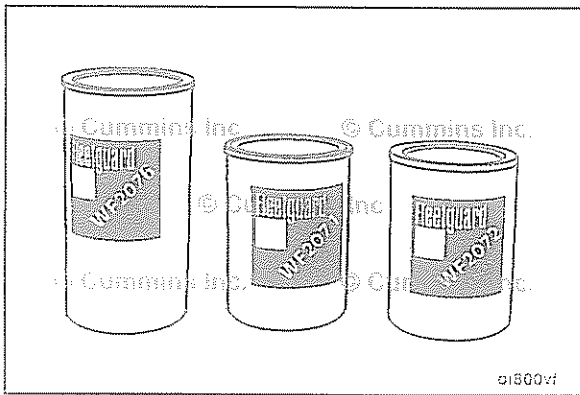




CAUTION

Engine oil pressure must be indicated on the gauge within 15 seconds after starting. If oil pressure is not registered within 15 seconds, shut off the engine immediately to avoid engine damage. Confirm the correct oil level in the oil pan.

Operate the engine at idle speed for three minutes to inspect for leaks at the filter(s).



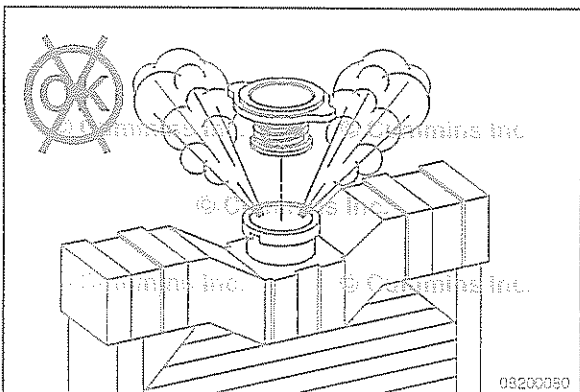
Coolant Filter

General Information

Change the coolant filter at every oil and filter change interval unless the supplemental coolant additive (SCA) level is over 3 units. See Supplemental Coolant Additive - Maintenance check in this section.

NOTE: The SCA level must be tested every six months.

The correct coolant filter to be used is determined by the total cooling system capacity and other operational factors. For the correct filter selection, Refer to Coolant Recommendations and Specifications in Maintenance, Procedure 018-004, in Section V.

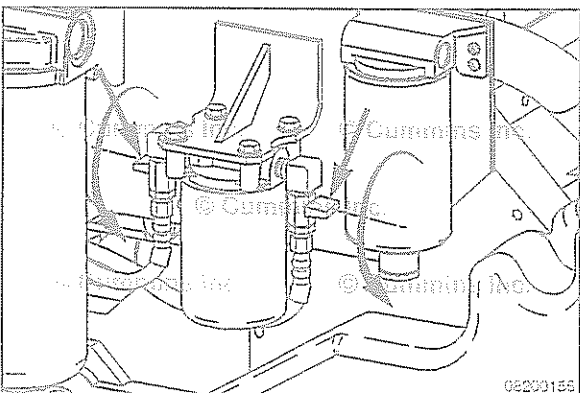


Remove

WARNING

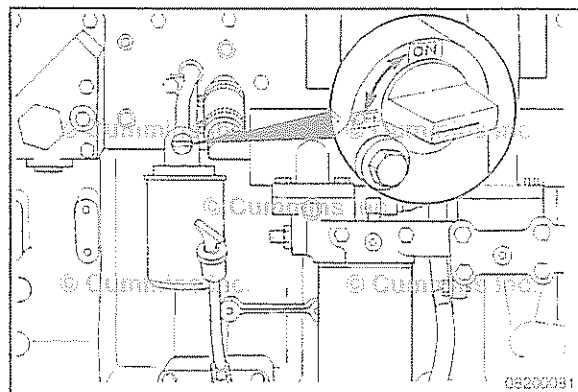
Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

Remove the cooling system pressure cap.



Turn the valve on the remote-mounted coolant filter head to the OFF position.

Turn the valve on the coolant filter head to the OFF position.

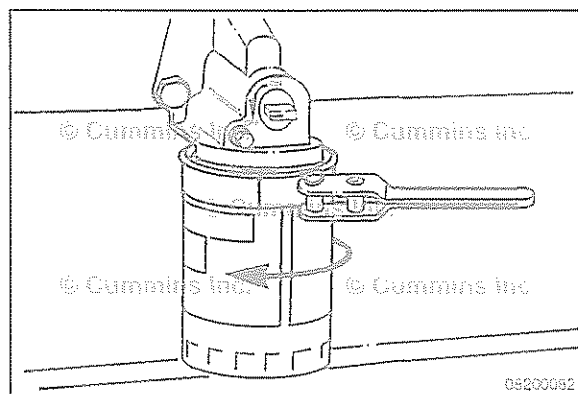


▲WARNING▲

A small amount of coolant can leak when servicing the coolant filter with the shutoff valve in the OFF position. To reduce the possibility of personal injury, avoid contact with hot coolant.

Remove and discard the coolant filter.

Using a lint free cloth, clean the coolant filter head gasket's surface.

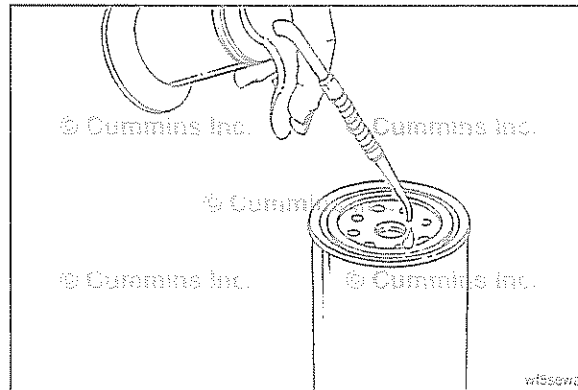


Install

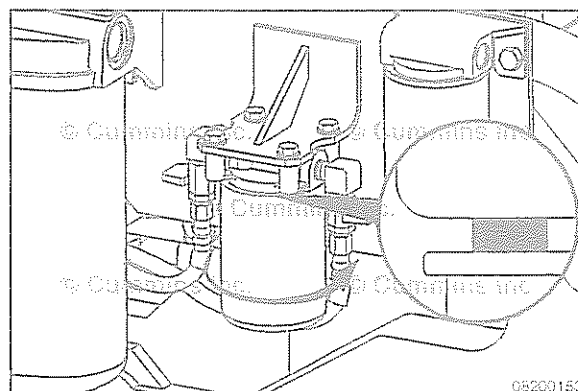
▲CAUTION▲

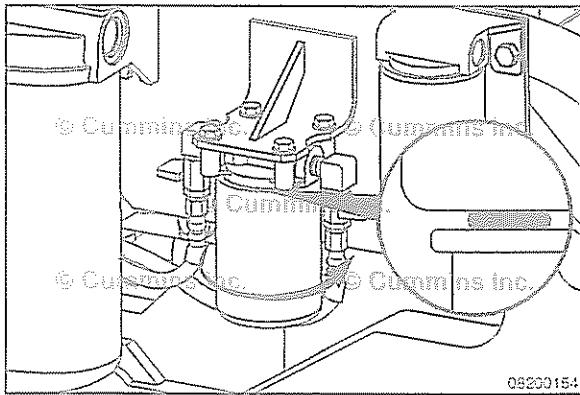
If the filter canister is damaged in any way, do not use it. Dents or scrapes can lead to a rupture or premature failure of the filter.

Apply a thin film of clean engine oil to the gasket sealing surface before installing the new coolant filter.



Install the new filter on the filter head. Tighten the filter until the gasket contacts the filter head surface.

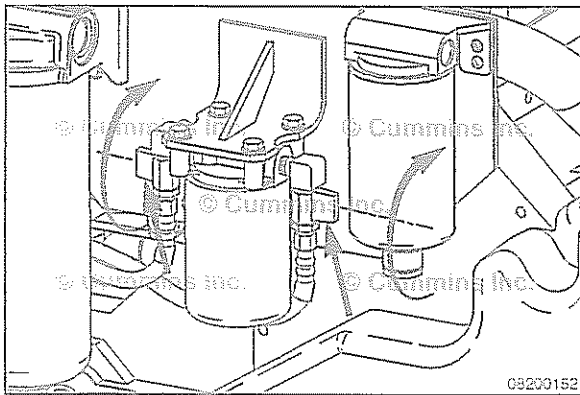




CAUTION

Mechanical overtightening can distort the filter threads or damage the filter head.

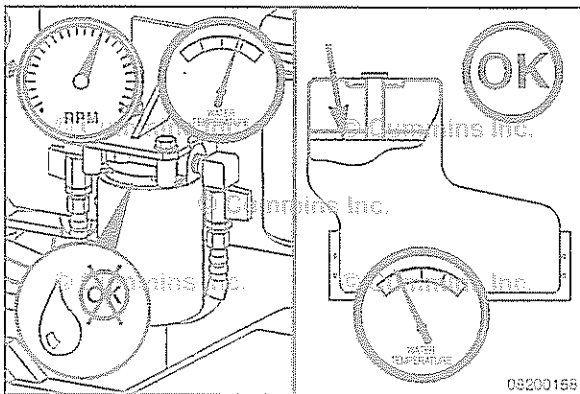
Tighten the filter an additional $\frac{1}{2}$ to $\frac{3}{4}$ of a turn, or as specified by the filter manufacturer.



CAUTION

The valve(s) must be in the ON position to reduce the possibility of engine damage.

Turn the valve on the filter head to the ON position, and install the coolant system pressure cap.



Operate the engine and check for leaks.

After the air has been purged from the system, check the coolant level again.

Supplemental Coolant Additive (SCA) and Antifreeze Concentration

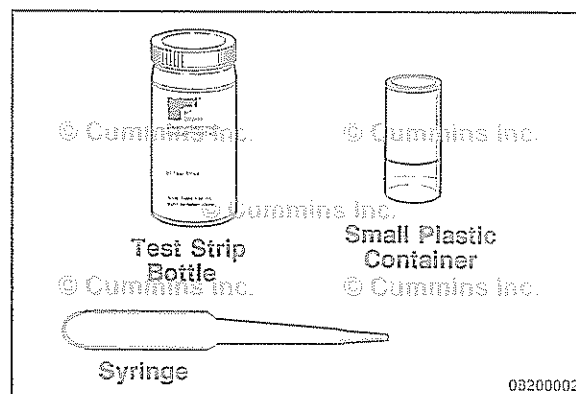
Maintenance Check

Supplemental Coolant Additive (SCA)

Check the SCA concentration level:

- At least twice a year
- At every subsequent oil drain interval if the concentration is above 3 units
- Whenever coolant is added to the cooling system between filter changes.

Use Fleetguard® coolant test kit, Part No. CC2602, to check the SCA concentration level. Instructions are included with the test kit. Refer to Coolant Recommendations and Specifications in Maintenance Specifications (Section V) for the correct SCA and antifreeze level.



Antifreeze



Overconcentration of antifreeze or use of high-silicate antifreeze can damage the engine.

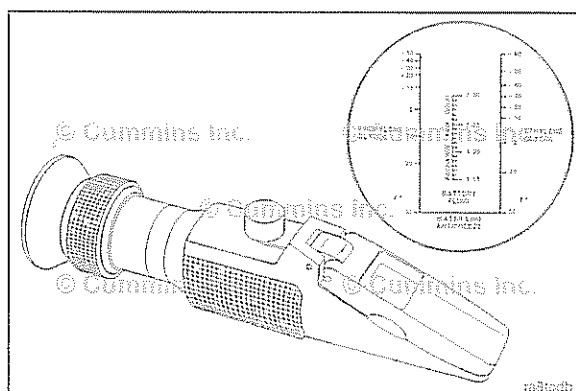
Check the antifreeze concentration. Use a mixture of 50-percent water and 50-percent ethylene glycol or propylene glycol-based antifreeze to protect the engine to -32°C [-26°F] year-around.

The Fleetguard® refractometer, Part Number C2800, provides a reliable, easy-to-read, and accurate measurement of freezing point protection and glycol (antifreeze) concentration.

Antifreeze is essential in every climate.

Antifreeze broadens the operating temperature range by lowering the coolant freezing point and by raising its boiling point.

The corrosion inhibitors also protect the cooling system components from corrosion and prolong component life.

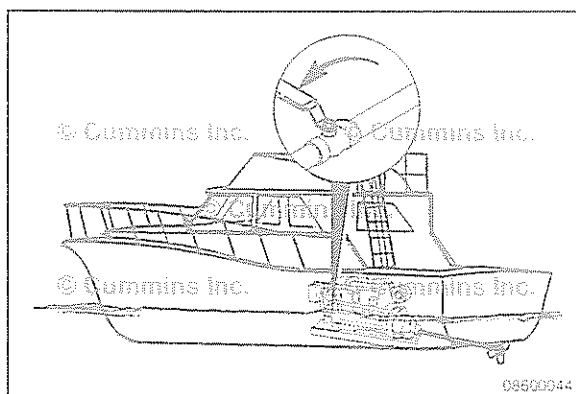


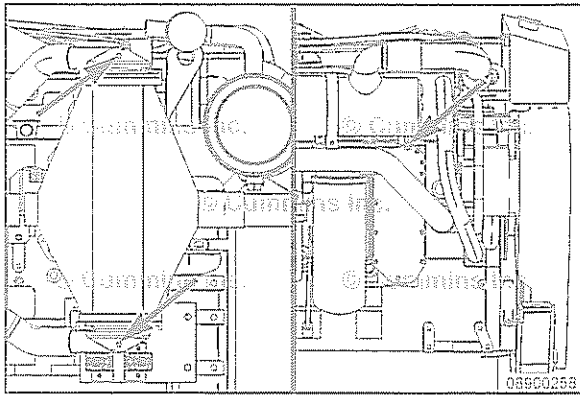
Zinc Anode

Remove

Marine Applications

Shut off the sea water inlet valve on the vessel hull.



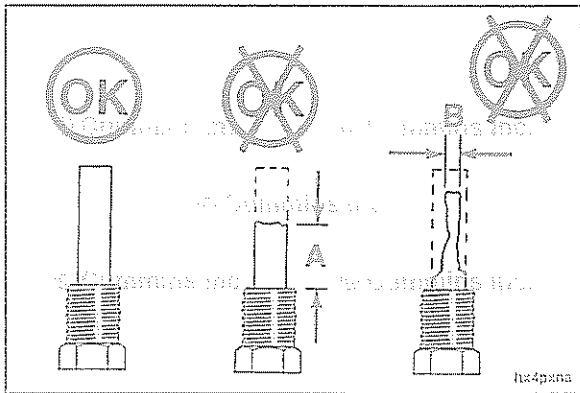


CAUTION

In some cases, it will be necessary to hold the welded fitting on the heat exchanger with an additional wrench to prevent damage to the heat exchanger.

Remove the zinc plug(s).

NOTE: The illustration shown is generic. For engine-specific locations, see to Section E.

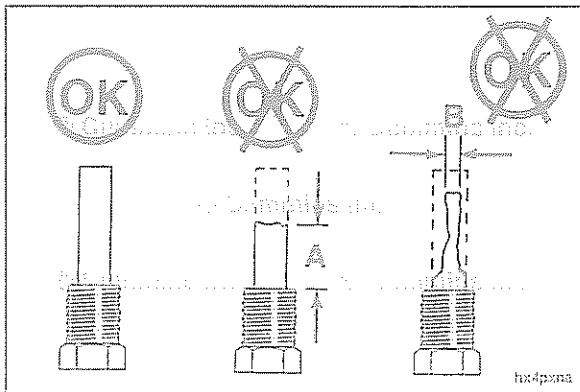


Inspect for Reuse

Marine Applications

Measure the length of the zinc plug(s) removed from the heat exchanger. Replace any zinc plug if it is 50 percent or more eroded.

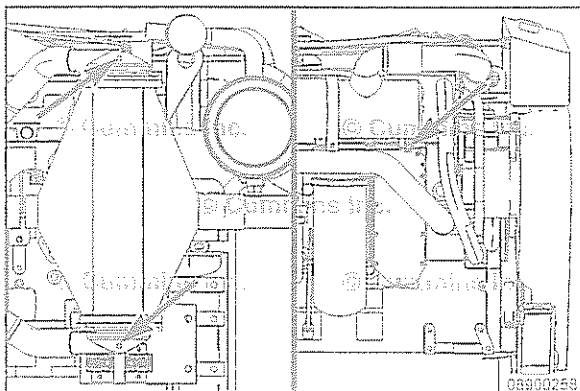
NOTE: The frequency of change depends on the chemical reaction of the sea water circulated through the heat exchanger.



Zinc Plug Erosion Limits

	mm		in
A	19.0	MIN	0.750
New	51.0	MAX	2.000
B	6.4	MIN	0.250
New	16.0	MAX	0.630

NOTE: If any zinc plug does not meet specification, replace it with a new zinc plug.



Install

Marine Applications

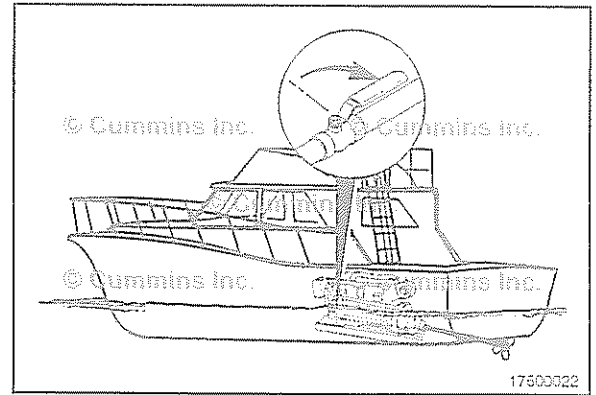
Install and tighten the zinc plug(s).

NOTE: Do not use Teflon™ tape or pipe sealant on the threads of the sacrificial zinc plugs. Use of any type of sealant will insulate the zinc anode and prevent a good ground.

Torque Value:

1/4-inch Plug	12 N•m	[9 ft-lb]
3/8-inch Plug	22 N•m	[17 ft-lb]
1/2-inch Plug	30 N•m	[22 ft-lb]

Open the sea water valve on the vessel hull.
Operate the engine and check for leaks.



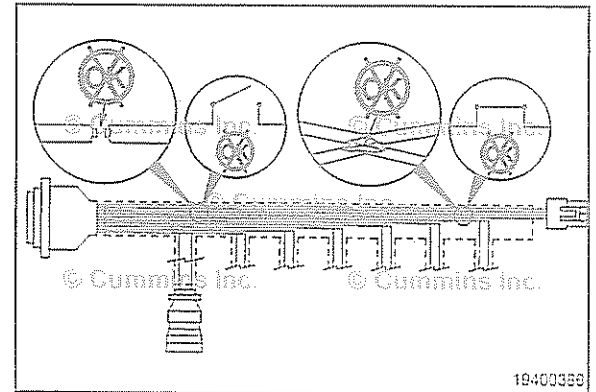
Engine Wiring Harness Maintenance Check

Marine and Industrial Applications



Never touch any wiring connection when the keyswitch is in the ON position. An electrical shock or personal injury can result.

Inspect all wiring connections and wiring harnesses for damage. Faulty wiring can cause improper engine operation and poor performance.



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Section 5 - Maintenance Procedures at 1500 Hours

Section Contents

	Page
Maintenance Procedures - Overview	5-1
General Information.....	5-1
Overhead Set	5-1
Measure.....	5-1



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Maintenance Procedures - Overview

General Information

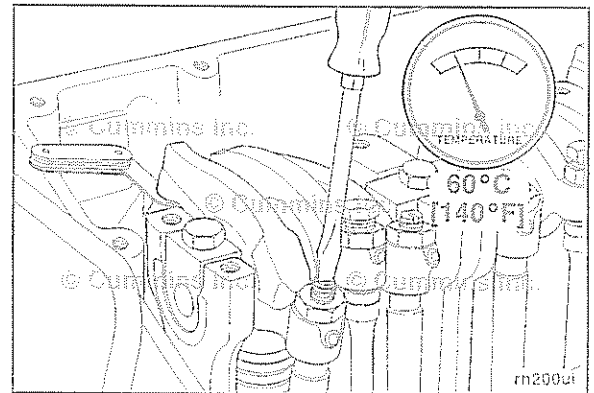
All maintenance checks and inspections listed in previous maintenance intervals must also be performed at this time, in addition to those listed under this maintenance interval.

Overhead Set

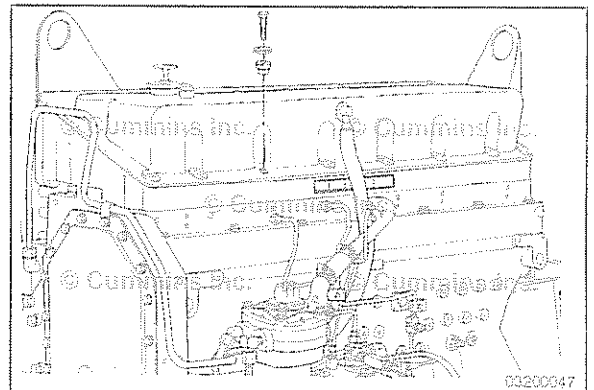
Measure

General Information

All overhead lash measurements must be made when the engine is cold. Stabilized coolant temperature must be at 60°C [140°F] or below.



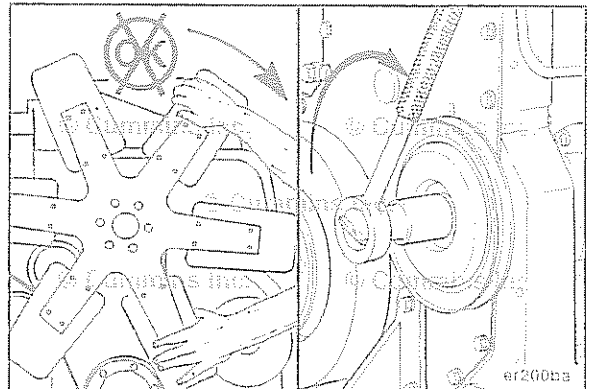
Remove the rocker lever cover. Refer to Procedure 003-011.

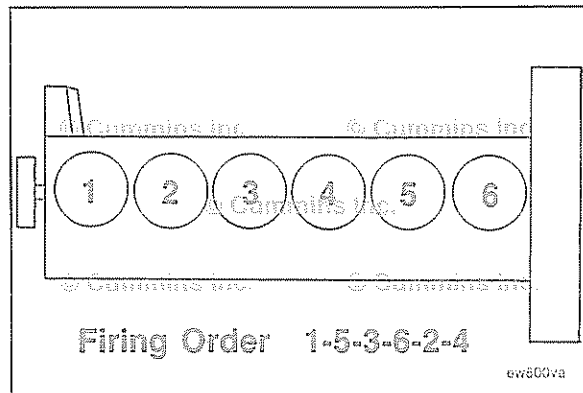


▲ WARNING ▲

Do not pull or pry on the fan blades to rotate the crankshaft. Doing so can damage the fan blades. Damaged fan blades can cause premature fan failures, which can result in serious personal injury or property damage.

The valve set marks are located on the accessory drive pulley. The marks align with a pointer on the gear housing. Use the accessory driveshaft to rotate the crankshaft.

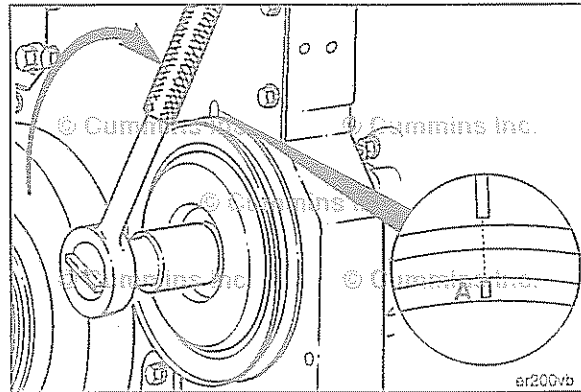




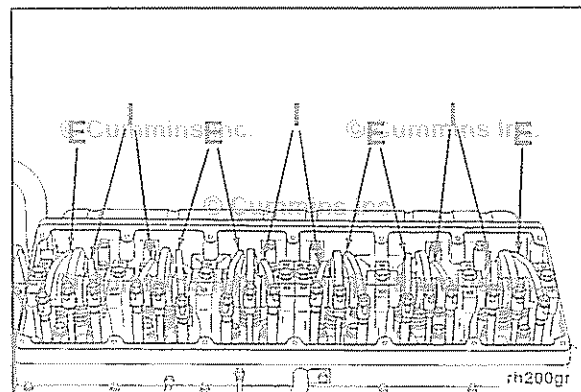
The crankshaft rotation is clockwise when viewed from the front of the engine.

The cylinders are numbered from the front end of the engine.

The firing order is 1-5-3-6-2-4.



Rotate the accessory drive clockwise until the A valve set mark on the accessory drive pulley is aligned with the pointer on the gear cover.



Each cylinder has three rocker levers:

- The long rocker lever (E) is the exhaust lever.
- The center rocker lever is the injector lever
- The short rocker lever (I) is the intake lever.

Valve and Injector Lash

Injector and Valve Measurement Sequence			
Bar engine in direction of rotation	Pulley position	Set Cylinder	
		Injector	Valve
Start	A	1	1
Advance to	B	5	5
Advance to	C	3	3
Advance to	A	6	6
Advance to	B	2	2
Advance to	C	4	4

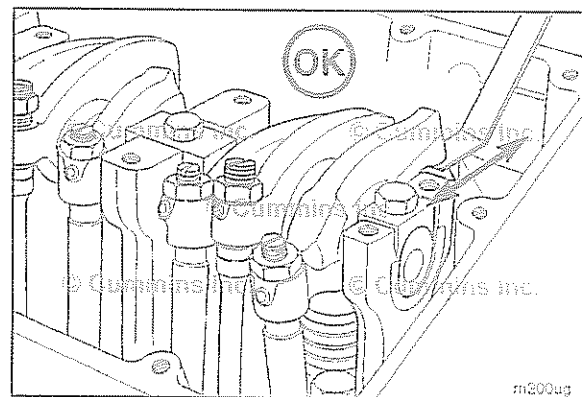
The firing order is 1-5-3-6-2-4.

When the A mark is aligned with the pointer the intake and exhaust valves for cylinder number 1 must be closed. The injector plunger for cylinder number 1 must be at the bottom of its stroke. If these conditions are not correct, cylinder number 6 must be ready to check. Check the injector and valves on the cylinder making sure both the intake and exhaust valve rocker levers can be rattled by hand or the push tubes can be freely rotated.

Valve Lash

Use a set of feeler gauges, measure the amount of clearance (lash) between the crosshead and the rocker lever nose. Measure and record the intake and exhaust valve lash. If the valve lash is not within specifications listed below, the valve must be adjusted. See the adjust step in this procedure.

ISM, M11 Series Valve Lash Recheck Limits			
Intake	0.10 mm	minimum	0.004 in
Intake	0.41 mm	maximum	0.016 in
Exhaust	0.46 mm	minimum	0.018 in
Exhaust	0.76 mm	maximum	0.030 in

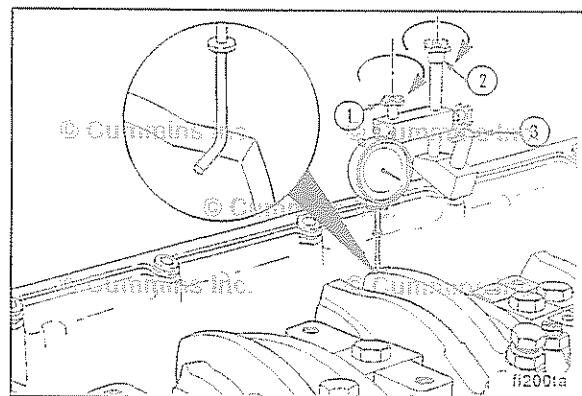


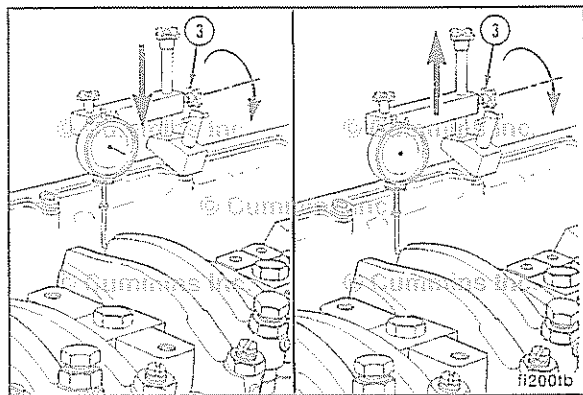
Injector Lash

Install the dial indicator and the support from injector travel measurement kit so the extension for the dial indicator is on top of the injector rocker lever, directly over the socket on the cylinder being checked.

Securely tighten the thumbscrew (1) and hold down the capscrews (2 and 3).

All adjusting screws must be tight to get an accurate measurement.

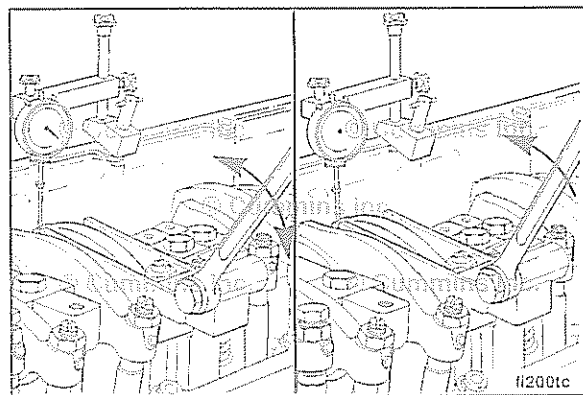




The tip of the dial indicator extension must rest on the rocker lever directly above the socket. If it does not, incorrect readings can result.

Loosen the thumb screw (3), and lower the indicator against the injector rocker lever until the stem is fully compressed.

Raise the indicator approximately 12.7 mm [0.5 in] and tighten the thumb screw (3) to hold the indicator in position.

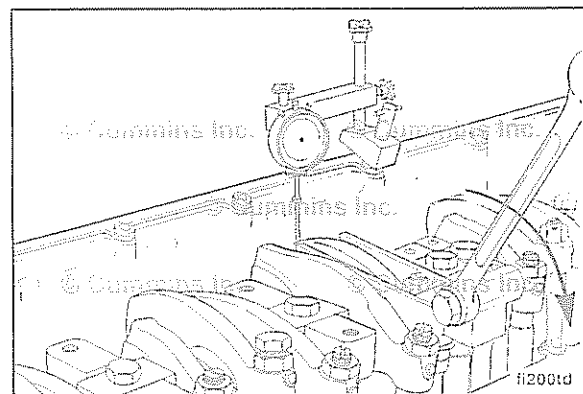


▲ WARNING ▲

The injector plunger is under spring-tension. Do not allow the tool to slip. Serious personal injury can result.

Actuate the injector plunger three or four times to make sure the fuel has been removed from the injector assembly. Allow the lever to return slowly to prevent damage to the dial indicator.

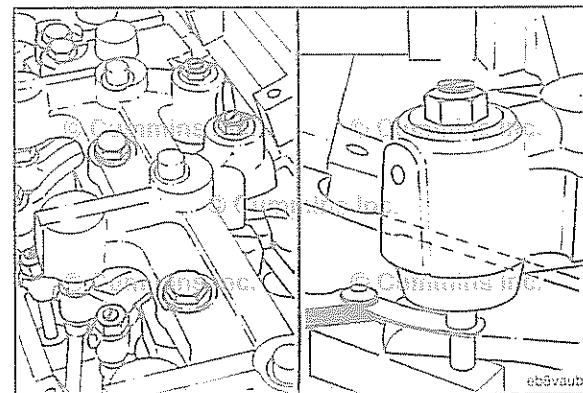
Actuate the lever again; set the dial indicator at zero while holding the injector plunger to the bottom of its travel.



Slowly release the actuator, and check the dial indicator travel. Record the measured value. If the injector setting is not within the specification listed below, the injector must be adjusted. See the adjust step listed in this procedure.



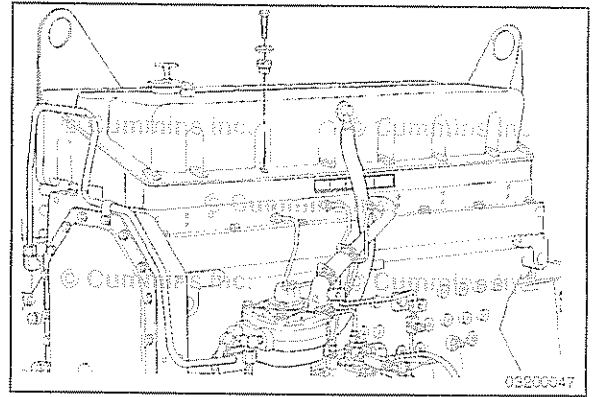
ISM, M11 Series Injector Lash Recheck Limits		
0.51 mm	minimum	0.020 in
2.04 mm	maximum	0.080 in



If the engine is equipped with engine brakes, check the brake piston clearance. Refer to Procedure 020-001.



Install the rocker lever cover. Refer to Procedure 003-011.



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Section 6 - Procedures at 6000 Hours or 2 Years

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Maintenance Procedures - Overview

General Information

All maintenance checks and inspections listed in previous maintenance intervals must also be performed at this time, in addition to those listed under this maintenance interval.

Air Cleaner Assembly (Engine-Mounted)

Maintenance Check

Marine and Industrial Applications



Holes, loose-end seals, dented sealing surfaces, and other forms of damage render the air cleaner inoperative and require immediate replacement.

Elements that have been cleaned several times will finally clog and airflow to the engine will be restricted. After cleaning the element, check the intake air restriction as previously described. Replace the element, if necessary.



To avoid damage to the element, pull the cover and the element straight out when removing them from the air cleaner housing.

Unsnap the latches (4) that secure the cover (1) to the air cleaner housing (5). Remove the cover (1).

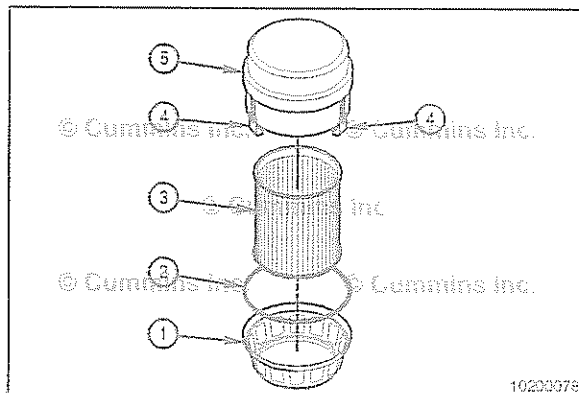
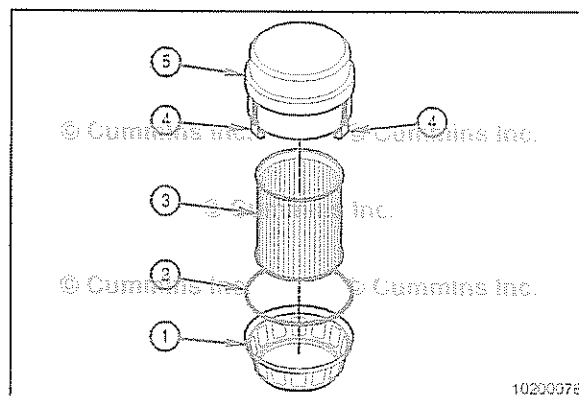
Pull the element (3) out of the housing.

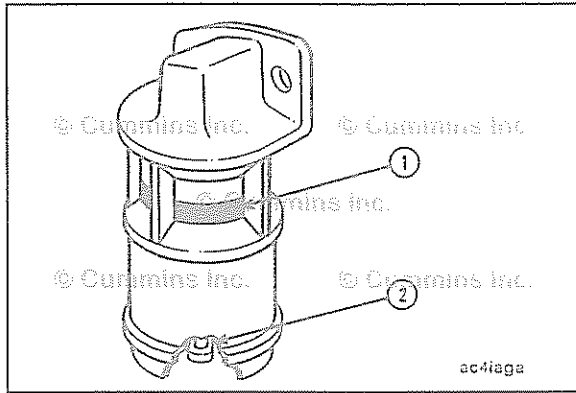
Remove the o-ring (2) from the cover (1) and inspect. If the o-ring (2) is damaged, it must be replaced.

Install the air cleaner element (3) into the housing.

Make sure that the o-ring (2) is seated properly on the cover (1). Install the cover (1).

Snap the latches (4) to secure the cover (1).





Air Cleaner Restriction Maintenance Check

Mechanical Indicator

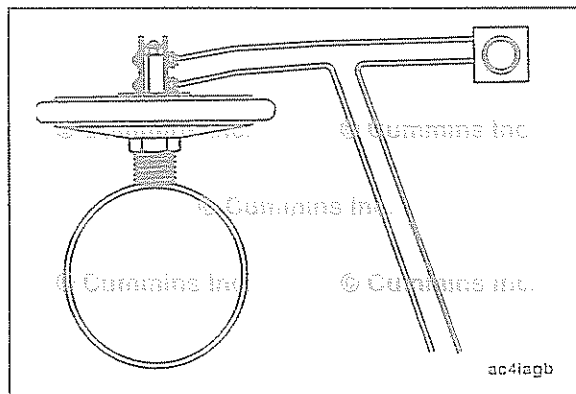
NOTE: Do not remove the felt washer from the indicator. The felt washer absorbs moisture.

A mechanical restriction indicator is available to indicate excessive air restriction through a dry-type air cleaner. This instrument can be mounted in the air cleaner outlet or on the instrument panel. The red flag (1) in the window gradually rises as the cartridge loads with dirt. After changing or replacing the cartridge, reset the indicator by pushing the reset button (2).

Restriction or vacuum indicators need to be installed as close as possible to the turbocharger air inlet in order to obtain a true indication of restrictions.

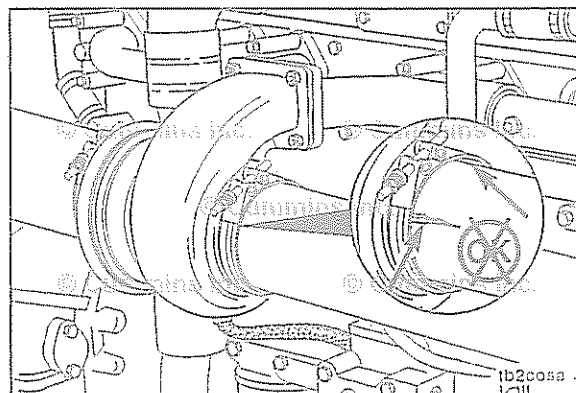
⚠CAUTION⚠

Never operate the engine without an air cleaner. Intake air must be filtered to prevent dirt and debris from entering the engine and causing premature wear.



Vacuum Indicator

Vacuum switches actuate a warning light on the instrument panel when the air restriction becomes excessive.



Air Leaks, Air Intake and Exhaust Systems

Maintenance Check

⚠CAUTION⚠

The engine's intake air must be filtered to prevent dirt and debris from entering the engine. If any of the intake air piping is damaged or loose, unfiltered air will enter the engine, which can cause premature wear and engine failure.

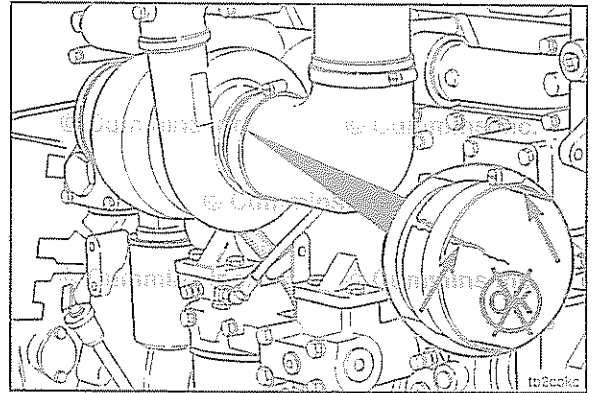
Inspect the intake air piping once a week for cracked hoses, damage, or loose clamps.

Replace damaged pipes and tighten loose clamps as necessary to make sure the air intake system does not leak.

Torque Value: 8 N•m [71 in-lb]

Check for corrosion of the intake system piping under the clamps and hoses. Corrosion can allow corrosive products and dirt to enter the intake system.

Disassemble and clean the intake air piping as required.

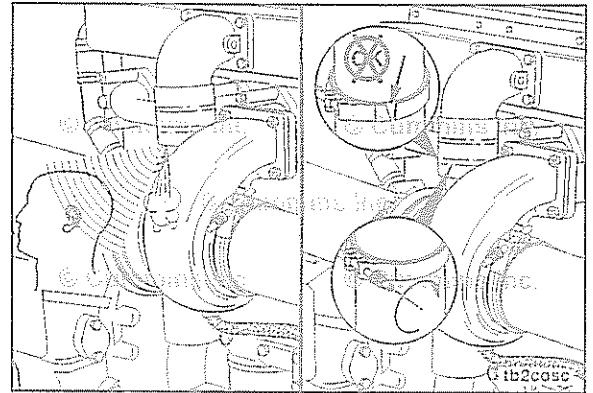


Noise can be caused by an air leak from the:

- Turbocharger-to-discharge-elbow connection

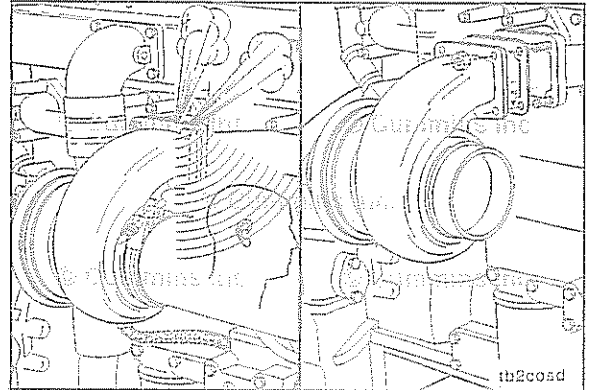
Inspect for damage. Tighten loose clamps.

Torque Value: 8 N•m [71 in-lb]



- Turbocharger-to-exhaust-manifold mounting gasket

Replace the gasket. Refer to Procedure 010-033 (Turbocharger) in Section A for turbocharger removal and installation.



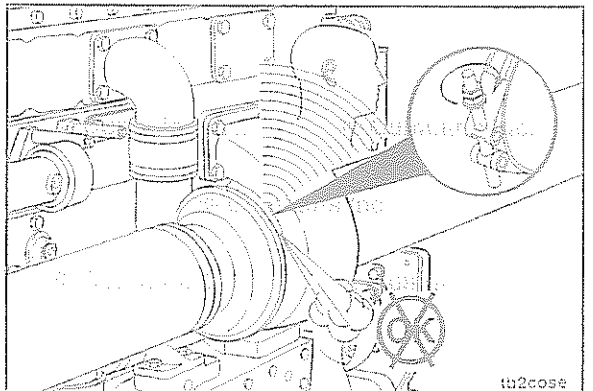
- Turbine housing sealing surface exhaust leak

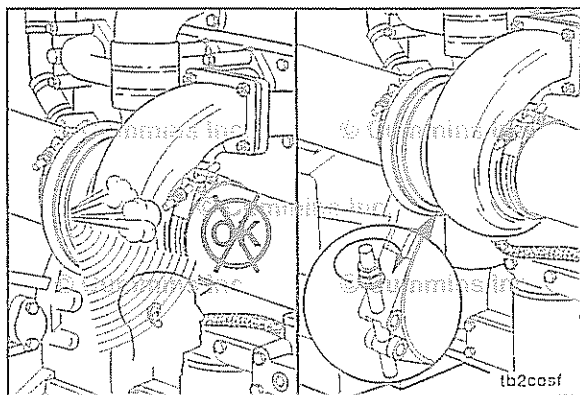
Tighten the turbine housing capscrews or V-band clamps.

Torque Value:

Capscrews 14 N•m [120 in-lb]

V-band 9 N•m [80 in-lb]





- Compressor housing sealing surface air leak.

Tighten the compressor housing V-band clamp nut.

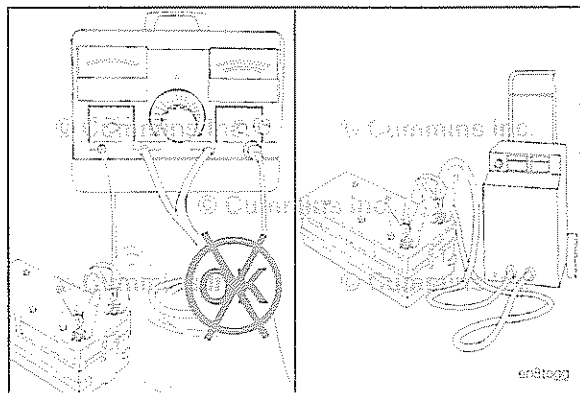


Torque Value:

Capscrews 7 N•m [62 in-lb]

V-band 9 N•m [80 in-lb]

If damage resulted in coolant, oil, excessive fuel or excessive black smoke entering the exhaust system, the aftertreatment system must be inspected. Refer to Procedure 014-013.



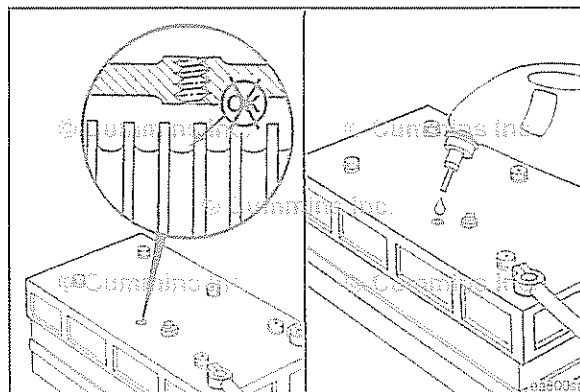
Batteries

Inspect



Use an inductive charging and cranking system analyzer to load-test the state of charge of maintenance-free batteries. If the state of charge is low, use a battery charger to charge the battery. Refer to the manufacturer's instructions.

Replace the battery if it will not charge to the manufacturer's specifications or the battery will not maintain a charge.



If conventional batteries are used, remove the cell caps or covers and check the electrolyte (water and sulfuric acid solution) level.



▲ WARNING ▲

Batteries can emit explosive gas. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the battery (-) negative cable first and attach the battery negative cable last.

NOTE: Maintenance-free batteries are sealed and do not require the addition of water.

Fill each battery cell with water. Refer to the manufacturer's specifications.

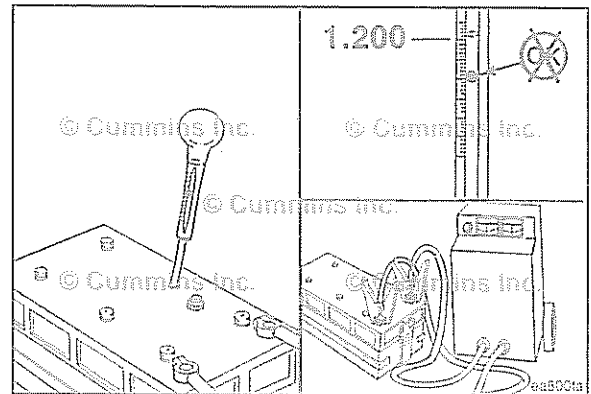
Refer to the accompanying table to determine the battery state of charge based on the specific-gravity readings.

Battery State of Charge	Specific Gravity @ 27°C [80°F]
100%	1.260 to 1.280
75%	1.230 to 1.250
50%	1.200 to 1.220
25%	1.170 to 1.190
Discharged	1.110 to 1.130

Use a hydrometer to measure the specific gravity of each cell.

NOTE: If the specific gravity of any cell is below 1.200, the battery must be charged.

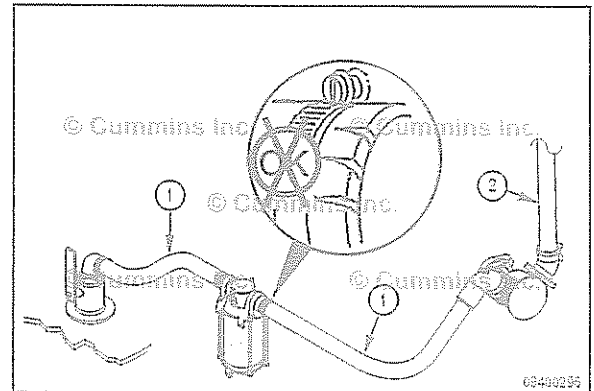
NOTE: Do not attempt to check the specific gravity of a battery immediately after adding water. If it is necessary to add water to allow use of the hydrometer, charge the battery several minutes at a high rate to mix the electrolyte.



Sea Water Hoses

Inspect

Start engine, and inspect all hoses and connections for deterioration or leaks. Replace damaged hoses and clamps, if necessary.

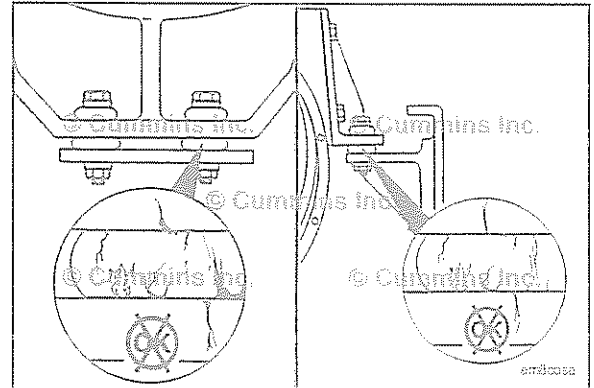


Engine Mounts

Inspect

Check the torque on the engine-mounting nuts and bolts. Tighten any that are loose. Refer to the OEM service manual or manufacturer for torque specifications.

Inspect the rubber for deterioration and age hardening. Replace any broken or lost bolts, capscrews, or damaged rubber.



Engine Steam Cleaning

Clean

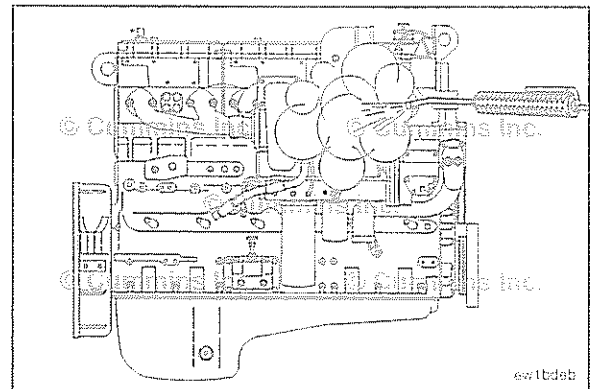


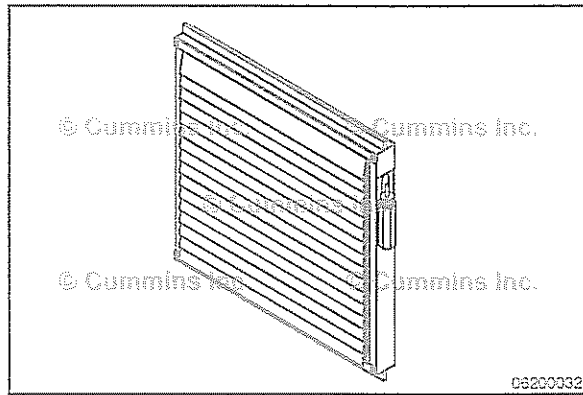
WARNING

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

Steam is the best method of cleaning a dirty engine or a piece of equipment. If steam is not available, use a solvent to wash the engine.

Protect all electrical components, openings, and wiring from the full force of the cleaner spray nozzle.



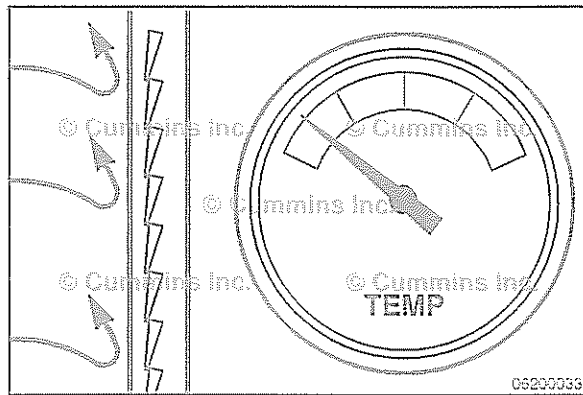


Radiator Shutter Assembly Maintenance Check



Industrial Applications

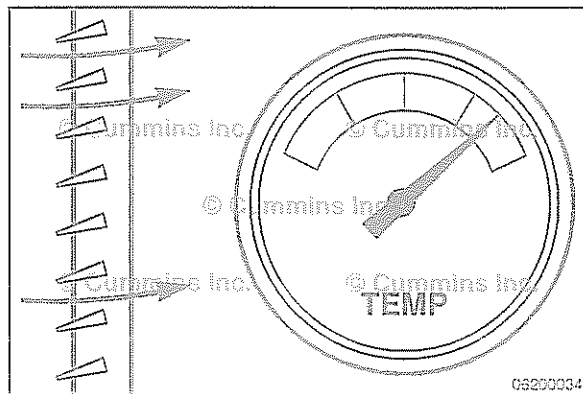
Check the shutters and the thermatic fan every 1500 hours or 1 year.



Check the shutters in the closed position to be sure they are completely closed.



NOTE: If the shutters are not closed, refer to the manufacturer's instructions.



Be sure the shutters open completely at the desired temperature setting.



NOTE: If the shutters are not open, refer to the manufacturer's instructions.



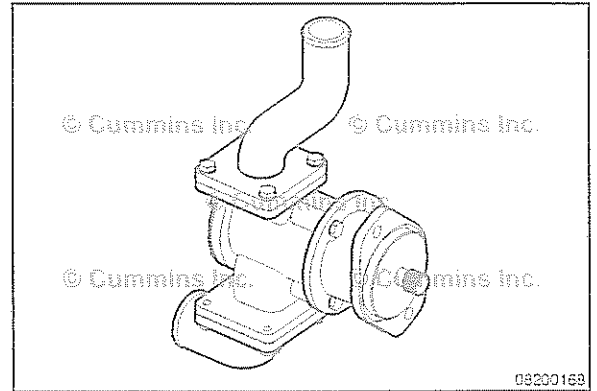
Shutters and thermatic fans must be set to operate in the same temperature range as the thermostat with which they are used. Refer to the Thermal Control Settings in Procedure 018-018 in Section V.

Sea Water Pump

General Information

Marine Applications

The QSM11 uses a vane-type, positive displacement, self-priming sea water pump.

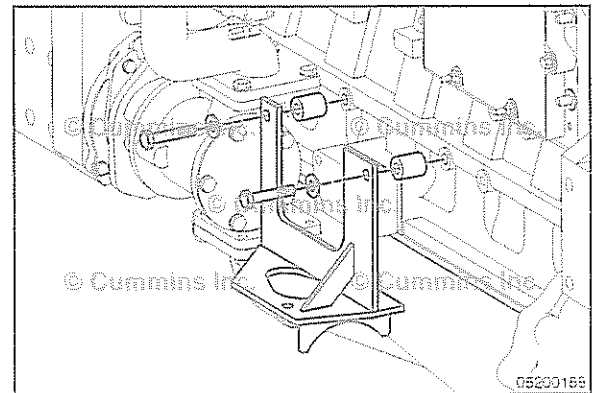


Preparatory Steps



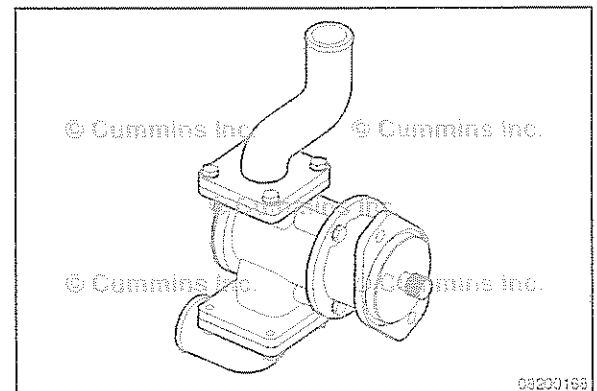
Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be disposed of in accordance with local environmental regulations

- Shut off the sea water inlet valves.
- Shut off the fuel supply and return valves (if a major kit repair is planned).
- Remove the fuel cooler (if a major kit repair is planned).
- Remove the fuel cooler mounting bracket for access to the sea water pump cover.



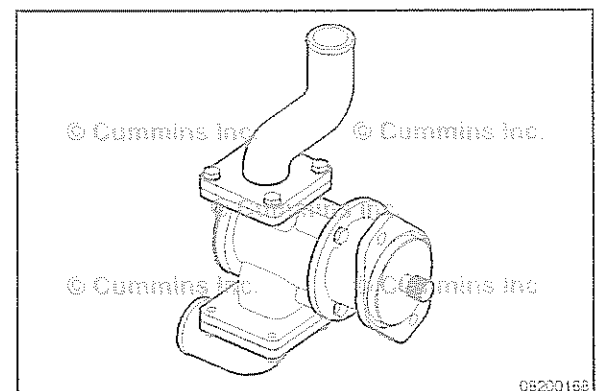
Remove

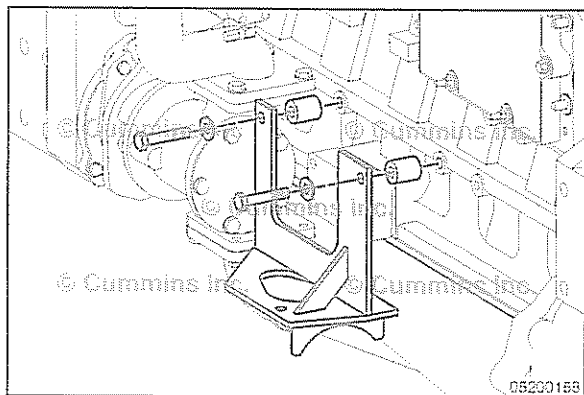
If a major kit repair is being performed, remove the sea water pump.



Install

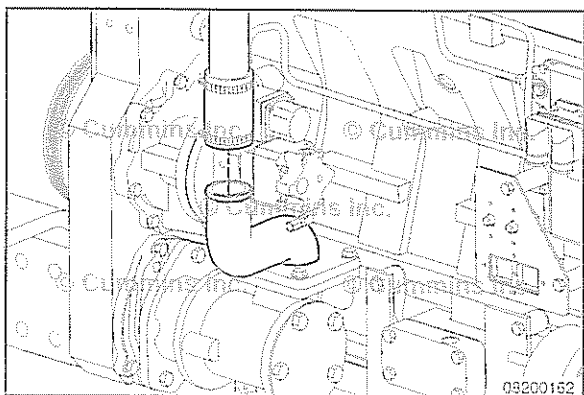
If a major kit repair was performed, install the sea water pump.





Finishing Steps

- Install the fuel cooler mounting bracket.
- Install the fuel cooler (if a major repair kit was installed).
- Open the fuel supply and return valves (if the fuel cooler was removed).
- Open the sea water inlet valve.
- Operate the engine and check for leaks.



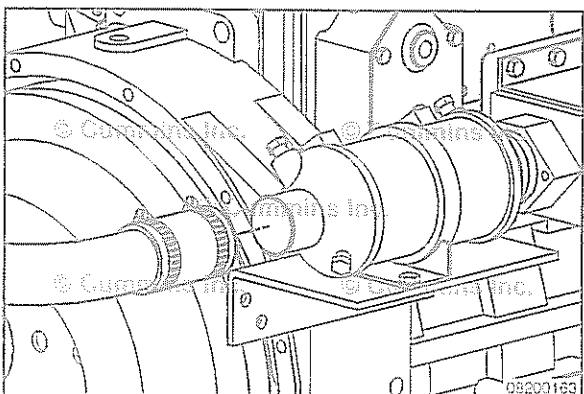
Sea Water System

Flush

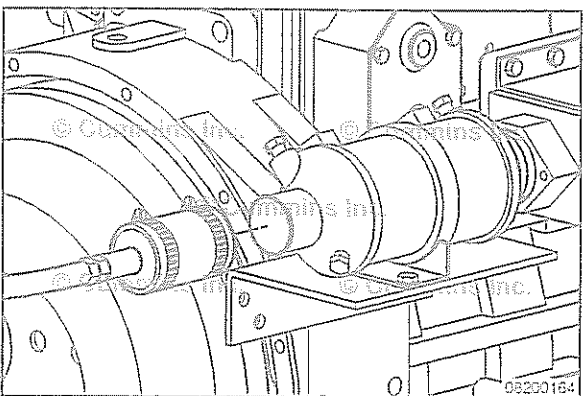
Marine Applications

Remove the hose from the sea water pump outlet.

NOTE: Place the hose into a container to catch the backflushed water.



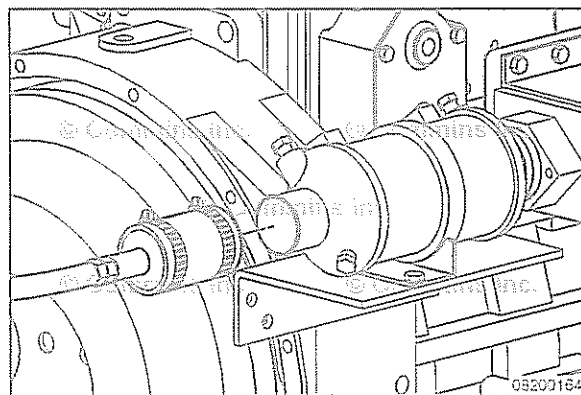
Remove the sea water discharge line from the gear cooler.



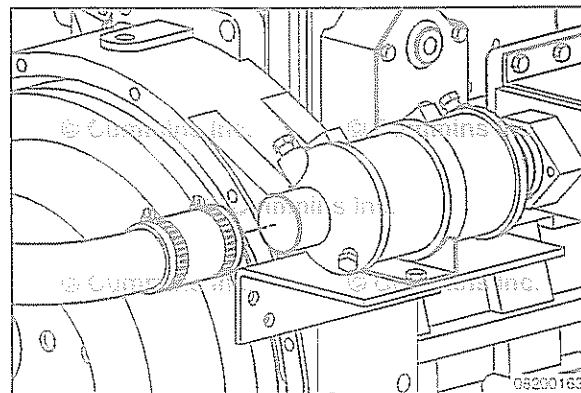
Using hose reducers, install a clean, fresh water supply to the outlet of the gear oil cooler.

Flush the sea water system.

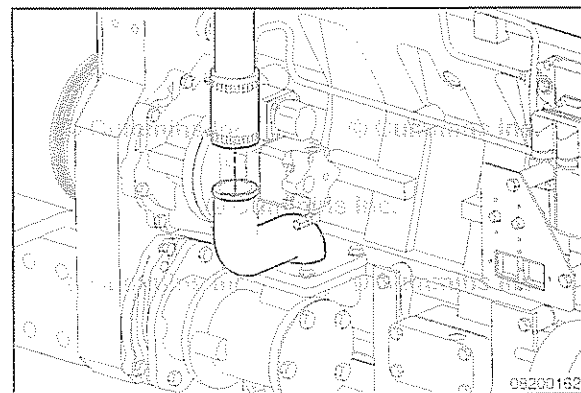
Remove the fresh water supply.



Install the sea water discharge line to the gear cooler.



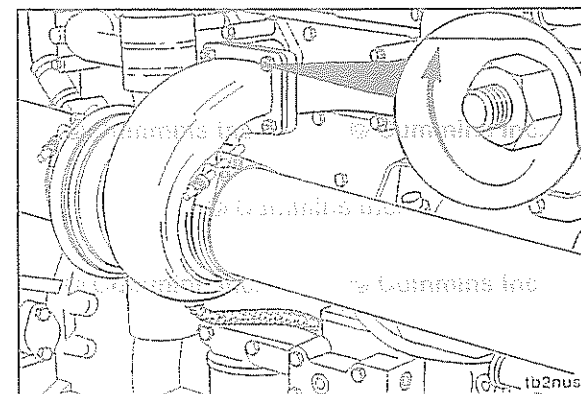
Install the sea water pump outlet line onto the sea water pump.

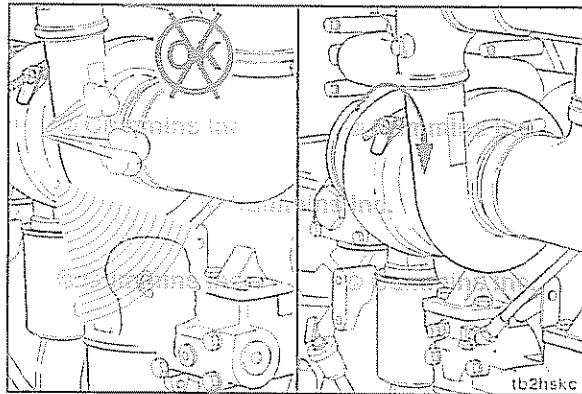


Turbocharger Maintenance Check

Tighten the turbocharger mounting nuts.

Torque Value: 61 N•m [45 ft-lb]



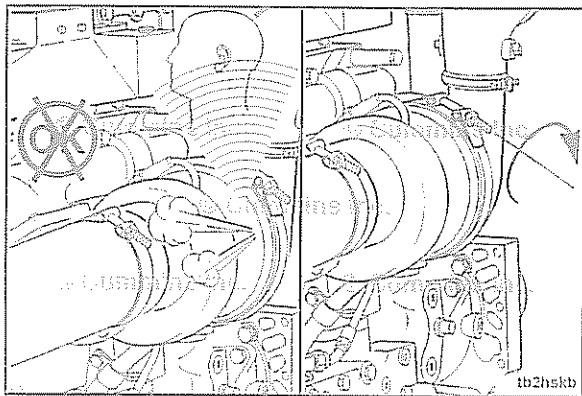


Check the turbine housing's sealing surface for exhaust leaks.

If a leak is found, tighten the V-band clamp nut.



Torque Value: 9 N•m [80 in-lb]

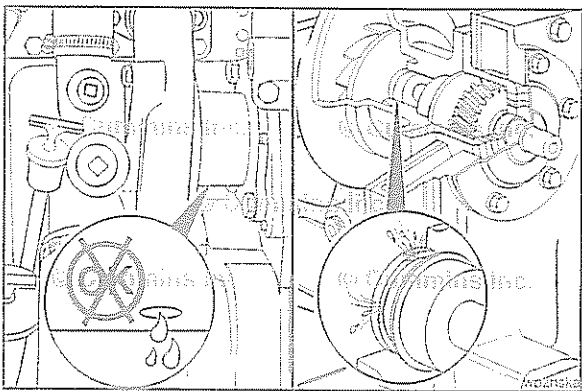


Check the compressor housing sealing surface for leaks.
If a leak is found, tighten the V-band clamp nut.

Torque Value: 9 N•m [80 in-lb]



If malfunction resulted in coolant, oil, excessive fuel or excessive black smoke entering the exhaust system, the aftertreatment system must be inspected. Refer to Procedure 014-013.

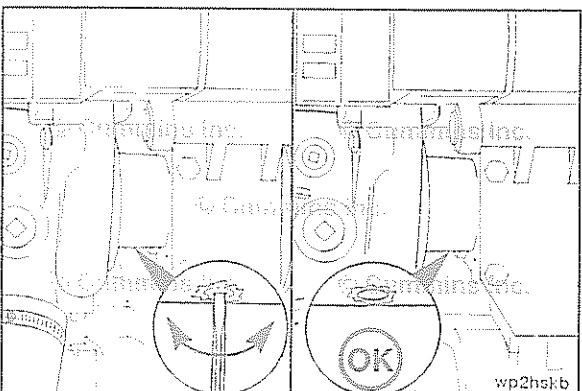


Water Pump Maintenance Check



A streak or chemical buildup at the weep hole is not justification for water pump replacement.

If a steady flow or drip of coolant or oil is observed at the weep hole, replace the water pump with a new or rebuilt unit. Refer to Procedure 008-062 (Water Pump) in Section A.



A small screwdriver or a similar tool can be used to remove any debris.

Make sure the weep hole is open.

Section 7 - Maintenance Procedures at 6000 Hours

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Vibration Damper, Viscous	7-1
Inspect.....	7-1

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Maintenance Procedures - Overview

General Information

All maintenance checks and inspections listed in previous maintenance intervals must also be performed at this time, in addition to those listed under this maintenance interval.

Vibration Damper, Viscous Inspect



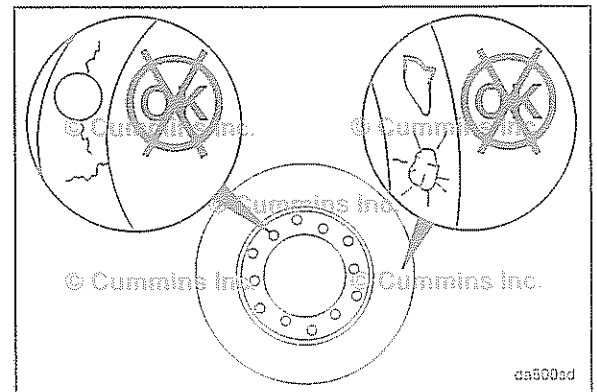
The silicone fluid in the vibration damper will become solid after extended service and will make the damper inoperative. An inoperative vibration damper can cause major engine or drivetrain failures.

Check the vibration damper for evidence of fluid loss, dents, and wobble. Inspect the vibration damper thickness for any deformation or raising of the damper cover plate.

If any of these conditions are identified, contact your local Cummins Authorized Repair Location to replace the vibration damper.

Viscous dampers have a limited life. The maximum damper life specifications are located in Maintenance Specifications (Section V).

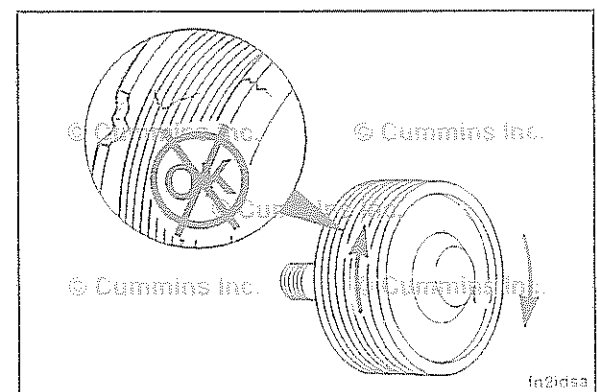
For vibration damper location, refer to Engine Diagrams in Engine Identification (Section E).

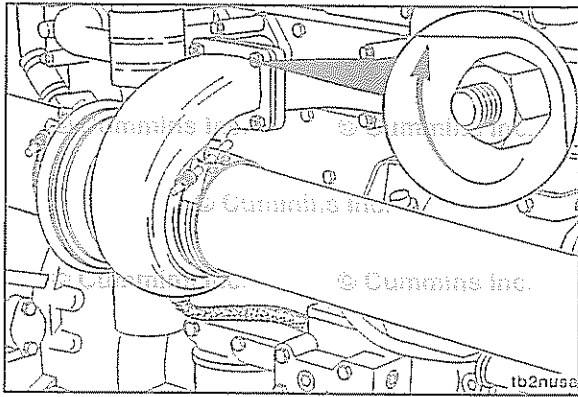


Fan Drive Idler Pulley Assembly Inspect

Inspect the idler pulley for:

- Freedom of rotation
- Cracked, chipped, or broken pulley grooves.

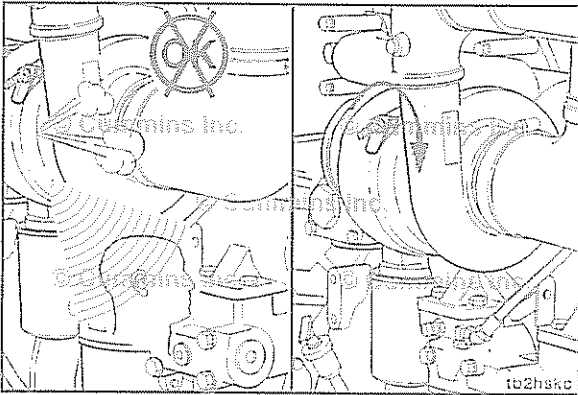




Turbocharger Maintenance Check

Tighten the turbocharger mounting nuts.

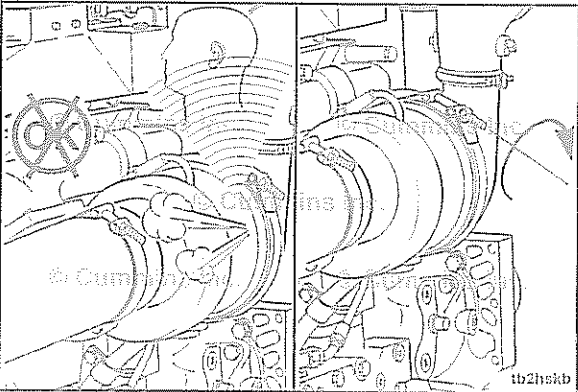
Torque Value: 61 N•m [45 ft-lb]



Check the turbine housing's sealing surface for exhaust leaks.

If a leak is found, tighten the V-band clamp nut.

Torque Value: 9 N•m [80 in-lb]



Check the compressor housing sealing surface for leaks.
If a leak is found, tighten the V-band clamp nut.

Torque Value: 9 N•m [80 in-lb]

If malfunction resulted in coolant, oil, excessive fuel or excessive black smoke entering the exhaust system, the aftertreatment system must be inspected. Refer to Procedure 014-013.



Cooling System General Information

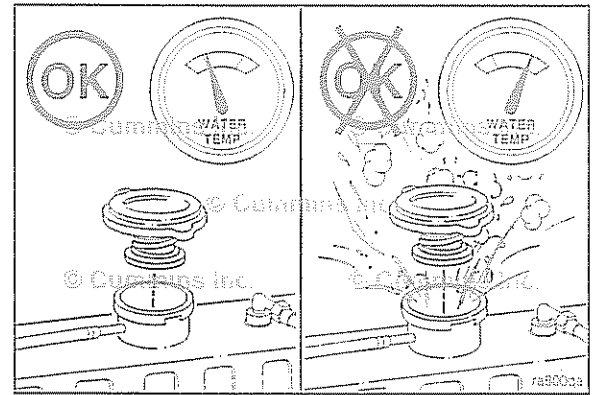
Restore™ is a heavy-duty cooling system cleaner that removes corrosion, silica gel, and other deposits. The performance of Restore™ is dependent on time, temperature, and concentration levels. An extremely scaled or flow-restricted system, for example, can require higher concentrations of cleaners, higher temperatures, or longer cleaning times, or the use of Restore Plus™. Up to twice the recommended concentration levels of Restore™ can be used safely. Restore Plus™ must be used only at its recommended concentration level. Extremely scaled or fouled systems can require more than one cleaning.

Clean

▲ WARNING ▲

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

Remove the radiator cap.

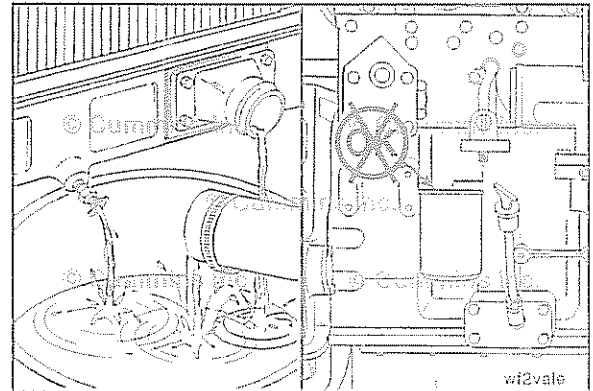


▲ WARNING ▲

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Drain the cooling system. Do not allow the cooling system to dry out.

Do not remove the coolant filter.

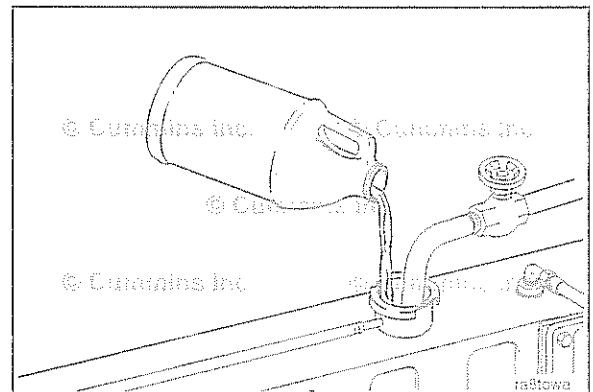


▲ CAUTION ▲

Fleetguard® Restore™ contains no antifreeze. Do not allow the cooling system to freeze during the cleaning operation.

Immediately add 3.8 liters [1 gal] of Fleetguard® Restore™, Restore™ Plus, or the equivalent, for each 38 to 57 liters [10 to 15 gal] of cooling system capacity, and fill the system with plain water.

Turn the cab heater temperature switch to high to allow maximum coolant flow through the heater core. The blower does not have to be on.

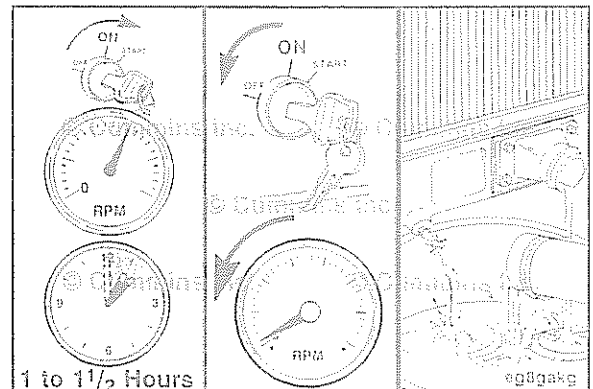


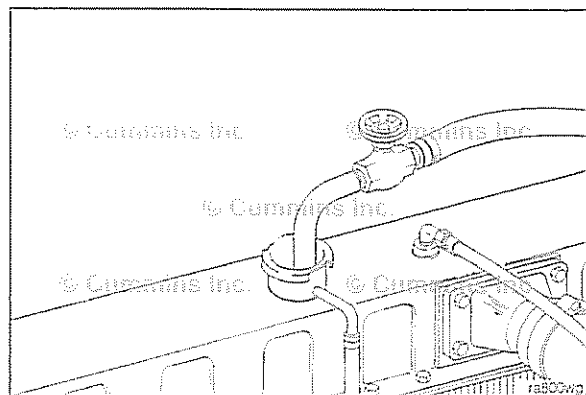
▲ WARNING ▲

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

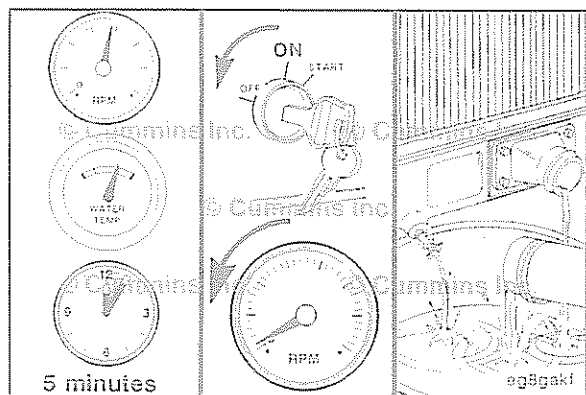
Operate the engine at normal operating temperatures, at least 85°C [185°F], for 1 to 1½ hours.

Shut off the engine and drain the cooling system.





Fill the cooling system with clean water.



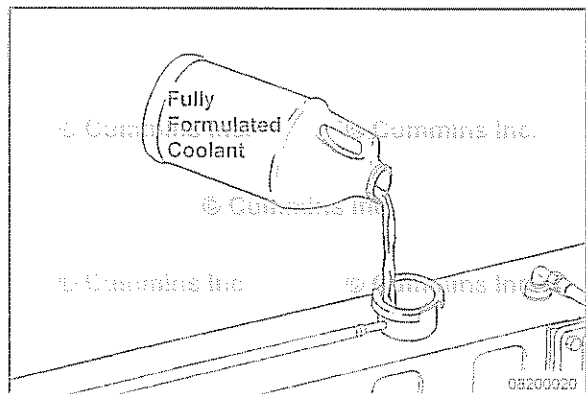
WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Operate the engine at high idle for five minutes with the coolant temperature above 85°C [185°F].

Shut off the engine and drain the cooling system.

If the water being drained is still dirty, the system must be continually flushed until the water is clean.



CAUTION

Do not add coolant to a hot engine. Engine castings can be damaged. Allow the engine to cool below 50°C [120°F] before adding coolant.



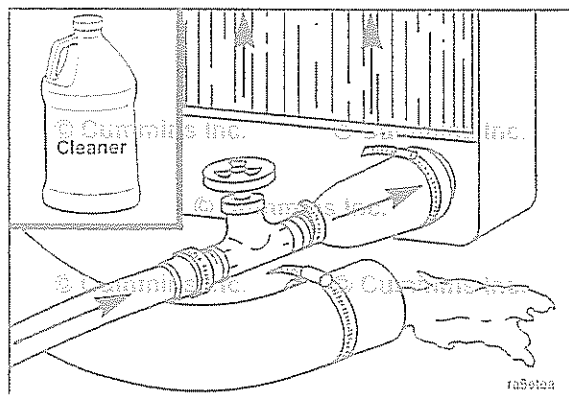
Fill the cooling system with fully formulated coolant or a 50/50 mixture of fully formulated antifreeze and good-quality water.



Use a service filter to bring the coolant to the correct SCA concentration level. Refer to Procedure 018-024 (Cummins/Fleetguard® Filter Specifications) in Section V for the correct filter to use.

Install the pressure cap. Operate the engine until it reaches a temperature of 80°C [180°F] and check for coolant leaks.

Flush



CAUTION

Do not use caustic cleaners in the cooling system. Aluminum components will be damaged.

The cooling system must be clean to work correctly and to eliminate buildup of harmful chemicals.

Restore™ is a heavy-duty cooling system cleaner that removes corrosion, silica gel, and other deposits. The performance of Restore™ is dependent on time, temperature, and concentration levels. An extremely scaled or flow-restricted system, for example, can require higher concentrations of cleaners, higher temperatures, or longer cleaning times, or the use of Restore Plus™. Up to twice the recommended concentration levels of Restore™ can be used safely. Restore Plus™ must be used **only** at its recommended concentration level. Extremely scaled or fouled systems can require more than one cleaning.

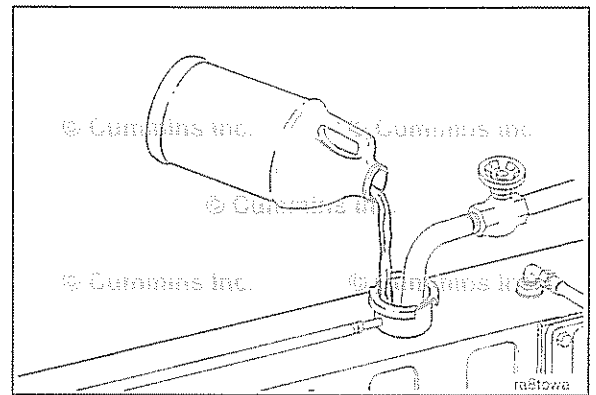


⚠ CAUTION ⚠

Fleetguard® Restore™ contains no antifreeze. Do not allow the cooling system to freeze during the cleaning operation.

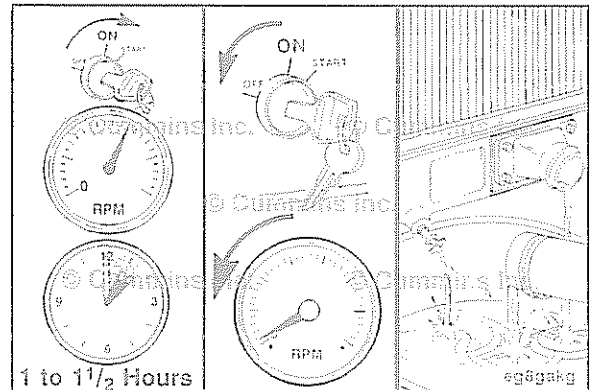
After draining the coolant, immediately add 3.8 liters [1 gal] of Fleetguard® Restore™, Restore Plus™, or equivalent, for each 38 to 57 liters [10 to 15 gal] of cooling system capacity, and fill the system with clean water.

Turn the heater temperature switch to high to allow maximum coolant flow through the heater core. The blower does not have to be ON.

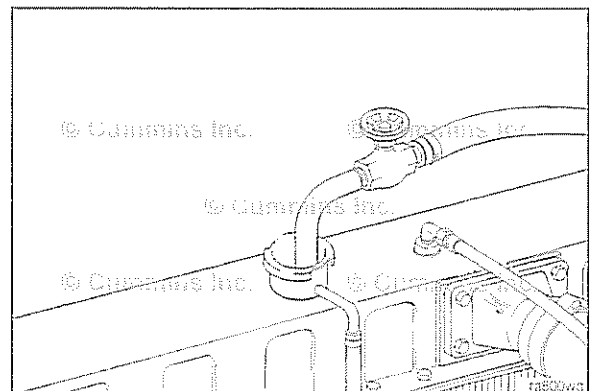


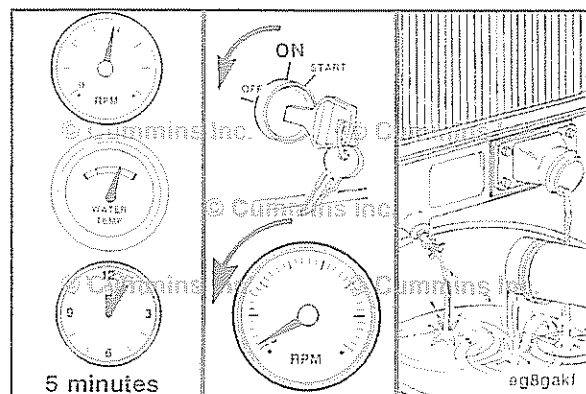
Operate the engine at normal operating temperatures, at least 85°C [185°F], for one to one and a half hours.

Shut off the engine and drain the cooling system.



Fill the cooling system with clean water.





▲ WARNING ▲

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

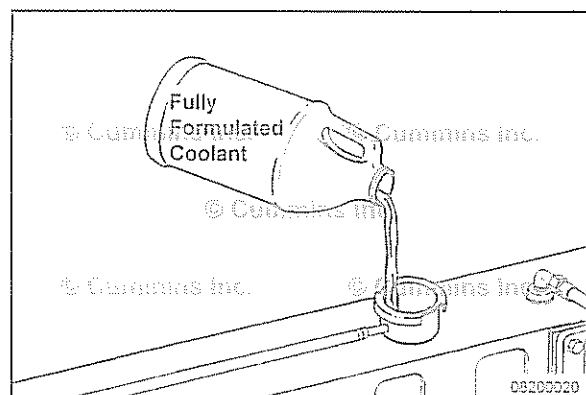
▲ WARNING ▲

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Operate the engine at high idle for five minutes with the coolant temperature above 85°C [185°F].

Shut off the engine and drain the cooling system.

If the water being drained is still dirty, the system must be flushed again until the water is clean.



▲ WARNING ▲

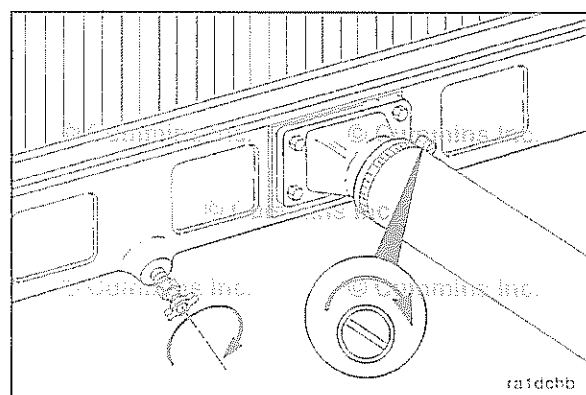
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

▲ CAUTION ▲

To reduce the possibility of engine damage, do not add coolant while the engine is hot.

Fill the cooling system with fully formulated coolant or a 50/50 mixture of fully formulated antifreeze and good quality water. Use a service filter to bring the coolant to the correct SCA concentration level. Refer to Procedure 018-004 (Coolant Recommendations and Specifications) in Section V.

Install the pressure cap. Operate the engine until it reaches a temperature of 80°C [180°F], and check for coolant leaks.



Fill

Close the radiator draincocks.



Install the lower radiator hose(s).

Tighten the hose clamps.

Torque Value: 5 N•m [40 in-lb]

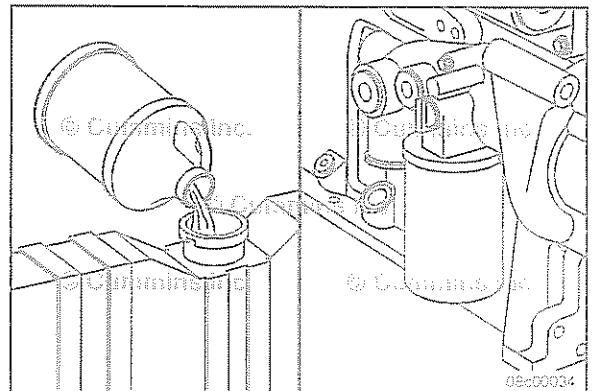
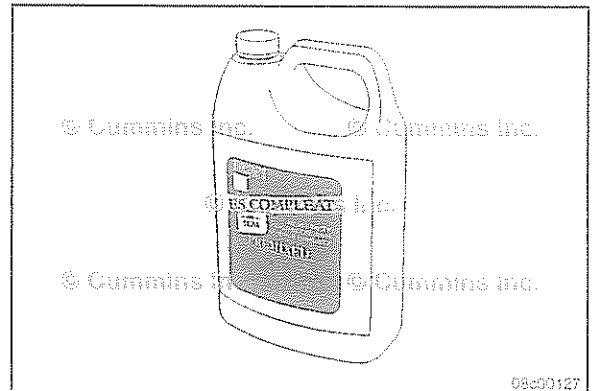
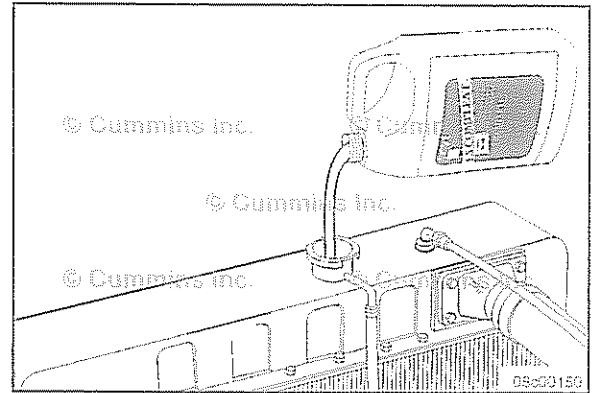
Cummins recommends using either a 50/50 mixture of good-quality water and fully-formulated antifreeze, or fully-formulated coolant when filling the cooling system. The fully-formulated antifreeze or coolant must meet TMC RP329 or TMC RP330 specifications.

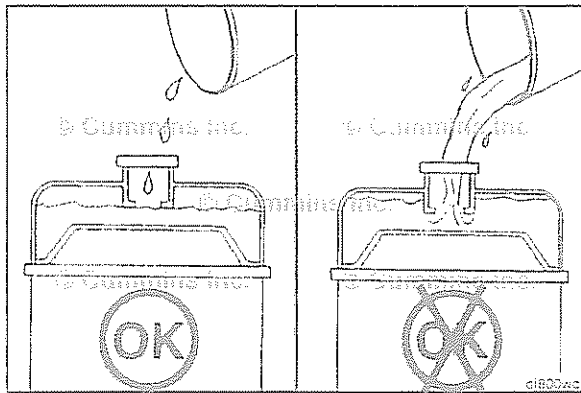
Good-quality water is important for cooling system performance. Excessive levels of calcium and magnesium contribute to scaling problems, and excessive levels of chlorides and sulfates cause cooling system corrosion.

Water Quality	
Calcium Magnesium (Hardness)	Maximum 170 ppm as (CaCO ₃ + MgCO ₃)
Chloride	40 ppm as (Cl)
Sulfate	100 ppm as (SO ₄)

Cummins recommends using Fleetguard® COMPLEAT ES. It is available in glycol forms (ethylene and propylene) and complies with TMC RP329 and RP330 standards.

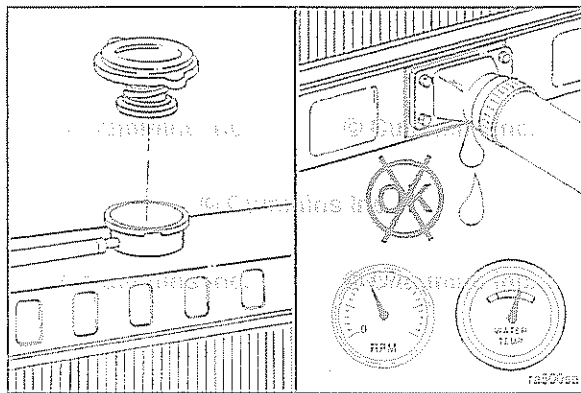
Fill the cooling system with heavy-duty coolant and install the correct service filter. Refer to Procedure 018-004 (Coolant Recommendations and Specifications) in Section V.





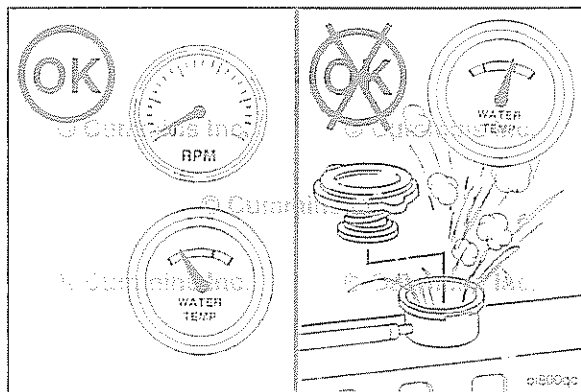
Fill the cooling system with coolant to the bottom of the fill neck in the radiator fill (or expansion) tank.

NOTE: Some radiators have two fill locations, both of which must be filled when the cooling system is drained.

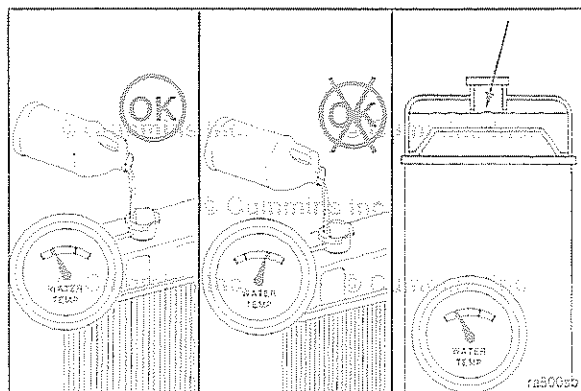


Install the radiator or expansion tank fill cap.

Operate the engine until it reaches a temperature of 82°C [180°F], and check for leaks.



Shut off the engine and allow to cool.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ CAUTION ⚠

Do not add cold coolant to a hot engine. This can cause engine casting damage. Allow the engine to cool to below 50°C [120°F] before adding coolant.

Remove the radiator or fill cap and check the coolant level. Refer to the OEM service manual instructions for recommendations. Add coolant if necessary. Do not overfill.

Air Compressor Discharge Lines

General Information

All air compressors have a small amount of lubricating oil carryover that lubricates the piston rings and moving parts. When this lubricating oil is exposed to normal air compressor operating temperatures over time, the lubricating oil will form varnish or carbon deposits. If the following maintenance check are not performed, the air compressor piston rings will wear and not seal correctly.

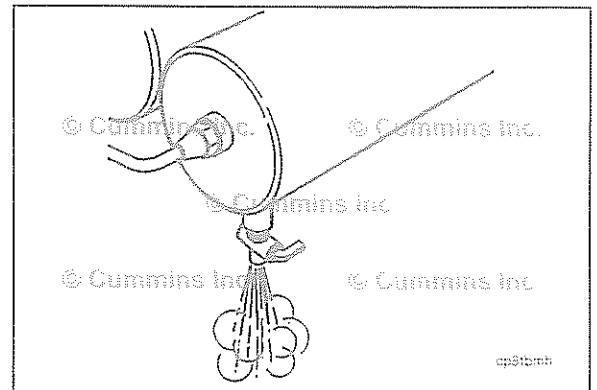
Maintenance Check



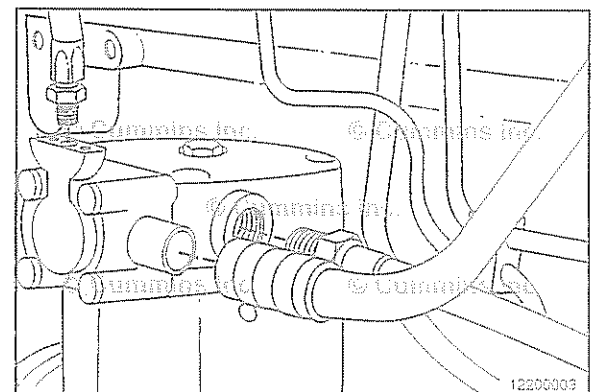
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Shut off the engine.

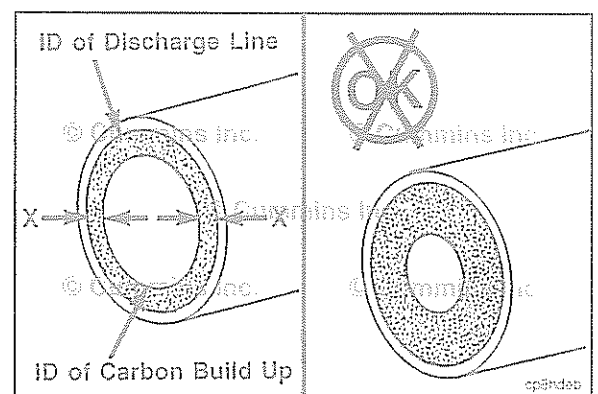
Open the drain valve on the wet tank to release the system air pressure.

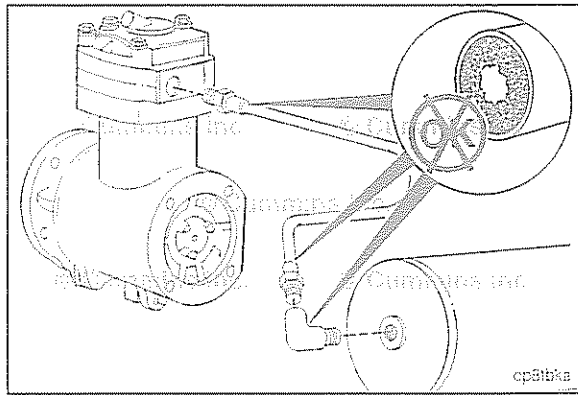


Remove the air compressor discharge line from the air compressor. Location of the air compressor discharge line can be found in Flow Diagram, Compressed Air System in System Diagrams (Section D).

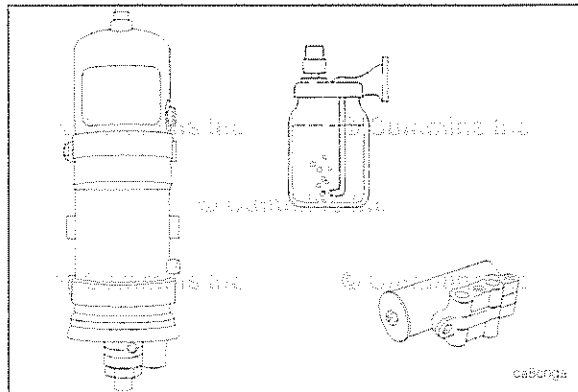


Measure the total carbon deposit thickness inside the air discharge line as shown. If the total carbon deposit ($X + X$) exceeds 2 mm [1/16 in], clean and inspect the cylinder head, the valve assembly, and the discharge line. Replace if necessary. Contact the Cummins Authorized Repair Location for procedures.





If the total carbon deposit exceeds specifications, continue checking the air discharge line connections up to the first tank until total carbon deposit is less than 2 mm [1/16 in]. Clean or replace any lines or connections that exceed this specification.



Inspect any air driers, splitter valves, pressure relief valves, and alcohol injectors for carbon deposits or malfunctioning parts. Inspect for air leaks. Maintain and repair the parts according to the manufacturer's specifications.

Section A - Adjustment, Repair, and Replacement

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Remove.....	A-7
With EGR.....	A-7

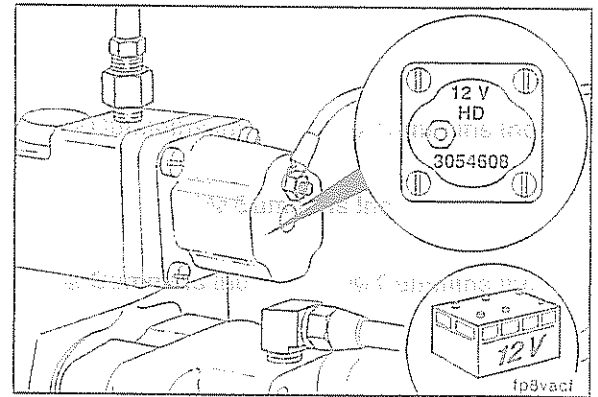
Fuel Shutoff Valve Voltage Check



To avoid damage to the ECM, connect only one wire to the fuel shut off solenoid.

Connect the wire. Make sure the shutoff valve coil is the correct voltage.

The coil voltage and part number are cast into the terminal connection end of the coil.

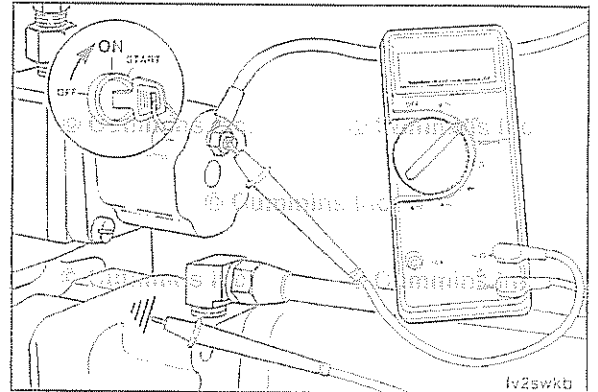


Turn the vehicle keyswitch to the ON position.

Check the voltage to the coil with a multimeter, Part Number 3377161, or equivalent.

The voltage must be the same as the battery voltage.

Turn the vehicle key switch to the OFF position.



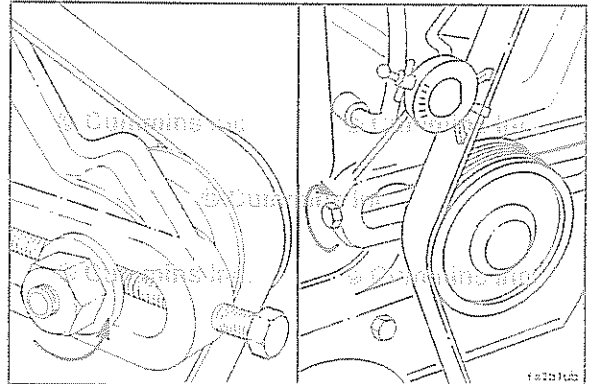
Drive Belt, Cooling Fan Adjust

NOTE: Consult the vehicle manufacturer for non-Cummins® fan hub installations.

Do not adjust belt tension to full value with the adjusting screw. Belt tension can increase when the locknut is tightened and cause reduced belt and bearing life.

Loosen the idler pulley shaft locknut.

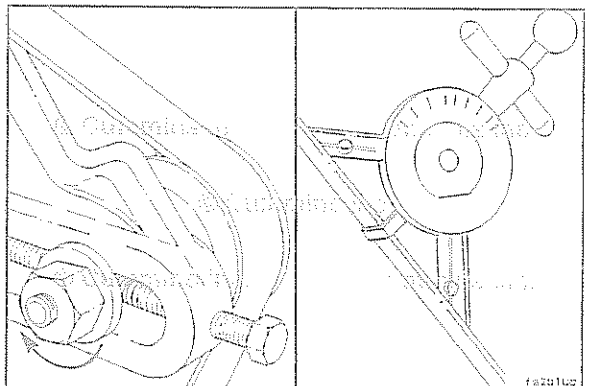
Use belt tension gauge, Part Number ST-1293, or equivalent, to adjust the belt to the correct tension. Refer to Procedure 018-005 (Drive Belt Tension Chart) in Section V for the belt tension chart.

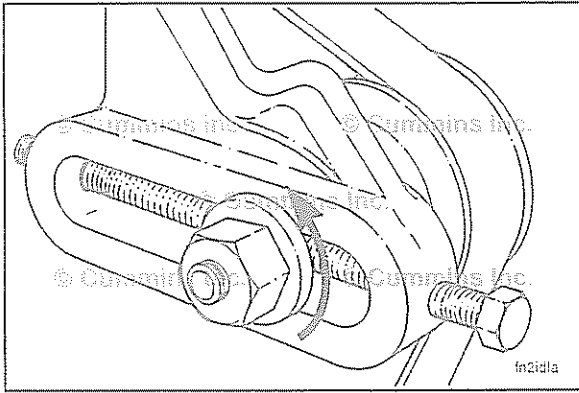


Tighten the idler pulley shaft locknut.

Torque Value: 190 N•m [140 ft-lb]

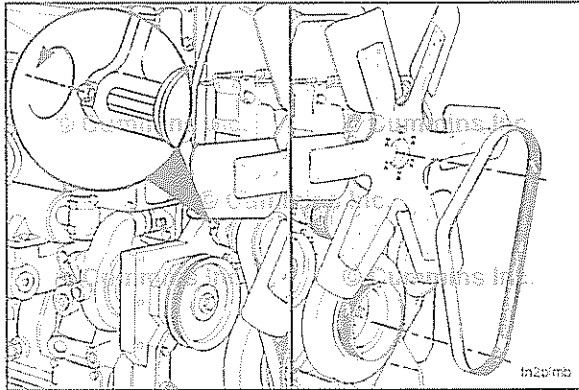
Check the belt tension again to make sure the belt is not too tight.





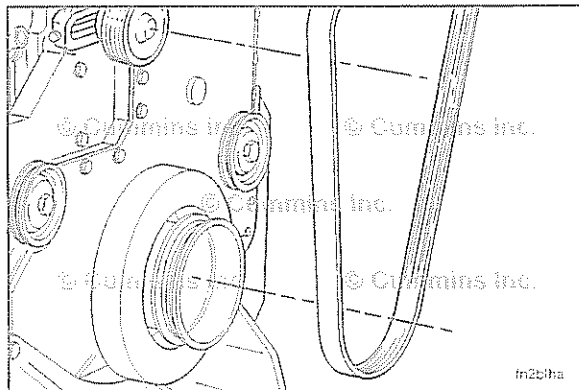
Remove

Loosen the fan idler pulley shaft locknut.



Turn the adjusting screw counterclockwise to release the belt tension.

Move the fan idler pulley and fan pulley centers as close as possible. The belt can then be removed without excessive force.

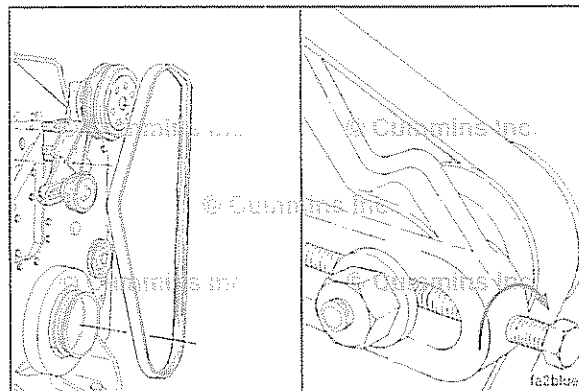


Install

⚠CAUTION⚠

To reduce the possibility of damage to the pulley and new belt, do not roll the belt over the pulley or pry it on with a tool.

Install a new fan belt on the pulleys.



Use belt tension gauge, Part Number ST-1293, or equivalent, to measure the belt tension. Refer to Procedure 018-005 (Drive Belt Tension) in Section V.



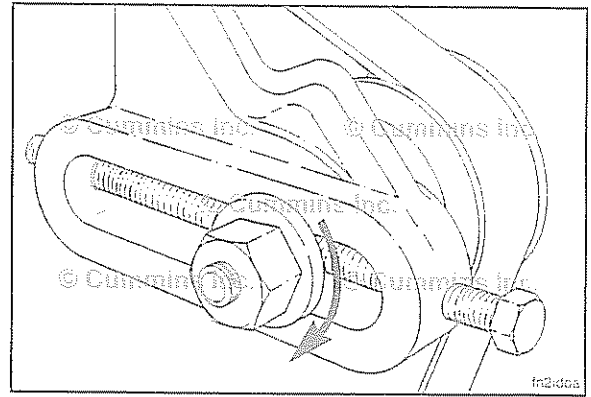
Turn the adjusting screw to adjust the belt tension.

NOTE: A belt is considered used if it has been in operation 10 minutes or longer. If used belt tension is below the minimum value, tighten the belt to the maximum value. Replace the belt if it will not maintain correct tension.

Tighten the idler pulley shaft locknut.

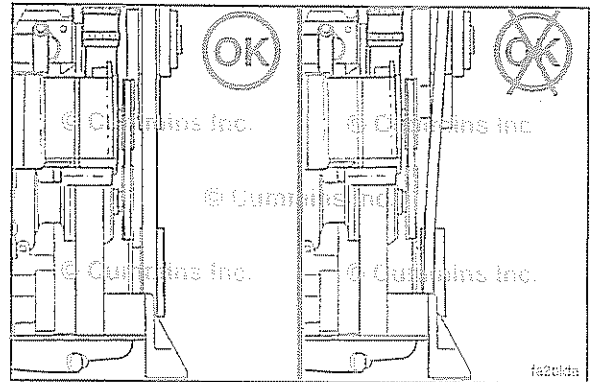
Torque Value: 190 N•m [140 ft-lb]

Check the belt tension again to make sure belt is not too tight.



Inspect the pulley misalignment must not exceed 6 mm for each meter [1/16 inch for each 12 inches] of distance between the pulley centers.

Replace the fan idler pulley and fan hub pulley if they are not in alignment. Refer to Procedure008-030 (Fan Drive Idler Pulley Assembly) in Section A.



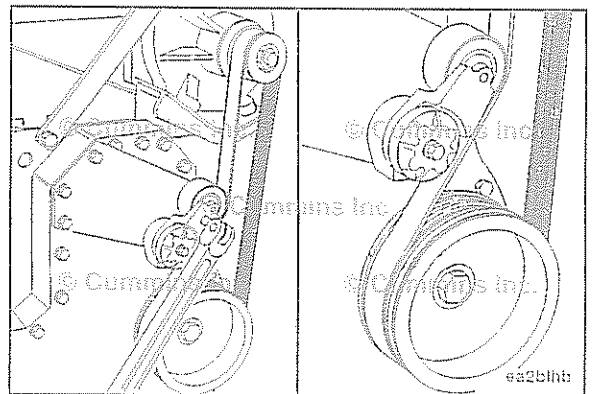
Drive Belt, Alternator

Install

Install a new belt over the pulleys while holding the tensioner back. Be careful not to damage the belt while working it over the flanged pulleys.

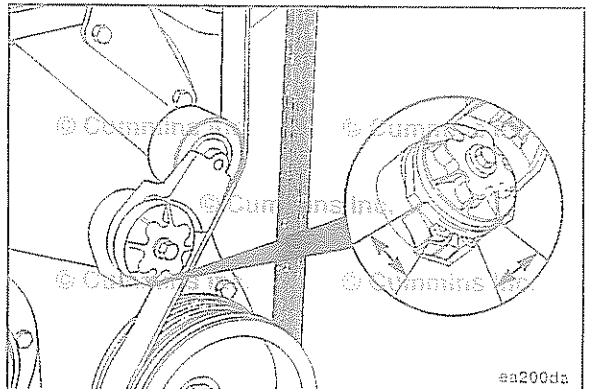
Release the tensioner, and remove the breaker bar.

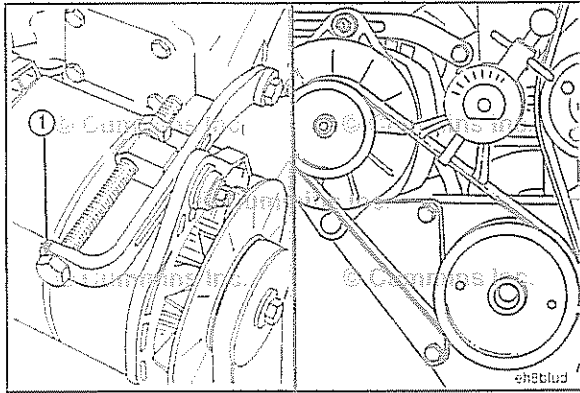
Belt drive systems equipped with an automatic belt tensioner can not be adjusted. A belt tension gauge will not give an accurate measure of the belt tension. The automatic belt tensioner is designed to maintain the proper belt tension over the life of the belt. Only inspection of the tensioner is required.



The belt tensioner is designed to operate within the limit of arm movement provided by the cast stops when the belt length and geometry are correct.

If the tensioner is hitting either of the limits during operation, check the mounting brackets and belt length. Loose brackets, bracket failure, alternator movement, incorrect belt length, or belt failure can cause the tensioner to hit the limits.





Install a new belt on the water pump and alternator pulleys. To prevent damage, do not roll a belt over the pulley or pry on it with a tool.

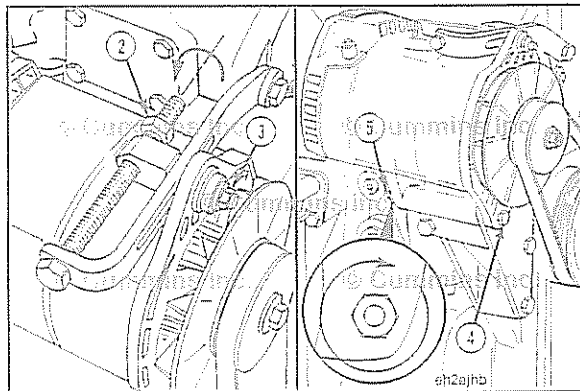


Turn the adjusting screw (1) clockwise to increase the belt tension.



Use belt tension gauge, Part Number ST-1293, or equivalent, to measure the belt tension. Refer to Procedure 018-005 (Drive Belt Tension) in Section V for the correct tension value for the belt is installed.

NOTE: A belt is considered used if it has been in operation for ten minutes or longer.



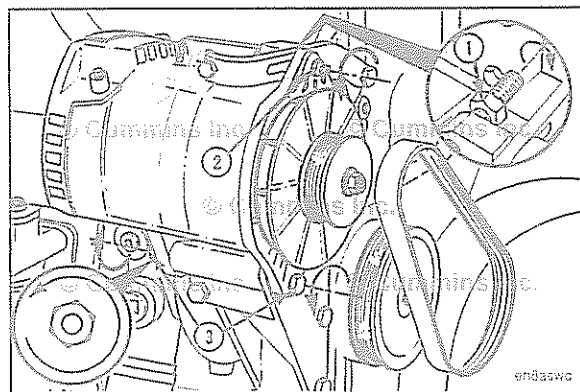
Tighten the adjusting screw locknut (2) against the retainer.

Tighten the adjustment link locking capscrew (3).

Torque Value: 80 N•m [60 ft-lb]

Tighten the pivot capscrew (4) and nut (5).

Torque Value: 47 N•m [35 ft-lb]



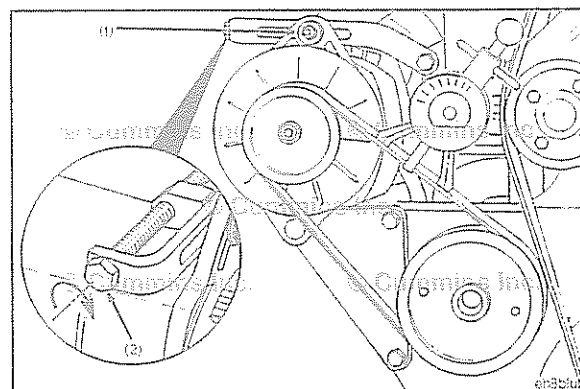
Adjust

Adjustment Link Type

Loosen the adjusting screw locknut (1).

Loosen the adjustment link locking capscrew (2).

Loosen the pivot capscrew and nut (3).



Use belt tension gauge, Part No. ST-1293, to measure belt tension.



Turn the alternator adjusting screw (1) clockwise to tighten the belt. Refer to Procedure 018-005 (Drive Belt Tension) in Section V.

NOTE: A belt is considered used if it has been in operation for ten minutes or longer.

Tighten the adjusting screw locknut (2) against the retainer.

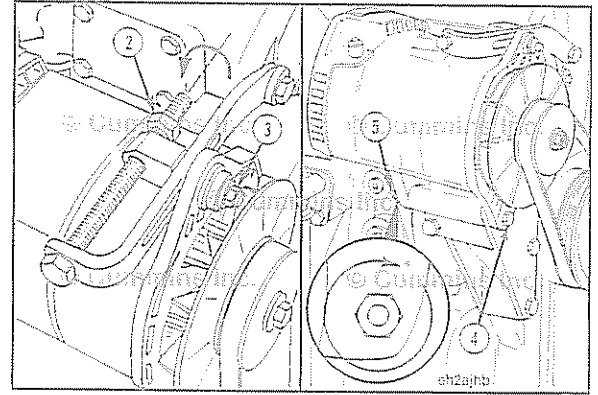


Tighten the adjustment link locking capscrew (3).

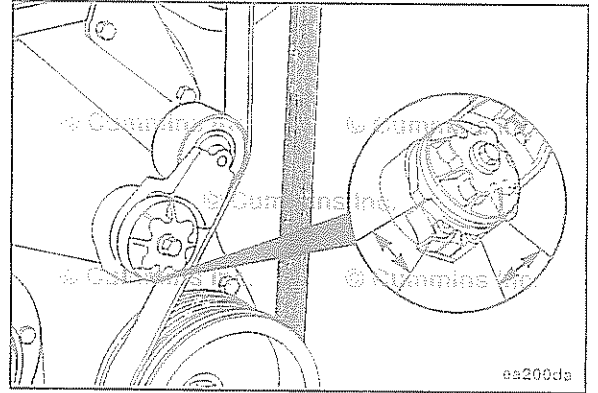
Torque Value: 80 N•m [60 ft-lb]

Tighten the pivot capscrew (4) and nut (5).

Torque Value: 47 N•m [35 ft-lb]



Belt drive systems equipped with an automatic belt tensioner can not be adjusted. A belt tension gauge will not give an accurate measure of the belt tension. The automatic belt tensioner is designed to maintain proper belt tension over the life of the belt. Only inspection of the tensioner is required.



The belt tensioner is designed to operate within the limit of arm movement provided by the cast stops when the belt length and geometry are correct.

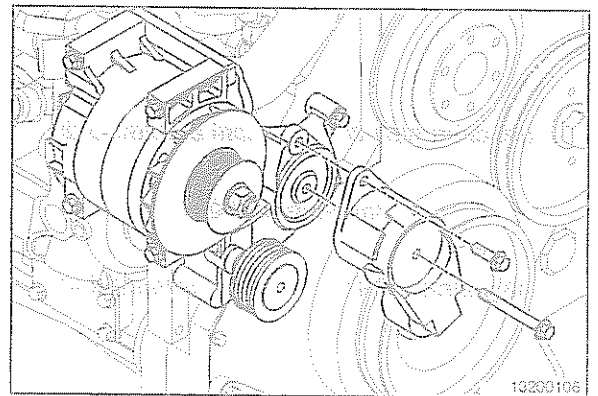
If the tensioner is hitting either of the limits during operation, check the mounting brackets and the belt length. Loose brackets, bracket failure, alternator movement, incorrect belt length, or belt failure can cause the tensioner to hit the limits.

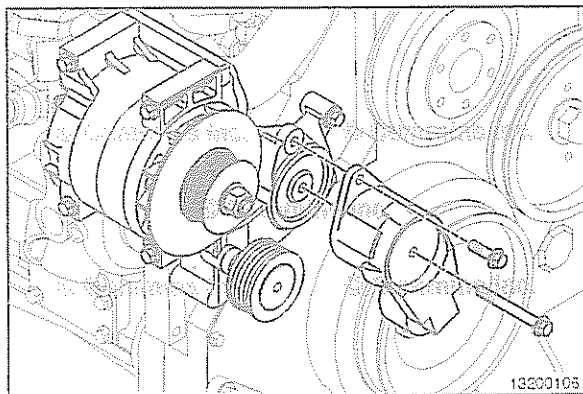
NOTE: If the engine is equipped with a secondary alternator, reference to the vehicle manufacturer for the secondary alternator belt adjustment procedure.

Belt Tensioner, Automatic (Alternator) General Information

A new automatic tensioning single belt fan and alternator drive is the standard option for the QSM industrial engine. The design uses an eight-rib poly-vee belt with an automatic tensioner and a fixed idler.

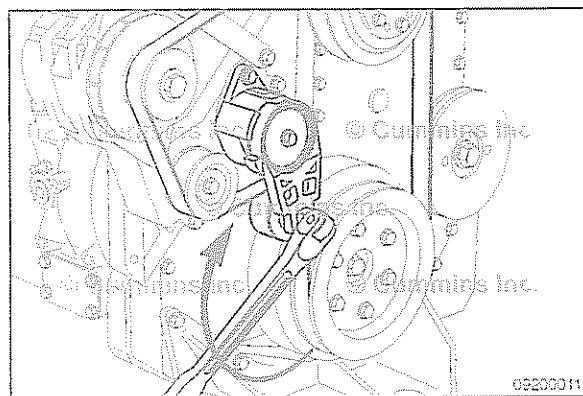
The belt tensioner is designed to operate within the limit of arm movement provided by the cast stops when the belt length and geometry are correct. If the tensioner is hitting either of the limits during operation, check the mounting brackets and the belt length. Loose brackets, bracket failure, alternator movement, incorrect belt length, or belt failure can cause the tensioner to hit the limits.





Remove

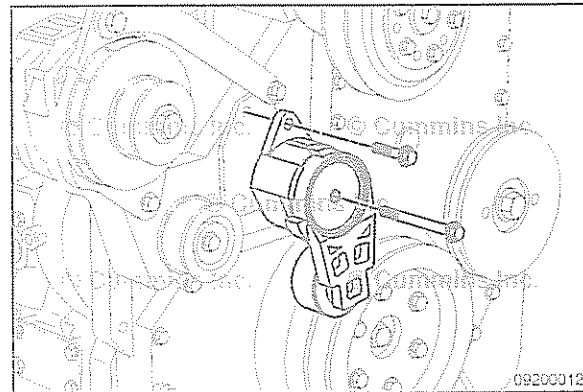
Remove the mounting capscrews and the belt tensioner from the bracket.



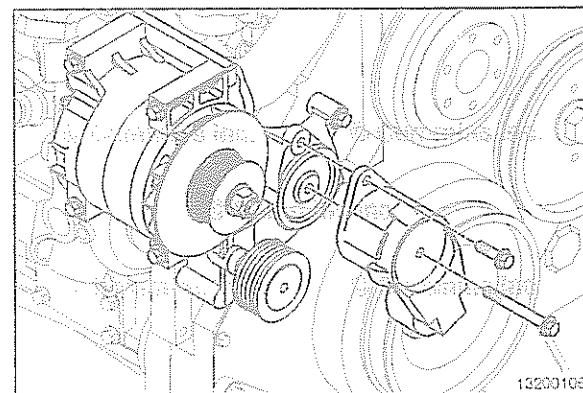
Use a 3/4-inch drive breaker bar to hold back the tensioner and relieve the belt tension.

Rotate the tensioner away from the belt until it stops.

Remove the alternator belt while holding the tensioner back.



Remove the belt tensioner from the engine.



Install

Install the belt tensioner.

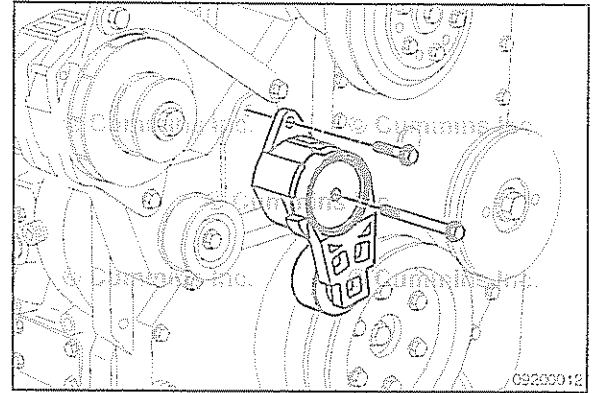


Install and tighten capscrews.

Torque Value: 47 N•m [35 ft-lb]

Tighten the tensioner mounting capscrew.

Torque Value: 47 N•m [35 ft-lb]

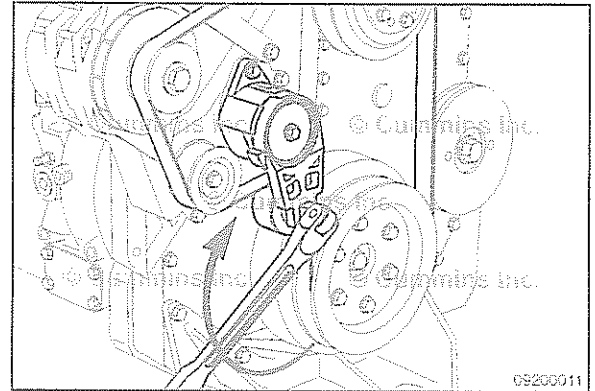


Install the alternator belt over the pulleys while holding the tensioner back with a 3/4-inch drive breaker bar. Use care in working belt over the edge of flanged pulleys.



Release the tensioner and remove the breaker bar.

The belt tensioner is designed to operate well within the limit of arm movement, provided by the cast stops, when the belt length and geometry are correct.



Water Pump

Remove

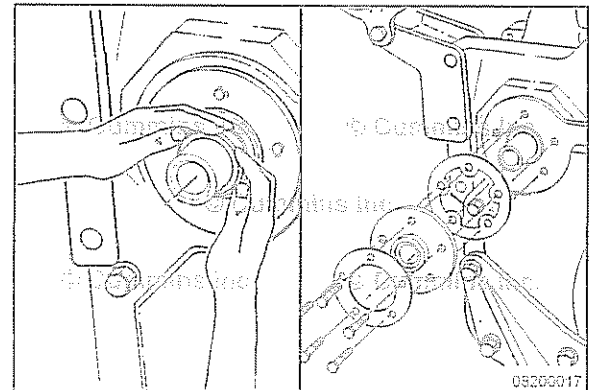
With EGR

Remove the dust seal.

Remove the five water pump oil seal capscrews, clamping ring, oil seal, and gasket.

Remove the dust seal as the seal carrier is removed; or use a heel bar, or similar tool, to pry the dust seal away from the seal case. Then remove the dust seal by hand.

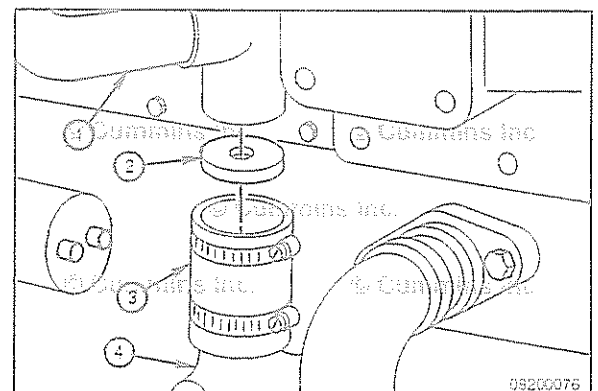
Discard the oil seal and dust seal.

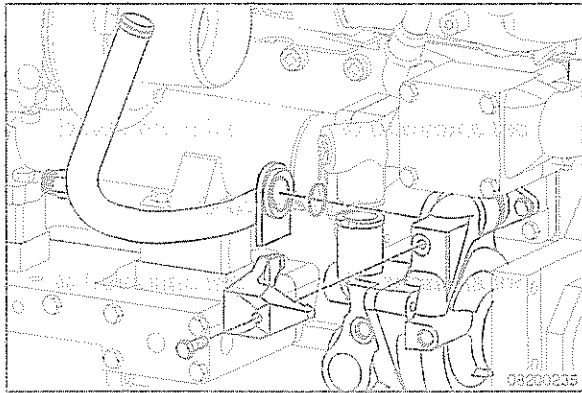


The coolant flow that provides cooling to the torque converter (if equipped) is achieved in different manners.

ISM and QSM Series engines use a torque converter cooler disc inside the coolant bypass hose to direct engine coolant to the inlet side of the torque converter cooler.

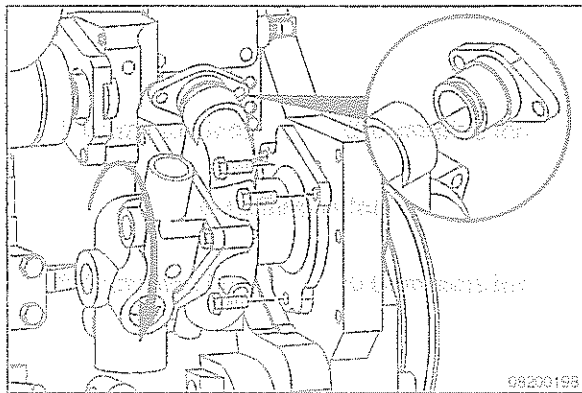
1. Torque converter coolant supply
2. Torque converter disc (orifice)
3. Bypass hose
4. Water pump.





Loosen the EGR cooler coolant supply lower tube retaining clamp capscrew. Remove the EGR cooler coolant supply lower tube retaining clamp and tube.

Remove and discard the o-ring.

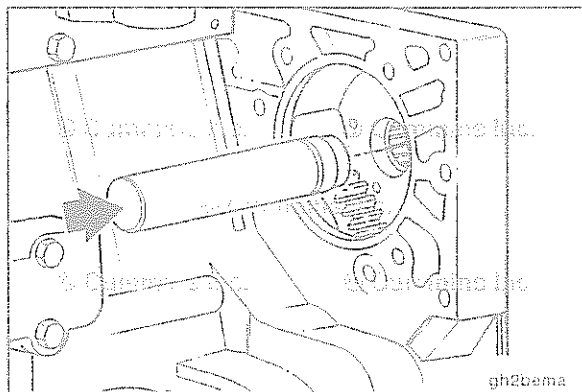


Remove the three water pump mounting capscrews.

Remove the two water pump water transfer connection capscrews and push the water pump transfer connection into the water pump as far as possible to facilitate water pump removal.

Remove the water pump. Twist the pump outward from the top, and angle the rear of the pump downward as it is being removed to allow the pump to pass the thermostat housing support.

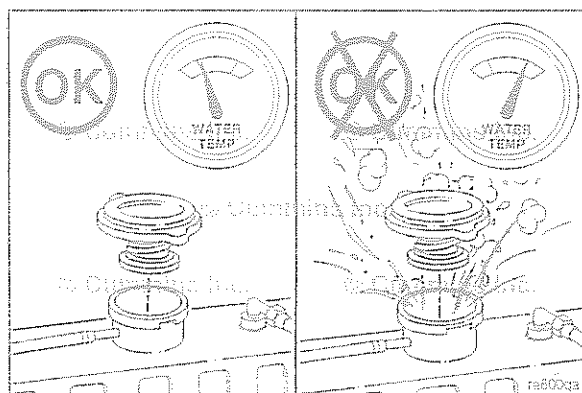
Remove the water pump water transfer connection from the water pump.



Check the needle bearing for damage and freedom of needle rotation. Replace the bearing if it is damaged or if it does not turn freely.



If the bearing is replaced, use bearing installation tool, Part Number 3824117, along with a cup driver to remove the needle bearing from the gear housing. Gently tap the bearing out from the rear side of the housing.



⚠ WARNING ⚠

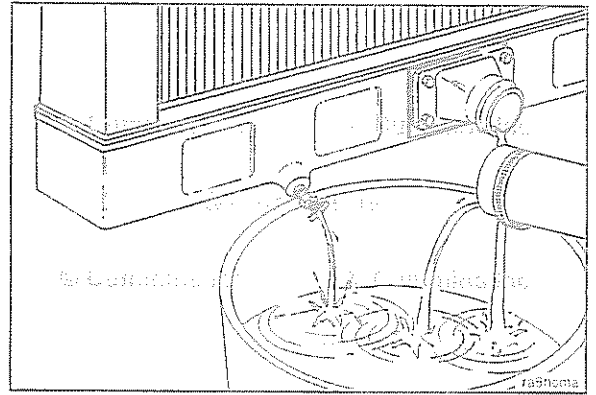
Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

Remove the pressure cap when the engine is cool.



Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Drain the cooling system. Refer to Procedure 008-018 (Cooling System) in Section 7.

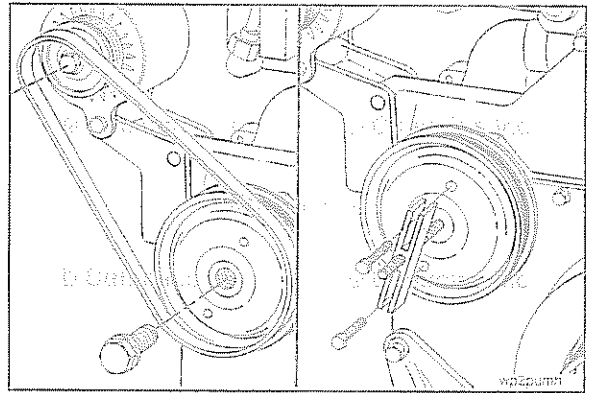


NOTE: Be sure the puller capscrews are threaded all the way through the puller before applying pressure to the puller screw.

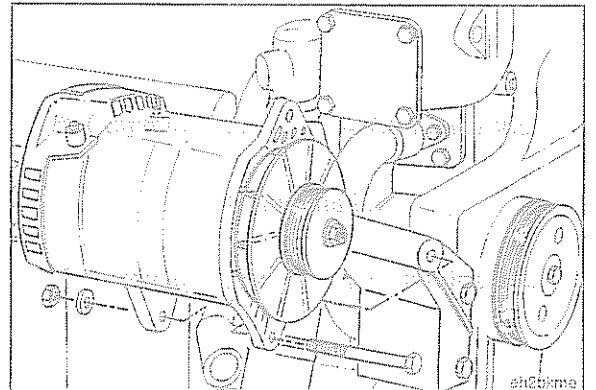
Remove the alternator drive belt. Refer to Procedure 013-005 (Drive Belt, Alternator) in Section A.

Remove the water pump pulley retaining capscrew.

Use the standard pulley puller, Part Number ST-647, or equivalent, and two 5/16 x 18 x 2 capscrews to remove the pulley.



Remove the alternator. Refer to a Cummins Authorized Repair Location.

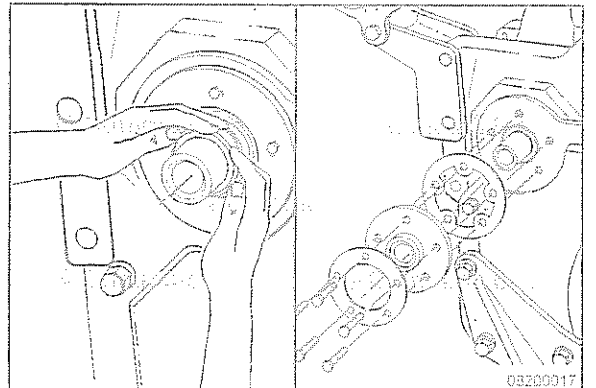


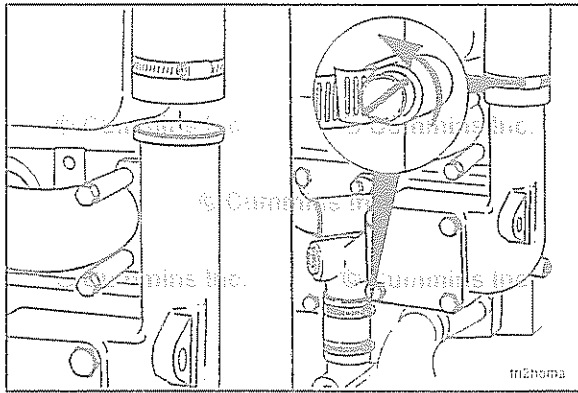
NOTE: Remove the dust seal as the seal carrier is removed, or use a heel bar, or similar tool, to pry the dust seal away from the seal case. Then remove the dust seal by hand.

Remove the dust seal.

Remove the five water pump oil seal capscrews, clamping ring, oil seal, and gasket.

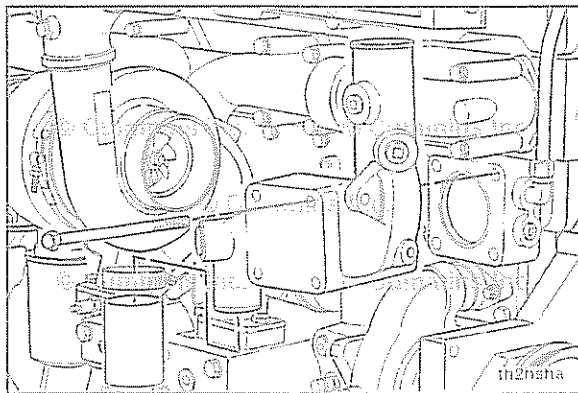
Discard the oil seal and dust seal.



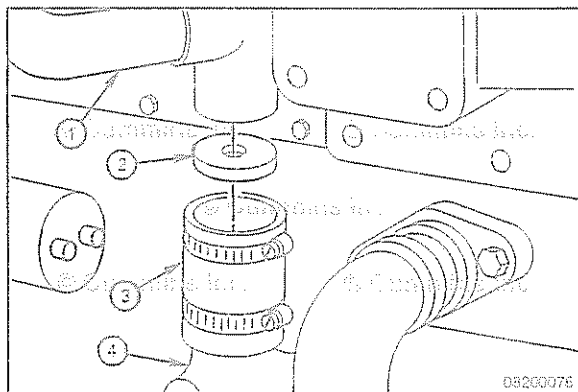


Loosen the coolant bypass hose clamps on both the upper and lower hoses.

Remove the upper coolant hose from the thermostat housing.



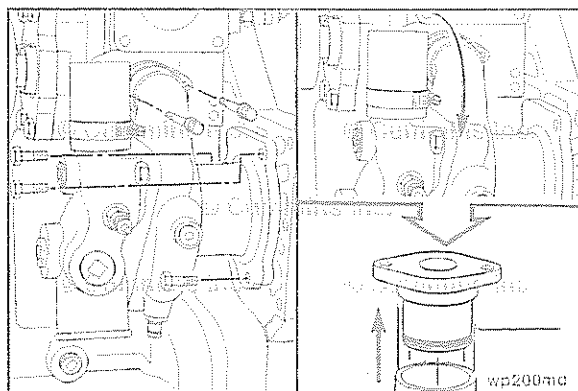
Remove the four thermostat housing mounting capscrews and the thermostat housing.



The coolant flow that provides cooling to the torque converter (if equipped) is achieved in different manners.

ISM and QSM Series engines use a torque converter cooler disc inside the coolant bypass hose to direct engine coolant to the inlet side of the torque converter cooler.

1. Torque converter coolant supply.
2. Torque converter disc (orifice).
3. Bypass hose.
4. Water pump.



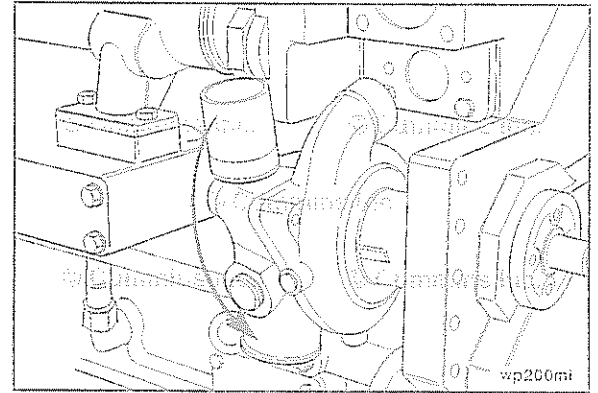
Remove the two water pump water transfer connection capscrews.

Remove the three water pump mounting capscrews.

Rotate the water pump outward so the water transfer connection can be removed from the water pump.

Remove the water transfer connection from the water pump.

Remove the water pump. Twist the pump outward from the top, and angle the rear of the pump downward as it is being removed to allow the pump to pass the thermostat housing support.



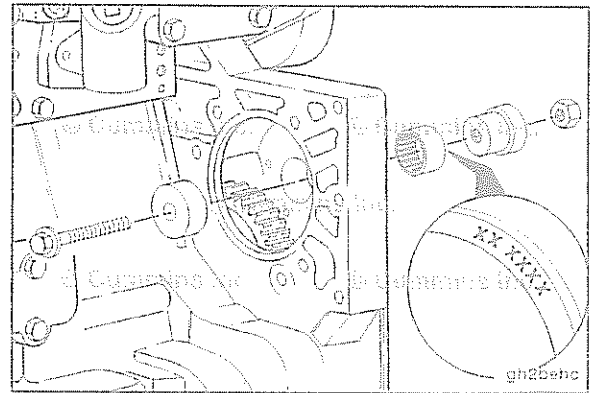
Install

With EGR

If the needle bearing was removed, use bearing installation tool, Part Number 3824117, to install a new needle bearing in the gear housing.

The bearing must be installed with the part number side of the bearing against the installation tool to prevent damage to the bearing during the installation.

Install the bearing from the front side of the gear housing until the bearing is flush with the front edge of the housing bore.



CAUTION

In order to reduce the possibility of pinching or rolling the o-rings during assembly do not lubricate the o-rings. Lubricate the o-ring connection mating surface with clean engine coolant, soapy water, or vegetable oil.

Install the water pump cover. Use a new o-ring seal.

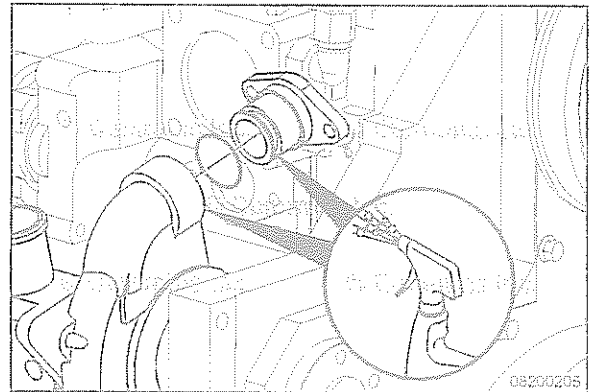
Torque Value: 47 N•m [35 ft-lb]

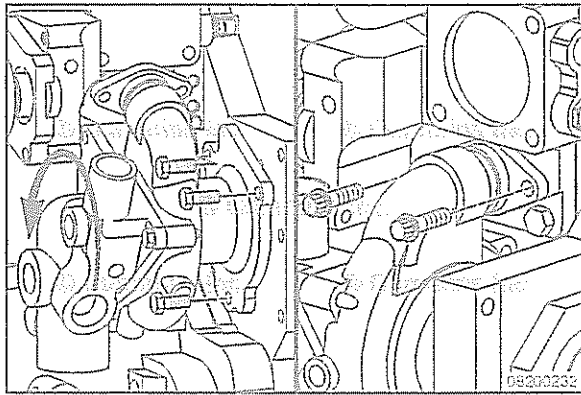
Install the water pump water transfer connection into the water pump.

Install a new o-ring on the water pump mounting flange.

The water pump must be twisted outward from the top until the transfer outlet clears the thermostat housing support during installation.

Install the water pump.





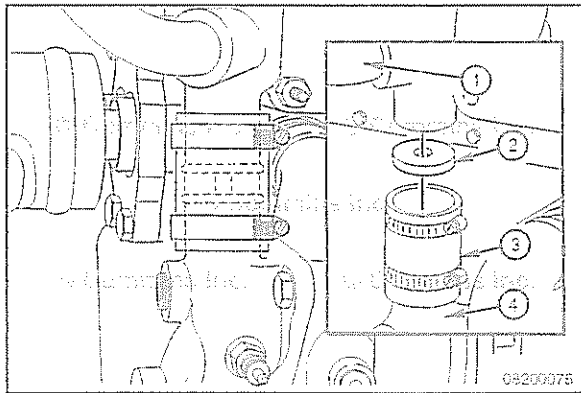
Twist the water pump inward and install the three water pump mounting capscrews.

Torque Value: 47 N•m [35 ft-lb]



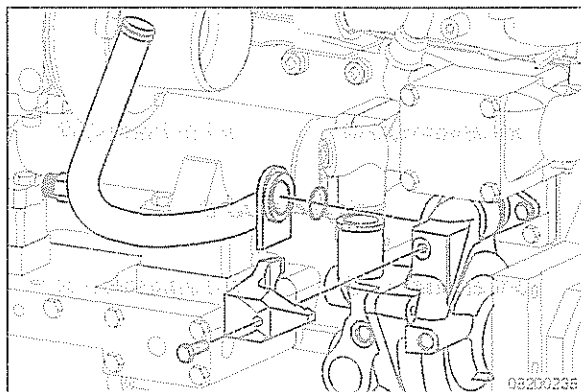
Install a new gasket on the water pump water transfer connection. Install and tighten the water transfer connection capscrews.

Torque Value: 25 N•m [18 ft-lb]



If the engine is equipped with a torque converter cooler, install the disc in the bypass hose before installing the thermostat housing.

1. Torque converter coolant supply
2. Torque converter disc (orifice)
3. Bypass hose
4. Water pump.



⚠CAUTION⚠

In order to reduce the possibility of pinching or rolling the o-rings during assembly do not lubricate the o-rings. Lubricate the o-ring connection mating surface with clean engine coolant, soapy water, or vegetable oil.

Install a new o-ring on the EGR cooler coolant supply lower tube.

Insert the EGR cooler coolant supply lower tube into the water pump body.

Install the EGR cooler coolant supply lower tube is seated completely in the water pump body before tightening the EGR cooler coolant supply lower tube retaining clamp capscrew to reduce the possibility of pinching the o-ring.

Tighten the capscrew.

Torque Value: 34 N•m [25 ft-lb]

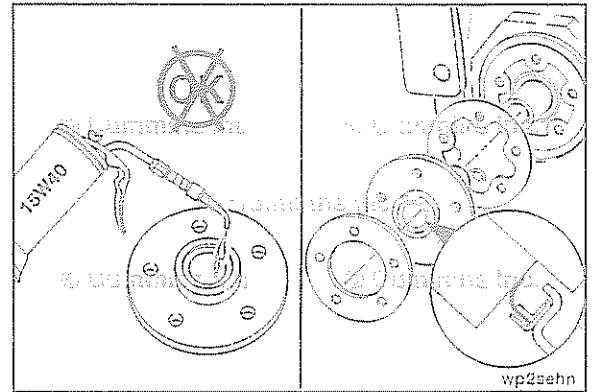
The oil seal must be installed with the lip of the seal and the shaft clean and dry. Do not lubricate. The yellow dust lip must be facing out.



Install the new gasket and oil seal. Use the installation sleeve provided with the new seal to install the seal.

The capscrew threads must be coated with thread sealant, Part Number 3823494, or equivalent, to prevent oil leakage.

Torque Value: Step1	7 N•m	[60 in-lb]
Step2	20 N•m	[180 in-lb]

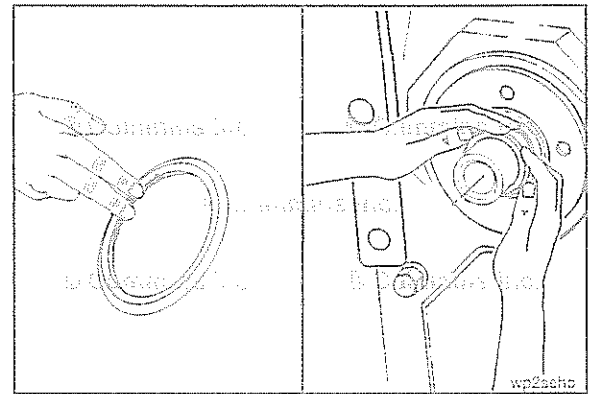


Place a light film of oil or antifreeze on the inside of the oil seal dust seal.



Install the dust seal onto the shaft with a larger outside surface facing the engine.

Push the dust seal back by hand on the shaft until the entire dust seal contacts the oil seal case.

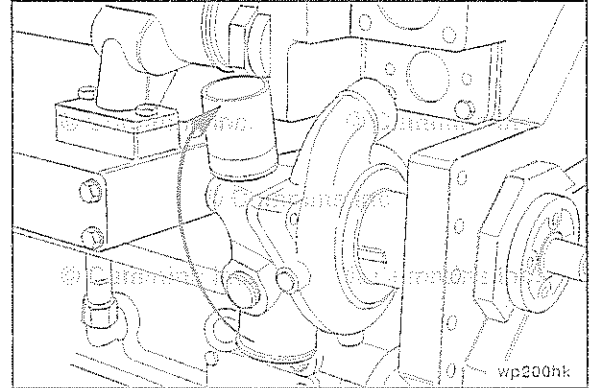


NOTE: The water pump must be twisted outward from the top until the transfer outlet clears the thermostat housing support during installation.



Install a new o-ring on the water pump mounting flange.

Install the water pump.



Install a new o-ring on the water pump water transfer tube.
Install the connection into the water pump.



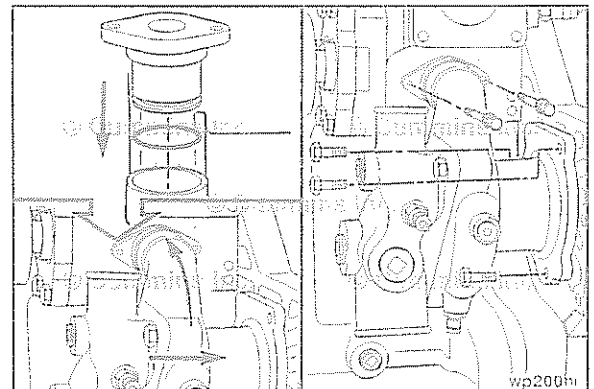
Twist the water pump inward, and install the three water pump mounting capscrews.

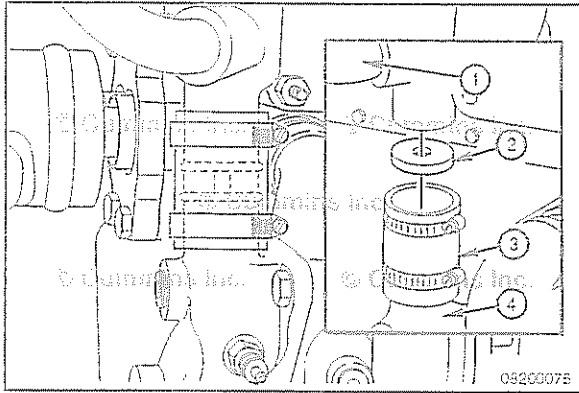


Torque Value: 47 N•m [35 ft-lb]

Install a new gasket on the water pump water transfer connection. Install and tighten the water transfer connection capscrews.

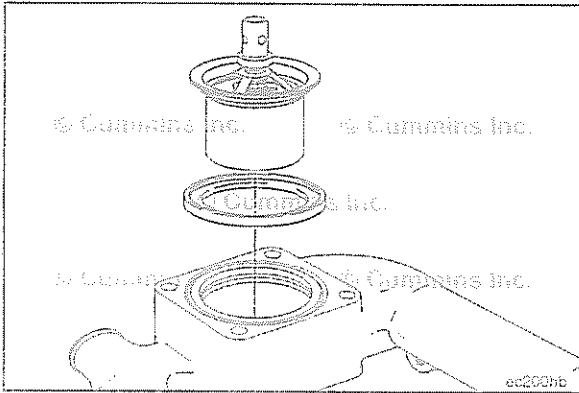
Torque Value: 25 N•m [221 in-lb]





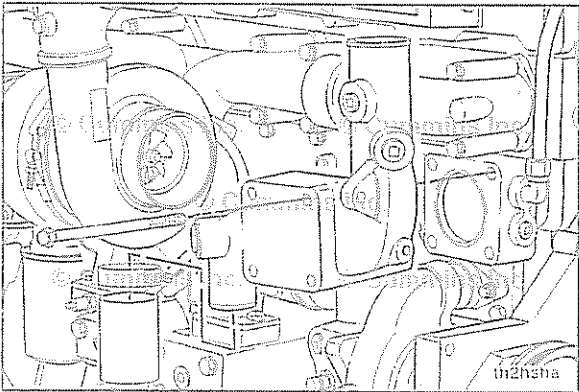
If the engine is equipped with a torque converter cooler, install the disc in the bypass hose before installing the thermostat housing.

1. Torque converter cooler coolant supply.
2. Torque converter cooler disc (orifice).
3. Bypass hose.
4. Water pump.



Install the thermostat in the housing.

Install a new seal in the groove on the thermostat housing mounting surface.

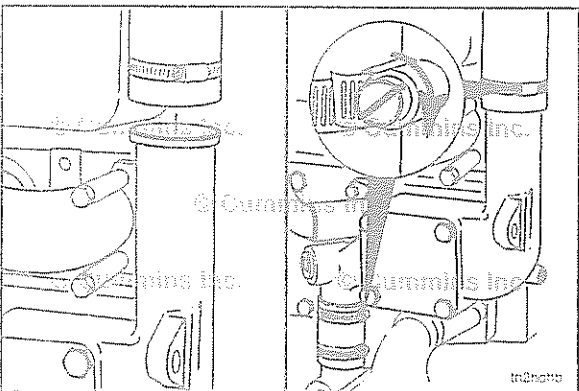


Install the hose on the thermostat housing bypass outlet.

Install the thermostat housing and four mounting capscrews.



Torque Value: 54 N•m [40 ft-lb]



Equally space the bypass hose over the water pump connection and thermostat housing connection, and tighten the bypass hose clamps.



Torque Value: 3 N•m [27 in-lb]

Install the upper and lower coolant hoses. Refer to the OEM's specifications for the correct torque value.

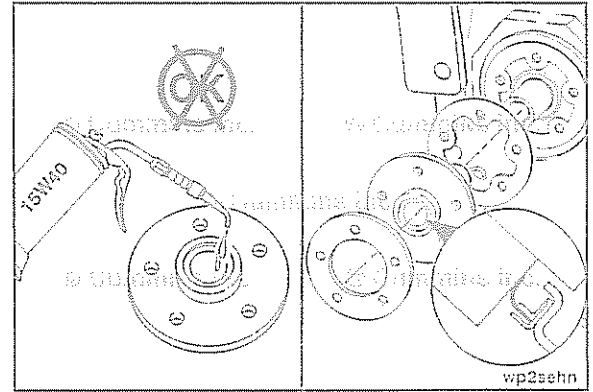


The oil seal must be installed with the lip of the seal and the shaft clean and dry. Do not lubricate. The yellow dust lip must be facing out.

Install the new gasket and oil seal. Use the installation sleeve provided with the new seal to install the seal.

The capscrew threads must be coated with thread sealant, Part No. 3823494, or equivalent, to prevent oil leakage.

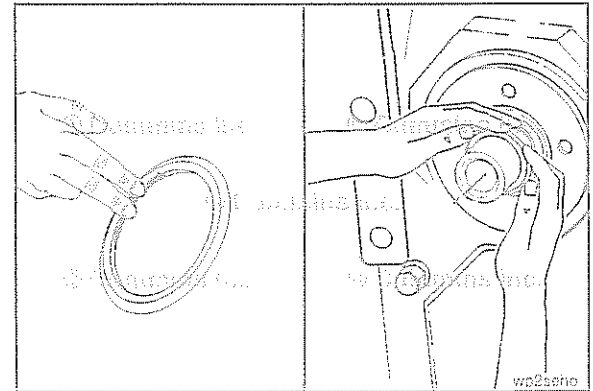
Torque Value: Step1	7 N•m	[62 in-lb]
Step2	20 N•m	[177 in-lb]



Place a light film of oil or antifreeze on the inside diameter of the dust seal.

Install the dust seal onto the shaft with the larger outside diameter facing the engine.

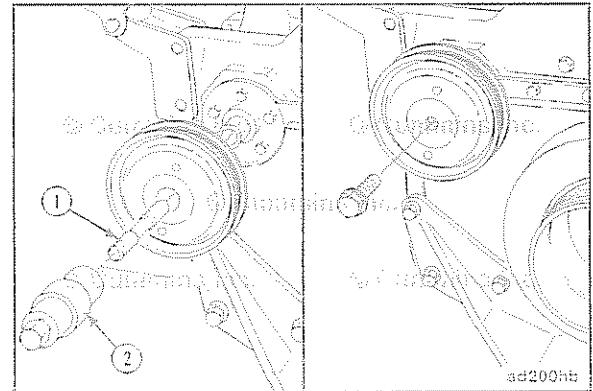
Push the dust seal back by hand on the shaft until the entire dust seal contacts the oil seal case.



Use pulley pusher adapter (1), Part Number 3377401, or equivalent, and pulley pusher (2), Part Number 3376326, or equivalent, to install the pulley.

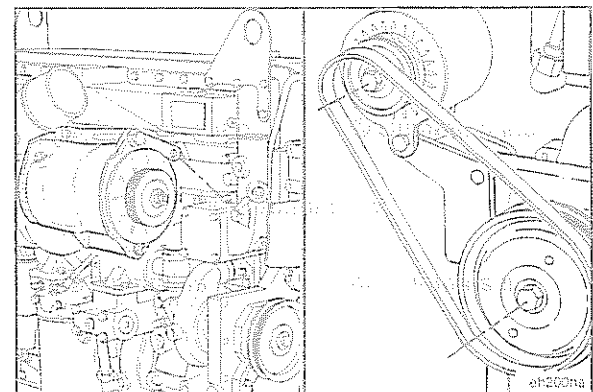
Install the capscrew in the shaft.

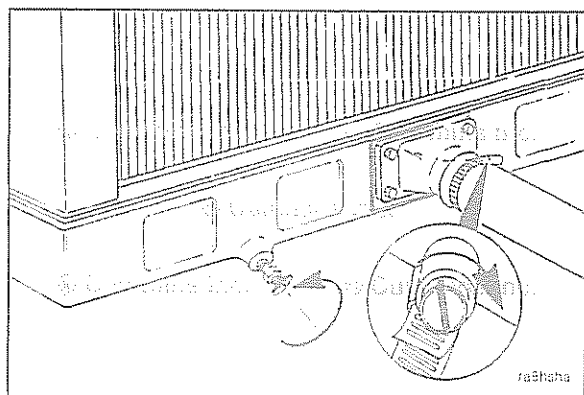
Torque Value: 75 N•m [55 ft-lb]



Install the alternator. Refer to a Cummins Authorized Repair Location.

Install and adjust the alternator drive belt. Refer to 013-005 (Drive Belt, Alternator) in Section A.

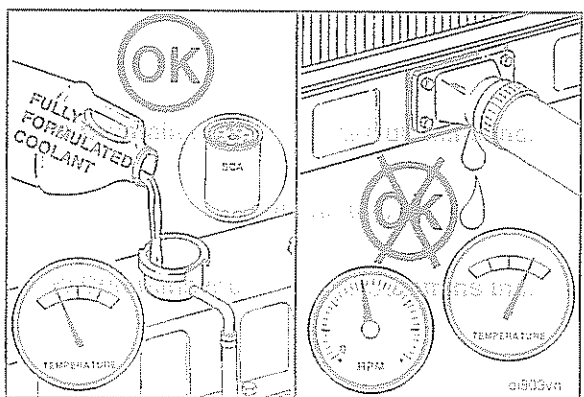




Close the cooling system draincock, and install the lower coolant hose.



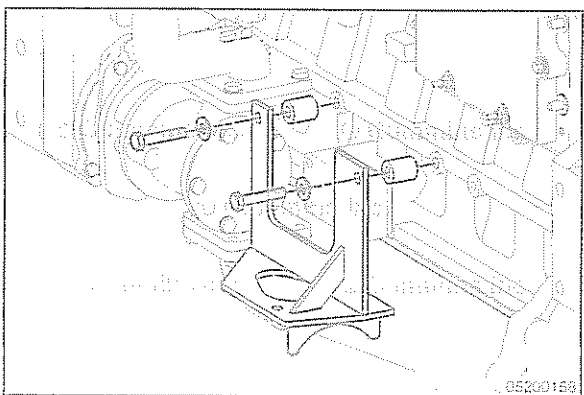
Tighten the hose clamp(s). Refer to the manufacturer's specifications for the correct torque value.



The correct concentration of coolant additives must be used in the cooling system. Refer to Procedure 018-004 (Coolant Recommendation and Specifications) in Section V.

Fill the cooling system. Refer to Procedure 008-018 (Cooling System) in Section 7.

Operate the engine until it reaches a temperature of 71° C [160°F] and check for coolant leaks.



Sea Water Pump Preparatory Steps

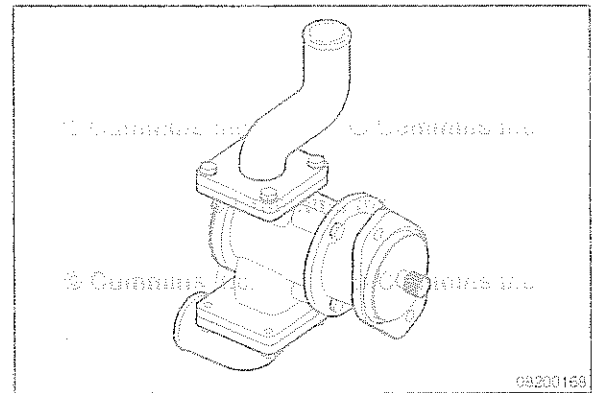
⚠CAUTION⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be disposed of in accordance with local environmental regulations

- Shut off the sea water inlet valves.
- Shut off the fuel supply and return valves (if a major kit repair is planned).
- Remove the fuel cooler (if a major kit repair is planned).
- Remove the fuel cooler mounting bracket for access to the sea water pump cover.

Remove

If a major kit repair is being performed, remove the sea water pump.



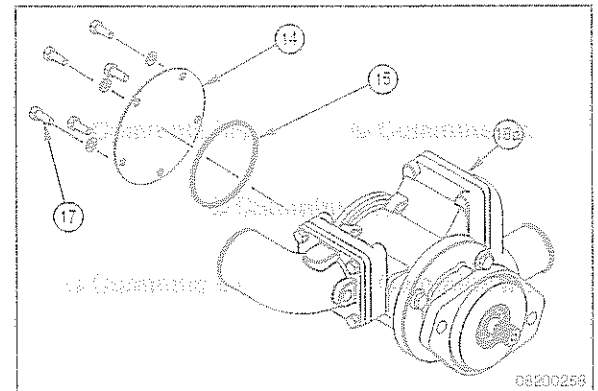
Disassemble

Marine Applications

If a minor kit repair is being performed, remove the cover plate.

Remove the five capscrews (17) and the washers and the cover (14) from the sea water pump.

Remove and discard the quad-ring (15).

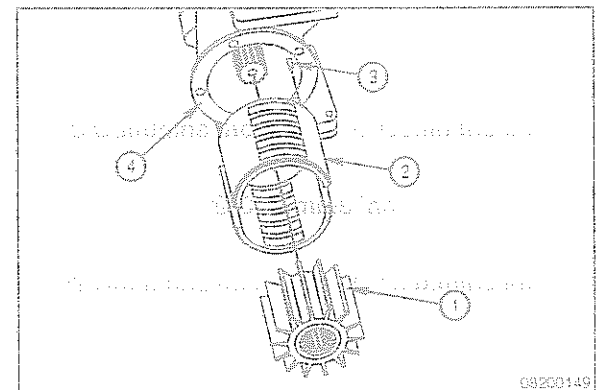


Do not pry against the cam liner to remove the impeller as this will break the cam liner.

NOTE: The two-piece cam liner will separate when removed. Do not drop the cam liner.

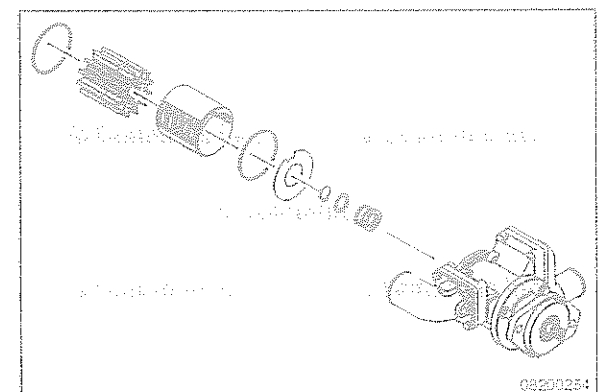
Check the direction of the impeller blades to make sure the impeller is properly installed at assembly.

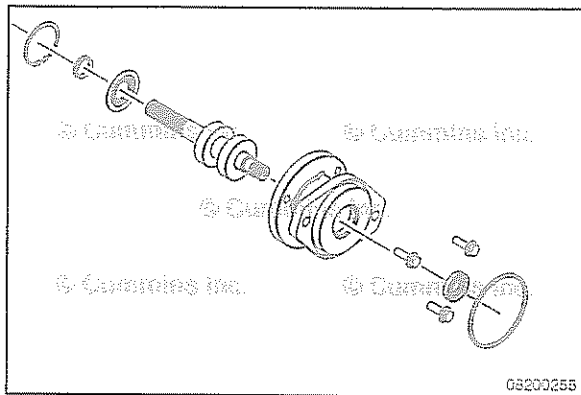
Remove the impeller (1), two-piece cam liner (2), and pin (3) from the sea water pump (4).



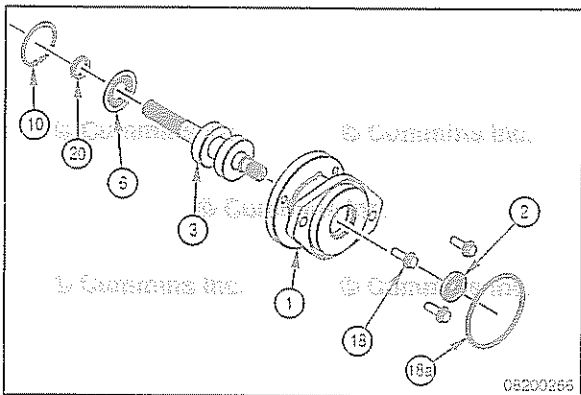
Remove the second quad ring and wearplate.

Remove the retaining ring from the shaft. Slide the washer and mechanical seal off of the shaft.





- Remove the bolts holding the body to the housing.
- Slide the housing off of the shaft and body assembly.
- Press the seal seat out of the housing.
- Remove the retaining ring from the body.
- Press the bearing and shaft assembly out of the body.
- Remove V-ring and bearing shield from the shaft.
- Remove the lip seal.



Assemble

Marine Applications

- Press the lip seal (2) into the pump body (1) with the backing ring of the lip seal toward the main pump housing.
- Align the lip seal so that it is uniformly pressed in the pump body. Press the shaft and bearing assembly (3) into the pump body (1).
- Slide the bearing shield (6) into the pump body and against the inner bearing.
- Place the V-ring (20) on the shaft against the bearing shield, with the open end of the V-ring toward the bearing assembly.
- Use snap ring pliers to install the retaining ring (10) into the body.



Do not use petroleum based lubricant. A petroleum based lubricant will cause damage to the impeller.

NOTE: Grease can be used to hold the pin in place during installation.

Press the mechanical seal (27) into the housing with the white ceramic seal face toward the impeller.

Lubricate the outside of the seat boot with a water-soluble lubricant or soapy water for easier assembly.

Slide the housing (22) over the shaft and body assembly.

Install the three bolts (18).

Torque Value: 34 N•m [25 ft-lb]

Install the washer (12) onto the shaft and use snap ring pliers to install the retaining ring (11) into the groove in the shaft (3) to compress the mechanical seal.

Install the wearplate (13) with the hole in the plate lined up with the slot in the housing.

Install the quad ring (15) into the housing on top of the wearplate.

Assemble the cam halves (9) together and install the pin (19) into the cam liner.

Place the cam assembly into the housing, align the pin with the slot in the housing.

Insert the impeller (8) in the housing (22) using a twisting motion.

Make sure the impeller blades are in the same orientation as when removed.

Apply a silicon lubricant to aid in the installation of the impeller.

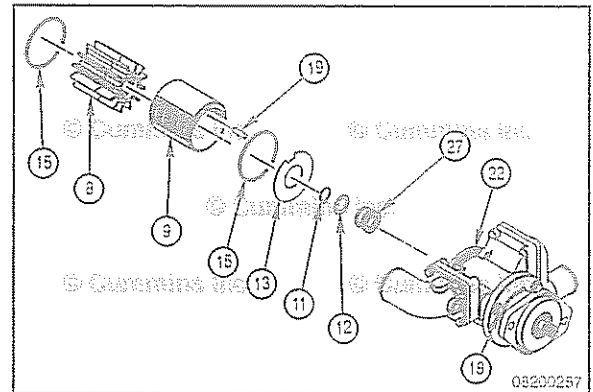
Insert the remaining quad ring (15) into the groove.

Install the cover (14) onto the housing (18a).

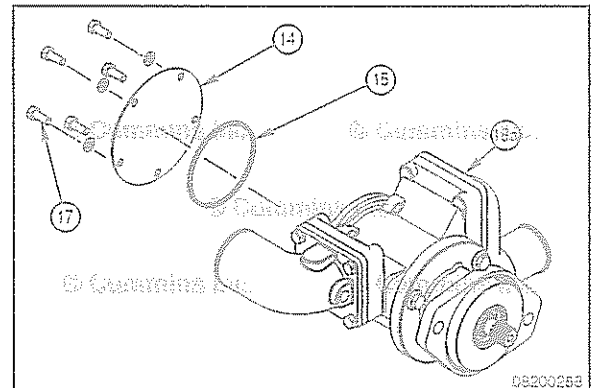
Install bolts (17).

Torque Value: 11 N•m [96 in-lb]

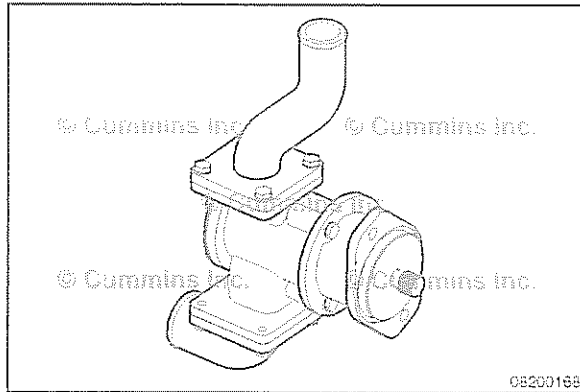
Install the front flange sealing o-ring onto the pump body.



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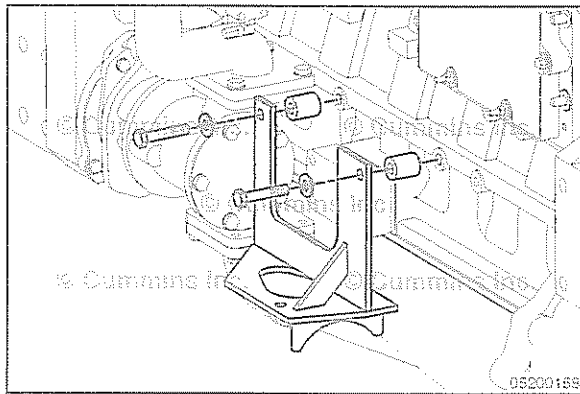


03200253



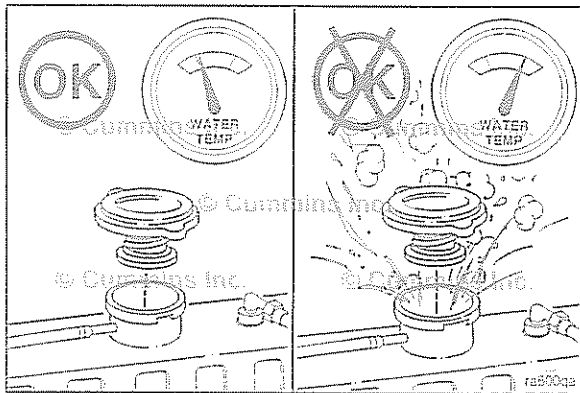
Install

If a major kit repair was performed, install the sea water pump.



Finishing Steps

- Install the fuel cooler mounting bracket.
- Install the fuel cooler (if a major repair kit was installed).
- Open the fuel supply and return valves (if the fuel cooler was removed).
- Open the sea water inlet valve.
- Operate the engine and check for leaks.

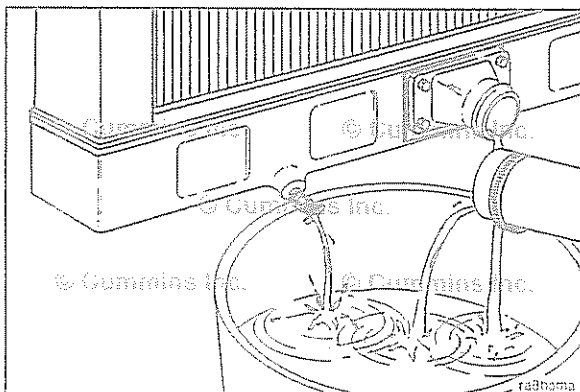


Coolant Thermostat Remove

▲ WARNING ▲

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the coolant system pressure cap. Heated coolant spray or steam can cause personal injury.

Remove the pressure cap when the engine is cool.



▲ WARNING ▲

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

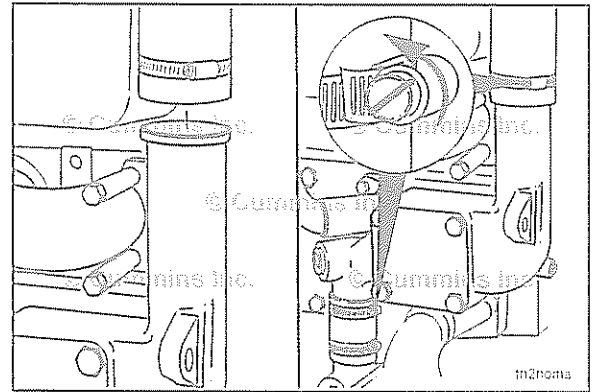
Drain the cooling system as follows:

- Open the radiator draincock.
- Remove the lower radiator hose.

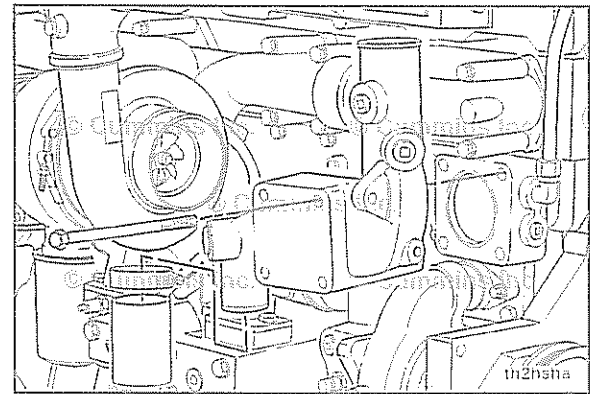
Remove the upper coolant hose from the thermostat housing.

Loosen the coolant bypass hose clamps.

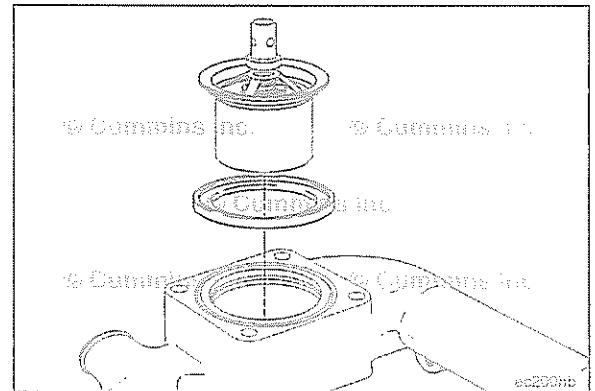
NOTE: Some models could have a converter cooler disc located in the bypass hose.



Remove the four thermostat housing mounting capscrews and the thermostat housing.



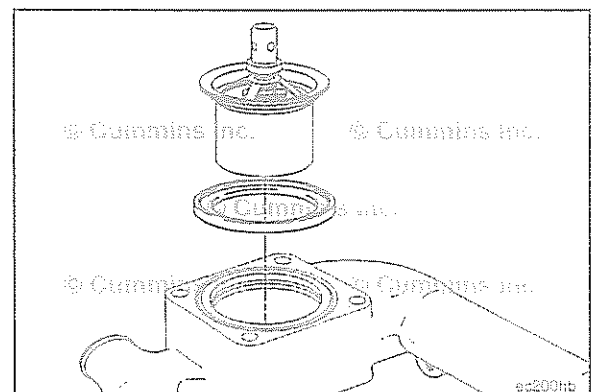
Remove the thermostat from the housing.

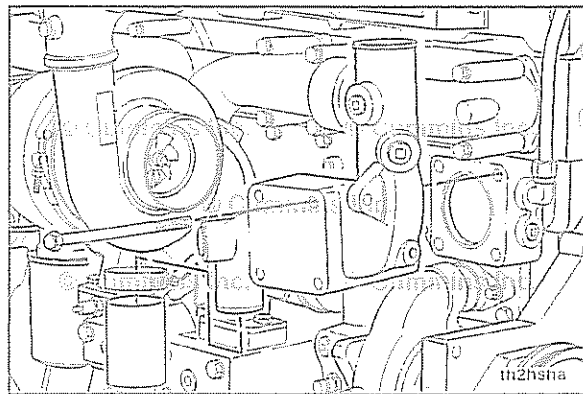


Install

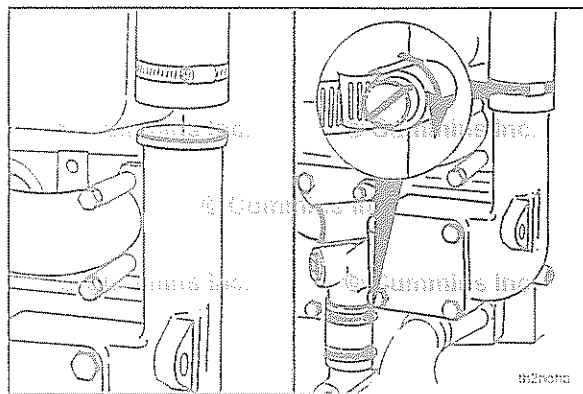
Install the thermostat in the housing.

Install a new o-ring seal in the groove on the thermostat housing mounting surface.

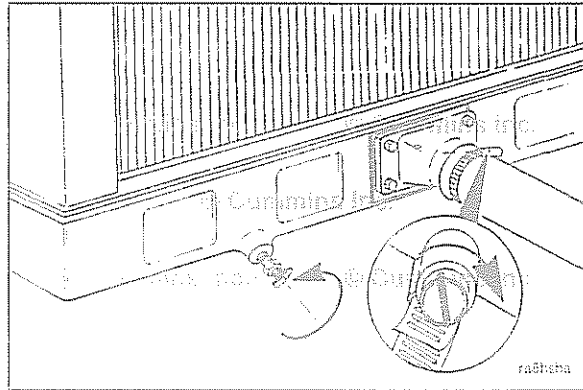




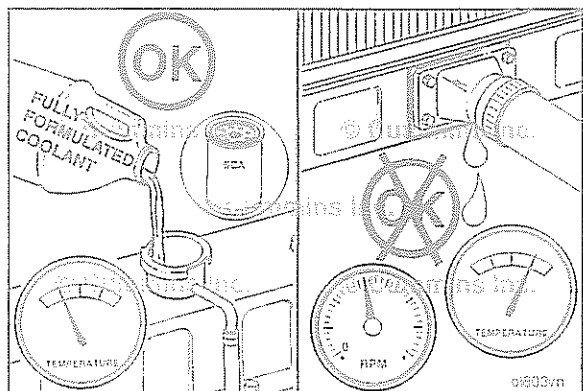
- Install the hose on the thermostat housing bypass outlet.
- Install the thermostat housing and four mounting capscrews.
- Tighten the mounting capscrews.
- Torque Value: 54 N•m [40 ft-lb]



- Install the upper coolant hose to the thermostat housing outlet.
- Torque Value: 3 N•m [30 in-lb]



- Close the cooling system draincock, and install the lower coolant hose.
- Tighten the hose clamp.
- Refer to the OEM's specifications for the correct torque value.



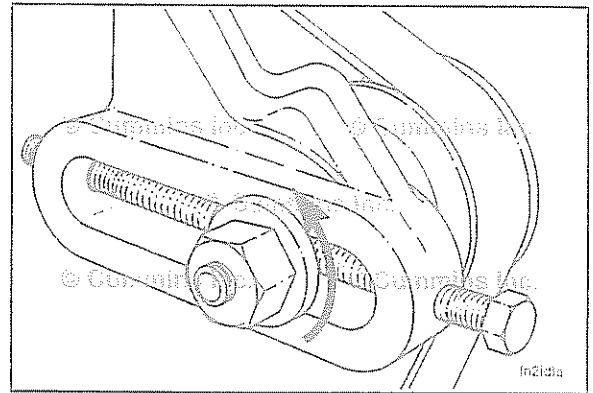
- The correct concentration of coolant additives must be used in the cooling system. Refer to Procedure 018-003 (Cummins Recommendations and Specifications) in Section V.
- Fill the cooling system. Refer to Procedure 008-018 (Cooling System) in Section 7.
- Operate the engine until it reaches 80°C [180°F], and check for coolant leaks.

Fan Drive Idler Pulley Assembly

Remove

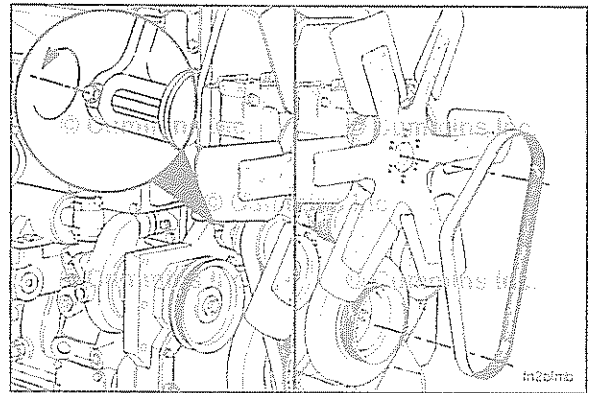
NOTE: For non-Cummins fan hub installations, contact the OEM.

Loosen the fan idler pulley shaft locknut.



Loosen the adjusting mechanism and move the fan idler pulley and fan pulley centers as close as possible.

Remove the fan belt.

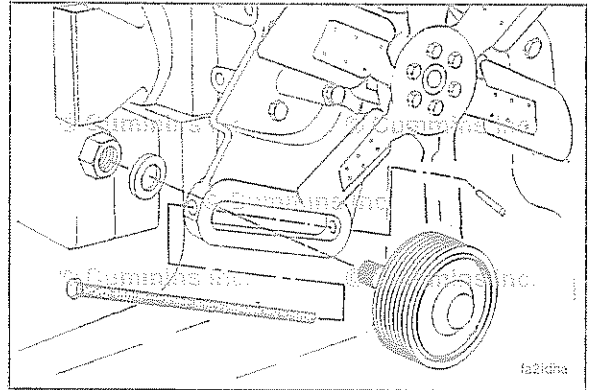


Remove the roll pin and washer from the idler pulley adjusting screw.

Remove the locknut and washer from the back of the idler pulley shaft.

Remove the adjusting screw.

Remove the fan idler pulley from the fan hub support bracket.

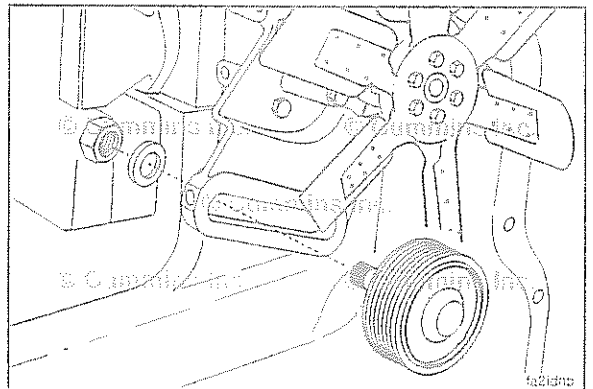


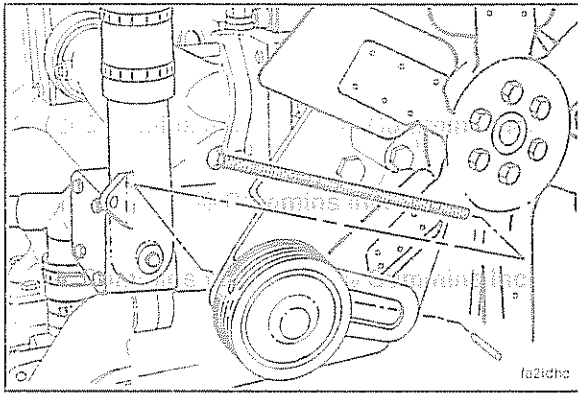
Install

NOTE: Do not tighten the locknut until the fan drive belt has been installed and adjusted.

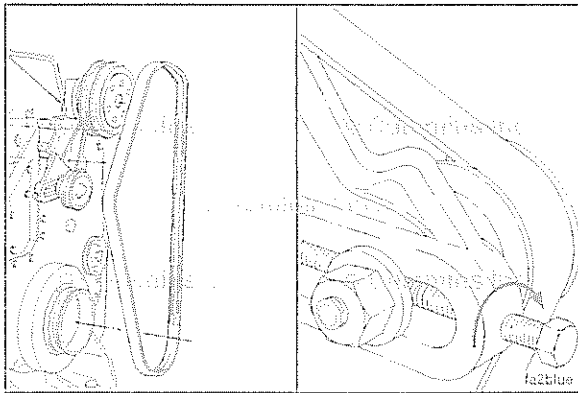
Install the idler pulley in the fan hub support bracket.

Install the washer and locknut on the idler pulley shaft.





Install the adjusting screw in the idler pulley shaft.
Turn the adjusting screw in far enough to install the washer and roll pin in the shaft at the bottom of the fan hub support bracket.



Install and adjust the fan drive belt. Refer to Procedure 008-002 (Drive Belt, Cooling Fan) in Section A.



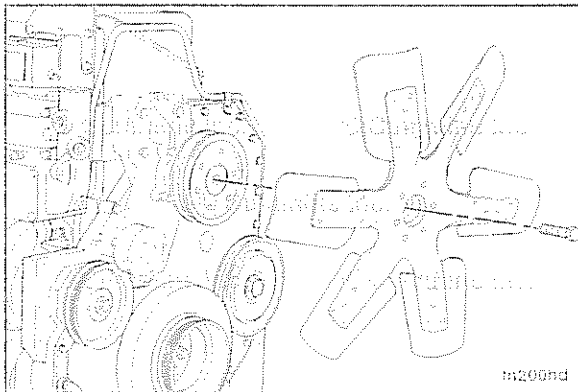
Refer to Procedure 018-005 (Drive Belt Tension) in Section V for proper belt tension value.



Tighten the idler pulley shaft locknut.

Torque Value: 190 N•m [140 ft-lb]

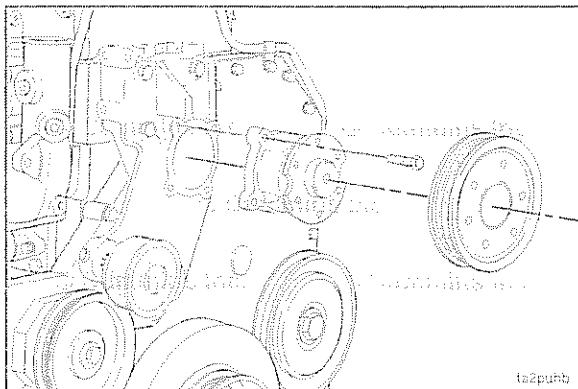
Check the belt tension again after the locknut is tightened.



Fan Hub, Belt Driven Remove

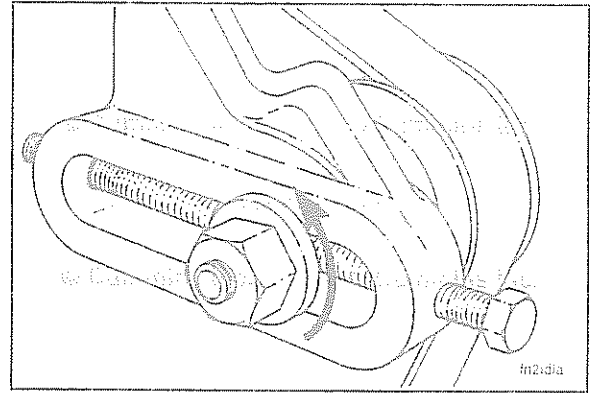


Remove the fan and clutch assembly. Refer to Procedure 008-040.

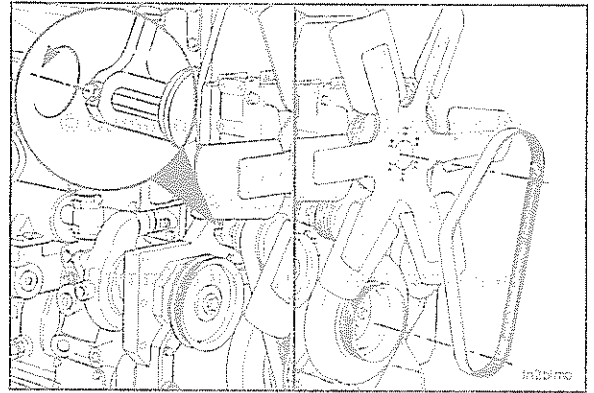


Remove the fan drive pulley.
Remove the four capscrews and the fan hub.

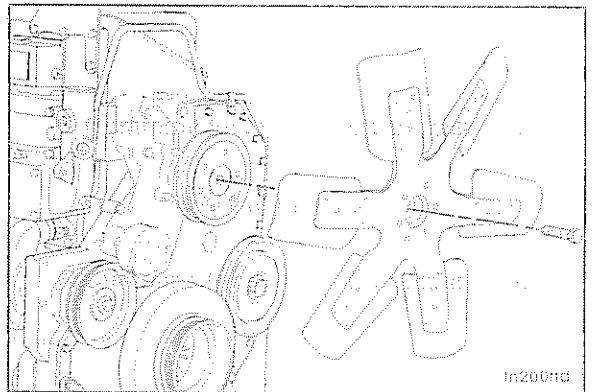
Loosen the fan idler pulley shaft locknut.



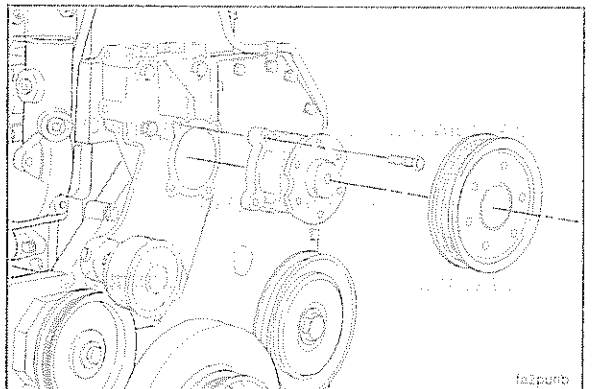
Loosen the adjusting mechanism, and move the fan idler pulley and fan pulley centers as close as possible.
Remove the fan belt.

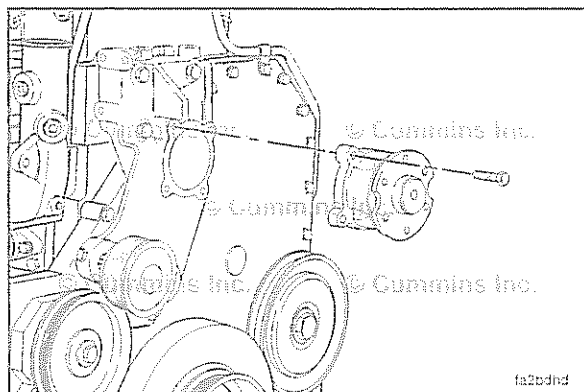


Remove the fan and clutch assembly. Refer to a Cummins Authorized Repair Location.



Remove the fan drive pulley.
Remove the four capscrews and the fan hub.



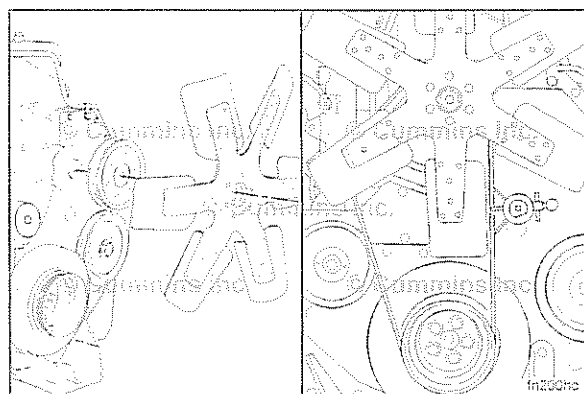


Install

Install the new fan hub and four capscrews.



Torque Value: 47 N•m [35 ft-lb]



Install the fan drive pulley.

Install the fan clutch and assembly. Refer to Procedure 008-040.

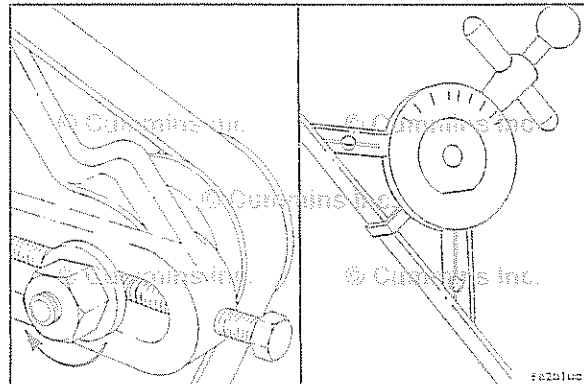


Install and tighten the mounting capscrews.

Torque Value: 68 N•m [50 ft-lb]



Install, adjust, and tighten the fan drive belt. Refer to Procedure 008-002.

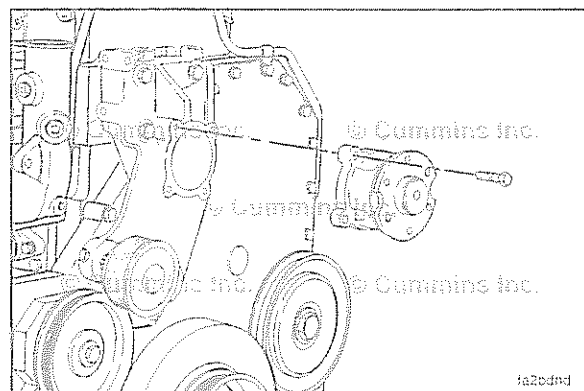


Tighten the fan idler pulley shaft locknut.

Torque Value: 190 N•m [140 ft-lb]



Check the belt tension again after the locknut is tightened.



Install the new fan hub and four capscrews.

Torque Value: 47 N•m [35 ft-lb]



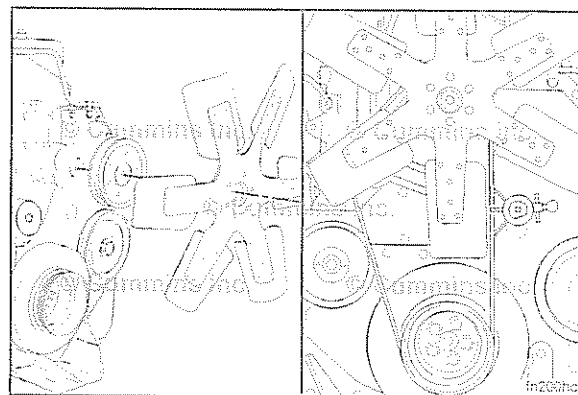
Install the fan drive pulley.

Install the fan and clutch assembly.

Install and tighten the mounting capscrews.

Torque Value: 68 N•m [50 ft-lb]

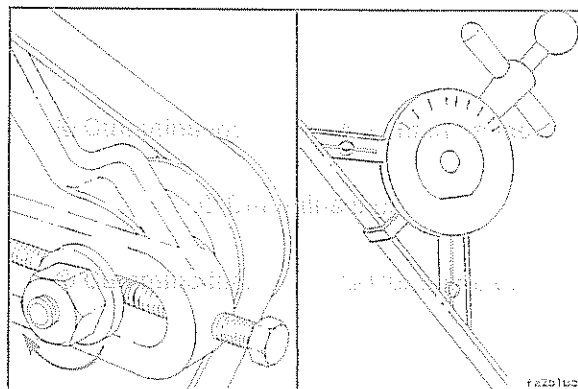
Install, adjust, and tighten the fan drive belt. Refer to Procedure 018-005 (Drive Belt Tension) in Section V for the correct belt tension.



Tighten the fan idler pulley shaft locknut.

Torque Value: 190 N•m [140 ft-lb]

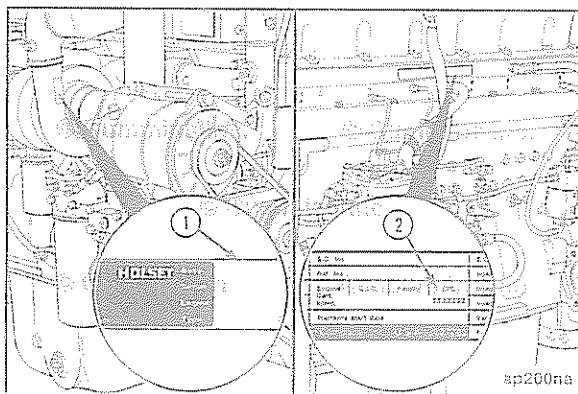
Check the belt tension again after the locknut is tightened.



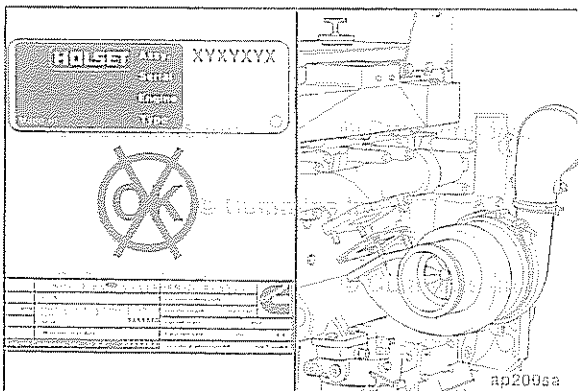
Turbocharger

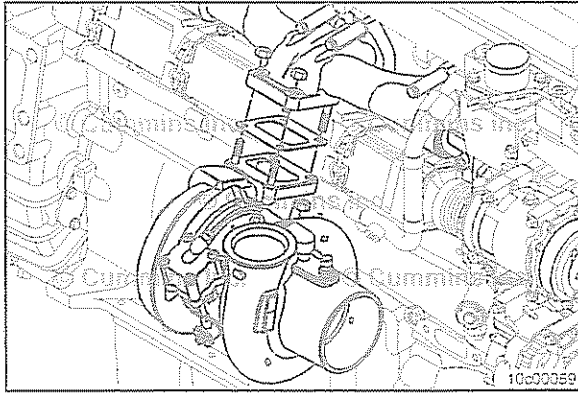
Check for Correct Component

Compare the assembly number (1) on the turbocharger data tag with the turbocharger specified in the engine control parts list (CPL) number (2) listed on the engine data tag.



If the correct turbocharger was not installed, remove it and install the correct turbocharger.





Remove Variable Geometry

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To avoid personal injury, use a hoist or get assistance to lift this component.

⚠ CAUTION ⚠

When installing the lifting eye, make sure the shoulder of the lifting eye is bottomed against the bearing housing. Failure to do so can result in failure of the lifting eye and personal injury.

⚠ CAUTION ⚠

If the lifting eye is installed in the turbocharger bearing housing, it is to be used exclusively for turbocharger removal and installation. It is not to be used in removal of the exhaust manifold, or engine. Doing so will cause damage the turbocharger.

⚠ CAUTION ⚠

The turbocharger actuator must not be used as a lifting mechanism. Doing so will cause damage to the turbocharger actuator.

NOTE: ISM CM870 - The capscrew in top of the turbocharger bearing housing can be removed and replaced with a lifting eye to aid in removal of the turbocharger.

NOTE: ISM CM876 - Thread a lifting strap around the turbocharger bearing housing. Take care not to bend the coolant lines when lifting the turbocharger.

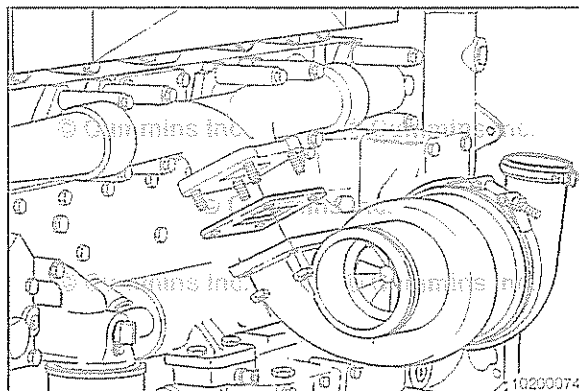
Care must be taken not to damage the turbocharger compressor inlet air temperature sensor, the turbocharger speed sensor, and the turbocharger actuator coolant lines when removing the turbocharger from the engine.

Remove the four turbocharger mounting nuts.

Remove the nuts from the turbocharger studs and remove the turbocharger.

NOTE: If the turbocharger mounting nuts do not loosen freely, split the nuts to avoid breaking a mounting stud.

Remove the turbocharger and discard the gaskets.



Wastegated

Remove the four turbocharger mounting nuts.

Remove the turbocharger and gasket.

Install

Variable Geometry

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

⚠ CAUTION ⚠

When installing lifting eye, make sure the shoulder of the lifting eye is bottomed against the bearing housing. Failure to do so can result in failure of lifting eye and personal injury.

⚠ CAUTION ⚠

If the lifting eye is installed in the turbocharger bearing housing, it is to be used exclusively for turbocharger removal and installation. It is not to be used in removal of the exhaust manifold, or engine. Doing so will cause damage to the turbocharger.

⚠ CAUTION ⚠

Do not rotate the turbocharger turbine housing. Loosening the turbine v-band and rotating the turbine housing may cause damage to an internal variable geometry mechanism.

⚠ CAUTION ⚠

The turbocharger actuator must not be used as a lifting mechanism. Doing so will cause damage to the turbocharger actuator.

NOTE: ISM CM870 - The capscrew in the top of the turbocharger bearing housing can be removed and replaced with a lifting eye to aid in removal of the turbocharger.

NOTE: ISM CM876 - Thread a lifting strap around the turbocharger bearing housing. Take care to not bend the coolant lines when lifting the turbocharger.

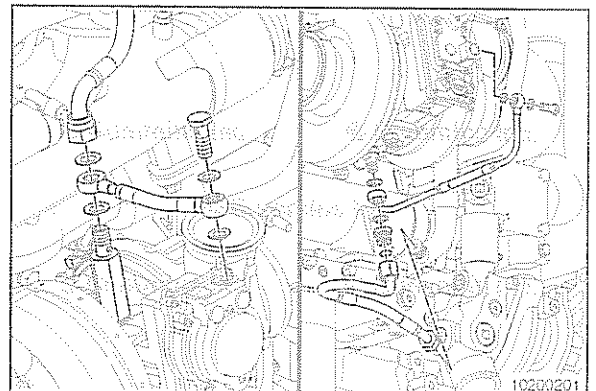
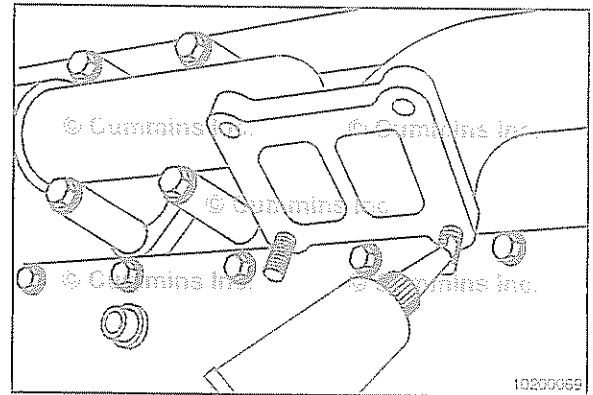
Care must be taken not to damage the turbocharger compressor inlet air temperature sensor, the turbocharger speed sensor and the turbocharger actuator coolant lines when installing the turbocharger.

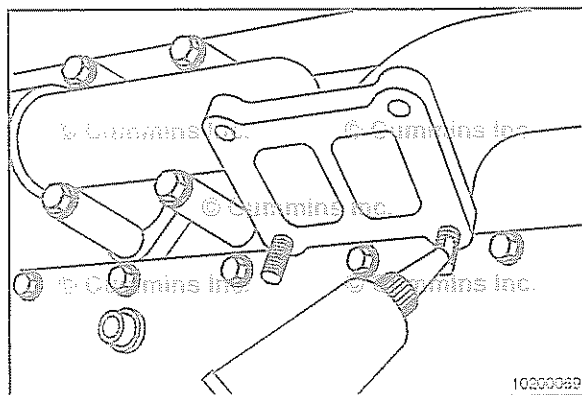
Install a new gasket, the turbocharger, and the four mounting nuts. Tighten the mounting nuts.

Torque Value: 88 N•m [65 ft-lb]

If installing a new turbocharger, install the coolant inlet and outlet fittings. Refer to Procedure 010-041.

If installing a new turbocharger, install the oil supply fitting. Refer to Procedure 010-046.



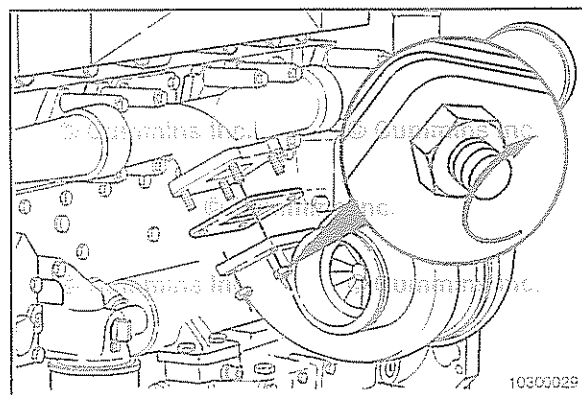


Wastegated

NOTE: For wastegated turbochargers, refer to Procedure 010-050 for wastegate actuator installation.



Apply a film of anti-seize compound, Part Number 3824879, or equivalent, to the turbocharger mounting studs.



Use a new gasket when installing the turbocharger.

Install and tighten the four mounting nuts.

Torque Value: 61 N•m [45 ft-lb]



Air Starting Motor

General Information

The air starting motor system (tanks, line sizes, and valves) is designed and installed by the OEM and starting motor suppliers. Refer any questions about the air starting systems to the manufacturer.

Air Starting Motor Maintenance

- Do not operate the air starting motor with air pressure lower than 480 kPa [70 psi].
- Maintain the air compressor according to the recommendations outlined in the manual.
- For maximum efficiency, the hoses, tubes, and lines must not leak.
- Refer to the OEM and starting motor manufacturer's manuals for specific information regarding the starting motors, valves, and systems.

Refrigerant Compressor Drive

General Information

Industrial Applications

A new refrigerant compressor drive system is offered on the QSM11 industrial. The new drive system has an accessory drive pulley, a six-rib poly-vee belt with automatic tensioning and a backside idler for use with a Sanden rotary compressor.

Install

Variable Geometry



WARNING

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.



CAUTION

When installing lifting eye, make sure the shoulder of the lifting eye is bottomed against the bearing housing. Failure to do so can result in failure of lifting eye and personal injury.



CAUTION

If the lifting eye is installed in the turbocharger bearing housing, it is to be used exclusively for turbocharger removal and installation. It is not to be used in removal of the exhaust manifold, or engine. Doing so will cause damage to the turbocharger.



CAUTION

Do not rotate the turbocharger turbine housing. Loosening the turbine v-band and rotating the turbine housing may cause damage to an internal variable geometry mechanism.



CAUTION

The turbocharger actuator must not be used as a lifting mechanism. Doing so will cause damage to the turbocharger actuator.

NOTE: ISM CM870 - The capscrew in the top of the turbocharger bearing housing can be removed and replaced with a lifting eye to aid in removal of the turbocharger.

NOTE: ISM CM876 - Thread a lifting strap around the turbocharger bearing housing. Take care to not bend the coolant lines when lifting the turbocharger.

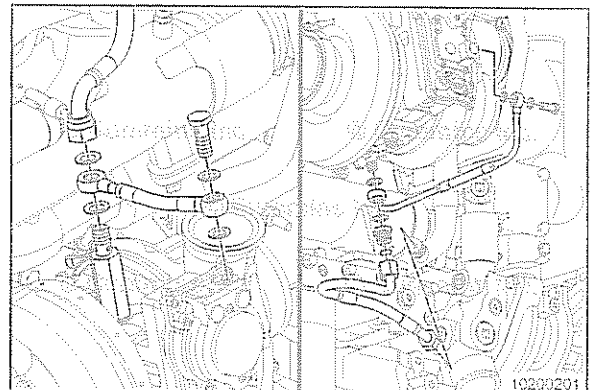
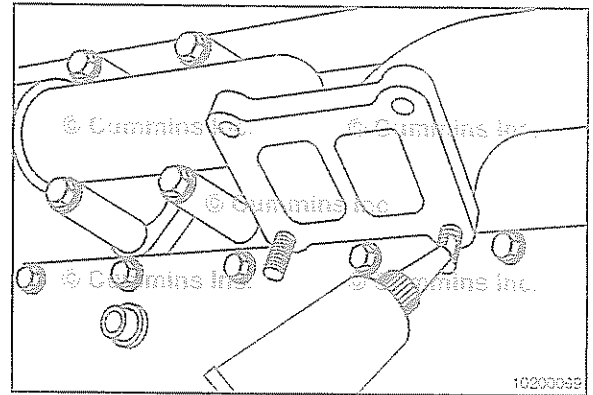
Care must be taken not to damage the turbocharger compressor inlet air temperature sensor, the turbocharger speed sensor and the turbocharger actuator coolant lines when installing the turbocharger.

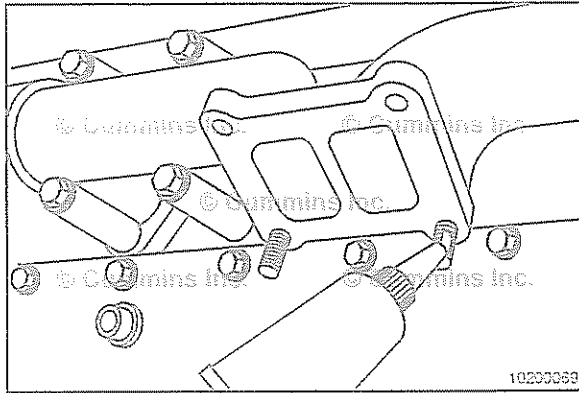
Install a new gasket, the turbocharger, and the four mounting nuts. Tighten the mounting nuts.

Torque Value: 88 N•m [65 ft-lb]

If installing a new turbocharger, install the coolant inlet and outlet fittings. Refer to Procedure 010-041.

If installing a new turbocharger, install the oil supply fitting. Refer to Procedure 010-046.



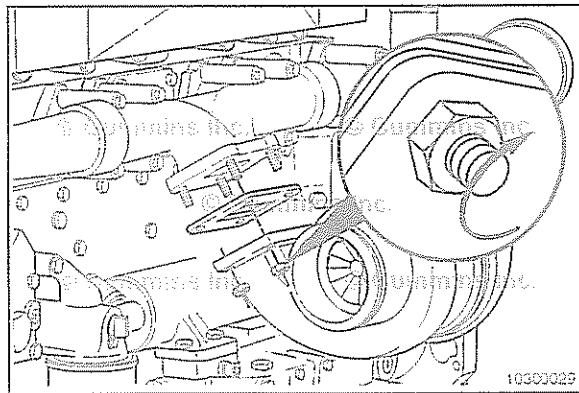


Wastegated

NOTE: For wastegated turbochargers, refer to Procedure 010-050 for wastegate actuator installation.



Apply a film of anti-seize compound, Part Number 3824879, or equivalent, to the turbocharger mounting studs.



Use a new gasket when installing the turbocharger.

Install and tighten the four mounting nuts.



Torque Value: 61 N•m [45 ft-lb]

Air Starting Motor

General Information

The air starting motor system (tanks, line sizes, and valves) is designed and installed by the OEM and starting motor suppliers. Refer any questions about the air starting systems to the manufacturer.

Air Starting Motor Maintenance

- Do not operate the air starting motor with air pressure lower than 480 kPa [70 psi].
- Maintain the air compressor according to the recommendations outlined in the manual.
- For maximum efficiency, the hoses, tubes, and lines must not leak.
- Refer to the OEM and starting motor manufacturer's manuals for specific information regarding the starting motors, valves, and systems.

Refrigerant Compressor Drive

General Information

Industrial Applications

A new refrigerant compressor drive system is offered on the QSM11 industrial. The new drive system has an accessory drive pulley, a six-rib poly-vee belt with automatic tensioning and a backside idler for use with a Sanden rotary compressor.

Remove

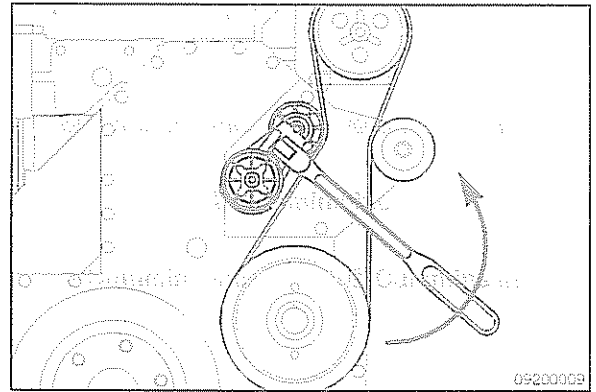
Industrial Applications

Use a 3/8-inch drive breaker bar to hold back the tensioner and relieve the belt tension.

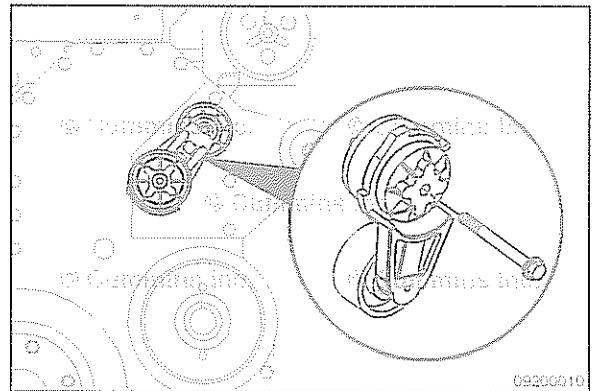
Rotate the tensioner away from the belt.

Remove the refrigerant drive belt while holding the tensioner back.

Relieve the tensioner to neutral position.



Remove the belt tensioner by removing the nut located on top of the tensioner.



Install

Industrial Applications

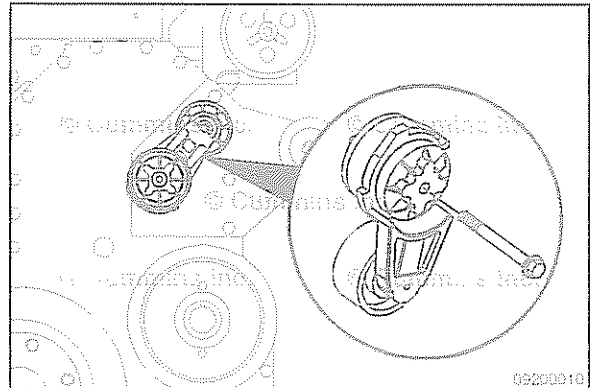
Position the enclosed subassembly (idler pulley) (1) and tensioner mounting bracket (2) on the gear cover and mount with the four capscrews (3) and washers.

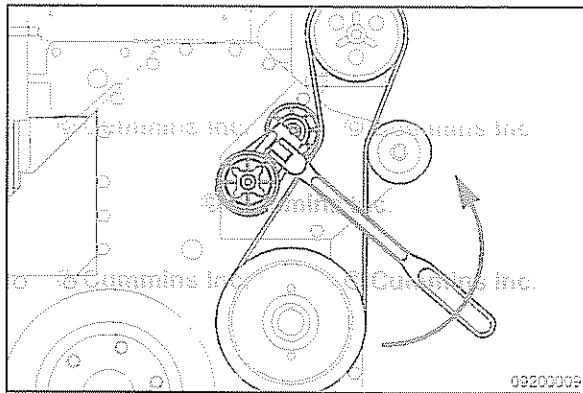
- The tensioner idler pulley will be located in front of two of the mounting capscrew locations in the freestate. The tensioner either will have to be removed completely for this process or the arm moved to one side to have access to all four capscrew mounting locations.
- Install the belt tensioner (4), and tighten the mounting nut.

Torque Value: 47 N•m [35 ft-lb]

Tighten the four mounting capscrews.

Torque Value: 20 N•m [177 in-lb]

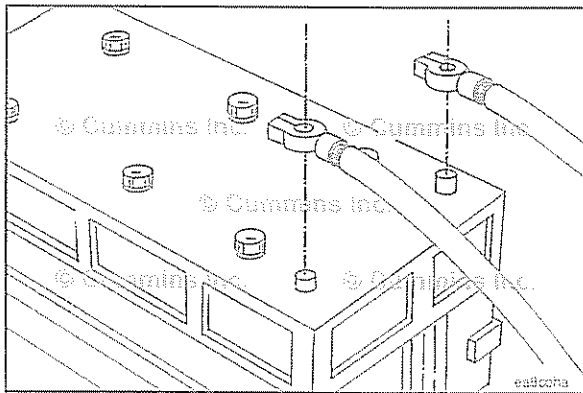




Use a 3/8-inch drive breaker bar to hold back the tensioner.

Install the drive belt while holding the tensioner back.

Allow the tensioner to rotate against the belt.



Starting Motor Remove

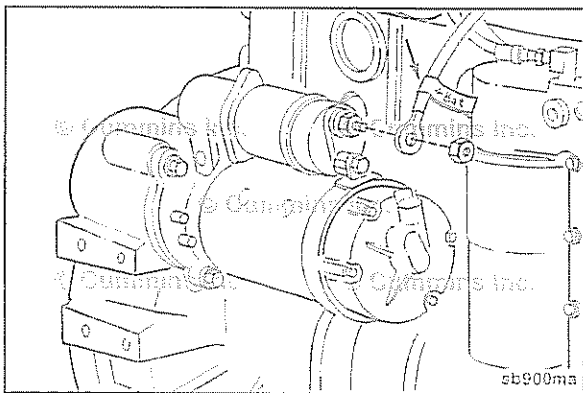


WARNING

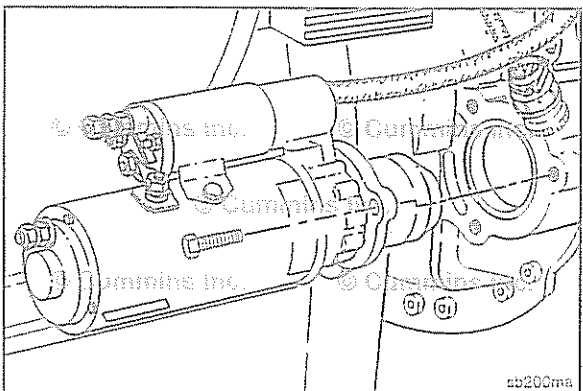
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first, and attach the negative (-) battery cable last.

NOTE: The starting motor mounting capscrews can be metric or standard thread sizes. Make sure to install the same-size capscrews that were removed.

Remove the electrical connections from the batteries.



Tag and remove the electrical connections from the starting motor.



Remove the three capscrews and the starting motor.

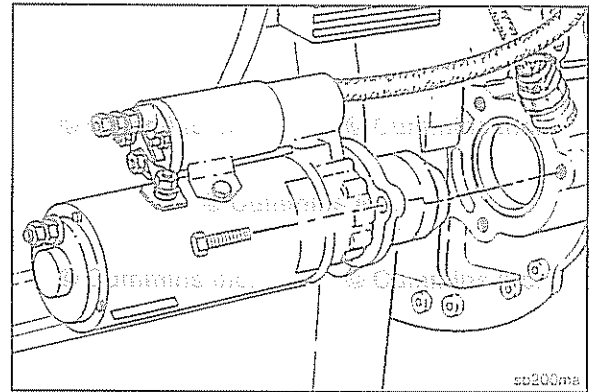
Install

Before installing the starting motor, make sure the capscrews are the correct size and grade.

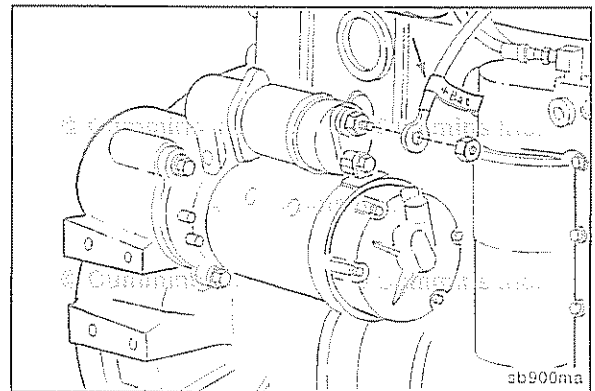
Install the starting motor and the three mounting capscrews.

Tighten the capscrews.

Torque Value: 90 N•m [66 ft-lb]



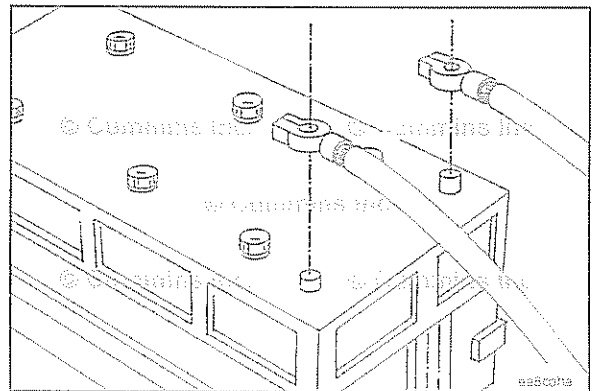
Install and tighten the electrical connections to the starting motor.



WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first, and attach the negative (-) battery cable last.

Install and tighten the battery electrical connections.



Engine Storage - Long Term

General Information

If the engine will be out of service longer than 6 months, take special precautions to prevent rust. Contact the nearest Cummins Authorized Repair Location for information concerning engine storage procedures.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Section D - System Diagrams

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System Diagrams - Overview

General Information

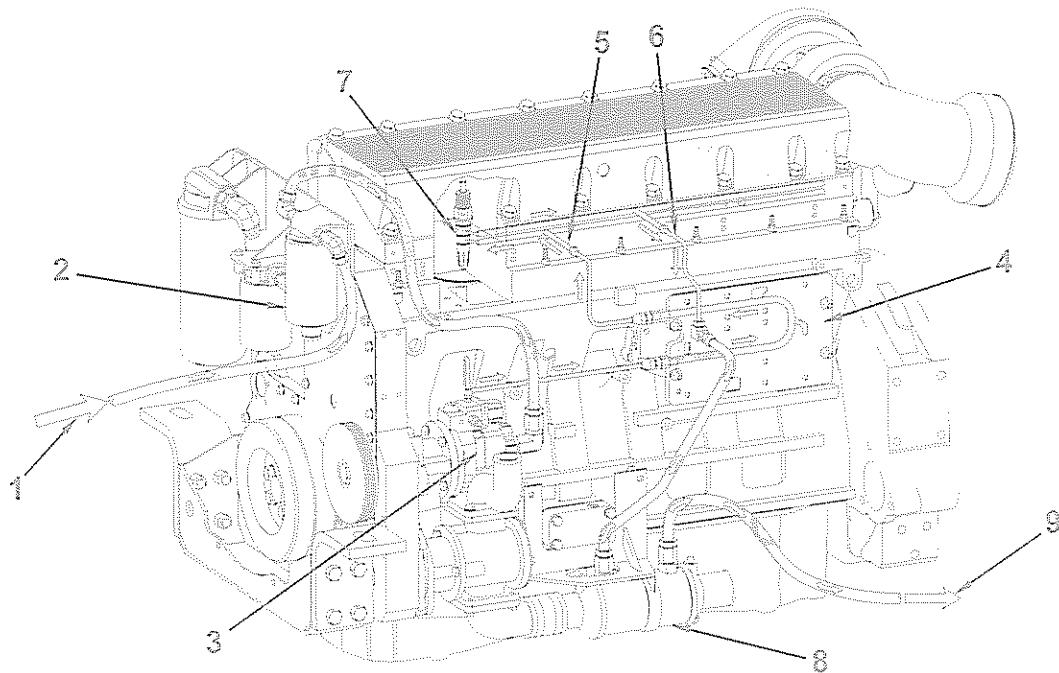
The following drawings show the flow through the engine systems. Although parts can change between different applications and installations, the flow remains the same. The systems shown are:

- Fuel System
- Lubricating Oil System
- Coolant System
- Intake Air System
- Exhaust System
- Compressed Air System.

Knowledge of the engine systems can help you in troubleshooting, service, and general maintenance of your engine.

Flow Diagram, Fuel System

Fuel System Marine Applications

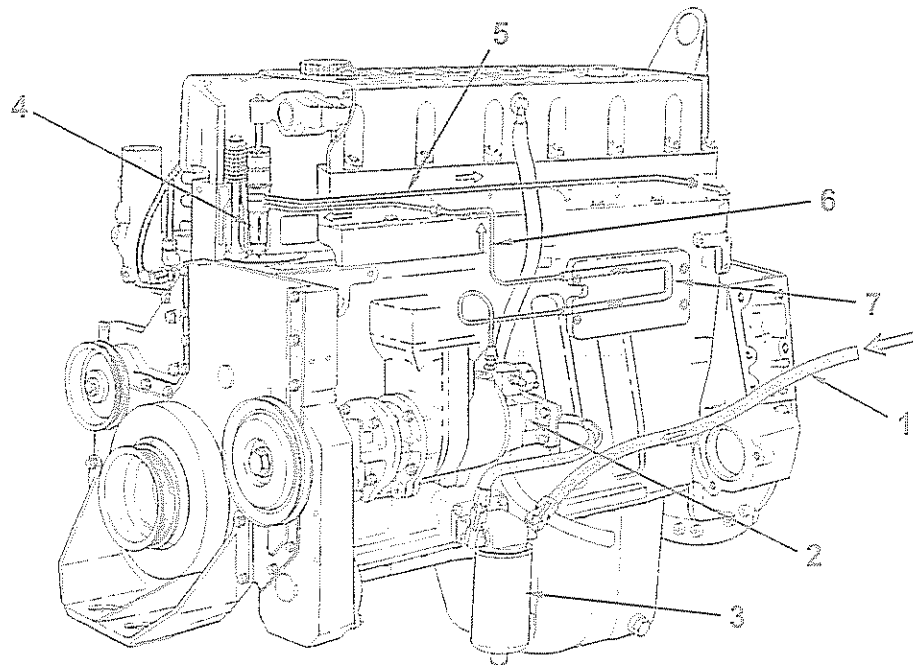


1. Fuel supply from tank
2. Fuel filter
3. Fuel pump
4. ECM cooling plate
5. Fuel to injector

6. Fuel drain to from injector
7. Injector
8. Fuel cooler
9. Fuel return to supply tank

05200155

Industrial and Generator Drive

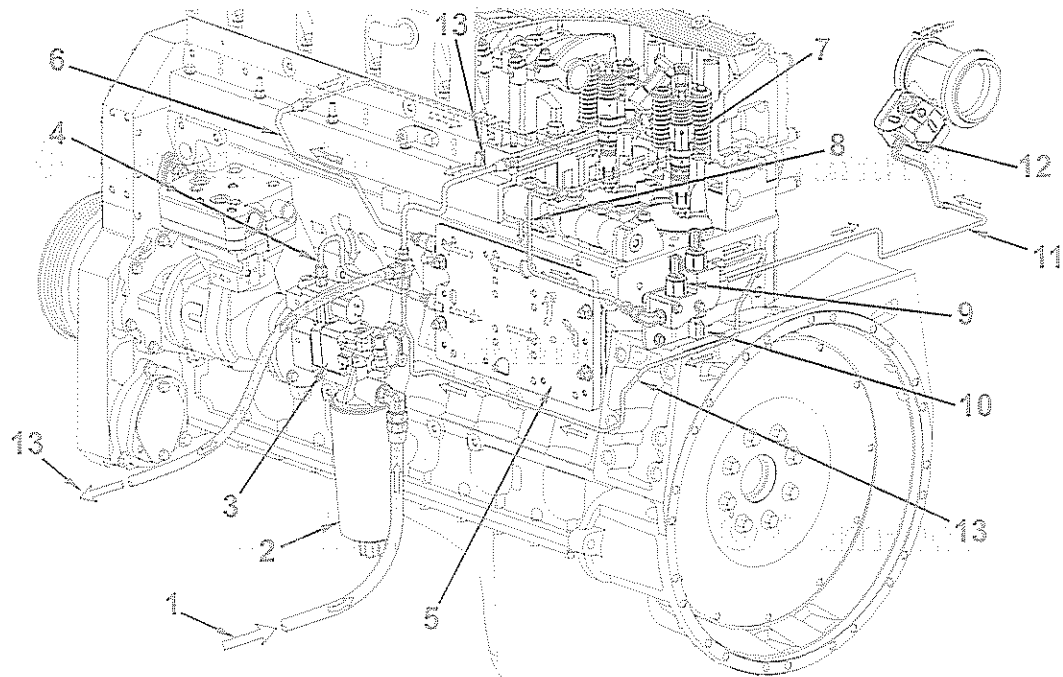


1. Fuel supply from tank
2. Fuel pump
3. Fuel filter
4. Injector

5. Fuel drain from injector
6. Fuel to injector
7. ECM cooling plate

05906034

Automotive with CM876

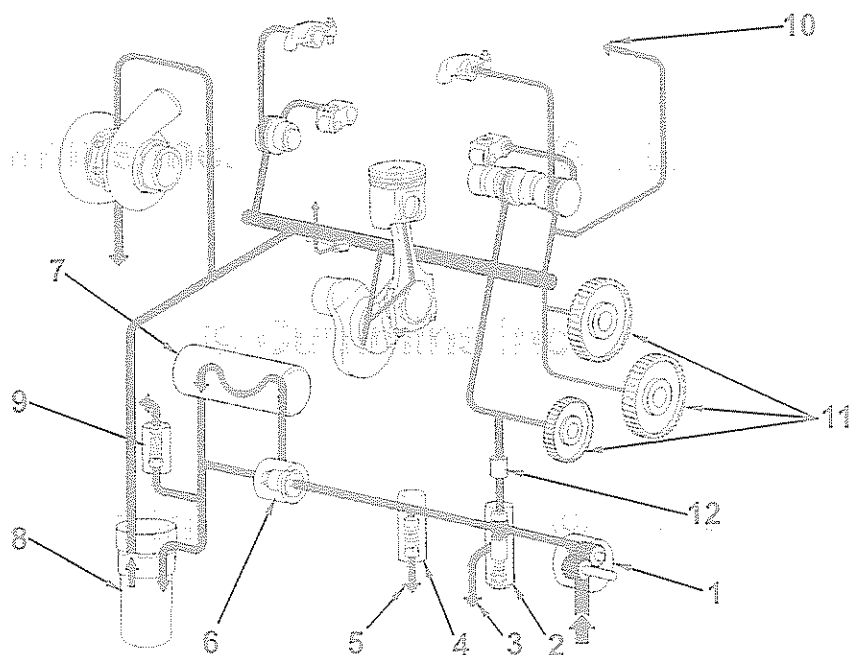


05200170

- | | |
|------------------------------|--|
| 1. Fuel supply from tank | 8. Fuel supply to aftertreatment injector system |
| 2. Fuel filter | 9. Fuel shutoff valve |
| 3. Fuel Pump | 10. Fuel pressure sensor |
| 4. Fuel supply to ECM cooler | 11. Fuel supply to aftertreatment fuel injector |
| 5. ECM cooling plate | 12. Aftertreatment fuel injector |
| 6. Fuel supply to injectors | 13. Fuel drain to tank |
| 7. Injector | |

Flow Diagram, Lubricating Oil System

Flow Diagram

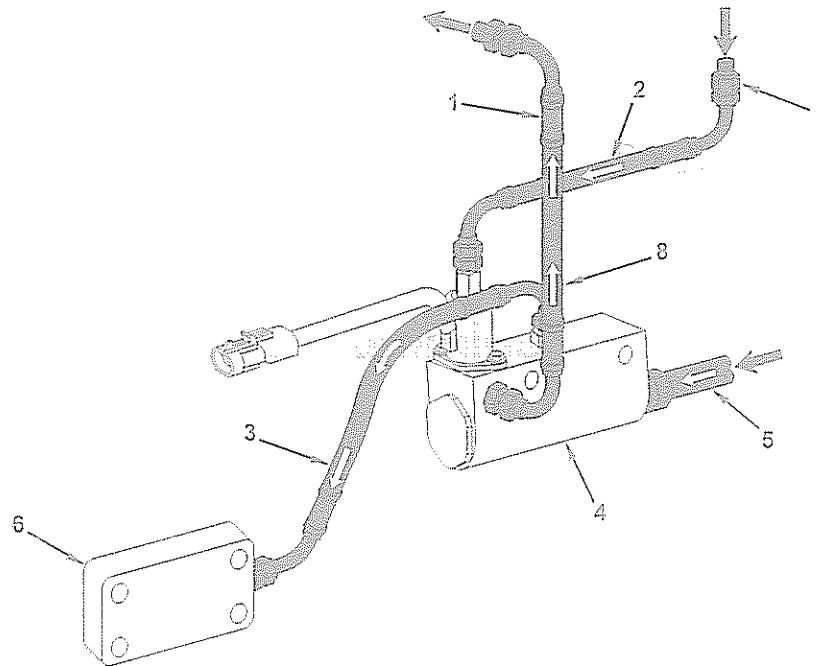


- | | |
|-------------------------------|---------------------------------------|
| 1. Oil pump | 7. Oil cooler |
| 2. Pressure regulator valve | 8. Combination oil filter |
| 3. Oil return to pan | 9. Filter bypass valve |
| 4. High pressure relief valve | 10. Accessory drive or air compressor |
| 5. Oil return to pan | 11. Idler gears |
| 6. Oil thermostat | 12. Viscosity sensor. |

07200041

Flow Diagram, Lubricating Oil System

Flow Diagram



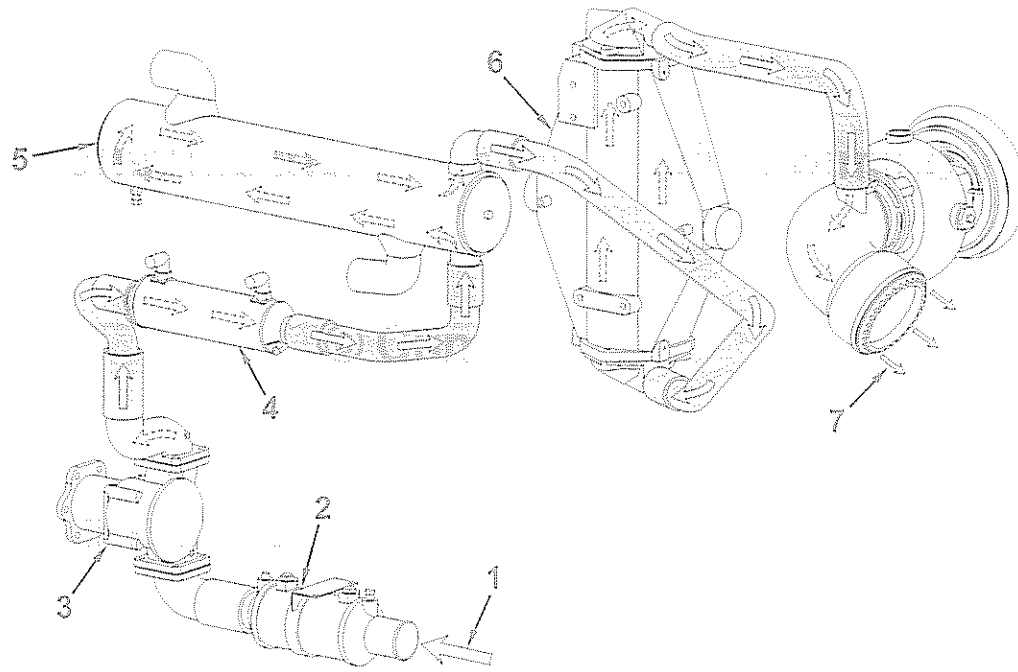
07200086

Centinel™ (optional)

- | | |
|--|--|
| 1. Oil burn line (used oil to fuel return) | 5. OEM fresh oil supply (from makeup tank) |
| 2. Oil rifle hose (used oil inlet) | 6. Handhole cover |
| 3. Oil replenishment hose (clean oil to pan) | 7. Rifle check valve |
| 4. Control valve | 8. Outlet check valve. |

Flow Diagram, Cooling System

Flow Diagram Marine Applications



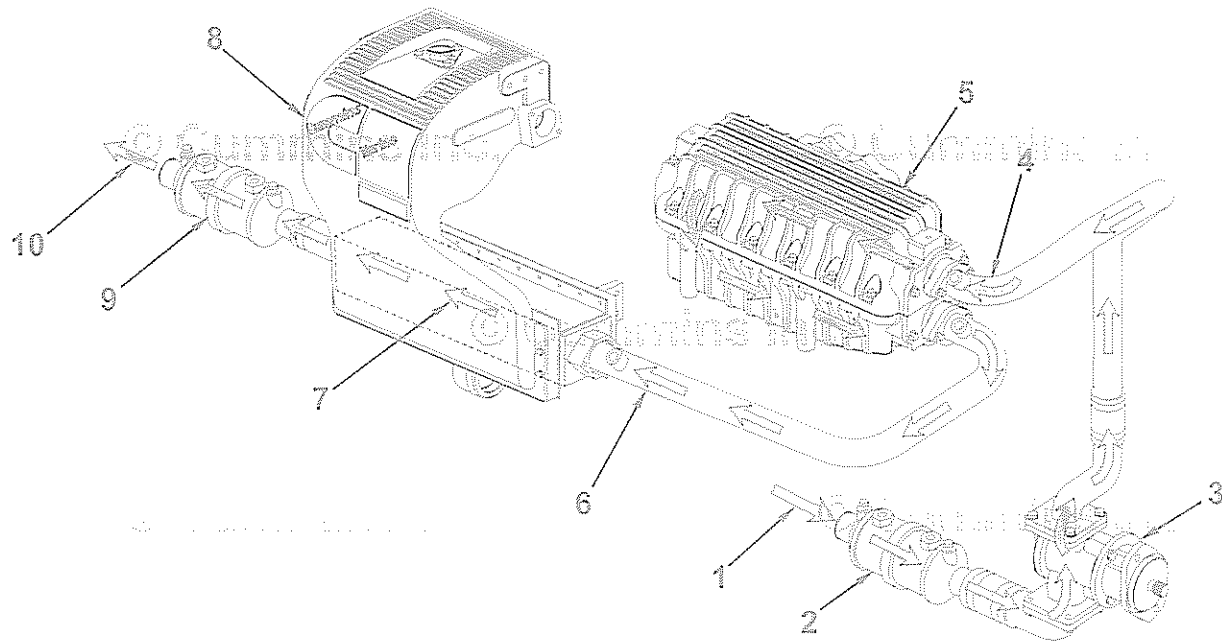
Sea Water Flow (Heat Exchanger)

- | | |
|--------------------|----------------------|
| 1. Sea water inlet | 5. Heat exchanger |
| 2. Fuel cooler | 6. Aftercooler |
| 3. Sea water pump | 7. Sea water outlet. |
| 4. Gear oil cooler | |

09200243

Flow Diagram, Cooling System

Flow Diagram Marine Applications

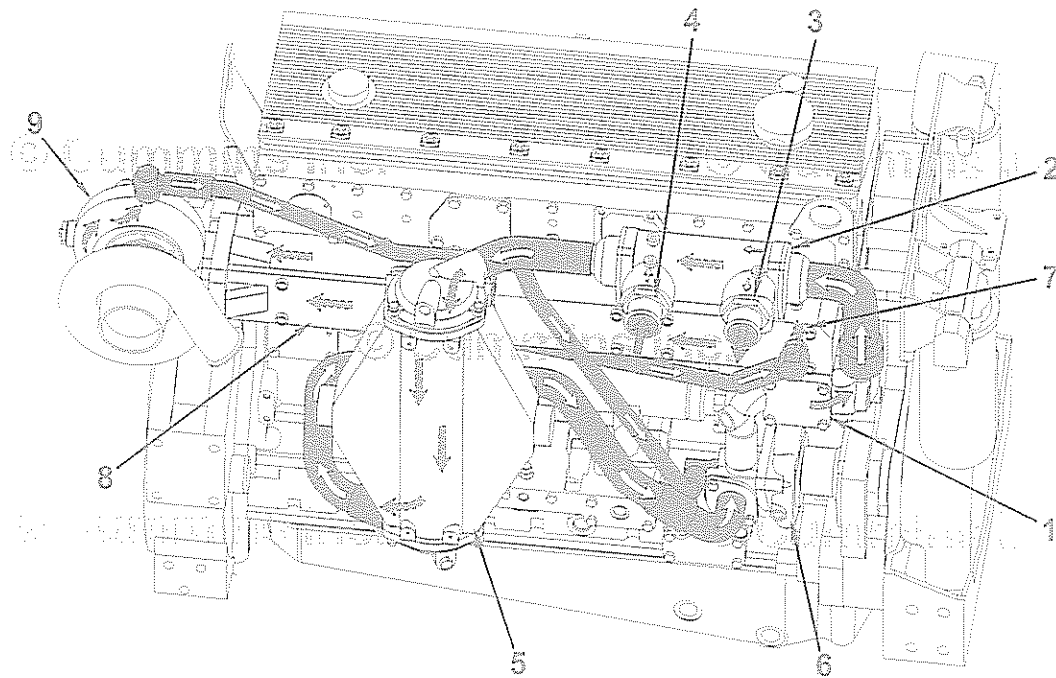


Sea Water Flow (Heat Exchanger)

- | | |
|----------------------------------|----------------------------------|
| 1. Sea water inlet | 6. Aftercooler to heat exchanger |
| 2. Fuel cooler | 7. Heat exchanger |
| 3. Sea water pump | 8. Expansion tank |
| 4. Sea water pump to aftercooler | 9. Gear oil cooler |
| 5. Aftercooler | 10. Sea water outlet. |

Flow Diagram, Cooling System

Flow Diagram Marine Applications



Coolant Flow, Keel Cooled (one of three)

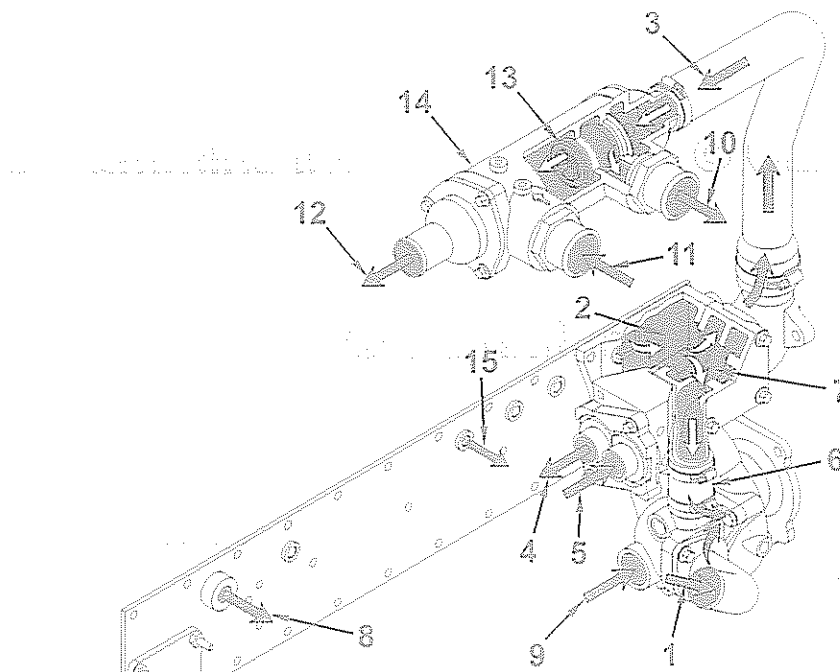
- | | |
|--|-------------------------------------|
| 1. Engine thermostat cavity (thermostat removed) | 6. Water pump |
| 2. Keel cooled thermostat housing | 7. Coolant flow to exhaust manifold |
| 3. Coolant out to keel cooler | 8. Water cooled exhaust manifold |
| 4. Coolant in from keel cooler | 9. Water cooled turbine housing. |
| 5. Aftercooler | |

08260242

Flow Diagram, Cooling System

Flow Diagram

Marine Applications



09200240

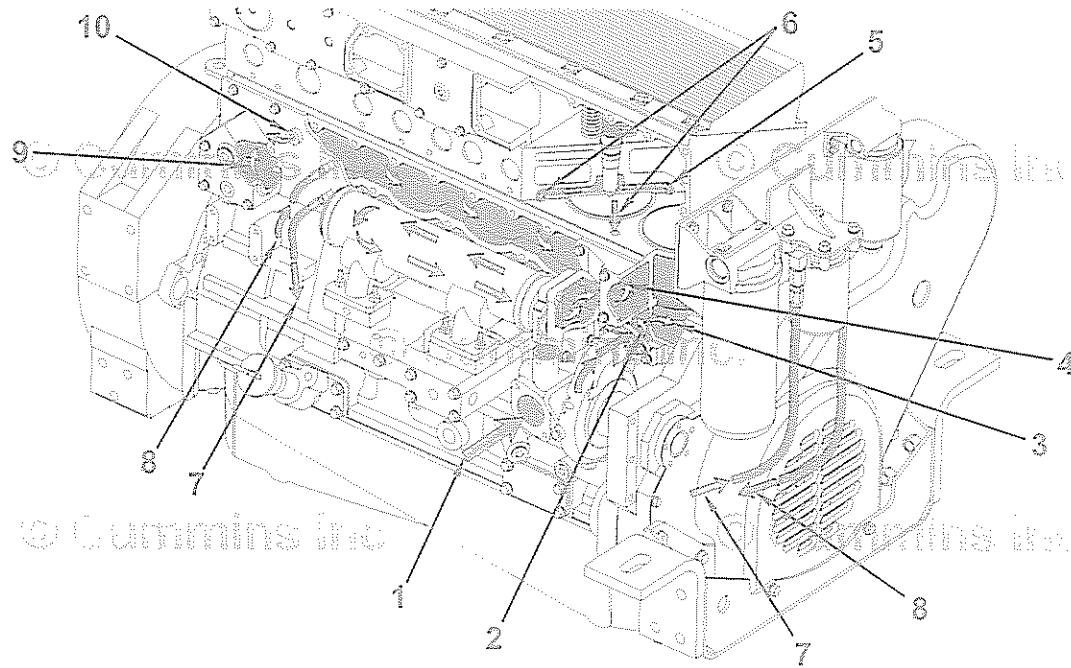
Coolant Flow, Keel Cooled (two of three)

- | | |
|--|--|
| 1. Water pump coolant inlet | 9. Turbocharger coolant return to water pump inlet |
| 2. Upper manifold cavity | 10. To keel cooler |
| 3. Coolant outlet | 11. From keel cooler |
| 4. Lower manifold cavity to cooler | 12. To aftercooler |
| 5. Cooler to upper manifold cavity (before thermostat) | 13. LTA thermostat |
| 6. Coolant bypass hose - plugged | 14. LTA thermostat housing |
| 7. Engine thermostat cavity (thermostat removed) | 15. To turbocharger. |
| 8. To exhaust manifold | |

Flow Diagram, Cooling System

Flow Diagram

Marine Applications



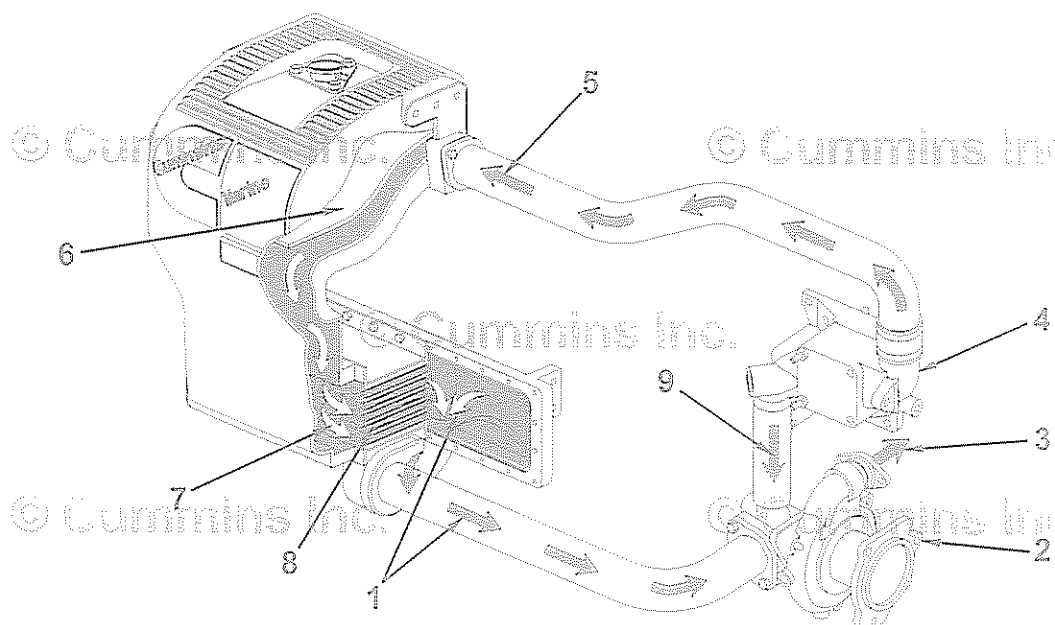
08200241

Coolant Flow (three of three)

- | | |
|---|---|
| 1. Flow to water pump | 6. Flow from cylinder head to upper manifold cavity |
| 2. Flow to oil cooler and lower manifold cavity | 7. Flow to coolant filter |
| 3. Flow from lower manifold cavity to cylinder liners | 8. Flow from coolant filter |
| 4. Flow from oil cooler to thermostat housing | 9. Coolant heater - optional |
| 5. Flow to cylinder head | 10. Upper manifold to exhaust manifold. |

Flow Diagram, Cooling System

Flow Diagram Marine Applications



03200161

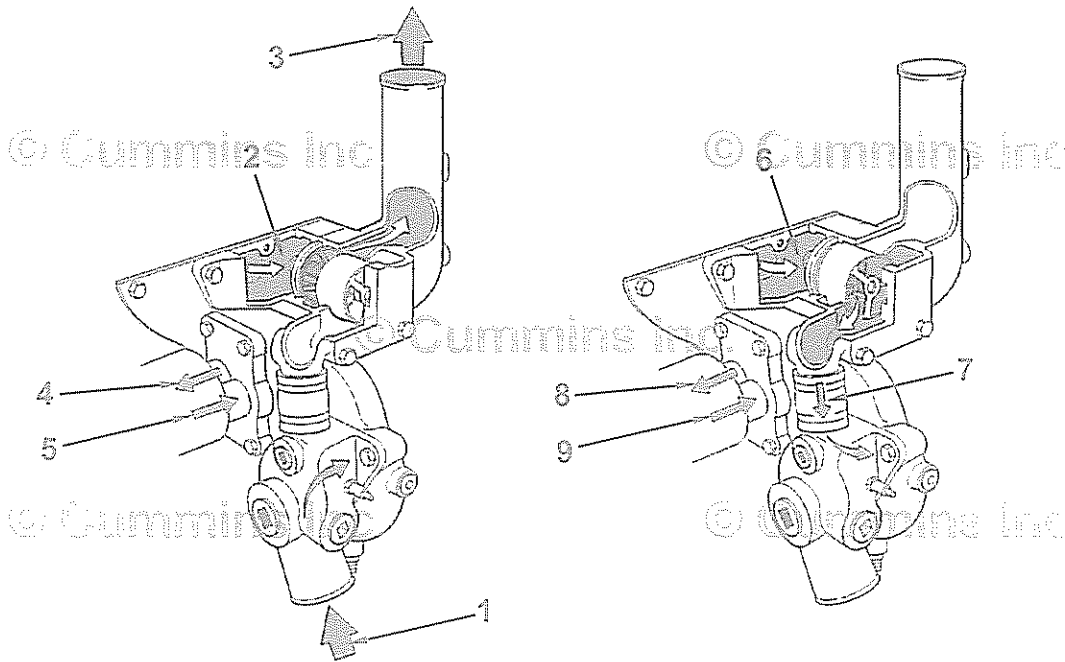
Coolant Flow, Heat Exchanger (one of three)

- | | |
|--|--|
| 1. Coolant flow to water pump | 6. Expansion tank |
| 2. Water pump | 7. Heat exchanger sea water core |
| 3. Coolant flow to lower manifold cavity | 8. Coolant flow through sea water core |
| 4. Thermostat housing | 9. Coolant bypass. |
| 5. Thermostat housing to heat exchanger | |

Flow Diagram, Cooling System

Flow Diagram

Marine Applications



Coolant Flow, Heat Exchanger (two of three)

08260087

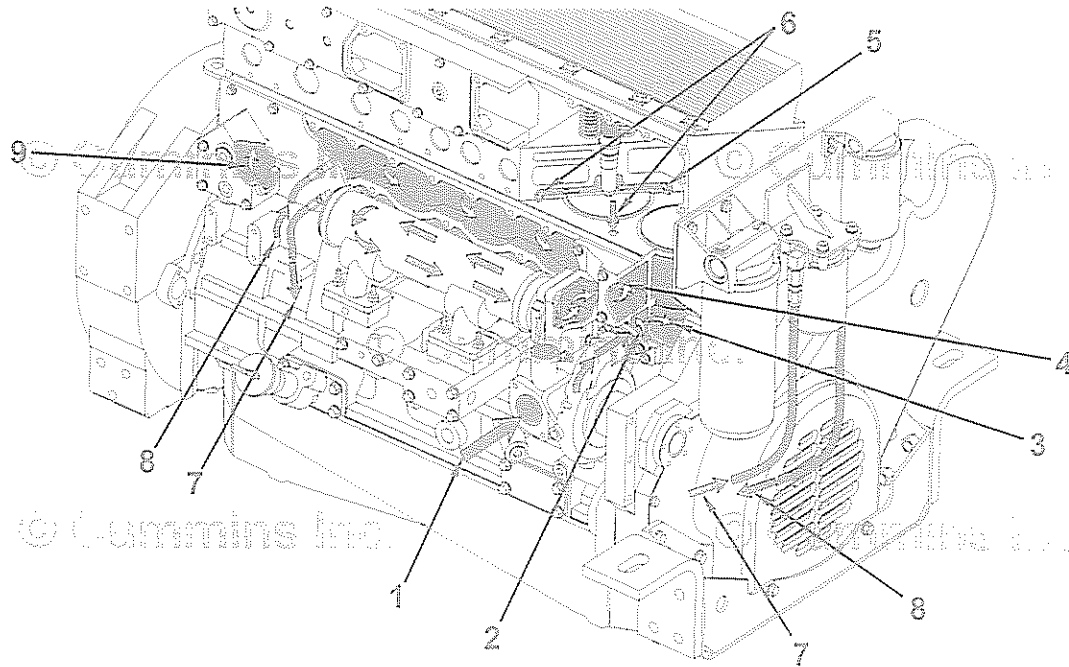
Thermostat Open (1-5)

Thermostat Closed (6-9)

- | | |
|---|---|
| 1. Water pump coolant inlet | 6. Upper manifold cavity (coolant to thermostat) |
| 2. Upper manifold cavity | 7. Coolant bypass (return to water pump) |
| 3. Coolant outlet | 8. Lower manifold cavity to cooler |
| 4. Lower manifold cavity to cooler | 9. Cooler to upper manifold cavity (before thermostat). |
| 5. Cooler to upper manifold cavity (before thermostat). | |

Flow Diagram, Cooling System

Flow Diagram Marine Applications

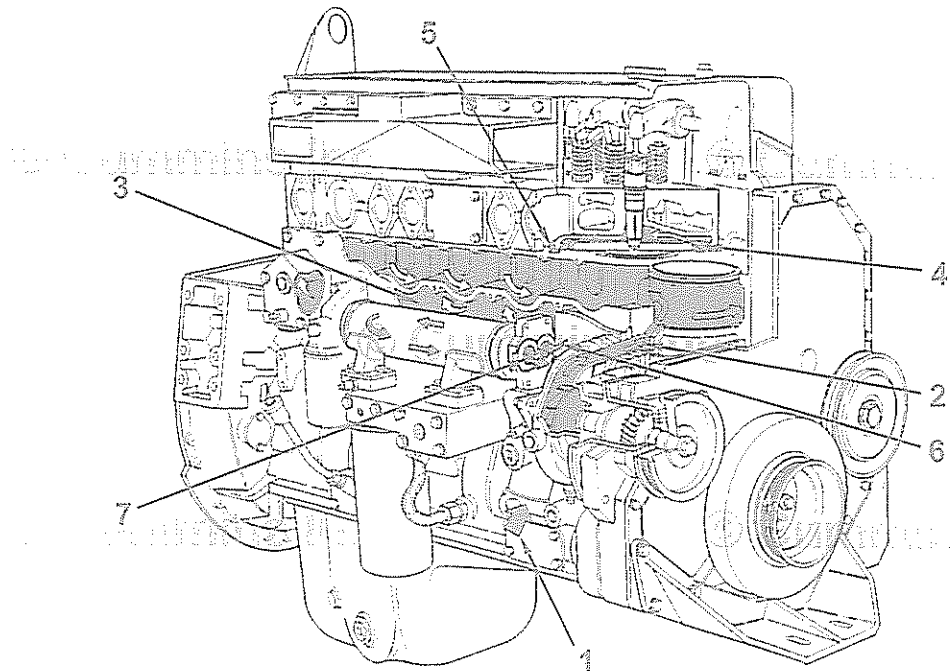


08200159

Coolant Flow, Heat Exchanger (three of three)

- | | |
|---|---|
| 1. Flow from heat exchanger to water pump | 6. Flow from cylinder head to upper manifold cavity |
| 2. Flow to oil cooler and lower manifold cavity | 7. Flow to coolant filter |
| 3. Flow from lower manifold cavity to cylinder liners | 8. Flow from coolant filter |
| 4. Flow from oil cooler to thermostat housing | 9. Coolant heater - optional. |
| 5. Flow to cylinder head | |

Industrial and Generator Drive

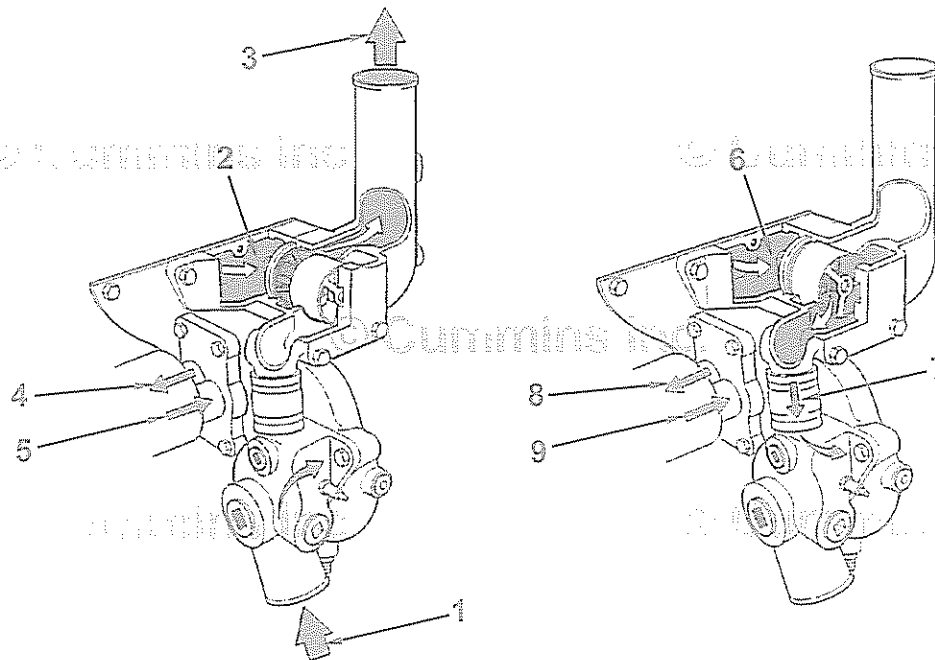


1. Water pump coolant inlet
2. Coolant to lower manifold cavity
3. Coolant to cylinder liner block cavity
4. Coolant to cylinder head

5. Coolant to upper manifold cavity
6. Lower manifold coolant to oil cooler
7. Oil cooler to upper manifold cavity.

08200086

Industrial and Generator Drive



03200087

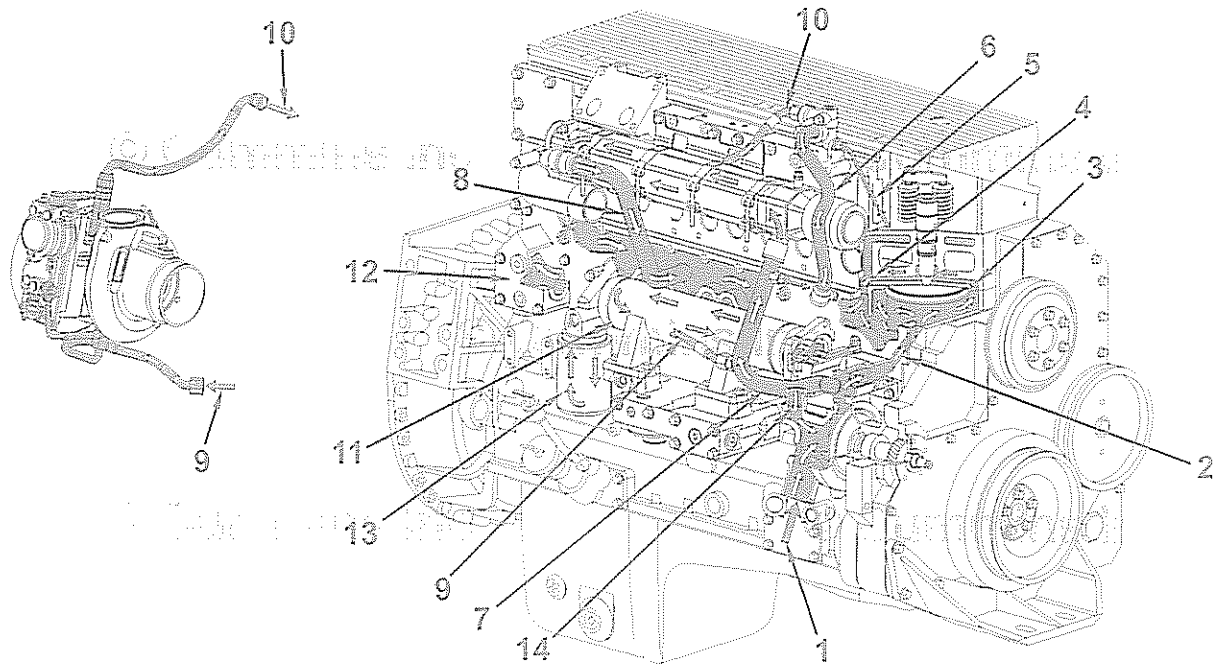
Open (1-5)

Closed (6-9)

1. Water pump coolant inlet
2. Upper manifold cavity (coolant to thermostat)
3. Coolant outlet
4. Lower manifold cavity to cooler
5. Cooler to upper manifold cavity (before thermostat).

6. Upper manifold cavity (coolant to thermostat)
7. Coolant bypass (return to water pump)
8. Lower manifold cavity to cooler
9. Cooler to upper manifold cavity (before thermostat).

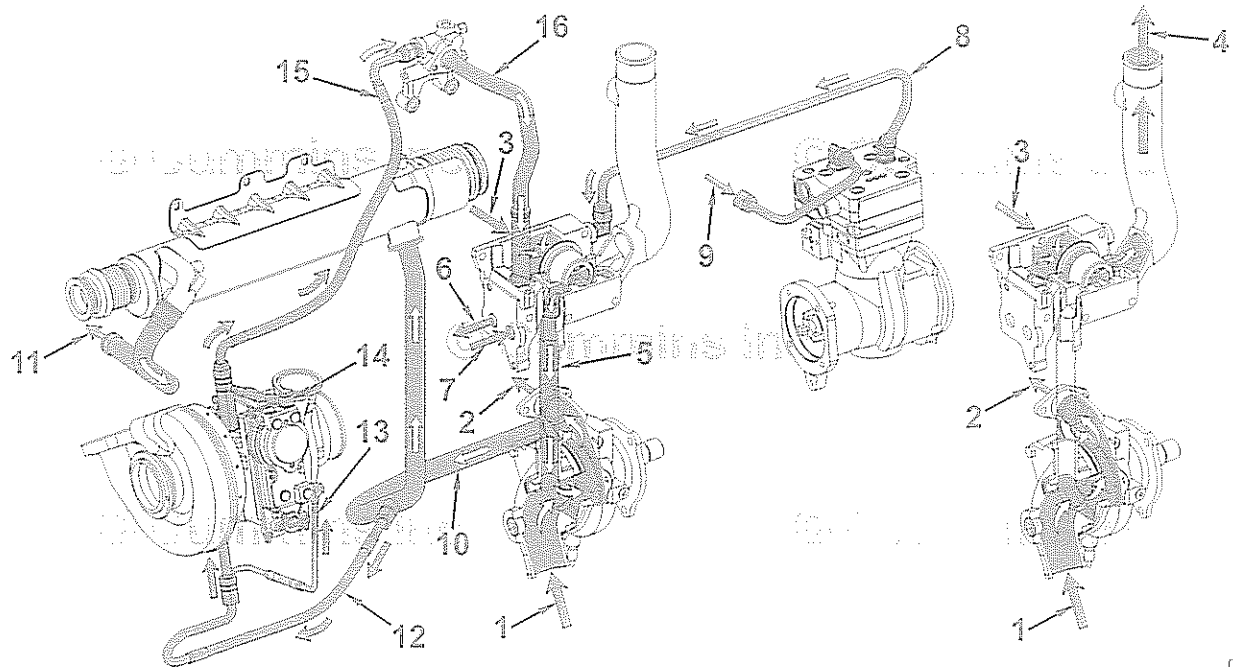
Automotive with CM876



05200247

- | | |
|---|---|
| 1. Water pump coolant inlet | 8. EGR cooler coolant return to low pressure cavity |
| 2. Coolant supply to block | 9. Coolant supply to turbocharger |
| 3. Coolant supply to cylinder head | 10. Turbocharger coolant return to water manifold |
| 4. Coolant return to low pressure cavity | 11. Oil cooler |
| 5. Cylinder head coolant vent to water manifold | 12. Heater housing |
| 6. Coolant return to thermostat housing | 13. Coolant filter |
| 7. Coolant supply to EGR cooler | 14. Coolant return from thermostat housing. |

Automotive with CM876

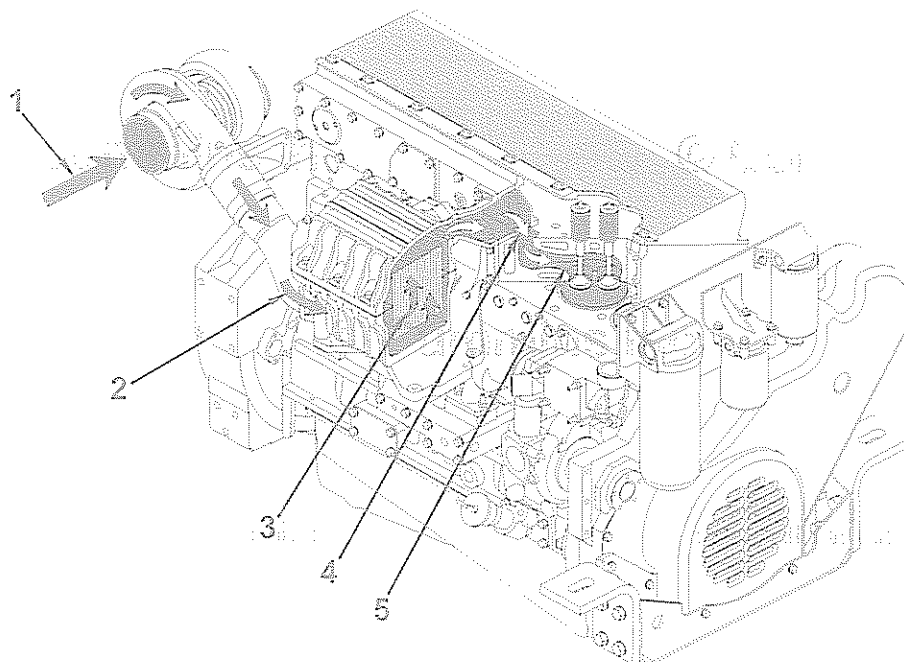


08260248

- | | |
|--|---|
| 1. Water pump coolant inlet | 9. Coolant to air compressor |
| 2. Coolant to block | 10. Coolant supply to EGR cooler |
| 3. Upper manifold cavity (coolant to thermostat) | 11. Coolant from EGR cooler to block |
| 4. Coolant outlet (coolant to radiator) | 12. Coolant supply to turbocharger |
| 5. Coolant bypass (return to water pump) | 13. Coolant supply to turbocharger actuator |
| 6. Lower manifold cavity to lubricating oil cooler | 14. Coolant return from turbocharger actuator |
| 7. Lubricating oil cooler to upper manifold cavity (before thermostat) | 15. Coolant from turbocharger to water manifold |
| 8. Coolant from air compressor | 16. Coolant return from water manifold to thermostat housing (before thermostat). |

Flow Diagram, Air Intake System

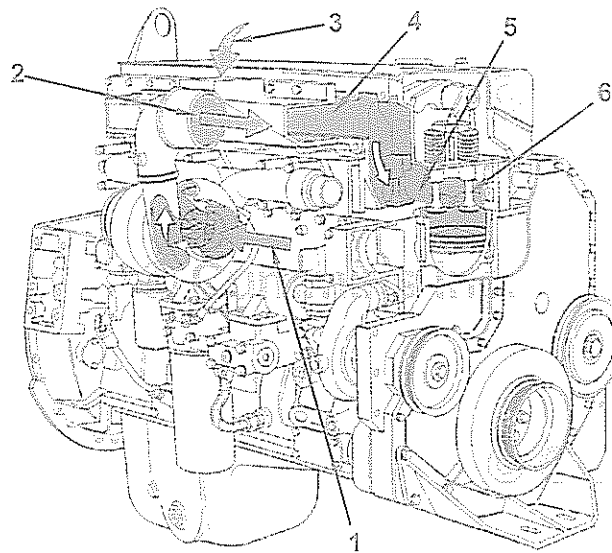
Flow Diagram Marine Applications



10200076

- | | |
|--|--|
| 1. Filtered intake air to turbocharger | 4. Aftercooled air through intake manifold |
| 2. Turbocharger air to aftercooler | 5. Intake valve ports. |
| 3. Intake air through the aftercooler | |

Industrial and Generator Drive

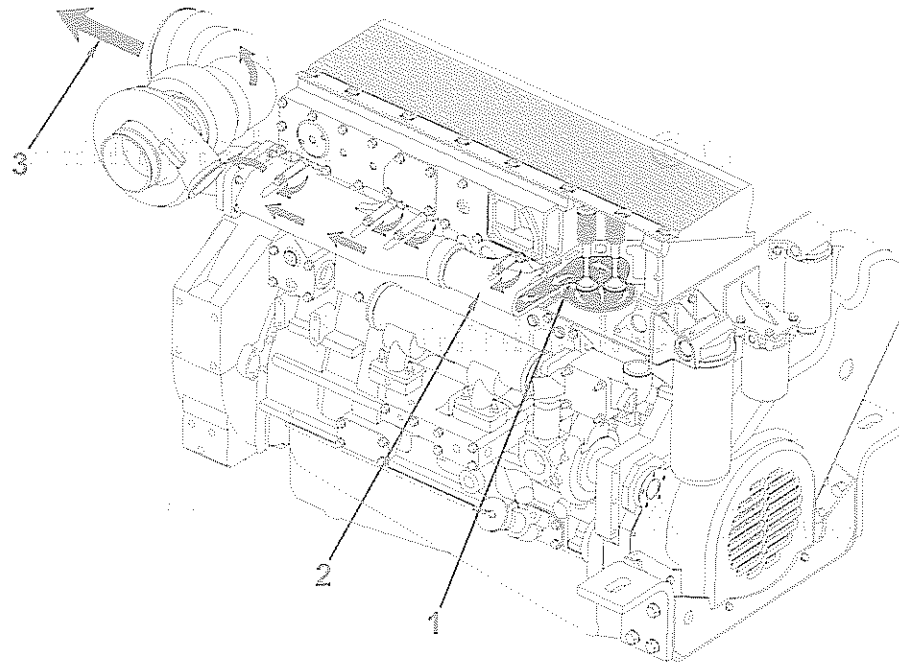


- | | |
|--|-----------------------|
| 1. Filtered intake air to turbocharger | 4. Intake manifold |
| 2. Turbocharger air to charge air cooler | 5. Intake valve ports |
| 3. Charge air cooler intake air to intake manifold | 6. Intake valves. |

10200035

Flow Diagram, Exhaust System

Flow Diagram Marine Applications

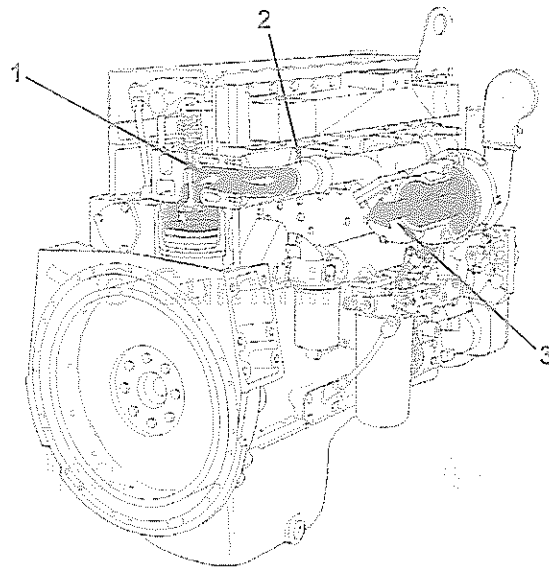


- 1. Exhaust valve ports
- 2. Exhaust manifold

- 3. Turbocharger exhaust outlet.

11200027

Industrial and Generator Drive

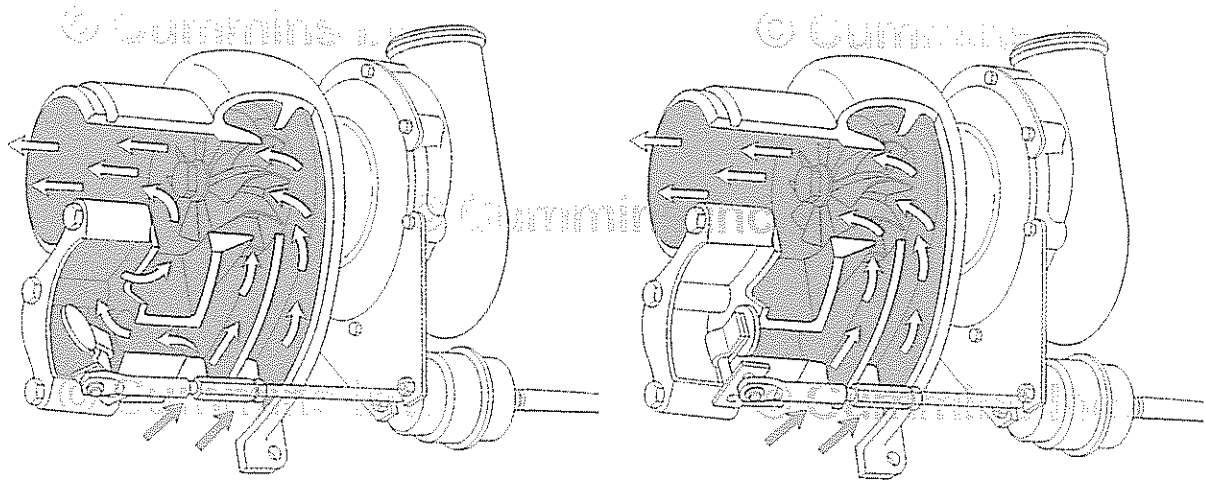


11200009

- 1. Exhaust valve ports
- 2. Exhaust manifold

- 3. Turbocharger exhaust outlet.

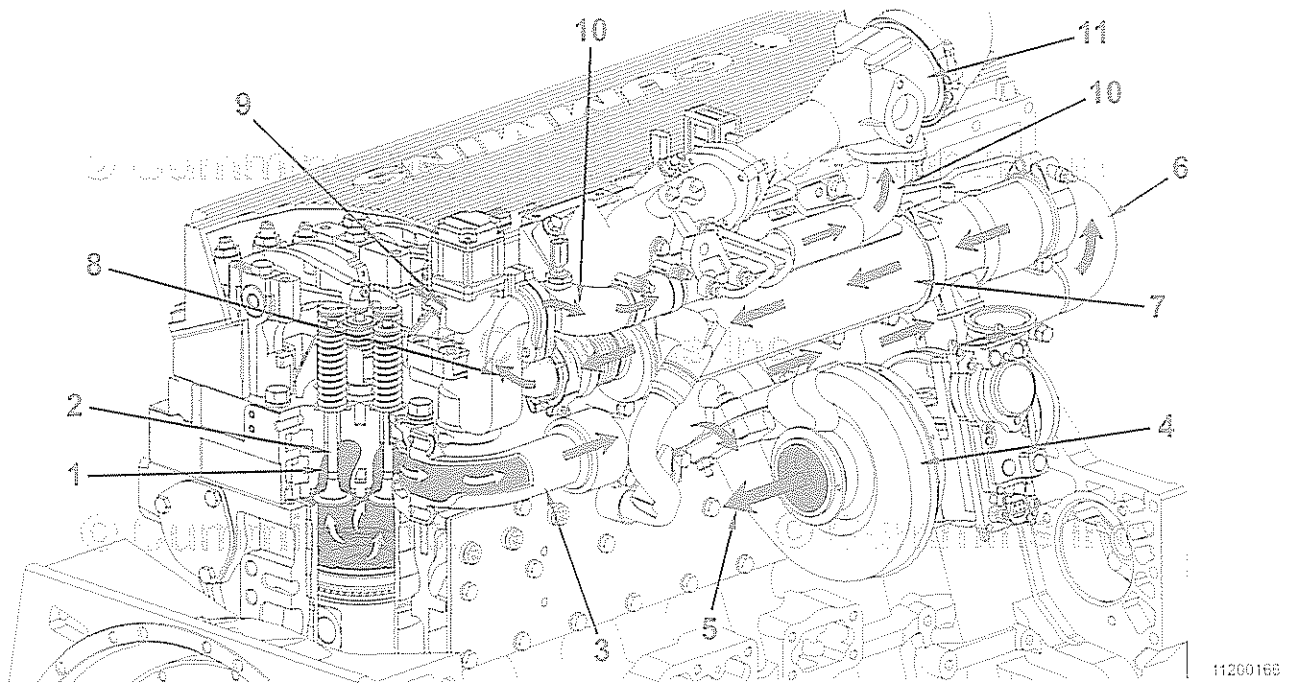
Industrial and Generator Drive



Wastegate Actuation OPEN and CLOSED

12200015

Automotive with CM876

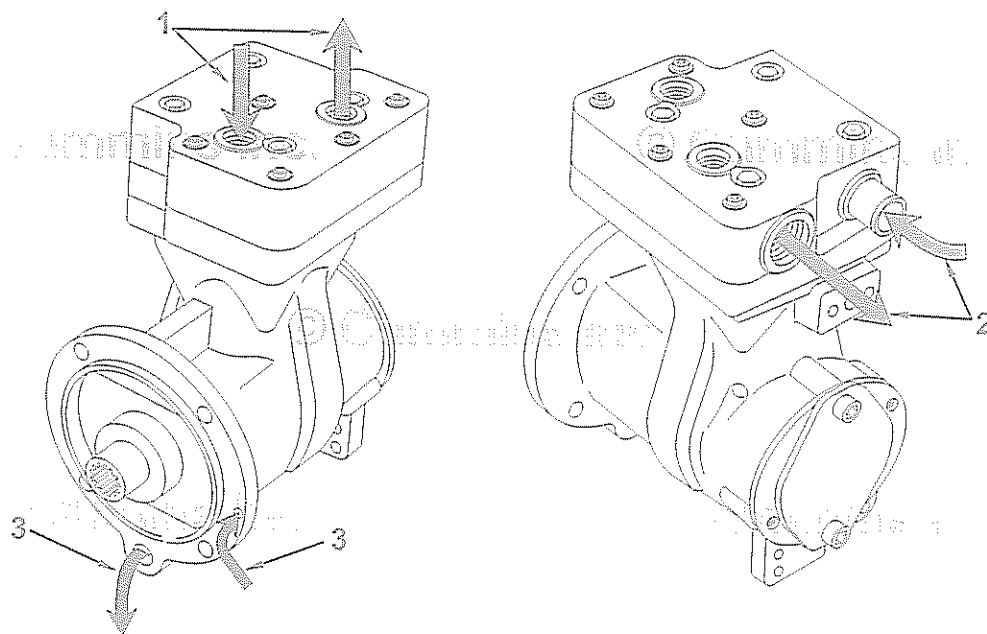


- | | |
|------------------------------------|--|
| 1. Exhaust gas to manifold | 7. EGR cooler |
| 2. Exhaust valve | 8. Cooled exhaust gas to EGR valve |
| 3. Exhaust gas to turbocharger | 9. EGR valve |
| 4. Variable geometry turbocharger | 10. Cooled exhaust gas from the EGR valve to the EGR mixer |
| 5. Turbocharger exhaust gas outlet | 11. EGR mixer |
| 6. Exhaust gas to EGR cooler | |

Flow Diagram, Compressed Air System

Compressed Air System

Cummins

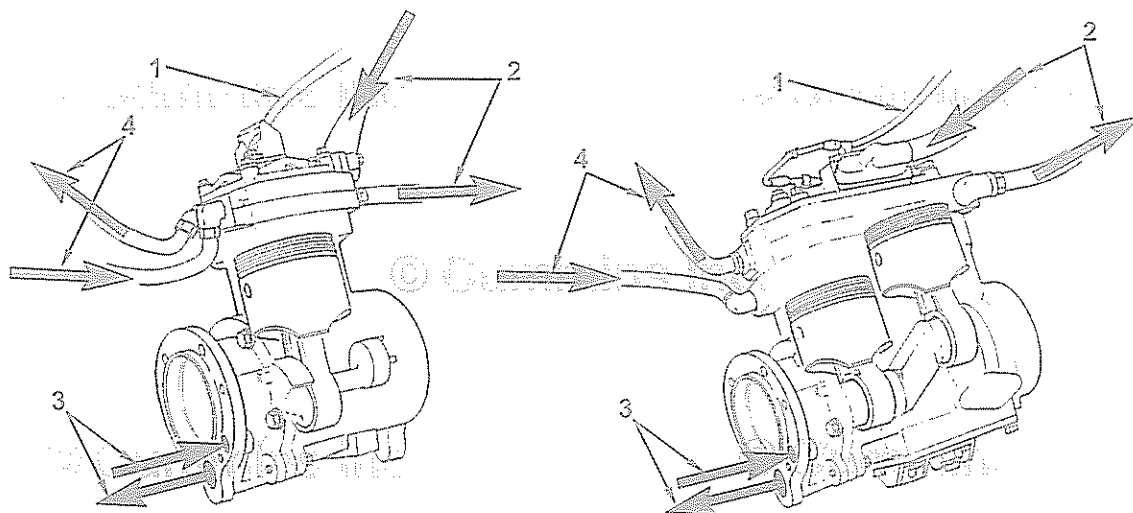


- 1. Coolant
- 2. Air

- 3. Lubricant.

10200072

Holset® Models



- 1. Air governor signal
- 2. Air

- 3. Lubricant
- 4. Coolant.

12200009

Cummins Service Publications Electronic Books on CD-ROM

Enclosed, we've included on CD-ROM, an electronic copy of this publication in Adobe® Acrobat® format (PDF) for your personal use and reference. Before you begin to use the CD-ROM, please read the "Cummins Service Publications Electronic Books on CD-ROM Information" printed section, beginning on the next page. This section includes information about this CD-ROM including System Requirements, Local Administrative Rights requirements, How to Use, Helpful Support Tips and other information.

Cummins Service Publications Electronic Books on CD-ROM Information

The enclosed CD-ROM provides you with the capability to view this publication electronically as well as perform searches. This CD-ROM is copy protected and will not allow Printing, Changing of Content, Extracting Content or Commenting / Annotating of the PDF file.

This CD-ROM is read only, and you will not be able to copy files from the CD-ROM and have them open or function in any manner. The only method to view the electronic publication is from the original CD-ROM.

This CD-ROM is intended for use by the owner of the printed publication only. Copying or distribution of content from this CD-ROM is expressly prohibited. This CD-ROM will not function from a network server CD-ROM drive. It will function only in a CD-ROM drive directly connected to your computer.

For Microsoft Windows NT 4.0, Microsoft Windows 2000 or Microsoft Windows XP, this CD-ROM requires that you have Local Administrative rights in order to use. See the System requirements section for additional details.

System Requirements:

Use of the Cummins Service Publications Electronic Books on CD-ROM requires Adobe Acrobat, Adobe Acrobat Reader, or Adobe Reader software to be pre-installed on your computer. This product has been tested for compatibility with Adobe Acrobat software versions 5.0, 6.0 and 7.0. It may function properly with versions later than Adobe Acrobat 7.0, but testing has not been performed. Acrobat Reader is available as a free download from www.adobe.com.

The Cummins Service Publications Electronic Books on CD-ROM product requires the following system components:

- Microsoft Windows Auto-Play must be enabled on your computer (it is enabled by default when Windows is installed)
- CD-ROM Drive
- Color Monitor with at least 800x600 resolution graphics
- Microsoft Windows XP Professional or Microsoft Windows 2000 Professional. This CD-ROM may be compatible with Microsoft Windows 98 SE, Microsoft Windows NT 4.0, or Microsoft Windows XP Home Edition, but testing has not been performed.
- Recommended memory, processor speed and available hard-disk space is in accordance with respective Microsoft Operating System Guidelines, as well as Adobe Acrobat system requirements listed later in these instructions.

Note: The Cummins Service Publications Electronic Books on CD-ROM product is not Macintosh compatible.

About required Local Administrative Rights:

The copy protection system used on this product requires direct access to CD-ROM drives in order to authenticate copy protected discs.

In order to allow the software to enable direct access to CD-ROM drives rights under Microsoft Windows, if you are using Microsoft Windows NT 4.0, Microsoft Windows 2000 or Microsoft Windows XP, your logon account/profile must have Local Administrative Rights on the computer you will be using your Cummins Service Publications Electronic Book on CD-ROM with. If this computer is set up in Restricted Mode, and your logon account/profile does not have Local Administrative Rights, then, with approval of your I.T. department or the owner of your computer, you can choose from the following options to allow the Cummins Service Publications Electronic Book on CD-ROM to function properly:

- Have your computer account enabled to have Local Administrative rights and then you can run the Cummins Service Publications Electronic Book on CD-ROM - OR -
- Have a user with Local Administrative Rights logon to your computer and run the CD-ROM one time. By doing this, the necessary files will be loaded to your computer and your computer will be enabled to have direct access to CD-ROM drives. Then restart this computer. It is very important to restart the computer after you have done this. For all future needs, you can run the Cummins Service Publications Electronic Book on CD-ROM under your own account, without requiring your logon account/profile to have Local Administrative Rights. (Note: This needs to be done only once for any Cummins Service Publications Electronic Book on CD-ROM and then does not need to be done again for other Cummins Service Publications Electronic Book on CD-ROM titles you may own.)

Cummins Service Publications Electronic Books on CD-ROM Information

(continued)

About the Copy Protection Software:

- This product may upgrade the proprietary copy protection software files it uses that are placed on your computer system to a new version, if an older version exists on your computer. If you have CD-ROM copy protected items from other companies and they use older versions of the same HexaLock copy protection software that the Cummins Service Publications Electronic Books on CD-ROM uses, it may render them incompatible for use on your computer.

Adobe Acrobat Reader 5.0 system requirements:

- PC Compatible computer w/ Intel® Pentium® processor
- One of the following Microsoft® Windows Operating Systems:
 - Microsoft Windows 98 SE
 - Microsoft Windows Millennium Edition
 - Microsoft Windows NT® 4.0 with Service Pack 5
 - Microsoft Windows 2000
 - Microsoft Windows XP
- 64MB of RAM
- 24MB of available hard-disk space
- Additional 70MB of hard-disk space for Asian fonts (optional)
(Asian fonts are not required for Cummins Service Publications Electronic Books on CD-ROM product.)

Adobe Reader 6.0.1 system requirements:

- PC Compatible computer w/ Intel Pentium processor
- One of the following Microsoft Windows Operating Systems:
 - Microsoft Windows 98 SE
 - Microsoft Windows Millennium Edition
 - Microsoft Windows NT 4.0 with Service Pack 6
 - Microsoft Windows 2000 with Service Pack 2
 - Microsoft Windows XP Professional or Home Edition
 - Microsoft Windows XP Tablet PC Edition
- 32MB of RAM (64MB recommended)
- 60MB of available hard-disk space
- Microsoft Internet Explorer 5.0.1, 5.5, 6.0, or 6.1

Adobe Reader 7.0 system requirements:

- PC Compatible computer w/ Intel Pentium processor
- One of the following Microsoft Windows Operating Systems:
 - Microsoft Windows 2000 with Service Pack 2
 - Microsoft Windows XP Professional or Home Edition
 - Microsoft Windows XP Tablet PC Edition
- 128MB of RAM
- Up to 90MB of available hard-disk space
- Microsoft Internet Explorer 5.5 or higher

How to Use the Cummins Service Publications Electronic Book on CD-ROM:

To use the CD-ROM, simply insert the CD-ROM into your computer and it should automatically run, open Adobe Acrobat and open the Electronic Book. (This happens with Microsoft Windows Auto-Play functionality, and Adobe Acrobat pre-loaded by you on your computer).

Cummins Service Publications Electronic Books on CD-ROM Information

(continued)

Helpful Support Tips:

- If Windows Auto-Play is not enabled, try re-enabling it, or navigate with Windows Explorer to your CD-ROM drive and double click on the CD-ROM drive letter to open it. Then double-click on the program named "clickhere". (Do not attempt to directly open the PDF file by clicking on the PDF document.)
- In rare occasions, after inserting the CD-ROM you may receive a notice from Adobe Acrobat that it "could not open the PDF because it is either not a supported file type or the file has been corrupted". This can occur in rare occasions due to application timing issues. If this occurs there are several options to remedy the situation: 1) Simply click OK on the notice, eject the CD-ROM and re-insert it to have it restart the program, or 2) Click OK on the notice and navigate with Windows Explorer to your CD-ROM drive and double click on the CD-ROM drive letter to open it. Then double-click on the program named "clickhere". (Do not attempt to directly open the PDF file by clicking on the PDF document) or 3) Click OK on the notice, eject the CD-ROM, restart your computer and after the system has restarted, re-insert the CD-ROM and have it automatically run.
- If the Cummins Service Publications Electronic Books on CD-ROM does not function, see the prior section on "About required Local Administrative Rights:" in this documentation and verify you have Local Administrative Rights on your computer in order to run the CD-ROM.

Electronic Book Technical Support:

Support for this product is available from Monday through Friday weekly, excluding Holidays, from 8 a.m. to 5 p.m. You may call (502) 540-4981 for telephone support. For e-mail support, please e-mail ebooksupport@merrickind.com.

CD-ROM Media Replacement Options:

Up to 90 days from date of purchase:

Within the first 90 days from your date of purchase, if your copy of an Electronic Book on CD-ROM does not function, and after a Cummins Electronic Book Technical Support Technician has confirmed the situation and authorized its replacement by providing you an RMA#, you may send back the CD-ROM for a free replacement. To do so, you must package the CD-ROM and ship/mail, with postage pre-paid by you, to the below address. You must also include a photocopy of the original invoice for proof of purchase of the publication clearly indicating the bulletin # and the purchase date. The RMA# must be on the address information of the package. If the proof of purchase copy of the invoice is not enclosed, your request will not be able to be processed and will not be returned. Delivery of the replacement will be shipped to you at no charge. Allow 3 to 4 weeks for your receipt of replacement copy. Note: Return only the non-functional CD-ROM, do not return the entire printed publication. The replacement CD-ROM maintains the original purchase/invoice date for the purposes of this replacement policy. This policy is subject to change at any time, without notice. For a copy of the most current replacement options policy, please e-mail ebooksupport@merrickind.com with your request.

Media Replacement Ship to Address:

Attn: Cummins Service Publications Electronic Books Technical Support
RMA#: XXXXXX (where XXXXXX is the RMA#)
808 E. Liberty Street
Louisville, KY 40204 U.S.A.

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Intel and Pentium are trademarks of Intel Corporation.
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All other names and products used herein are trademarks for their respective owner.

Section L - Service Literature

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Additional Service Literature

General Information

The following publications can be purchased:

Bulletin No.	Title of Publication
3666322	ISM, ISMe, and QSM11 Engines Service Manual
3666266	ISM and QSM11 Engines, Electronic Control System, Troubleshooting and Repair Manual
4021477	ISM CM875 Electronic Control System, Troubleshooting and Repair Manual
4021560	ISM CM876 / ISX CM871 Electronic Control System, Troubleshooting and Repair Manual
3666075	M11 Series Engine Shop Manual
3666269	ISM Wiring Diagram
4021478	ISM CM875 Wiring Diagram
4021574	ISM CM876 Wiring Diagram
4021342	ISM and ISMe Owners Manual
3379000	Air for Your Engines
3679001	Fuel for Cummins Engines
3810340	Cummins Engine Oil Recommendations
3387622	Cold Weather Operation
3666132	Coolant Requirements and Maintenance
3666209	Cooling System Maintenance Extended Interval

Service Literature Ordering Location Contact Information

Region

United States and Canada

Ordering Location

Cummins Distributors

or

Credit Cards at 1-800-646-5609

or

Order online at www.powerstore.cummins.com

All Other Countries

Cummins Distributors or Dealers

Cummins Customized Parts Catalog

General Information

Cummins is pleased to announce the availability of a parts catalog compiled specifically for you. Unlike the generic versions of parts catalogs that support general high volume parts content; Cummins Customized catalogs contains only the new factory parts that were used to build your engine.

The catalog cover, as well as the content, is customized with you in mind. You can use it in your shop, at your worksite, or as a coffee table book in your RV or boat. The cover contains your name, company name, address, and telephone number. Your name and engine model identification even appears on the catalog spine. Everybody will know that Cummins created a catalog specifically for you.

This new catalog was designed to provide you with the exact information you need to order parts for your engine. This will be valuable for customers that do not have easy access to the Cummins Electronic Parts Catalog or the Cummins Parts Microfilm System.

Additional Features of the Customized Catalog include:

- Engine Configuration Data
- Table of Contents
- Separate Option and Parts Indexes
- Service Kits (when applicable)
- ReCon Part Numbers (when applicable)

Ordering the Customized Parts Catalog

Ordering by Telephone

North American customers can contact their Cummins Distributor or call Gannett Direct Marketing Services at 1-800-646-5609 and order by credit card. Outside North America order on-line or make an International call to Gannett at (+ +)502-454-6660.

Ordering On-Line

The Customized Parts Catalog can be ordered On-Line from the Cummins Powerstore by credit card. Contact the Powerstore at WWW.POWERSTORE.CUMMINS.COM

Contact GDMS or the CUMMINS POWERSTORE for the current price; Freight may be an additional expense.

Information we need to take your Customized Parts Catalog Order. This information drives the cover content of the CPC.

- Customer Name
- Street Address
- Company Name (optional)
- Telephone no.
- Credit Card No.
- Cummins Engine Serial Number (located on the engine data plate)
- Please identify the required media: Printed Catalog, CD-ROM, or PDF File

Unfortunately not all Cummins Engines can be supported by this parts catalog. Engines older than 1984 or newer than 3 months may not have the necessary parts information to compile a catalog. We will contact you if this occurs and explain why we are unable to fill your order.

Customized Parts Catalogs are produced specifically for a single customer. This means they are not returnable for a refund. If we make an error and your catalog is not useable, we will correct that error by sending you a new catalog.

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Section M - Component Manufacturers

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Component Manufacturers' Addresses

NOTE: The following list contains addresses and telephone numbers of suppliers of accessories used on Cummins engines. Suppliers can be contacted directly for any specifications not covered in this manual.

Air Compressors

Bendix Heavy Vehicles Systems
Div. of Allied Automotive
901 Cleveland Street
Elyria, OH 44036
Telephone: (216) 329-9000

Holset Engineering Co., Inc.
1320 Kemper Meadow Drive
Suite 500
Cincinnati, OH 45240
Telephone: (513) 825-9600

Midland-Grau
Heavy Duty Systems
Heavy Duty Group Headquarters
10930 N. Pamona Avenue
Kansas City, MO 64153
Telephone: (816) 891-2470

Air Cylinders

Bendix Ltd.
Douglas Road
Kingswood
Bristol
England
Telephone: 0117-671881

Catching Engineering
1733 North 25th Avenue
Melrose Park, IL 60160
Telephone: (708) 344-2334

TEC - Hackett Inc.
8909 Rawles Avenue
Indianapolis, IN 46219
Telephone: (317) 895-3670

Air Heaters

Fleetguard, Inc.
1200 Fleetguard Road
Cookeville, TN 38502
Telephone: (615) 526-9551

Kim Hotstart Co.
P.O. Box 11245
Spokane, WA 99211-0245
Telephone: (509) 534-6171

Air Starting Motors

Ingersoll Rand
Chorley New Road
Horwich
Bolton
Lancashire
England
BL6 6JN
Telephone: 01204-65544

Ingersoll-Rand Engine

Starting Systems
888 Industrial Drive
Elmhurst, IL 60126
Telephone: (708) 530-3875

StartMaster
Air Starting Systems
A Division of Sycon Corporation
9595 Cheney Avenue
P. O. Box 491
Marion, OH 43302
Telephone: (614) 382-5771

Alternators

Robert Bosch Ltd.
P.O. Box 98
Broadwater Park
North Orbital Road
Denham
Uxbridge
Middlesex UD9 5HG
England
Telephone: (0)1895-838383

Prestolite Electrics
Cleveland Road
Leyland
PR5 1XB
England
Telephone: (0)1772-421663

C. E. Niehoff & Co.
2021 Lee Street
Evanston, IL 60202
Telephone: (708) 866-6030

Delco-Remy America
2401 Columbus Avenue
P.O. Box 2439
Anderson, IN 46018
Telephone: (317) 646-3528

Leece-Neville Corp.
400 Main Street
Arcade, NY 14009
Telephone: (716) 492-1700

Auxiliary Brakes

The Jacobs Manufacturing Company
Vehicle Equipment Division
22 East Dudley Town Road
Bloomfield, CT 06002
Telephone: (203) 243-1441

Belts

T.B.A. Belting Ltd.
P.O. Box 77
Wigan
Lancashire
WN2 4XQ
England
Telephone: (0)1942-259221

Dayco Mfg.
Belt Technical Center
1955 Enterprize
Rochester Hills, MI 48309
Telephone: (810) 853-8300

Gates Rubber Company
900 S. Broadway
Denver, CO 80217

Goodyear Tire and
Rubber Company
Industrial Products Div.
2601 Fortune Circle East
Indianapolis, IN 46241
Telephone: (317) 898-4170

Catalytic Converters

Donaldson Company, Inc.
1400 West 94th Street
P.O. Box 1299
Minneapolis, MN 55440
Telephone: (612) 887-3835

Nelson Division
Exhaust and Filtration Systems
1801 U.S. Highway 51 P.O. Box 428
Stoughton, WI 53589
Telephone: (608) 873-4200

Walker Manufacturing
3901 Willis Road
P.O. Box 157
Grass Lake, MI 49240
Telephone: (517) 522-5500

Coolant Level Switches

Robertshaw Controls Company
P.O. Box 400
Knoxville, TN 37901
Telephone: (216) 885-1773

Clutches

Twin Disc International S.A.
Chaussee de Namur
Nivelles
Belguim
Telephone: 067-224941

Twin Disc Incorporated
1328 Racine Street
Racine, WI 53403
Telephone: (414) 634-1981

Coolant Heaters

Fleetguard, Inc.
1200 Fleetguard Road
Cookeville, TN 38502
Telephone: (615) 526-9551

Drive Plates

Detroit Diesel Allison
Division of General Motors
Corporation
P.O. Box 894
Indianapolis, IN 46206-0894
Telephone: (317) 242-5000

Electric Starting Motors

Prestolite Electric
Cleveland Road
Leyland
PR5 1XB
England
Telephone: 01772-421663

Delco-Remy America
2401 Columbus Avenue
P.O. Box 2439
Anderson, IN 46018
Telephone: (317) 646-3528

Leece-Neville Corp.
400 Main Street
Arcade, NY 14009
Telephone: (716) 492-1700

Nippondenso Inc.
2477 Denso Drive
P.O. Box 5133
Southfield, MI 48086
Telephone: (313) 350-7500

Electronic Switches

Cutler-Hammer Products
Eaton Corporation
4201 N. 27th Street
Milwaukee, WI 53216
Telephone: (414) 449-6600

Engine Protection Controls

Flight Systems Headquarters
Hempt Road
P.O. Box 25
Mechanicsburg, PA 17055
Telephone: (717) 697-0333

The Nason Company
2810 Blue Ridge Blvd.
West Union, SC 29696
Telephone: (803) 638-9521

Teddington Industrial
Equipment
Windmill Road
Sunbury on Thames
Middlesex
TW16 7HF
England
Telephone: (0)9327-85500

Fan Clutches

Kysor Cooling Systems N.A.
6040 West 62nd Street
Indianapolis, IN 46278
Telephone: (317) 328-3330

Holset Engineering Co. Ltd.
ST Andrews Road
Huddersfield, West Yorkshire
England HD1 6RA
Telephone: (0)1484-22244

Horton Industries, Inc.
P.O. Box 9455
Minneapolis, MN 55440
Telephone: (612) 378-6410

Rockford Clutch Company
1200 Windsor Road
P.O. Box 2908
Rockford, IL 61132-2908
Telephone: (815) 633-7460

Fans

Truflo Ltd.
Westwood Road
Birmingham
B6 7JF
England
Telephone: (0)121-3283041

Hayes-Albion Corporation
Jackson Manufacturing Plant
1999 Wildwood Avenue
Jackson, MI 49202
Telephone: (517) 782-9421

Engineered Cooling Systems, Inc.
201 W. Carmel Drive
Carmel, IN 46032
Telephone: (317) 846-3438

Brookside Corporation
P.O. Box 30
McCordsville, IN 46055
Telephone: (317) 335-2014

TCF Aerovent Company
9100 Purdue Rd., Suite 101
Indianapolis, IN 46268-1190
Telephone: (317) 872-0030

Kysor-Cadillac
1100 Wright Street
Cadillac, MI 49601
Telephone: (616) 775-4681

Schwitzer
6040 West 62nd Street
P.O. Box 80-B
Indianapolis, IN 46206
Telephone: (317) 328-3010

Fault Lamps

Cutler-Hammer Products
Eaton Corporation
4201 N. 27th Street
Milwaukee, WI 53216
Telephone: (414) 449-6600

Filters

Fleetguard International Corp.
Cavalry Hill Industrial Park
Weedon
Northampton NN7 4TD
England
Telephone: 01327-341313

Fleetguard, Inc.
1200 Fleetguard Road
Cookeville, TN 38502
Telephone: 1-800-22-Filters
(1-800-223-4583)

Flexplates

Corrugated Packing and
Sheet Metal
Hamsterley
Newcastle Upon Tyne
England
Telephone: (0)1207-560-505

Allison Transmission
Division of General Motors
Corporation
P.O. Box 894
Indianapolis, IN 46206-0894
Telephone: (317) 242-5000

Midwest Mfg. Co.
29500 Southfield Road, Suite 122
Southfield, MI 48076
Telephone: (313) 642-5355

Wohler Corporation
708 East Grand River Avenue
P.O. Box 20217
Lansing, MI 48901
Telephone: (517) 485-3750

Fuel Coolers

Hayden, Inc.
1531 Pomona Road
P.O. Box 848
Corona, CA 91718-0848
Telephone: (909) 736-2665

Fuel Pumps

Robert Bosch Corp.
Automotive Group
2800 South 25th Ave.
Broadview, IL 60153

Fuel Warmers

Fleetguard, Inc.
1200 Fleetguard Road
Cookeville, TN 38502
Telephone: (615) 526-9551

Gauges

Grasslin U.K. Ltd.
Vale Rise
Tonbridge
Kent
TN9 1TB
England
Telephone: (0)1732-359888

Datcon Instruments
P.O. Box 128
East Petersburg, PA 17520
Telephone: (717) 569-5713

Rochester Gauges, Inc.
11616 Harry Hines Blvd.
P.O. Box 29242
Dallas, TX 75229
Telephone: (214) 241-2161

Governors

Woodward Governor Co.
P.O. Box 1519
Fort Collins, CO 80522
Telephone: (303) 482-5811
(800) 523-2831

Barber Colman Co.
1354 Clifford Avenue
Loves Park, IL 61132
Telephone: (815) 637-3000

United Technologies
Diesel Systems
1000 Jorie Blvd.
Suite 111
Oak Brook, IL 69521
Telephone: (312) 325-2020

Heat Sleeves

Bentley Harris Manufacturing Co.
100 Bentley Harris Way
Gordonville, TN 38563
Telephone: (313) 348-5779

**Hydraulic and Power Steering
Pumps**

Honeywell Control Systems Ltd.
Honeywell House
Arlington Business Place
Bracknell
Berks RG12 1EB
Telephone: (0)1344-656000

Sperry Vickers
P.O. Box 302
Troy, MI 48084
Telephone: (313) 280-3000

Z.F.
P.O. Box 1340
Grafvonsoden Strasse
5-9 D7070
Schwaebisch Gmuend
Germany
Telephone: 7070-7171-31510

In-Line Connectors

Pioneer-Standard Electronics, Inc.
5440 Neiman Parkway
Solon, OH 44139
Telephone: (216) 349-1300

Deutsch
Industrial Products Division
37140 Industrial Avenue
Hemet, CA 92343
Telephone: (714) 929-1200

Oil Heaters

Fleetguard, Inc.
1200 Fleetguard Road
Cookeville, TN 38502
Telephone: (615) 526-9551

Kim Hotstart Co.
P.O. Box 11245
Spokane, WA 99211-0245
Telephone: (509) 534-6171

Prelubrication Systems

RPM Industries, Inc.
Suite 109
55 Hickory Street
Washington, PA 15301
Telephone: (412) 228-5130

Radiators

JB Radiator Specialties, Inc.
P.O. Box 292087
Sacramento, CA 95829-2087
Telephone: (916) 381-4791

The G&O Manufacturing Company
100 Gando Drive
P.O. Box 1204
New Haven, CT 06505-1204
Telephone: (203) 562-5121

Young Radiator Company
2825 Four Mile Road
Racine, WI 53404
Telephone: (910) 271-2397

L and M Radiator, Inc.
1414 East 37th Street
Hibbing, MN 55746
Telephone: (218) 263-8993

Throttle Assemblies

Williams Controls, Inc.
14100 SW 72nd Avenue
Portland, OR 97224
Telephone: (503) 684-8600

Torque Converters

Twin Disc International S.A.
Chaussee de Namur
Nivelles
Belgium
Telephone: 067-224941

Twin Disc Incorporated
1328 Racine Street
Racine, WI 53403-1758
Telephone: (414) 634-1981

Rockford Powertrain, Inc.
Off-Highway Systems
1200 Windsor Road
P.O. Box 2908
Rockford, IL 61132-2908
Telephone: (815) 633-7460

Modine Mfg. Co.
1500 DeKoven Avenue
Racine, WI 53401
Telephone: (414) 636-1640

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Routine Service and Parts

General Information

Personnel at Cummins Authorized Repair Locations can assist you with the correct operation and service of your engine. Cummins has a worldwide service network of more than 5,000 Distributors and Dealers who have been trained to provide sound advice, expert service, and complete parts support. Check the telephone directory yellow pages or refer to the directory in this section for the nearest Cummins Authorized Repair Location.

Emergency and Technical Service

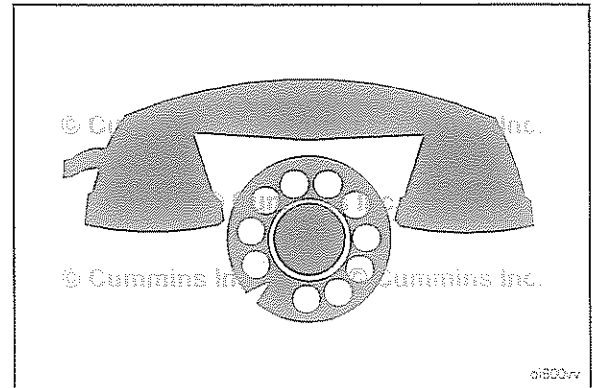
General Information

The Cummins Customer Assistance Center provides a 24-hour, toll free telephone number to aid in technical and emergency service when a Cummins Authorized Repair Location can not be reached or is unable to resolve an issue with a Cummins product.

If additional assistance is required, call Toll-Free:

1-800-DIESELS
(1-800-343-7357)

- Includes all 50 states, Bermuda, Puerto Rico, Virgin Islands, and the Bahamas.
- Outside of North America contact your Regional Office. Telephone numbers and addresses are listed in the International Directory.



Problem Solving

General Information

Normally, any problem that arises with the sale, service, or repair of your engine can be handled by a Cummins Authorized Repair Location in your area. Refer to the telephone directory yellow pages for the one nearest you. If the problem has not been handled satisfactorily, follow the steps outlined below:

1. If the disagreement is with a Dealer, talk to the Cummins Distributor with whom he has his service agreement.
2. If the disagreement is with a Distributor, call the nearest Cummins Division or Regional Office; however, most problems are solved below the Division or Regional office level. Telephone numbers and addresses are listed in this section. Before calling, write down the following information:
 - a. Engine model and serial number
 - b. Type and make of equipment
 - c. Total kilometers [miles] or hours of operation
 - d. Warranty start date
 - e. Nature of problem
 - f. Summary of the current problem arranged in the order of occurrence
 - g. Name and location of the Cummins Distributor or Dealer
12. If a problem can not be resolved satisfactorily through your Cummins Authorized Repair Location or Division Office, write to:

Cummins Customer Assistance Center - 41403, Cummins Inc., Box 3005, Columbus, IN 47202-3005

Division and Regional Offices - Locations

Australia Regional Office	Cummins Engine Company Pty. Ltd.	2 Caribbean Drive Scoresby, Victoria 3179 Australia Telephone: (61-3) 9765-3222 FAX: (61-3) 9763-0079 NOTE: This office also serves New Zealand.
Cummins Americas Regional Office	Cummins Americas Inc.	3350 SW 148 Avenue Suite 205 Miramar, FL 33027 Telephone: [1-954] 431-5511 Fax: [1-954] 433-5797 NOTE: This office serves Puerto Rico and South America excluding Brazil.
China Beijing	Cummins (China) Investment Co. Ltd	28F, Tower A, GATEWAY, No.18, Xiaguangli North Road, East Third Ring, Chaoyang District Beijing, Beijing ZIP / Postal Code: 100027 China Telephone: [86-10] 84548888 Telefax: [86-10] 67876347
Brazil	Cummins Brasil Ltda.	Rua Jati, 266 07180-900 Guarulhos Sao Paulo Brazil Phone: [55-11] 6465-9811 Fax: [55-11] 6412-1483
Daventry	Cummins Engine Company Ltd (Africa)	Royal Oak Way South Daventry, Northants ZIP / Postal Code: NN11 5NU United Kingdom Telephone: [44-1327] 886000 Telefax: [44-1327] 886106
Daventry	Cummins Engine Co. Ltd (Serving Czech Republic)	Royal Oak Way South Daventry, Northants ZIP / Postal Code: NN11 5NU United Kingdom Telephone: [44-1327] 886000 Telefax: [44-1327] 886106
Daventry	Cummins Engine Company Ltd (Middle East)	Royal Oak Way South Daventry, Northants ZIP / Postal Code: NN11 5NU United Kingdom Telephone: [44-1327] 886000 Telefax: [44-1327] 886106
Dubai United Arab Emirates	Cummins Middle East FZE	Units ZF 5/6 Jebel Ali Free Zone P.O.Box No 17636 Dubai United Arab Emirates Phone: [971-4] 883 8998 Fax: [971-4] 883 7971

India Pune	Cummins India Ltd.	Kothrud Pune, Maharashtra ZIP / Postal Code: 411029 India Telephone: [91-20] 2538-5435 / 0240 / 1105 Telefax: [91-20] 2538-0125
Korea Seoul	Cummins Korea Ltd.	25th floor, ASEM tower, 159-1, Samsung-Dong Kangnam-ku, Seoul ZIP / Postal Code: 135-798 South Korea Telephone: [82-2] 3420-0901 Telefax: [82-2] 3452-4113 / 539-6569
SLP Mexico	Cummins, S. de R.L. de C.V.	Arquimedes No. 209 Col. Polanco Mexico, Distrito Federal ZIP / Postal Code: 11560 Mexico Telephone: [52-5] 254-3822 / 3783 / 3622 Telefax: [52-5] 254-3645
Russia Moscow	Cummins Engine Company, Inc.	Park Place Office E708, 113/1 Leninskiy Prospekt Moscow ZIP / Postal Code: 117198 Russia Telephone: [7-495] 956-51-22 / 23 Telefax: [7-495] 956-53-62
Singapore	Cummins Diesel Sales Corporation	8 Tanjong Penjuru Singapore ZIP / Postal Code: 609019 Singapore Telephone: [65] 6265-0155

Distributors and Branches - United States

Alabama	Birmingham	Cummins Mid-South, LLC 2200 Pinson Highway P.O. Box 1147 Birmingham, AL 35217 Telephone: (205) 841-0421 FAX: (205) 849-5926
Alabama	Mobile	Cummins Mid-South, LLC 1924 N. Beltline Hwy. Mobile, AL 36617 Telephone: (334) 456-2236 FAX: (334) 452-6419
Alaska	Anchorage	Cummins Northwest, Inc. 2618 Commercial Drive Anchorage, AK 99501-3095 Telephone: (907) 279-7594 FAX: (907) 276-6340
Arizona	Phoenix	Cummins Rocky Mountain, LLC 2239 N. Black Canyon Hgwy Phoenix, AZ 85009 Telephone: (602) 252-8021 FAX: (602) 253-6725
Arkansas	Little Rock	Cummins Mid-South, Inc. 6600 Interstate 30 Little Rock, AR 72209 Telephone: Sales: (501) 569-5600 Service: (501) 569-5656 Parts: (501) 569-5613 FAX: (501) 565-2199
California	San Leandro	Cummins West, Inc. 14775 Wicks Blvd. San Leandro, CA 94577-6779 Telephone: (510) 351-6101 FAX: (510) 352-3925
California	Arcata	Cummins West, Inc. 4751 West End Road Arcata, CA 95521 Telephone: (707) 822-7392 FAX: (707) 822-7585
California	Bakersfield	Cummins West, Inc. 4601 East Brundage Lane Bakersfield, CA 93307 Telephone: (805) 325-9404 FAX: (805) 861-8719
California	Fresno	Cummins West, Inc. 5333 N Cornelia Ave Fresno, CA 93722 Telephone: (559) 277-6760 FAX: (559) 277-6769

California	Redding	Cummins West, Inc. 20247 Charlanne Drive Redding, CA 96002 Telephone: (530) 222-4070 FAX: (530) 224-4075
California	Stockton	Cummins West, Inc. 5250 Claremont Ave Suite 204 Stockton, California 95207, USA Telephone: (209) 472-3460 FAX: (209) 472-3450
California	West Sacramento	Cummins West, Inc. 875 Riverside Parkway West Sacramento, CA 95605-1502 Telephone: (916) 371-0630 FAX: (916) 371-2849
California	Los Angeles	Cummins Cal Pacific Inc. 1939 Deere Avenue (Irvine) Irvine, CA 92606 Telephone: (949) 253-6000 FAX: (949) 253-6070
California	Montebello	Cummins Cal Pacific Inc. 1105 South Greenwood Avenue Montebello, CA 90640 Telephone: (323) 728-8111 FAX: (323) 889-7499
California	Bloomington	Cummins Cal Pacific Inc. 3061 S. Riverside Avenue Bloomington, CA 92316 Telephone: (909) 877-0433 FAX: (909) 877-3787
California	San Diego	Cummins Cal Pacific Inc. 310 N. Johnson Avenue El Cajon, CA 92020 Telephone: (619) 593-3093 FAX: (619) 593-0600
California	Ventura	Cummins Cal-Pacific Inc. 3958 Transport St. Ventura, CA 93003 Telephone: (805) 644-7281 FAX: (805) 644-7284
Colorado	Denver	Cummins Rocky Mountain, Inc. 8211 East 96th Ave Henderson, Colorado 80640 Telephone: (303) 287-0201 FAX: (303) 288-7080
Colorado	Grand Junction	Cummins Rocky Mountain, Inc. 2380 U.S. Highway 6 & 50 P.O. Box 339 Grand Junction, CO 81501 Telephone: (970) 242-5776 FAX: (970) 243-5494

Connecticut	Rocky Hill	Cummins Metropower, Inc. 914 Cromwell Ave. Rocky Hill, CT 06067 Telephone: (860) 529-7474 FAX: (860) 529-7524
Florida	Ft. Myers	Cummins Power South, LLC 2671 Edison Avenue Ft. Myers, FL 33916 Telephone: (941) 337-1211 FAX: (941) 337-5374
Florida	Jacksonville	Cummins Power South 755 Pickettville Rd. Jacksonville, FL 32220 Telephone: (904) 378-1902 FAX: (904) 378-1904
Florida	Hialeah (Miami)	Cummins Power South, LLC 9900 N.W. 77th Avenue Hialeah Gardens, FL 33016 Telephone: (305) 821-4200 FAX: (305) 557-2992
Florida	Ocala	Cummins Power South, LLC 321 Southwest 52nd Ave. Ocala, FL 34474-1892 Telephone: (352) 861-1122 FAX: (352) 861-1130
Florida	Orlando	Cummins Power South, LLC 4020 North Orange Blossom Trail Orlando, FL 32810 Telephone: (407) 298-2080 FAX: (407) 290-8727
Florida	Tampa	Cummins Power South, LLC 5421 N. 59th Street Tampa, FL 33610 Telephone: (813) 621-7202 FAX: (813) 621-8250
Florida		Cummins Power South, LLC 5906 Breckenridge Parkway Suite J Tampa, FL 33610 Telephone: (813) 664-5868 FAX: (813) 623-5442
Florida		Cummins Power South, LLC 5912 E. Hillsborough Avenue Tampa, FL 33610 Telephone: (813) 626-1101 FAX: (813) 628-8888
Florida		Cummins Power South, LLC 6606 N. 56th Street Tampa, FL 33610 Telephone: (813) 623-3330 FAX: (813) 628-4162

Georgia	Atlanta	Cummins South, Inc. 100 University Ave. S.W. Atlanta, Georgia 30315-2202 Telephone: (404) 527-7800 FAX: (404) 527-7832
Georgia		Cummins South, Inc. 5125 Georgia Highway 85 College Park, GA 30349 Telephone: (404) 763-0151 FAX: (404) 766-2132
Georgia	Albany	Cummins South, Inc. 1915 W. Oakridge Drive Albany, GA 31707-4938 Telephone: (912) 888-6210 FAX: (912) 883-1670
Georgia	Augusta	Cummins South, Inc. 1255 New Savannah Road Augusta, GA 30901-3891 Telephone: (706) 722-8825 FAX: (706) 722-7553
Georgia	Savannah	Cummins South, Inc. 8 Interchange Court Savannah, GA 31401-1627 Telephone: (912) 232-5565 FAX: (912) 232-5145
Hawaii	Kapolei	Cummins West Inc. 91-230 Kalaeloa Blvd. Kapolei, HI 96707 Telephone: (808) 682-8110 FAX: (808) 682-8477
Idaho	Boise	Cummins Rocky Mountain, LLC 8949 So. Federal Way City Boise, Idaho 84716 Telephone: (208) 336-5000 FAX: (208) 338-5436
Illinois	Chicago	Cummins Power, LLC 7145 Santa Fe Drive Hodgkins, IL 60525 Telephone: (708) 579-9222 FAX: (708) 352-7547
Illinois	Bloomington	Cummins Mid-States Power, Inc. (at U.S. 51 N and I-55) 414 W. Northtown Road Bloomington-Normal, IL 61761 Telephone: (309) 452-4454 FAX: (309) 452-1642
Illinois	Onan Branch	Cummins/Onan Northern Illinois 8745 W. 82nd Place Justin, IL 60458 Telephone: (708) 563-7070 FAX: (708) 563-7095

Illinois	Madison	Cummins Mid-South, LLC 222 SR-203 Madison, Illinois 62060 Telephone: (618) 798-9512 FAX: (618) 798-9521
Illinois	Rock Island	Cummins Central Power, LLC 7820 - 42nd Street West Rock Island, IL 61201 Telephone: (309) 787-4300 FAX: (309) 787-4397
Indiana	Indianapolis	Cummins Mid-States Power, Inc. P.O. Box 42917 3762 West Morris Street Indianapolis, IN 46242-0917 Telephone: (317) 243-7979 FAX: (317) 240-1925
Indiana		Cummins Mid-States Power, Inc. P.O. Box 42917 3661 West Morris St Indianapolis, IN 46241 Telephone: (317) 486-5287 FAX: (317) 486-5281
Indiana		Cummins Mid-States Power, Inc. 4301 W. Morris St. 3762 West Morris Street Indianapolis, IN 46241 Telephone: (317) 240-1967 FAX: (317) 240-1975
Indiana	Evansville	Cummins Cumberland, Inc. 7901 Highway 41 North Evansville, IN 47725 Telephone: (812) 867-4400 FAX: (812) 867-4411
Indiana	Ft. Wayne	Cummins Mid-States Power, Inc. 3415 Coliseum Blvd. West (At Jct. I-69 & 30/33) Ft. Wayne, IN 46808 Telephone: (219) 482-3691 FAX: (219) 484-8930
Indiana	Gary	Cummins Northern Illinois, Inc. 1440 Texas Street Gary, IN 46402 Telephone: (219) 885-5591 FAX: (219) 883-4817
Indiana	Indianapolis	Cummins Mid-States Power, Inc. 3661 West Morris St Indianapolis, IN 46241 Telephone: (317) 486-5287 FAX: (317) 486-5281
Iowa	Cedar Rapids	Cummins Central Power, LLC 625 - 33rd Avenue SW Cedar Rapids, IA 52404 Telephone: (319) 366-7537 (24 hours) FAX: (319) 366-7562

Iowa	Des Moines	Cummins Central Power, LLC 1680 N.E. 51st Avenue Des Moines, IA 50313 Telephone: (515) 262-9591 Faxes: (515) 262-9744 FAX: (515) 262-0626
Kansas	Colby	Cummins Central Power, LLC 1880 South Range Colby, KS 67701 Telephone: (785) 462-3945 FAX: (785) 462-3970
Kansas	Garden City	Cummins Central Power, LLC 1285 Acraway Garden City, KS 67846 Telephone: (316) 275-2277 FAX: (316) 275-2533
Kansas	Wichita	Cummins Central Power, LLC 5101 North Broadway Wichita, KS 67219 Telephone: (316) 838-0875 FAX: (316) 838-0704
Kentucky	Louisville	Cummins Cumberland, Inc. (Corporate Office) 2301 Nelsonville Parkway Louisville, KY 40223 Telephone: (502) 254-7375 FAX: (502) 254-1215
Kentucky	Hazard	Cummins Cumberland, Inc. Highway 15 South Hazard, KY 41701 Telephone: (606) 436-5718 FAX: (606) 436-5038
Kentucky	Louisville	Cummins Cumberland, Inc. 9820 Bluegrass Parkway Louisville, KY 40299 Telephone: (502) 491-4263 FAX: (502) 499-0896
Louisiana	Morgan City	Cummins Mid-South, LLC 9508 Highway 90 E Amelia, LA 70340 Telephone: (504) 631-0576 FAX: (504) 631-0081
Louisiana	New Orleans	Cummins Mid-South, LLC 110 E. Airline Highway Kenner, LA 70062 Telephone: (504) 465-3412 FAX: (504) 465-3408
Maine	Scarborough	Cummins Northeast, Inc. 10 Gibson Road Scarborough, ME 04074 Telephone: (207) 883-8155 FAX: (207) 883-5526

Maryland	Baltimore	Cummins Power Systems, Inc. 1907 Park 100 Drive MD 21061 Telephone: (410) 590-8700 FAX: (410) 590-8731
Massachusetts	Boston	Cummins Northeast, Inc. 100 Allied Drive Dedham, MA 02026 Telephone: (781) 329-1750 FAX: (781) 329-4428
Massachusetts	Springfield	Cummins Northeast, Inc. 177 Rocus Street Springfield, MA 01104 Telephone: (413) 737-2659 FAX: (413) 731-1082
Michigan	Detroit (New Hudson)	Standby Power, Div. of Cummins Bridgeway, LLC 21810 Clessie Ct New Hudson, Michigan 48165 Telephone: (248) 573-1600
Michigan		Cummins Bridgeway, LLC 54240 Grand River Ave New Hudson, Michigan 48165 Telephone: (517) 573-1900
Michigan	Grand Rapids	Cummins Bridgeway, LLC 3715 Clay Avenue, S.W. Grand Rapids, MI 49508 Telephone: (616) 538-2250 FAX: (616) 538-3830
Michigan	Grand Rapids	Standby Power, Inc. 7580 Expressway Drive S.W. Grand Rapids, MI 49548 Telephone: (616) 281-2211 FAX: (616) 281-3177
Michigan	Iron Mountain	Cummins NPower, LLC 1901 Stevenson Avenue Iron Mountain, MI 49801 Telephone: (906) 774-2424 FAX: (906) 774-1190
Michigan	Saginaw	Cummins Bridgeway, LLC 722 N. Outer Drive Saginaw, MI 48605 Telephone: (989) 752-5200 FAX: (989) 752-4194
Minnesota	St. Paul	Cummins NPower, LLC 3030 Centre Pointe Drive Suite 500 Roseville, MN 55113 Telephone: (651) 636-1000 FAX: (651) 638-2442
Minnesota	Duluth	Cummins NPower, LLC 3115 Truck Center Drive Duluth, MN 55806-1786 Telephone: (218) 628-3641 FAX: (218) 628-0488

Mississippi	Jackson	Cummins Mid-South, LLC 325 New Highway 49 South Jackson, MS 39288-4224 Telephone: (601) 939-7016 FAX: (601) 932-7399
Missouri	Kansas City	Cummins Central Power, LLC 8201 NE Parvin Road Kansas City, MO 64161 Telephone: (816) 414-8200 FAX: (816) 414-8299
Missouri	Joplin	Cummins Central Power, LLC 3507 East 20th Street Joplin, MO 64801 Telephone: (417) 623-1661 FAX: (417) 623-1817
Missouri	Springfield	Cummins Central Power, LLC 3637 East Kearney Springfield, MO 65803 Telephone: (417) 862-0777 FAX: (417) 862-4429
Missouri	Columbia	Cummins Mid-South, LLC 5221 Highway 763 North Columbia, MO 65205 Telephone: (314) 449-3711 FAX: (314) 449-3712
Missouri	Sikeston	Cummins Mid-South, LLC 101 Keystone Drive Sikeston, MO 63801 Telephone: (573) 472-0303 FAX: (573) 472-0306
Missouri	Industrial Power Branch	Cummins Mid-South, LLC 3256 E. Outer Road Scott City, MO 63780 Telephone: (573) 335-7399 FAX: (573) 335-7062
Montana	Billings	Cummins Rocky Mountain, LLC 5151 Midland Road Billings, MT 59101 Telephone: (406) 245-4194 FAX: (406) 245-7923 Toll Free #: (800) 332-7788
Montana	Missoula	Cummins Northwest, Inc. 4950 North Reserve Street Missoula, MT 59802-1498 Telephone: (406) 728-1300 FAX: (406) 728-8523
Nebraska	Omaha	Cummins Great Plains Diesel, Inc. 5515 Center Street Omaha, NE 68106 Telephone: (402) 551-7678 (24 Hours) FAX: (402) 551-1952

Nebraska	Kearney	Cummins Central Power, LLC 515 Central Avenue Kearney, NE 68847 Telephone: (308) 234-1994 FAX: (308) 234-5776
Nevada	Elko	Cummins Rocky Mountain, LLC 5370 East Idaho Street Elko, NV 89801 Telephone: (702) 738-6405 FAX: (702) 738-1719
Nevada	Las Vegas	Cummins Rocky Mountain, LLC 2750 Losee Road North Las Vegas, NV 89030 Telephone: (702) 399-2614 FAX: (702) 399-7457
Nevada	Sparks	Cummins Rocky Mountain, LLC 150 Glendale Avenue Sparks, NV 89431 Telephone: (775) 331-4983 FAX: (775) 331-7429
New Jersey	Newark	Cummins Metropower, Inc. 41-85 Doremus Ave. Newark, NJ 07105 Telephone: (973) 491-0100 FAX: (973) 578-8873
New Mexico	Albuquerque	Cummins Rocky Mountain, LLC 1921 Broadway N.E. Albuquerque, NM 87102 Telephone: (505) 247-2441 FAX: (505) 842-0436
New Mexico	Farmington	Cummins Rocky Mountain, LLC 1101 North Troy King Road Farmington, NM 87401 Telephone: (505) 327-7331 FAX: (505) 326-2948
New York	Bronx	Cummins Metropower, Inc. 890 Zerega Avenue Bronx, NY 10473 Telephone: (718) 892-2400 FAX: (718) 892-0055
New York	Albany	Cummins Northeast, Inc. 101 Railroad Avenue Albany, NY 12205 Telephone: (518) 459-1710 FAX: (518) 459-7815
New York	Buffalo	Cummins Northeast, Inc. 480 Lawrence Bell Dr. Williamsville, NY 14221-7090 Telephone: (716) 631-3211 FAX: (716) 626-0799

New York	Syracuse	Cummins Northeast, Inc. 6193 Eastern Avenue Syracuse, NY 13211 Telephone: (315) 437-2751 FAX: (315) 437-8141
North Carolina	Charlotte	Cummins Atlantic, Inc. 11101 Nations Ford Road Charlotte, NC 28273 Telephone: (704) 588-1240 FAX: (704) 587-4870
North Carolina	Charlotte	Cummins Atlantic, Inc. 3700 North Interstate 85 Charlotte, NC 28206 Telephone: (704) 596-7401 FAX: (704) 596-3038
North Carolina	Greensboro	Cummins Atlantic, Inc. 513 Preddy Boulevard Greensboro, NC 27406 Telephone: (336) 275-4531 FAX: (336) 275-8304
North Dakota	Fargo	Cummins NPower, LLC 3801 - 34th Ave. SW Fargo, ND 58104 Telephone: (701) 282-2466 FAX: (701) 277-5399
North Dakota	Grand Forks	Cummins NPower, LLC 4728 Gateway Drive Grand Forks, ND 58201 Telephone: (701) 775-8197 FAX: (701) 775-4833
North Dakota	Minot	Cummins NPower, LLC 1501 - 20th Avenue, S.E. Minot, ND 58701 Telephone: (701) 852-3585 FAX: (701) 852-3588
Ohio	Columbus	Cummins Bridgeway, LLC 4000 Lyman Drive Hilliard (Columbus), OH 43026 Telephone: (614) 771-1000 FAX: (614) 771-0769
Ohio	Cincinnati	Cummins Interstate Power, Inc. 10470 Evendale Drive Cincinnati, OH 45241 Telephone: (513) 563-6670 FAX: (513) 563-0594
Ohio	Cleveland	Cummins Bridgeway, LLC 7585 Northfield Road Cleveland, OH 44146 Telephone: (440) 439-6800 FAX: (440) 439-2131 Toll Free: (800) 243-6885

Ohio	Toledo	Cummins Bridgeway, LLC 801 Illinois Avenue Maumee , OH 43537 Telephone: (419) 893-8711 FAX: (419) 893-5362
Ohio	Youngstown	Cummins Bridgeway, LLC 7145 Masury Road Hubbard (Youngstown), OH 44425 Telephone: (216) 534-1935 FAX: (216) 534-5606
Oklahoma	Oklahoma City	Cummins Southern Plains ,Ltd. 5800 West Reno Oklahoma City, OK 73127 Telephone: (405) 946-4481 (24 hours) FAX: (405) 946-3336
Oklahoma	Tulsa	Cummins Southern Plains, Ltd. 16525 East Skelly Drive Tulsa, OK 74116 Telephone: (918) 234-3240 FAX: (918) 234-2342
Oregon	Coburg/Eugene	Cummins Northwest, Inc. 91201 Industrial Parkway Coburg, OR 97401 (Telephone: (541) 687-0000 FAX: (541) 687-1977 Toll Free Telephone (800)777-0336
Oregon	Medford	Cummins Northwest, Inc. 4045 Crater Lake Highway Medford, OR 97504-9796 Telephone: (541) 779-0151 FAX: (541) 772-2395 Toll Free Telephone (800)826-9414
Oregon	Pendleton	Cummins Northwest, Inc. 223 S.W. 23rd Street Pendleton, OR 97801-1810 Telephone: (541) 276-2561 FAX: (541) 276-2564 Toll Free Telephone (800)666-2561
Oregon	Portland	Cummins Northwest, Inc. 4711 N. Basin Avenue Portland, OR 97217-3557 Telephone: (503)286-5938 FAX: (503)286-5938 Toll Free Telephone: ((800)283-0336 FAX: (503)240-5553
Pennsylvania	Philadelphia	Cummins Power Systems, Inc. 2727 Ford Road Bristol, PA 19007 Telephone: (215) 785-6005 FAX: (215) 785-4085

Pennsylvania	Pittsburgh	Cummins Power Systems, Inc. 3 Alpha Drive Pittsburgh, PA 15138-2901 Telephone: (412) 820-8300 FAX: (412) 820-8308
Pennsylvania	Harrisburg	Cummins Power Systems, Inc. 4499 Lewis Road Harrisburg, PA 17111-2541 Telephone: (717) 564-1344 FAX: (717) 558-8217
Puerto Rico		Cummins de Puerto Rico, Inc. Calle 1 G1 Urb. Industrial, Barrio Palmas Cataño ZIP / Postal Code: 00962 Puerto Rico Telephone: (787) 275-2000 FAX: (787) 275-2030
South Carolina	Charleston	Cummins Atlantic Inc. 231 Farmington Road Charleston, SC 29483 Telephone: (843) 851-9819 FAX: (843) 875-4338
South Carolina	Columbia	Cummins Atlantic, Inc. 2791 Shop Road Ext Columbia, South Carolina 29209 Telephone: (803) 799-2410 FAX: (803) 779-3427
South Dakota	Sioux Falls	Cummins Central Power, LLC 701 East 54th Street North Sioux Falls, SD 57104 Telephone: (605) 336-1715 FAX: (605) 336-1748
Tennessee	Memphis	Cummins Mid-South, LLC 670 Riverside Drive Memphis, TN 38173 Telephone: (901) 577-0600 FAX: (901) 522-8758
Tennessee	Chattanooga	Cummins South, Inc. 1509 East 26th Street Chattanooga, TN 37407-1095 Telephone: (423) 629-1447 FAX: (423) 629-1494
Tennessee	Knoxville	Cummins Cumberland, Inc. 1211 Ault Road Knoxville, TN 37914 Telephone: (423) 523-0446 FAX: (423) 523-0343
Tennessee	Memphis	Cummins Mid-South, LLC 1784 E. Brooks Road Memphis, TN 38116 Telephone: (901) 577-0666 FAX: (901) 522-8758

Tennessee	Nashville	Cummins Cumberland, Inc. 706 Spence Lane Nashville, TN 37217 Telephone: (615) 366-4341 FAX: (615) 366-5693
Texas	Arlington	Cummins Southern Plains, Ltd. 600 N Watson Road Arlington, TX 76004-76011 Telephone: (817) 640-6801 FAX: (817) 640-6852
Texas	Amarillo	Cummins Southern Plains, Ltd. 5224 Interstate 40 - Expressway East Amarillo, TX 79120-1570 Telephone: (806) 373-3793 (24 hours) FAX: (806) 372-8547
Texas	Dallas	Cummins Southern Plains, Ltd. 3707 Irving Boulevard Dallas, TX 75247 Telephone: (214) 631-6400 (24 hours) FAX: (214) 631-2322
Texas	El Paso	Cummins Rocky Mountain, LLC 14333 Gateway West El Paso, TX 79927 Telephone: (915) 852-4200 FAX: (915) 852-3295
Texas	Fort Worth	Cummins Southern Plains, Ltd. 3250 North Freeway Fort Worth, TX 76111 Telephone: (817) 624-2107 (24 hours) FAX: (817) 624-3296
Texas	Houston	Cummins Southern Plains, Ltd. 7045 North Loop East Houston, Texas 77028 Telephone: (713) 679-2220 FAX: (713) 679-7774
Texas	Mesquite	Cummins Southern Plains, Ltd. 2615 Big Town Blvd. Mesquite, TX 75150 Telephone: (214) 321-5555 (24 hours) FAX: (214) 328-2732
Texas	Odessa	Cummins Southern Plains, Ltd. 1210 South Grandview P.O. Box 633 Odessa, Texas 79761 Telephone: (915) 332-9121 (24 hours) FAX: (915) 333-4655
Texas	San Antonio	Cummins Southern Plains, Ltd. 6226 Pan Am Expressway North San Antonio, TX 78218-0385 Telephone: (512) 655-5420 (24 hours) FAX: (512) 655-3865

Utah	Salt Lake City	Cummins Rocky Mountain, LLC 1030 South 300 West Salt Lake City, UT 84101 Telephone: (801) 524-1321 FAX: (801) 524-1351
Virginia		
Virginia	Richmond	Cummins Atlantic, Inc. 3900 Deepwater Terminal Road Richmond, VA 23234 Telephone: (804) 232-7891 FAX: (804) 232-7428
Virginia	Tidewater	Cummins Atlantic, Inc. 3729 Holland Blvd. Chesapeake, VA 23323 Telephone: (757) 485-4848 FAX: (757) 485-5085
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Section TS - Troubleshooting Symptoms

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Troubleshooting Procedures and Techniques

General Information

This guide describes some typical engine operating problems, their causes, and some acceptable corrections to those problems. Unless noted otherwise, the problems listed are those which an operator can diagnose and repair.



Performing troubleshooting procedures NOT outlined in this section can result in equipment damage or personal injury or death. Troubleshooting must be performed by trained, experienced technicians. Consult a Cummins Authorized Repair Location for diagnosis and repair beyond that which is outlined, and for symptoms not listed in this section. Before beginning any troubleshooting, refer to General Safety Instructions in Section I of this manual.

Follow the suggestions below for troubleshooting:

- Study the complaint thoroughly before acting
- Refer to the engine system diagrams
- Do the easiest and most logical things first
- Find and correct the cause of the complaint

Troubleshooting Symptoms Charts

General Information

Use the charts on the following pages of this section to aid in diagnosing specific engine symptoms. Read each row of blocks from top to bottom. Follow through the chart to identify the corrective action.



Troubleshooting presents the risk of equipment damage, personal injury or death. Troubleshooting must be performed by trained, experienced technicians.

Air Compressor Air Pressure Rises Slowly

Cause

Correction

STEP 1

Air intake system restriction to air compressor is excessive

Replace the air compressor air cleaner (if installed). Check the air intake piping. Check engine air intake restriction if the air compressor inlet is plumbed to the vehicle or equipment intake system. Refer to Procedure 012-015 (Air Compressor Carbon Buildup) in Section 7.

OK

Go To Next Step

STEP 2

Air system leaks

Block the vehicle wheels and check the air system for leaks with spring brakes applied and released. Check for leaks from the air compressor gaskets and the air system hoses, fittings, tanks, and valves. Refer to the OEM service manual.

OK

Go To Next Step

STEP 3

Carbon buildup is excessive in the air discharge line, downstream air valves, or cylinder head

Check for carbon buildup. Replace the air compressor discharge line and cylinder head assembly if necessary. Refer to Procedure 012-015 (Air Compressor Carbon Buildup) in Section 7.

OK

Go To Next Step

STEP 4

Contact a Cummins Authorized Repair Facility

Air Compressor Cycles Frequently

Cause	Correction
<u>STEP 1</u> Air system leaks	Block the vehicle wheels and check the air system for leaks with spring brakes applied and released. Check for leaks from the air compressor gaskets and the air system hoses, fittings, tanks, and valves. Refer to the OEM service manual.
OK Go To Next Step	
<u>STEP 2</u> Carbon buildup is excessive in the air discharge line, check valve, or cylinder head	Check for carbon buildup. Replace the air compressor discharge line, if necessary. Refer to Procedure 012-015 (Air Compressor Carbon Buildup) in Section 7.
OK Go To Next Step	
<u>STEP 3</u> Air compressor pumping time is excessive	Replace the desiccant cartridge on the air dryer (if equipped). Refer to the OEM service manual. Check the air compressor duty cycle. Install a larger air compressor, if necessary. Refer to the Master Repair Manual, Holset® Air Compressors, Bulletin 3666121.
OK Go To Next Step	
<u>STEP 4</u> Contact a Cummins Authorized Repair Facility	

Air Compressor Noise is Excessive

Cause	Correction
<u>STEP 1</u> Carbon buildup is excessive in the air discharge line, downstream air valves, or cylinder head	Check for carbon buildup. Replace the air compressor discharge line and cylinder head assembly if necessary. Refer to Procedure 012-015 (Air Compressor Carbon Buildup) in Section 7.
OK Go To Next Step	
<u>STEP 2</u> Air compressor is sending air pulses into the air tanks	Install a ping tank between the air dryer and the wet tank. Refer to the OEM service manual.
OK Go To Next Step	
<u>STEP 3</u> Ice buildup in the air system components	For all models, check for ice in low spots of the air discharge line, dryer inlet, and elbow fittings. On Holset® models, also check the Econ valve (if equipped). Refer to the OEM service manual.
OK Go To Next Step	
<u>STEP 4</u> Contact a Cummins Authorized Repair Facility	

Air Compressor Pumping Excess Lubricating Oil into the Air System

Cause

Correction

STEP 1

Lubricating oil drain interval is excessive

Verify the correct lubricating oil drain interval. Refer to Procedure 102-002 (Maintenance Schedule) in Section 2.

OK

Go To Next Step

STEP 2

Air intake system restriction to air compressor is excessive

Replace the air compressor air cleaner (if installed). Check the air intake piping. Check engine air intake restriction if the air compressor inlet is plumbed to the vehicle or equipment intake system. Refer to Procedure 010-059 (Air Cleaner Restriction) in Section 6.

OK

Go To Next Step

STEP 3

Air compressor pumping time is excessive

Replace the desiccant cartridge on the air dryer (if equipped). Refer to the OEM service manual. Check the air compressor duty cycle. Install a larger air compressor, if necessary. Refer to a Cummins Authorized Repair Location.

OK

Go To Next Step

STEP 4

Carbon buildup is excessive in the air discharge line, check valve, or cylinder head

Check for carbon buildup. Replace the air compressor discharge line, if necessary. Refer to Procedure 012-015 (Air Compressor Discharge Lines) in Section 7.

OK

Go To Next Step

STEP 5

Contact a Cummins Authorized Repair Facility

Air Compressor Will Not Maintain Adequate Air Pressure (Not Pumping Continuously)

Cause

Correction

STEP 1

Air system leaks

Block the vehicle wheels and check the air system for leaks with spring brakes applied and released. Check for leaks from the air compressor gaskets and the air system hoses, fittings, tanks, and valves. Refer to the OEM service manual.

OK

Go To Next Step

STEP 2

Contact a Cummins Authorized Repair Facility

Air Compressor Will Not Stop Pumping

Cause

Correction

STEP 1

Air system leaks

Block the vehicle wheels and check the air system for leaks with spring brakes applied and released. Check for leaks from the air compressor gaskets and the air system hoses, fittings, tanks, and valves. Refer to the OEM service manual.

OK

Go To Next Step

STEP 2

Contact a Cummins Authorized Repair Facility

Alternator Not Charging or Insufficient Charging

Cause

Correction

STEP 1

Alternator belt is loose

Check the alternator belt tension. Refer to Procedure 018-005 (Drive Belt Tension) in Section V.

OK

Go To Next Step

STEP 2

Battery cables or connections are loose, broken, or corroded (excessive resistance)

Check the battery cables and connections.

OK

Go To Next Step

STEP 3

Batteries have failed

Check the condition of the batteries. Replace the batteries, if necessary. Refer to Procedure 013-007 (Batteries) in Section 6.

OK

Go To Next Step

STEP 4

Electrical system is "open" (blown fuses, broken wires, or loose connections)

Check the fuses, wires, and connections. Refer to the OEM service manual and the manufacturer's wiring diagrams.

OK

Go To Next Step

STEP 5

Alternator pulley is loose on the shaft

Tighten the pulley. Refer to OEM service manual.

OK

Go To Next Step

STEP 6

Contact a Cummins Authorized Repair Facility

Coolant Loss

Cause	Correction
<u>STEP 1</u> Coolant level is above specification OK Go To Next Step	Check the coolant level. Refer to Procedure 008-066 (Coolant Level) in Section 3.
<u>STEP 2</u> Cooling system is contaminated with dirt, scale, or sludge OK Go To Next Step	Clean the cooling system. Refer to the OEM service manual.
<u>STEP 3</u> Contact a Cummins Authorized Repair Facility	

Coolant Temperature Above Normal - Gradual Overheat

Cause

Correction

STEP 1

Coolant level is below specification

Inspect the engine and cooling system for external coolant leaks. Repair if necessary. Add coolant. Refer to Procedure 008-066 (Coolant Level) in Section 3.

OK

Go To Next Step

STEP 2

Charge air cooler fins, radiator fins, or air conditioner condenser fins are damaged or obstructed with debris

Inspect the charge air cooler, air conditioner condenser, and radiator fins. Clean, if necessary. Refer to Procedure 010-028 (Charge Air Piping) in Section 3.

OK

Go To Next Step

STEP 3

Cold weather radiator cover or winterfront is closed

Open the cold weather radiator cover or the winterfront. Maintain a minimum of 784 cm² [122 in²] or approximately 28 x 28 cm [11 x 11 in] of opening at all times. Refer to Procedure 101-015 (Operating the Engine) in Section 1.

OK

Go To Next Step

STEP 4

Cooling system hose is collapsed, restricted, or leaking

Inspect the hoses. Refer to Procedure 008-086 (Coolant System Hoses) in Section 3 or Procedure 008-104 (Seawater Hoses) in Section 6.

OK

Go To Next Step

STEP 5

Fan drive belt or water pump belt is loose

Check the belt tension and tighten if necessary. Refer to Procedure 103-002 (Drive Belts) in Section 3.

OK

Go To Next Step

STEP 6

Lubricating oil level is above or below specification

Check the oil level. Add or drain oil, if necessary. Refer to Procedure 007-043 (Lubricating Oil Level) in Section 3.

OK

Go To Next Step

STEP 7

Fan shroud is damaged or missing or the air recirculation baffles are damaged or missing

Inspect the shroud and the recirculation baffles. Repair, replace, or install, if necessary. Refer to the OEM service manual.

OK

Go To Next Step

STEP 8

Radiator cap is not correct, is malfunctioning, or has low-pressure rating

Check the radiator pressure cap. Refer to the OEM service manual.

OK

Go To Next Step

Coolant Temperature Above Normal - Gradual Overheat

Cause

STEP 9

Supplemental coolant additive (SCA) level is above specification or the coolant is overconcentrated with antifreeze

Correction

Check the SCA level. Verify the antifreeze concentration. Refer to Procedure 008-046 (Supplemental Coolant Additive (SCA) and Antifreeze Concentration) in Section 4.

OK

Go To Next Step

STEP 10

Contact a Cummins Authorized Repair Facility

Coolant Temperature Above Normal - Sudden Overheat

Cause

Correction

STEP 1

Electronic fault codes are active

For instructions on how to read active fault codes, refer to Procedure 101-007 (Electronic Controlled Fuel System) in Section 1 for diagnostic fault codes.. If fault codes are active, contact a Cummins Authorized Repair Facility.

OK

Go To Next Step

STEP 2

Coolant level is below specification

Inspect the engine and cooling system for external coolant leaks. Repair if necessary. Add coolant. Refer to Procedure 008-066 (Coolant Level) in Section 3.

OK

Go To Next Step

STEP 3

Charge air cooler fins, radiator fins, or air conditioner condenser fins are damaged or obstructed with debris

Inspect the charge air cooler, air conditioner condenser, and radiator fins. Clean, if necessary. Refer to Procedure 010-028 (Charge Air Piping) in Section 3.

OK

Go To Next Step

STEP 4

Cooling system hose is collapsed, restricted, or leaking

Inspect the hoses. Refer to Procedure 008-086 (Coolant System Hoses) in Section 3 or Procedure 008-104 (Sewater Hoses) in Section 6.

OK

Go To Next Step

STEP 5

Fan drive belt is broken

Check the fan drive belt. Replace the belt, if necessary. Refer to Procedure 008-002 (Drive Belt, Cooling Fan) in Section A.

OK

Go To Next Step

STEP 6

Radiator cap is not correct, is malfunctioning, or has low-pressure rating

Check the radiator pressure cap. Refer to the OEM service manual.

OK

Go To Next Step

STEP 7

Cold weather radiator cover or winterfront is closed

Open the cold weather radiator cover or the winterfront. Maintain a minimum of 784 cm² [122 in²] or approximately 28 x 28 cm [11 x 11 in] of opening at all times. Refer to Procedure 101-015 (Operating the Engine) in Section 1.

OK

Go To Next Step

STEP 8

Fill line or vent lines are restricted, obstructed, or not routed correctly

Check the vent lines and the fill line for correct routing and for restriction. Refer to the OEM service manual.

OK

Go To Next Step

Coolant Temperature Above Normal - Sudden Overheat
Cause

Correction

STEP 9

Contact a Cummins Authorized Repair Facility

Coolant Temperature Below Normal

Cause

Correction

STEP 1

Electronic fault codes are active

For instructions on how to read active fault codes, refer to Procedure 101-007 (Electronic Controlled Fuel System) in Section 1 for diagnostic fault codes.. If fault codes are active, contact a Cummins Authorized Repair Facility.

OK

Go To Next Step

STEP 2

Engine is operating at low ambient temperature

Check the winterfront, shutters, and under-the-hood air. Use under-the-hood intake air in cold weather. Refer to Procedure 101-015 (Operating the Engine) in Section 1.

OK

Go To Next Step

STEP 3

Radiator shutters are stuck open or opening early

Check the shutter operation. Repair or replace the shutters if necessary. Refer to Procedure 101-015 (Operating the Engine) in Section 1 and the OEM service manual.

OK

Go To Next Step

STEP 4

Coolant temperature gauge or sensor is malfunctioning

Test the gauge and the sensor. Repair or replace, if necessary. Refer to the OEM service manual.

OK

Go To Next Step

STEP 5

Contact a Cummins Authorized Repair Facility

Crankcase Gases (Blowby) Excessive

Cause

Correction

STEP 1

Crankcase ventilation system is plugged

Check and clean the crankcase breather and vent tube. Refer to Procedure 003-018 (Crankcase Breather Tube) in Section 3.

OK

Go To Next Step

STEP 2

Contact a Cummins Authorized Repair Facility

Engine Acceleration or Response Poor

Cause

Correction

STEP 1

Electronic fault codes are active

For instructions on how to read active fault codes, refer to Procedure 101-007 (Electronic Controlled Fuel System) in Section 1. If fault codes are active, contact a Cummins Authorized Repair Facility.

OK

Go To Next Step

STEP 2

Engine is operating at low ambient temperature

Check the winterfront, shutters, and under-the-hood air. Use under-the-hood intake air in cold weather. Refer to Procedure 101-015 (Operating the Engine) in Section 1.

OK

Go To Next Step

STEP 3

Air intake or exhaust leaks

Check for loose or damaged piping connections and missing pipe plugs. Check the turbocharger and exhaust manifold mounting. Refer to Procedure 010-024 (Air Leaks, Air Intake and Exhaust Systems) in Section 6.

OK

Go To Next Step

STEP 4

Fuel temperature is above specification

Fill the fuel tanks. Turn off or bypass the fuel heater, if equipped. Refer to the OEM service manual.

OK

Go To Next Step

STEP 5

Fuel grade is not correct for the application or the fuel quality is poor

Operate the engine from a tank of high-quality fuel. Refer to Procedure 018-002 (Fuel Recommendations and Specifications) in Section V for specifications.

OK

Go To Next Step

STEP 6

Drivetrain is not correctly matched to the engine

Check for correct gearing and drivetrain components. Make sure the propeller size is correct.

OK

Go To Next Step

STEP 7

Engine is operating at high ambient temperature

Use outside air to the turbocharger in hot weather.

OK

Go To Next Step

STEP 8

Contact a Cummins Authorized Repair Facility

Engine Decelerates Slowly

Cause	Correction
STEP 1 Electronic fault codes are active	For instructions on how to read active fault codes, refer to Procedure 101-007 (Electronic Controlled Fuel System) in Section 1. If fault codes are active, contact a Cummins Authorized Repair Facility.
OK Go To Next Step	
STEP 2 Accelerator pedal or lever is sticking	Check for a sticking accelerator pedal. Refer to the OEM service manual.
OK Go To Next Step	
STEP 3 Contact a Cummins Authorized Repair Facility	

Engine Difficult to Start or Will Not Start (Exhaust Smoke)

Cause

Correction

STEP 1

Electronic fault codes are active

For instructions on how to read active fault codes, refer to Procedure 101-007 (Electronic Controlled Fuel System) in Section 1 for diagnostic fault codes. If fault codes are active, contact a Cummins Authorized Repair Facility.

OK

Go To Next Step

STEP 2

Battery voltage is low

Check the batteries and the unswitched battery supply circuit. Refer to the OEM service manual.

OK

Go To Next Step

STEP 3

Engine-driven units are engaged

Disengage engine-driven units.

OK

Go To Next Step

STEP 4

Engine cranking speed is too slow

If the cranking speed is slower than 150 rpm, refer to the Engine Will Not Crank or Cranks Slowly symptom tree.

OK

Go To Next Step

STEP 5

Fuel shutoff valve(s) closed (electronically controlled injection)

Check the fuel shutoff valve and circuit. Contact a Cummins Authorized Repair Facility.

OK

Go To Next Step

STEP 6

Fuse(s) malfunctioning

Replace the fuse(s) in the OEM interface harness.

OK

Go To Next Step

STEP 7

Exhaust system restriction

Check the exhaust system for any restrictions. Refer to Procedure 018-020 (Exhaust System Specifications) in Section V.

OK

Go To Next Step

STEP 8

Fuel grade is not correct for the application or the fuel quality is poor

Operate the engine from a tank of high-quality fuel. Refer to Procedure 018-002 (Fuel Recommendations and Specifications) in Section V.

OK

Go To Next Step

STEP 9

Starting aid, if necessary for cold weather, is malfunctioning

Check for correct operation of the cold weather starting aid. Refer to Procedure 101-004 (Cold Weather Starting) in Section 1.

OK

Go To Next Step

Engine Difficult to Start or Will Not Start (Exhaust Smoke)

Cause	Correction
<u>STEP 10</u> Fuel filter or fuel suction line is restricted	Replace the fuel filter. Refer to Procedure 006-015 (Fuel Filter (Spin-on) in Section 4. Check the fuel suction line for restriction Procedure 006-015 (Fuel Filter (Spin-on) in Section 4.
OK Go To Next Step	
<u>STEP 11</u> Air intake system restriction is above specification	Check the air intake system for restriction. Clean or replace the air filter and inlet piping as necessary. Refer to Procedure 010-059 (Air Cleaner Restriction) in Section 6.
OK Go To Next Step	
<u>STEP 12</u> Contact a Cummins Authorized Repair Facility	

Engine Difficult to Start or Will Not Start (No Exhaust Smoke)

Cause

Correction

STEP 1

Electronic fault codes are active

For instructions on how to read active fault codes, refer to Procedure 101-007 (Electronic Controlled Fuel System) in Section 1 for diagnostic fault codes. If fault codes are active, contact a Cummins Authorized Repair Facility.

OK

Go To Next Step

STEP 2

Fuel level is low in the tank

Fill the supply tank. Refer to the OEM service manual.

OK

Go To Next Step

STEP 3

Starting motor rotation is not correct

Check the direction of crankshaft rotation. Replace the starting motor if necessary.

OK

Go To Next Step

STEP 4

Engine cranking speed is too slow

If the cranking speed is slower than 150 rpm, refer to the Engine Will Not Crank or Cranks Slowly symptom tree.

OK

Go To Next Step

STEP 5

Exhaust system leaking hot air into engine compartment

Check the exhaust plumbing for leaks or broken components. Refer to Procedure 010-024 (Air Leaks, Air Intake and Exhaust Systems) in Section 6.

OK

Go To Next Step

STEP 6

Fuel shutoff valve is closed

Check for loose wires. Check for power to the fuel shutoff valve solenoid.

OK

Go To Next Step

STEP 7

Battery voltage supply to the electronic control system is low, interrupted, or open

Check the battery connections. Check the fuses and the unswitched battery supply circuit. Refer to the OEM service manual.

OK

Go To Next Step

STEP 8

Fuel inlet restriction

Check for fuel inlet restriction. Contact a Cummins Authorized Repair Facility.

OK

Go To Next Step

STEP 9

Fuel connections on the suction side of the pump are loose

Tighten all the fuel fittings and connections between the fuel tanks and fuel pump.

OK

Go To Next Step

Engine Difficult to Start or Will Not Start (No Exhaust Smoke)

Cause	Correction
STEP 10 Fuel drain line restriction	Check the fuel drain lines for restriction. Clear or replace the fuel lines, check valves, or tank vents as necessary.
OK Go To Next Step	
STEP 11 Fuel suction standpipe in the fuel tank is broken	Check and repair the standpipe, if necessary. Refer to the OEM service manual.
OK Go To Next Step	
STEP 12 Contact a Cummins Authorized Repair Facility	

Engine Noise Excessive

Cause	Correction
<u>STEP 1</u> Lubricating oil pressure is below specification OK Go To Next Step	Check the oil pressure. If the pressure is low, refer to a Cummins Authorized Repair Location.
<u>STEP 2</u> Lubricating oil is thin or diluted OK Go To Next Step	Refer to Procedure 018-003 (Lubricating Oil Recommendations and Specifications) in Section V and the Lubricating Oil Contaminated symptom tree.
<u>STEP 3</u> Coolant temperature is above specification OK Go To Next Step	Refer to the Coolant Temperature Is Above Normal - Sudden Overheat or the Coolant Temperature Is Above Normal - Gradual Overheat symptom tree.
<u>STEP 4</u> Fan drive belt is loose, tight, or not in alignment OK Go To Next Step	Check the fan drive belt. Refer to Procedure 008-002 (Drive Belt, Cooling Fan) in Section A.
<u>STEP 5</u> Vibration damper is damaged OK Go To Next Step	Inspect the vibration damper. Refer to Procedure 001-052 (Vibration Damper, Viscous) in Section 7.
<u>STEP 6</u> Engine mounts are worn, damaged, or not correct OK Go To Next Step	Check the engine mounts. Refer to the OEM service manual.
<u>STEP 7</u> Contact a Cummins Authorized Repair Facility	

Engine Noise Excessive — Combustion Knocks

Cause

Correction

STEP 1

Engine is operating at low ambient temperature

Check the winterfront, shutters, and under-the-hood air. Use under-the-hood intake air in cold weather. Refer to Procedure 101-015 (Operating the Engine) in Section 1.

OK

Go To Next Step

STEP 2

Ether starting aid is malfunctioning

Repair or replace the ether starting aids. Refer to the OEM service manual.

OK

Go To Next Step

STEP 3

Fuel grade is not correct for the application or the fuel quality is poor

Operate the engine from a tank of high-quality fuel. Refer to Procedure 018-002 (Fuel Recommendations and Specifications) in Section V for specifications.

OK

Go To Next Step

STEP 4

Contact a Cummins Authorized Repair Facility

Engine Power Output Low

Cause	Correction
<u>STEP 1</u> Electronic fault codes are active OK Go To Next Step	For instructions on how to read active fault codes, refer to Procedure 101-007 (Electronic Controlled Fuel System) in Section 1 for diagnostic fault codes. If fault codes are active, contact a Cummins Authorized Repair Facility.
<u>STEP 2</u> Drivetrain is not correctly matched to the engine OK Go To Next Step	Check for correct gearing and drivetrain components. Make sure the propeller size is correct.
<u>STEP 3</u> Engine is operating above recommended altitude OK Go To Next Step	Engine power decreases above recommended altitude. Refer to the Engine Data Sheet for specifications.
<u>STEP 4</u> Turbocharger is not correct OK Go To Next Step	Check the turbocharger part number and compare it to the Control Parts List (CPL), Bulletin 3379133 or 4021327. Replace the turbocharger if necessary. Refer to Procedure 010-033 (Turbocharger) in Section A.
<u>STEP 5</u> Fuel filter or fuel suction line is restricted OK Go To Next Step	Replace the fuel filter. Refer to Procedure 006-015 (Fuel Filter (Spin-on)) in Section 4. Check the fuel suction line for restriction Procedure 006-015 (Fuel Filter (Spin-on)) in Section 4.
<u>STEP 6</u> Lubricating oil level is above specification OK Go To Next Step	Check the oil level. Verify the dipstick calibration and oil pan capacity. Fill the system to the specified level. Refer to Procedure 007-043 (Lubricating Oil Level) in Section 3.
<u>STEP 7</u> Air intake system restriction is above specification OK Go To Next Step	Check the air intake system for restriction. Clean or replace the air filter and inlet piping as necessary. Refer to Procedure 010-024 (Air Leaks, Air Intake and Exhaust Systems) in Section 6.

Engine Power Output Low

Cause

Correction

STEP 8

Air intake or exhaust leaks

Check for loose or damaged piping connections and missing pipe plugs. Check the turbocharger and exhaust manifold mounting. Refer to Procedure 010-024 (Air Leaks, Air Intake and Exhaust Systems) in Section 6.

OK

Go To Next Step

STEP 9

Air in the fuel system

Check for air in the fuel system. Tighten or replace the fuel connections, fuel lines, fuel tank standpipe, and fuel filters as necessary.

OK

Go To Next Step

STEP 10

Fuel drain line restriction

Check the fuel drain lines for restriction. Clear or replace the fuel lines, check valves, or tank vents as necessary.

OK

Go To Next Step

STEP 11

Fuel grade is not correct for the application or the fuel quality is poor

Operate the engine from a tank of high-quality fuel. Refer to Procedure 018-002 (Fuel Recommendations and Specifications) in Section V.

OK

Go To Next Step

STEP 12

Intake manifold air temperature is above specification

Refer to the Intake Manifold Air Temperature Above Specification symptom tree.

OK

Go To Next Step

STEP 13

Fuel inlet temperature to pump is above specification

Fill the fuel tank, turn off or bypass the fuel heaters, and check the fuel cooler. Refer to the OEM service manual.

OK

Go To Next Step

STEP 14

Contact a Cummins Authorized Repair Facility

Engine Runs Rough at Idle

Cause	Correction
<u>STEP 1</u> Engine is operating at low ambient temperature OK Go To Next Step	Check the winterfront, shutters, and under-the-hood air. Use under-the-hood intake air in cold weather. Refer to Procedure 101-015 (Operating the Engine) in Section 1.
<u>STEP 2</u> Electronic fault codes are active OK Go To Next Step	For instructions on how to read active fault codes, refer to Procedure 101-007 (Electronic Controlled Fuel System) in Section 1 for the diagnostic fault codes. If fault codes are active, contact a Cummins Authorized Repair Facility.
<u>STEP 3</u> Engine mounts are worn, damaged, or not correct OK Go To Next Step	Check the engine mounts. Refer to the OEM service manual.
<u>STEP 4</u> Engine idle speed is set too low (electronically controlled fuel systems) OK Go To Next Step	Verify the correct idle speed setting. Increase the idle speed with the idle increment switch or an electronic service tool. Refer to a Cummins Authorized Repair Location.
<u>STEP 5</u> Air in the fuel system OK Go To Next Step	Check for air in the fuel system. Tighten or replace the fuel connections, fuel lines, fuel tank standpipe, and fuel filters as necessary.
<u>STEP 6</u> Fuel grade is not correct for the application or the fuel quality is poor OK Go To Next Step	Operate the engine from a tank of high-quality fuel. Refer to Procedure 018-002 (Fuel Recommendations and Specifications) in Section V.
<u>STEP 7</u> Overhead adjustments are not correct OK Go To Next Step	Measure and adjust the overhead settings. Refer to Procedure 003-004 (Overhead Set) in Section 5.
<u>STEP 8</u> Contact a Cummins Authorized Repair Facility	

Engine Runs Rough or Misfires

Cause	Correction
<u>STEP 1</u> Condition occurs only at idle OK Go To Next Step	Refer to the Engine Runs Rough at Idle symptom tree.
<u>STEP 2</u> Engine is operating at low ambient temperature OK Go To Next Step	Check the winterfront, shutters, and under-the-hood air. Use under-the-hood intake air in cold weather. Refer to Procedure 101-015 (Operating the Engine) in Section 1.
<u>STEP 3</u> Electronic fault codes are active OK Go To Next Step	For instructions on how to read active fault codes, refer to Procedure 101-007 (Electronic Controlled Fuel System) in Section 1 for diagnostic fault codes. If fault codes are active, contact a Cummins Authorized Repair Facility.
<u>STEP 4</u> Fuel leak OK Go To Next Step	Check the fuel lines, fuel connections, and fuel filters for leaks. Check the fuel lines to the supply tanks. Refer to the OEM service manual.
<u>STEP 5</u> Fuel grade is not correct for the application or the fuel quality is poor OK Go To Next Step	Operate the engine from a tank of high-quality fuel. Refer to Procedure 018-002 (Fuel Recommendations and Specifications) in Section V.
<u>STEP 6</u> Air in the fuel system OK Go To Next Step	Check for air in the fuel system. Tighten or replace the fuel connections, fuel lines, fuel tank standpipe, and fuel filters as necessary.
<u>STEP 7</u> Fuel inlet restriction OK Go To Next Step	Check for fuel inlet restriction. Contact a Cummins Authorized Repair Facility.
<u>STEP 8</u> Engine mounts are worn, damaged, or not correct OK Go To Next Step	Check the engine mounts. Refer to the OEM service manual.
<u>STEP 9</u> Overhead adjustments are not correct OK Go To Next Step	Measure and adjust the overhead settings. Refer to Procedure 003-004 (Overhead Set) in Section 5.

Engine Runs Rough or Misfires

Cause

Correction

STEP 10

Contact a Cummins Authorized Repair Facility

Engine Shuts Off Unexpectedly or Dies During Deceleration

Cause

Correction

STEP 1

Electronic fault codes are active

For instructions on how to read active fault codes, refer to Procedure 101-007 (Electronic Controlled Fuel System) in Section 1 or diagnostic fault codes. If fault codes are active, contact a Cummins Authorized Repair Facility.

OK

Go To Next Step

STEP 2

Battery voltage supply to the electronic control module (ECM) has been lost

Check the battery connections. Check the fuses and the unswitched battery supply circuit. Refer to the OEM service manual.

OK

Go To Next Step

STEP 3

Idle shutdown or PTO shutdown features are activated

Check the time limit on idle and PTO shutdowns with an electronic service tool. Refer to a Cummins Authorized Repair Location.

OK

Go To Next Step

STEP 4

Engine will not restart

Refer to Procedure 101-014 (Normal Starting Procedure) in Section 1.

OK

Go To Next Step

STEP 5

Fuel inlet restriction

Check for fuel inlet restriction. Contact a Cummins Authorized Repair Facility.

OK

Go To Next Step

STEP 6

Air in the fuel system

Check for air in the fuel system. Tighten or replace the fuel connections, fuel lines, fuel tank standpipe, and fuel filters as necessary.

OK

Go To Next Step

STEP 7

Battery voltage is low

Check the batteries and the unswitched battery supply circuit. Refer to the OEM service manual.

OK

Go To Next Step

STEP 8

OEM engine protection system is malfunctioning

Isolate the OEM engine protection system. Follow the OEM service manuals to check for a malfunction.

OK

Go To Next Step

STEP 9

Fuel shutoff valve is closed

Check for loose wires. Check for power to the fuel shutoff valve solenoid.

OK

Go To Next Step

Engine Shuts Off Unexpectedly or Dies During Deceleration

Cause

Correction

STEP 10

Fuel grade is not correct for the application or the fuel quality is poor

Operate the engine from a tank of high-quality fuel. Refer to Procedure 018-002 (Fuel Recommendations and Specifications) in Section V.

OK

Go To Next Step

STEP 11

Contact a Cummins Authorized Repair Facility

Engine Speed Surges at Low or High Idle

Cause

Correction

STEP 1

Electronic fault codes are active

For instructions on how to read active fault codes, refer to Procedure 101-007 (Electronic Controlled Fuel System) in Section 1 diagnostic fault codes. If fault codes are active, contact a Cummins Authorized Repair Facility.

OK

Go To Next Step

STEP 2

Fuel level is low in the tank

Fill the supply tank. Refer to the OEM service manual.

OK

Go To Next Step

STEP 3

Contact a Cummins Authorized Repair Facility

Engine Speed Surges Under Load or in Operating Range

Cause

Correction

STEP 1

Electronic fault codes are active

For instructions on how to read active fault codes, refer to Procedure 101-007 (Electronic Controlled Fuel System) in Section 1 for diagnostic fault codes. If fault codes are active, contact a Cummins Authorized Repair Facility.

OK

Go To Next Step

STEP 2

Fuel level is low in the tank

Fill the supply tank. Refer to the OEM service manual.

OK

Go To Next Step

STEP 3

Contact a Cummins Authorized Repair Facility

Engine Starts But Will Not Keep Running

Cause

Correction

<u>STEP 1</u> Electronic fault codes are active	For instructions on how to read active fault codes, refer to Procedure 101-007 (Electronic Controlled Fuel System) in Section 1 for diagnostic fault codes. If fault codes are active, contact a Cummins Authorized Repair Facility.
OK Go To Next Step	
<u>STEP 2</u> Idle shutdown or PTO shutdown features are activated	Turn OFF the keyswitch for 5 seconds. Turn ON the keyswitch and check fault lamp operation.
OK Go To Next Step	
<u>STEP 3</u> Fuel level is low in the tank	Fill the supply tank. Refer to the OEM service manual.
OK Go To Next Step	
<u>STEP 4</u> Air in the fuel system	Check for air in the fuel system. Tighten or replace the fuel connections, fuel lines, fuel tank standpipe, and fuel filters as necessary.
OK Go To Next Step	
<u>STEP 5</u> Fuel filter or fuel suction line is restricted	Replace the fuel filter. Refer to Procedure 006-015 (Fuel Filter (Spin-on) in Section 4. Check the fuel suction line for restriction Procedure 006-015 (Fuel Filter (Spin-on) in Section 4.
OK Go To Next Step	
<u>STEP 6</u> Engine-driven units are engaged	Disengage engine-driven units.
OK Go To Next Step	
<u>STEP 7</u> Fuel is waxing due to cold weather	Check the fuel heater, if installed. Weather conditions sometimes require a fuel heater.
OK Go To Next Step	
<u>STEP 8</u> Fuel inlet restriction	Check for fuel inlet restriction. Contact a Cummins Authorized Repair Facility.
OK Go To Next Step	
<u>STEP 9</u> Fuel shutoff valve is closed	Check for loose wires. Check for power to the fuel shutoff valve solenoid.
OK Go To Next Step	

Engine Starts But Will Not Keep Running

Cause

Correction

STEP 10
Exhaust system restriction

Check the exhaust system for any restrictions. Refer to Procedure 018-020 (Exhaust System Specifications) in Section V.

OK

Go To Next Step

STEP 11
Fuel grade is not correct for the application or the fuel quality is poor

Operate the engine from a tank of high-quality fuel. Refer to Procedure 018-002 (Fuel Recommendations and Specifications) in Section V for specifications.

OK

Go To Next Step

STEP 12
Air intake system restriction is above specification

Check the air intake system for restriction. Clean or replace the air filter and inlet piping as necessary. Refer to Procedure 010-059 (Air Cleaner Restriction) in Section 6.

OK

Go To Next Step

STEP 13
Contact a Cummins Authorized Repair Facility

Engine Vibration Excessive

Cause	Correction
<u>STEP 1</u> Electronic fault codes are active OK Go To Next Step	For instructions on how to read active fault codes, refer to Procedure 101-007 (Electronic Controlled Fuel System) in Section 1 for diagnostic fault codes. If fault codes are active, contact a Cummins Authorized Repair Facility.
<u>STEP 2</u> Engine is misfiring OK Go To Next Step	Refer to the Engine Runs Rough or Misfires symptom tree.
<u>STEP 3</u> Engine idle speed is set too low (electronically controlled fuel systems) OK Go To Next Step	Verify the correct idle speed setting. Increase the idle speed with the idle increment switch or an electronic service tool. Refer to a Cummins Authorized Repair Location.
<u>STEP 4</u> Fan is loose, damaged, or not balanced OK Go To Next Step	Check the fan. Refer to Procedure 008-040 (Fan, Cooling) in Section 3.
<u>STEP 5</u> Belt-driven accessories are malfunctioning OK Go To Next Step	Check the fan hub, alternator, refrigerant compressor, and hydraulic pump for interference. Isolate belt-driven accessories and check for vibration. Refer to Procedure 008-036 (Fan Hub, Belt Driven) in Section A.
<u>STEP 6</u> Engine mounts are worn, damaged, or not correct OK Go To Next Step	Check the engine mounts. Refer to the OEM service manual.
<u>STEP 7</u> Vibration damper is damaged OK Go To Next Step	Inspect the vibration damper. Refer to Procedure 001-052 (Vibration Damper, Viscous) in Section 7.
<u>STEP 8</u> Contact a Cummins Authorized Repair Facility	

Engine Will Not Crank or Cranks Slowly (Air Starter)

Cause

Correction

STEP 1

Air pressure is low in the air tanks

..... Increase air pressure with an external air source. Refer to the OEM service manual.

OK

Go To Next Step

STEP 2

Engine-driven units are engaged

..... Disengage engine-driven units.

OK

Go To Next Step

STEP 3

Lubricating oil temperature switch, gauge, or sensor malfunctioning or not in the correct location

..... Check the oil temperature switch, gauge, or sensor for correct operation and location. Refer to a Cummins Authorized Repair Location.

OK

Go To Next Step

STEP 4

Lubricating oil does not meet specifications for operating conditions

..... Change the oil and filters. Refer to Procedures 007-002 (Lubricating Oil and Filters) and 007-013 (Lubricating Oil Filters (Spin-on)) in Section 4. Use the oil recommended in Section V.

OK

Go To Next Step

STEP 5

Lubricating oil level is above specification

..... Check the oil level. Verify the dipstick calibration and oil pan capacity. Fill the system to the specified level. Refer to Procedure 007-043 (Lubricating Oil Level) in Section 3.

OK

Go To Next Step

STEP 6

Contact a Cummins Authorized Repair Facility

Engine Will Not Crank or Cranks Slowly (Electric Starter)

Cause

Correction

STEP 1

Battery cables or connections are loose, broken, or corroded (excessive resistance)

Check the battery cables and connections.

OK

Go To Next Step

STEP 2

Engine-driven units are engaged

Disengage engine-driven units.

OK

Go To Next Step

STEP 3

Lubricating oil temperature is below specification

Install an oil pan heater, or drain the oil and fill the system with warm oil.

OK

Go To Next Step

STEP 4

Lubricating oil does not meet specifications for operating conditions

Change the oil and filters. Refer to Procedures 007-002 (Lubricating Oil and Filters) and 007-013 (Lubricating Oil Filters (Spin-on) in Section 4. Use the oil type recommended in Section V.

OK

Go To Next Step

STEP 5

Contact a Cummins Authorized Repair Facility

Engine Will Not Reach Rated Speed (RPM)

Cause	Correction
<u>STEP 1</u> Keyswitch circuit is malfunctioning OK Go To Next Step	Check the vehicle, equipment, or vessel keyswitch circuit. Refer to a Cummins Authorized Repair Location.
<u>STEP 2</u> Tachometer is not calibrated or is malfunctioning OK Go To Next Step	Compare the tachometer reading with a handheld tachometer or an electronic service tool reading. Calibrate or replace the tachometer as necessary. Refer to the OEM service manual.
<u>STEP 3</u> Electronic fault codes are active OK Go To Next Step	For instructions on how to read active fault codes, refer to Procedure 101-007 (Electronic Controlled Fuel System) in Section 1 diagnostic fault codes. If fault codes are active, contact a Cummins Authorized Repair Facility.
<u>STEP 4</u> Drivetrain is not correctly matched to the engine OK Go To Next Step	Check for correct gearing and drivetrain components. Make sure the propeller size is correct.
<u>STEP 5</u> Engine power output is low OK Go To Next Step	Refer to the Engine Power Output Low symptom tree.
<u>STEP 6</u> Fuel grade is not correct for the application or the fuel quality is poor OK Go To Next Step	Operate the engine from a tank of high-quality fuel. Refer to Procedure 018-002 (Fuel Recommendations and Specifications) in Section V.
<u>STEP 7</u> Fuel inlet restriction OK Go To Next Step	Check for fuel inlet restriction. Contact a Cummins Authorized Repair Facility.
<u>STEP 8</u> Overhead adjustments are not correct OK Go To Next Step	Measure and adjust the overhead settings. Refer to Procedure 003-004 (Overhead Set) in Section 5.
<u>STEP 9</u> Vessel is overloaded OK Go To Next Step	Load the vessel correctly. Refer to the OEM service manual.

Engine Will Not Reach Rated Speed (RPM)

Cause

Correction

STEP 10

Programmable parameters or selected features
are not correct

.....

Check the programmable parameters and the
selected features with an electronic service tool.
Set the parameters and features again if
necessary. Refer to a Cummins Authorized Repair
Location.

OK

Go To Next Step

STEP 11

Contact a Cummins Authorized Repair Facility

Engine Will Not Shut Off

Cause	Correction
<u>STEP 1</u> Keyswitch circuit is malfunctioning	Check the vehicle, equipment, or vessel keyswitch circuit. Refer to the OEM service manual.
OK Go To Next Step	
<u>STEP 2</u> Engine is running on fumes drawn into the air intake	Check the air intake ducts. Locate and isolate the source of the fumes. Repair as necessary. Refer to the OEM service manual.
OK Go To Next Step	
<u>STEP 3</u> Contact a Cummins Authorized Repair Facility	

Fuel Consumption Excessive

Cause	Correction
<u>STEP 1</u> Verify the complaint OK Go To Next Step	If low power is relevant, refer to the Engine Power Output Low symptom tree. If acceleration is poor, refer to the Engine Acceleration or Response Poor symptom tree. If fuel consumption is relevant, continue with this tree.
<u>STEP 2</u> Lubricating oil level is above specification OK Go To Next Step	Check the oil level. Verify the dipstick calibration and oil pan capacity. Fill the system to the specified level. Refer to Procedure 007-043 (Lubricating Oil Level) in Section 3.
<u>STEP 3</u> Air intake system restriction is above specification OK Go To Next Step	Check the air intake system for restriction. Clean or replace the air filter and inlet piping as necessary. Refer to Procedure 010-059 (Air Cleaner Restriction) in Section 6.
<u>STEP 4</u> Fuel leak OK Go To Next Step	Check the fuel lines, fuel connections, and fuel filters for leaks. Check the fuel lines to the supply tanks. Refer to the OEM service manual.
<u>STEP 5</u> Operator technique is not correct OK Go To Next Step	Refer to Procedure 101-015 (Operating the Engine) in Section 1.
<u>STEP 6</u> Equipment and environmental factors are affecting fuel consumption OK Go To Next Step	Consider ambient temperatures, wind, tire size, axle alignment, routes, and use of aerodynamic aids when evaluating fuel consumption.
<u>STEP 7</u> Hour meter is miscalibrated OK Go To Next Step	Check the hour meter. Calibrate or replace the hour meter if necessary. Calculate fuel consumption with new figures.
<u>STEP 8</u> Contact a Cummins Authorized Repair Facility	

Fuel in Coolant

Cause	Correction
<u>STEP 1</u> Bulk coolant supply is contaminated	Check the bulk coolant supply. Drain the coolant and replace with noncontaminated coolant. Replace the coolant filters. Refer to Procedure 008-018 (Cooling System) in Section 7 and Procedure 008-006 (Coolant Filter) in Section 4.
OK Go To Next Step	
<u>STEP 2</u> Fuel heater is malfunctioning (if equipped)	Check the fuel heater and replace, if necessary. Refer to the OEM service manual.
OK Go To Next Step	
<u>STEP 3</u> Contact a Cummins Authorized Repair Facility	

Fuel in the Lubricating Oil

Cause	Correction
<u>STEP 1</u> Bulk oil supply is contaminated OK Go To Next Step	Check the bulk oil supply. Drain the oil and replace with noncontaminated oil. Replace the oil filter(s). Refer to Procedures 007-002 (Lubricating Oil and Filters) and 007-013 (Lubricating Oil Filters (Spin-on)) in Section 4.
<u>STEP 2</u> Engine idle time is excessive OK Go To Next Step	Low oil and coolant temperatures can be caused by long idle time (greater than 10 minutes). Shut off the engine rather than idle for long periods. If idle time is necessary, raise the idle speed. Procedures 007-002 (Lubricating Oil and Filters) and 007-013 (Lubricating Oil Filters (Spin-on)) in Section 4
<u>STEP 3</u> Contact a Cummins Authorized Repair Facility	

Intake Manifold Air Temperature Above Specification

Cause

Correction

STEP 1

Vehicle speed is too low for adequate cooling with high engine load

Reduce the engine load. Increase the engine (fan) rpm by downshifting.

OK

Go To Next Step

STEP 2

Fan shroud is damaged or missing or the air recirculation baffles are damaged or missing

Inspect the shroud and the recirculation baffles. Repair, replace, or install, if necessary. Refer to the OEM service manual.

OK

Go To Next Step

STEP 3

Fan drive belt is broken

Check the fan drive belt. Replace the belt, if necessary. Refer to Procedure 103-002 (Drive Belts) in Section 3.

OK

Go To Next Step

STEP 4

Cold weather radiator cover or winterfront is closed

Open the cold weather radiator cover or the winterfront. Maintain a minimum of 784 cm² [122 in²] or approximately 28 x 28 cm [11 x 11 in] of opening at all times. Refer to Procedure 101-015 (Operating the Engine) in Section 1.

OK

Go To Next Step

STEP 5

Radiator shutters are not opening completely or the shutterstat setting is wrong

Inspect the radiator shutters. Repair or replace if necessary. Refer to the OEM service manual. Check the shutterstat setting.

OK

Go To Next Step

STEP 6

Charge air cooler fins, radiator fins, or air conditioner condenser fins are damaged or obstructed with debris

Inspect the charge air cooler, air conditioner condenser, and radiator fins. Clean, if necessary. Refer to Procedure 010-028 (Charge Air Piping) in Section 3.

OK

Go To Next Step

STEP 7

Contact a Cummins Authorized Repair Facility

Intake Manifold Pressure (Boost) is Below Normal

Cause

Correction

STEP 1

Turbocharger is not correct

Check the turbocharger part number and compare it to the Control Parts List (CPL), Bulletin 3379133 or 4021327. Replace the turbocharger if necessary. Refer to Procedure 010-033 (Turbocharger) in Section A.

OK

Go To Next Step

STEP 2

Air intake or exhaust leaks

Check for loose or damaged piping connections and missing pipe plugs. Check the turbocharger and exhaust manifold mounting. Refer to Procedure 010-024 (Air Leaks, Air Intake and Exhaust Systems) in Section 6.

OK

Go To Next Step

STEP 3

Air compressor connection is loose or damaged

Check the connection between the manifold and the air compressor. Repair or replace if necessary.

OK

Go To Next Step

STEP 4

Contact a Cummins Authorized Repair Facility

Lubricating Oil Consumption Excessive

Cause

Correction

STEP 1

Verify the oil consumption rate

Check the amount of oil added versus the mileage.

OK

Go To Next Step

STEP 2

Lubricating oil leak (external)

Inspect the engine for external oil leaks. Tighten the capscrews, pipe plugs, and fittings. Replace gaskets, if necessary. Refer to Procedure 018-009 (Capscrew Markings and Torque Values) in Section V.

OK

Go To Next Step

STEP 3

Lubricating oil drain interval is excessive

Verify the correct lubricating oil drain interval. Refer to Procedure 102-002 (Maintenance Schedule) in Section 2.

OK

Go To Next Step

STEP 4

Lubricating oil does not meet specifications for operating conditions

Change the oil and filters. Refer to Procedure 007-013 (Lubricating Oil Filters (Spin-on) in Section 4. Use the oil type recommended in Section V.

OK

Go To Next Step

STEP 5

Crankcase ventilation system is plugged

Check and clean the crankcase breather and vent tube. Refer to Procedure 003-018 (Crankcase Breather Tube) in Section 4.

OK

Go To Next Step

STEP 6

Contact a Cummins Authorized Repair Facility

Lubricating Oil Contaminated

Cause	Correction
<u>STEP 1</u> Identify lubricating oil contamination OK Go To Next Step	Perform an oil analysis to determine the contaminants.
<u>STEP 2</u> Bulk oil supply is contaminated OK Go To Next Step	Check the bulk oil supply. Drain the oil and replace with noncontaminated oil. Replace the oil filter(s). Refer to Procedure 007-002 (Lubricating Oil and Filters) and Procedure 007-013 (Lubricating Oil Filters) in Section 4.
<u>STEP 3</u> Lubricating oil sludge is excessive OK Go To Next Step	Change the oil and filters. Refer to the Lubricating Oil Sludge in the Crankcase Excessive symptom tree.
<u>STEP 4</u> Fuel in the lubricating oil OK Go To Next Step	Refer to the Fuel in the Lubricating Oil symptom tree.
<u>STEP 5</u> Contact a Cummins Authorized Repair Facility	

Lubricating Oil Pressure High

Cause	Correction
<u>STEP 1</u> Lubricating oil pressure switch, gauge, or sensor is malfunctioning or is not in the correct location OK Go To Next Step	Check the oil pressure switch, gauge, or sensor for correct operation and location. Refer to the OEM service manual.
<u>STEP 2</u> Coolant temperature is below specification OK Go To Next Step	Refer to the Coolant Temperature Below Normal symptom tree.
<u>STEP 3</u> Lubricating oil does not meet specifications for operating conditions OK Go To Next Step	Change the oil and filters. Refer to Procedure 007-013 (Lubricating Oil Filters (Spin-on) in Section 4. Use the oil type recommended in Section V.
<u>STEP 4</u> Contact a Cummins Authorized Repair Facility	

Lubricating Oil Pressure Low

Cause	Correction
<u>STEP 1</u> Electronic fault codes are active OK Go To Next Step	For instructions on how to read active fault codes, refer to Procedure 101-007 (Electronic Controlled Fuel System) in Section 1 for diagnostic fault codes. If fault codes are active, contact a Cummins Authorized Repair Facility.
<u>STEP 2</u> Lubricating oil level is above or below specification OK Go To Next Step	Check the oil level. Add or drain oil, if necessary. Refer to Procedure 007-043 (Lubricating Oil Level) in Section 3.
<u>STEP 3</u> Lubricating oil leak (external) OK Go To Next Step	Inspect the engine for external oil leaks. Tighten the capscrews, pipe plugs, and fittings. Replace gaskets, if necessary. Refer to Procedure 018-009 (Capscrew Markings and Torque Values) in Section V for specifications.
<u>STEP 4</u> Lubricating oil pressure switch, gauge, or sensor is malfunctioning or is not in the correct location OK Go To Next Step	Check the oil pressure switch, gauge, or sensor for correct operation and location. Refer to the OEM service manual.
<u>STEP 5</u> Lubricating oil does not meet specifications for operating conditions OK Go To Next Step	Change the oil and filters. Refer to Procedure 007-013 (Lubricating Oil Filters (Spin-on) in Section 4. Use the oil type recommended in Section V.
<u>STEP 6</u> Lubricating oil is contaminated with coolant or fuel OK Go To Next Step	Contact a Cummins Authorized Repair Facility.
<u>STEP 7</u> Lubricating oil filter is plugged OK Go To Next Step	Change the oil and filter. Refer to Procedure 007-013 (Lubricating Oil Filters (Spin-on) in Section 4. Review the oil change interval.
<u>STEP 8</u> Lubricating oil viscosity sensor is loose OK Go To Next Step	Check for proper assembly. Refer to a Cummins Authorized Repair Location.
<u>STEP 9</u> Contact a Cummins Authorized Repair Facility	

Lubricating Oil Sludge in the Crankcase Excessive

Cause

Correction

STEP 1

Bulk oil supply is contaminated

Check the bulk oil supply. Drain the oil and replace with noncontaminated oil. Replace the oil filter(s). Refer to Procedure 007-002 (Lubricating Oil and Filters) and Procedure 007-013 (Lubricating Oil Filter- Spin-on) in Section 4.

OK

Go To Next Step

STEP 2

Lubricating oil does not meet specifications for operating conditions

Change the oil and filters. Refer to Procedure 007-013 (Lubricating Oil Filter- Spin-on) in Section 4. Use the oil type recommended in Section V.

OK

Go To Next Step

STEP 3

Lubricating oil drain interval is excessive

Verify the correct lubricating oil drain interval. Refer to Procedure 102-002 (Maintenance Schedule) in Section 2.

OK

Go To Next Step

STEP 4

Fuel grade is not correct for the application or the fuel quality is poor

Operate the engine from a tank of high-quality fuel. Refer to Procedure 018-002 (Fuel Recommendations and Specifications) in Section V for specifications.

OK

Go To Next Step

STEP 5

Lubricating oil is contaminated with coolant

Check the lubricating oil for coolant. Refer to the Coolant Loss - Internal symptom tree.

OK

Go To Next Step

STEP 6

Crankcase ventilation system is plugged

Check and clean the crankcase breather and vent tube. Refer to Procedure 003-018 (Crankcase Breather Tube) in Section 3.

OK

Go To Next Step

STEP 7

Contact a Cummins Authorized Repair Facility

Lubricating Oil Temperature Above Specification

Cause

Correction

STEP 1

Lubricating oil level is above or below specification

Check the oil level. Add or drain oil, if necessary. Refer to Procedure 007-043 (Lubricating Oil Level) in Section 3.

OK

Go To Next Step

STEP 2

Coolant temperature is above specification

Refer to the Coolant Temperature Is Above Normal - Sudden Overheat or the Coolant Temperature Is Above Normal - Gradual Overheat symptom tree.

OK

Go To Next Step

STEP 3

Lubricating oil temperature switch, gauge, or sensor malfunctioning or not in the correct location

Check the oil temperature switch, gauge, or sensor for correct operation and location. Refer to a Cummins Authorized Repair Location.

OK

Go To Next Step

STEP 4

Contact a Cummins Authorized Repair Facility

Lubricating or Transmission Oil in the Coolant

Cause	Correction
<u>STEP 1</u> Bulk coolant supply is contaminated	Check the bulk coolant supply. Drain the coolant and replace with noncontaminated coolant. Replace the coolant filters. Refer to Procedure 008-018 (Cooling System) in Section 7 and Procedure 008-006 (Coolant Filters) in Section 4.
OK Go To Next Step	
<u>STEP 2</u> Contact a Cummins Authorized Repair Facility	

PTO or Cruise Control Does Not Operate

Cause	Correction
STEP 1 Cruise control/PTO on/off switch is off OK Go To Next Step	Turn on the switch.
STEP 2 Cruise control/PTO switch wires or connectors are loose OK Go To Next Step	Check the cruise control/PTO on/off switch, selector switch, clutch switch, and brake switch for loose wires and connections.
STEP 3 Electronic fault codes are active OK Go To Next Step	For instructions on how to read active fault codes, refer to Procedure 101-007 (Electronic Controlled Fuel System) in Section 1 for diagnostic fault codes. If fault codes are active, contact a Cummins Authorized Repair Facility.
STEP 4 Contact a Cummins Authorized Repair Facility	

Smoke, Black — Excessive

Cause	Correction
<u>STEP 1</u> Turbocharger wheel clearance is out of specification OK Go To Next Step	Check the radial bearing clearance and axial clearance. Inspect the turbocharger. Repair or replace the turbocharger if necessary.
<u>STEP 2</u> Air intake or exhaust leaks OK Go To Next Step	Check for loose or damaged piping connections and missing pipe plugs. Check the turbocharger and exhaust manifold mounting. Refer to Procedure 010-024 (Air Leaks, Air Intake and Exhaust Systems) in Section 6.
<u>STEP 3</u> Fuel grade is not correct for the application or the fuel quality is poor OK Go To Next Step	Operate the engine from a tank of high-quality fuel. Refer to Procedure 018-002 (Fuel Recommendations and Specifications) in Section V.
<u>STEP 4</u> Fuel drain line restriction OK Go To Next Step	Check the fuel drain lines for restriction. Refer to Procedure 018-016 (Fuel System Specifications) in Section V.
<u>STEP 5</u> Contact a Cummins Authorized Repair Facility	

Smoke, White — Excessive

Cause	Correction
<u>STEP 1</u> Electronic fault codes are active OK Go To Next Step	For instructions on how to read active fault codes, refer to Procedure 101-007 (Electronic Controlled Fuel System) in Section 1 for diagnostic fault codes. If fault codes are active, contact a Cummins Authorized Repair Facility.
<u>STEP 2</u> Engine block heater is malfunctioning (if equipped) OK Go To Next Step	Check the electrical sources and wiring to the cylinder block heater. Replace the block heater, if necessary. Refer to the OEM service manual.
<u>STEP 3</u> Coolant temperature is below specification OK Go To Next Step	Refer to the Coolant Temperature Below Normal symptom tree.
<u>STEP 4</u> Engine is cold OK Go To Next Step	Allow the engine to warm to operating temperature. If the engine will not reach operating temperature, refer to the Coolant Temperature Below Normal symptom tree.
<u>STEP 5</u> Fuel grade is not correct for the application or the fuel quality is poor OK Go To Next Step	Operate the engine from a tank of high-quality fuel. Refer to Procedure 018-002 (Fuel Recommendations and Specifications) in Section V.
<u>STEP 6</u> Overhead adjustments are not correct OK Go To Next Step	Measure and adjust the overhead settings. Refer to Procedure 003-004 (Overhead Set) in Section 5.
<u>STEP 7</u> Contact a Cummins Authorized Repair Facility	

Turbocharger Leaks Engine Oil or Fuel

Cause	Correction
<u>STEP 1</u> Engine is operating for extended periods under light or no-load conditions (slobbering) OK Go To Next Step	Review the engine operating instructions. Refer to Procedure 101-015 (Operating the Engine) in Section 1.
<u>STEP 2</u> Lubricating oil or fuel is entering the turbocharger OK Go To Next Step	Remove the intake and exhaust piping, and check for oil or fuel.
<u>STEP 3</u> Turbocharger oil drain line is restricted OK Go To Next Step	Remove the turbocharger oil drain line and check for restriction. Clean or replace the oil drain line. Refer to Procedure 010-033 (Turbocharger) in Section A.
<u>STEP 4</u> Contact a Cummins Authorized Repair Facility	

Notes

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Section V - Maintenance Specifications

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General Engine

Specifications

Industrial and Generator Drive

Type.....Four-cycle, In-line six-cylinder, four-valves per cylinder
Horsepower.....Refer to engine dataplate
Engine Speed at Maximum Output.....Refer to engine dataplate
Bore and Stroke.....125 mm [4.921 in] x 147 mm [5.787 in]
Displacement.....10.8 liters [659 C.I.D.]
Firing Order.....1-5-3-6-2-4
Engine Weight (with standard accessories):.....940 kg [2072 lb]

Industrial Applications

Generator Drive Applications

Crankshaft Rotation (viewed from the front of the engine)..... Clockwise

Marine Applications

Type.....Four-cycle, In-line six-cylinder, four-valves per cylinder
Horsepower.....Refer to engine dataplate
Engine Speed at Maximum Output.....Refer to engine dataplate
Bore and Stroke.....125 mm [4.921 in] x 147 mm [5.787 in]
Displacement.....10.8 liters [659 C.I.D.]
Firing Order.....1-5-3-6-2-4
Engine Weight (with standard accessories):.....1124 kg [2478 lb]
Dry Weight.....1124 kg [2478 lb]
Crankshaft Rotation (viewed from the front of the engine)..... Clockwise

Fuel System

Specifications

Maximum Fuel Inlet Temperature.....	71°C [160°F]
Fuel Inlet Maximum Restriction:	
Marine Applications	
Clean Fuel Filter.....	152 mm Hg [6 in Hg]
Dirty Fuel Filter.....	203 mm Hg [8 in Hg]
Industrial and Generator Drive Applications.....	
Clean Fuel Filter.....	102 mm Hg [4 in Hg]
Dirty Fuel Filter.....	203 mm Hg [8 in Hg]
Fuel Drain Maximum Restriction:	
Without Check Valves.....	63mm Hg [2.5 in Hg]
Engine Minimum Cranking Speed.....	63mm Hg [2.5 in Hg]
	120 rpm

Lubricating Oil System

Specifications

Oil Pressure	
Minimum at Idle.....	69 kPa [10 psi]
Operating Range.....	207 to 448 kPa [30 to 65 psi]
at torque peak or 1200 rpm (minimum allowable).....	207 kPa [30 psi]
Oil Temperature	
Industrial and Marine	
Maximum.....	121°C [250°F]
Generator Drive	
Maximum.....	135°C [275°F]
Oil Pan Capacity	
Marine Applications	
Low.....	26.5 liters [7 gal]
High.....	30.3 liters [8 gal]
Generator Drive Applications	
Low.....	30.3 liters [8 gal]
High.....	34 liters [9 gal]
Industrial Applications	
Low.....	26.5 liters [7 gal]
High.....	34 liters [9 gal]
Total System Capacity:.....	37 liters [9.8 gal]
With engine Mounted Filter (combination filter holds 2.6 liters [0.7 gal]).....	37 liters [9.8 gal]
With Remote Mounted Filter.....	38 liters [10 gal]
Maximum Operational Angularity of Oil Pan:	
Marine Applications.....	
Front Down:	
Rear Sump.....	30 degrees
Front Sump.....	45 degrees
Rear Down:	
Rear Sump.....	25 degrees
Front Sump.....	27 degrees
Side to Side:	
Rear Sump.....	30 degrees
Front Sump.....	42 degrees
Industrial and Generator Drive (Rear Sump).....	
Varies by oil pan option - example numbers are based on stamped steel pan.....	
Front Down:	
Rear Sump.....	42 degrees
Front Sump.....	45 degrees
Rear Down:	
Rear Sump.....	45 degrees
Front Sump.....	45 degrees
Side to Side:	
Rear Sump.....	45 degrees
Front Sump.....	35 degrees

Cooling System

Specifications

Marine Applications

Coolant Capacity:

Engine with Heat Exchanger.....29.4 liters [8 gal]

Minimum Drawdown.....15.2 liters [4 gal]

Standard Thermostat Modulating Range.....71 to 79.4°C [106 to 175°F]

Block coolant Pressure with Closed Thermostat (pressure cap removed)

Maximum.....246 kPa [36 psi]

Coolant Inlet Pressure with Open Thermostat and No Cap:

Minimum.....64 mm Hg [2.5 in Hg]

Maximum Coolant Temperature.....96°C [205°F]

Minimum Fill Rate.....18.9 liters/min [5 gpm]

Minimum Initial Fill Time.....2 minutes

Maximum Deaeration Time.....10 minutes

Minimum Coolant Expansion Space:

Percent of Cooling system.....5%

External Pressure Loss in Cooling System:

Maximum.....35 kPa [5 psi]

Static Pressure of Coolant:

Maximum - Exclusive of Pressure Cap.....103 kPa [15 psi]

Maximum Sea Water Pressure.....103 kPa [15 psi]

Maximum Sea Water Pump Inlet Restriction.....127 mm Hg [5 psi]

Maximum Sea Water Pump Initial Suction Lift.....3 m [10 ft]

Pressure Cap Rating (with heat exchanger).....103 kPa [15 psi]

Maximum Allowable Coolant Flow to Accessories.....37.9 liters/min [10 gal/min]

Industrial Applications

Coolant Capacity.....9.5 liters [2.5 gal]

Standard Modulating Thermostat Range.....82 to 93°C [180 to 200°F]

Cylinder block Coolant Pressure (pressure cap removed):

Minimum:

Closed Thermostat (1800 rpm no load).....138 kPa [20 psi]

Maximum:

Closed Thermostat.....275 kPa [40 psi]

Maximum Allowable Operating Coolant Temperature.....100°C [212°F]

Minimum Recommended Operating Coolant Temperature.....70°C [160°F]

Maximum Allowable Deaeration Time.....35 minutes

Minimum Recommended Pressure Cap.....48 kPa [7 psi]

Generator Set Applications

Coolant Capacity.....9.5 liters [2.5 gal]

Standard Modulating Thermostat Range.....82 to 93°C [180 to 200°F]

Minimum Pressure Cap.....48 kPa [7 psi]

Maximum Coolant Friction Head External to Engine

1500 rpm.....55 kPa [8 psi]

1800 rpm.....69 kPa [10 psi]

Maximum Static Head of Coolant Above Engine Crank Centerline.....14 m [46 ft]

Maximum Top Tank Temperature for Standby/Prime Power.....104/100°C [220/212°F]

Air Intake System

Specifications

Marine Applications

Maximum Intake Restriction (Clean Element).....	254 mm H ₂ O [10 in H ₂ O]
Maximum Intake Restriction (Dirty Element).....	507 mm H ₂ O [20 in H ₂ O]
Maximum Intake Manifold Temperature.....	49°C [120°F]
Maximum Engine Room Temperature:	
Intake air Versus Ambient Air at Rated Condition.....	17°C [63°F] difference

Industrial Applications

Maximum Intake Restriction (Clean Element).....	381 mm H ₂ O [15 in H ₂ O]
Maximum Intake Restriction (Dirty Element).....	635 mm H ₂ O [25 in H ₂ O]
Maximum Temperature Rise Between Ambient Air and Engine Inlet Air:	
Ambient above 0°C [32°F].....	35°C [63°F] difference

Generator Set Applications

Maximum Intake Air Restriction Including Air Filter Plumbing	
Clean Filter Element.....	635 mm H ₂ O [25 in H ₂ O]
Dirty Filter Element.....	381 mm H ₂ O [15 in H ₂ O]
Maximum Allowable Pressure Drop from Turbocharger Outlet to Intake Manifold	
1500 rpm.....	2.5 in Hg [64 mm Hg]
1800 rpm.....	4 in Hg [102 mm Hg]
Maximum Temperature Rise Between Engine Air Inlet and Intake Manifold.....	25°C [45°F]

Exhaust System

Specifications

Marine Applications

Exhaust Pipe Size (normally acceptable inside diameter).....102 mm [4 in]

Maximum Exhaust Pipe Restriction

H₂O.....1016 mm H₂O [40 in H₂O]

Hg.....75 mm Hg [3 in Hg]

Industrial and Generator Drive

Exhaust Pipe Size (normally acceptable inside diameter).....102 mm [4 in]

Maximum Exhaust Pipe Restriction

H₂O.....1016 mm H₂O [40 in H₂O]

Hg.....75 mm Hg [3 in Hg]

Electrical System

Specifications

Starting Motor Circuit:

Maximum resistance"

12 VDC.....	0.00075 ohm
24 VDC.....	0.002 ohm

Maximum Voltage Drop:

12 VDC.....	1.95 VDC
24 VDC.....	1.7 VDC

Marine Applications

Unaided Cold Start:.....	-7°C [19°F]
Maximum Ambient Temperature.....	-7°C [19°F]
Minimum Cranking Speed Required.....	120 rpm

Industrial Applications

Unaided Cold Start:.....	-12°C [10°F]
Maximum Ambient Temperature.....	-12°C [10°F]
Minimum Cranking Speed Required.....	120 rpm

Generator Drive Applications

Minimum Ambient Temperature for Cold Start With Coolant Heated to Rated Speed.....	
Minimum Ambient Temperature for Unaided Cold Start to Low Idle Speed.....	
Minimum Ambient Temperature for NFPA 110 Cold Start (90°F minimum coolant temperature).....	0°C [32°F]

Minimum recommended battery capacity:

System Voltage	Ambient Temperature			
	-18°C [0°F]		0°C [32°F]	
	Cold Cranking Amperes	Reserve Capacity ¹ Minutes	Cold Cranking Amperes	Reserve Capacity ¹ Minutes
12 VDC	1800	640	1280	480
24 VDC ²	900	320	640	240

* The number of plates within a given battery size determines reserve capacity. Reserve capacity determines the length of time that sustained cranking can occur.
 * Cold Cranking Amperes rating are based on two 12-VDC batteries in series.

A minimum of 6.5 VDC at the power connector is required to power up the ECM.

Compressed Air System

Specifications

Cummins 18.7 CFM Model

Cylinders.....	1
Compressor Swept Volume at 1250.....	6.6 l/sec. [140 cfm]
Piston Displacement.....	318 cc [19.405 C.I.D.]
Bore.....	85 mm [3.346 in]
Stroke.....	56 mm [2.205 in]
Speed.....	Engine speed
Cooling.....	Engine coolant
Lubrication.....	Engine lubricating oil
Plumbing Line Sizes:	
Coolant Inlet and Outlet.....	3/4 x 16 STOR
Air Inlet.....	1-in nominal hose
Air Outlet.....	M27 x 2 STOR
Unloader Port.....	M10 x 1 STOR
Governor Mounting Direct.....	M8 x .25
Height, Overall (approximate).....	217.4 mm [8.559 in]
Width, Overall (approximate).....	142 mm [5.591 in]
Length, Overall (approximate).....	216 mm [8.504 in]
Weight (approximate).....	15 kg [33.0 lb]

Cummins 37.4 CFM Model

Cylinders.....	1
Compressor Swept Volume at 1250.....	15.5 l/sec. [35.5 cfm]
Piston Displacement.....	636 cc [39 C.I.D.]
Bore.....	85 mm [3.35 in]
Stroke.....	56 mm [2.2 in]
Speed.....	Engine speed
Cooling.....	Engine coolant
Lubrication.....	Engine lubricating oil
Plumbing Line Sizes:	
Coolant Inlet and Outlet.....	3/4 x 16 STOR
Air Inlet.....	1-in nominal hose
Air Outlet.....	M27 x 2 STOR
Unloader Port.....	M10 x 1 STOR
Governor Mounting Direct.....	M8 x .25
Height, Overall (approximate).....	306 mm [12 in]
Width, Overall (approximate).....	142 mm [5.6 in]
Length, Overall (approximate).....	286.7 mm [11.29 in]
Weight (approximate).....	17.24 kg [38 lb]

Cummins/Fleetguard® Filter Specifications

Specifications

Fleetguard® is a subsidiary of Cummins Inc. Fleetguard® filters are developed through joint testing at Cummins Inc. and Fleetguard®. Fleetguard® filters are standard on new Cummins Inc. engines. Cummins Inc. recommends their use.

Fleetguard® products met all Cummins Inc. Source Approval Test standards to provide the quality filtration necessary to achieve the engine's design life. If other brands are substituted, the purchaser must insist on products that the supplier has tested to meet Cummins Inc. high-quality standards.

Cummins Inc. can not be responsible for problems caused by non-genuine filters that do not meet Cummins Inc. performance or durability requirements.

Filter Types

Lubricating Oil Filter

Cummins Part Number	3101869 and 3101868
Fleetguard® Part Number	LF9001 ¹ or LF9000

Fuel Filter

Cummins Part Number	3329289
Fleetguard® Part Number	FS 1000

Turbocharger Control Shutoff Valve Filter

Cummins Part Number	3682958
Fleetguard® Part Number	

Coolant Filter²

Cummins Part Number	4058965
Fleetguard® Part Number	WF2076
Cummins Part Number	3100304
Fleetguard® Part Number	WF2071
Cummins Part Number	3100305
Fleetguard® Part Number	WF2072

Note:

- Centinel™ equipped automotive applications must use Fleetguard® filter LF9031, Cummins Part Number 4906633. Non-automotive Centinel™ equipped applications can use Fleetguard® filter LF9001, Cummins Part Number 3406809.
- Use the correct Fleetguard® coolant filter to maintain the recommended SCA concentration in the system. The correct filter for each application is determined by the total cooling system capacity and oil drain interval. Refer to the Coolant Capacity Specifications in Section V.

Fuel Recommendations and Specifications

Fuel Recommendations



Do not bleed the fuel system of a hot engine; this can result in fuel spilling onto a hot exhaust manifold, which can cause a fire.



Due to the precise tolerances of diesel injection systems, it is extremely important that the fuel be kept clean and free of dirt or water. Dirt or water in the system can cause severe damage to both the fuel pump and the fuel injectors.



Do not use diesel fuel blended with lubricating oil in engines equipped with an aftertreatment device. Service intervals for aftertreatment devices will be reduced.

Fuel Recommended Properties

Cummins Inc. recommends the use of fuels that meet the following specifications:

Property	Recommended Specifications
Viscosity (ASTM D445)	1.3 to 4.1 centistokes [1.3 to 4.1 mm per second] at 40°C [104°F].
Cetane Number (ASTM D613)	40 Minimum above 0°C [32°F]. 45 Minimum below 0°C [32°F].
Sulfur Content (ASTM D129 or 1552)	Not to exceed 0.05 mass percent for engines without aftertreatment devices. For engines with aftertreatment devices, sulfur content can not exceed 0.0015 (15 ppm) mass percent.
Active Sulfur (ASTM D130)	Copper Strip Corrosion not to exceed Number 3 rating after three hours at 50°C [122°F].
Water and Sediment (ASTM D1796)	Not to exceed 0.05 volume percent.
Carbon Residue (Rams bottom, ASTM D524 or Conradson, ASTM D189)	Not to exceed 0.35 mass percent on 10 volume percent residuum.
Density (ASTM D287)	42 to 30° API gravity at 0.816 to 0.876 g/cc at 15°C [60°F].
Cloud Point (ASTM D97)	6°C [10°F] below lowest ambient temperature at which the fuel is expected to operate.
Ash (ASTM D482)	Not to exceed 0.02 mass percent (0.05 mass percent with lubricating oil blending). Oil blending is prohibited on engines with aftertreatment devices.
Distillation (ASTM D86)	The distillation curve must be smooth and continuous.
Acid Number (ASTM D664)	Not to exceed 0.1 mg KOA per 100 ml.

1. For additional information on fuel recommendations and specifications, refer to Fuel for Cummins Engines, Bulletin 3379001. See ordering information in the back of this manual.

Lubricating Oil Recommendations and Specifications

General Information

Industrial Applications

The use of high-quality engine lubricating oils and appropriate oil drain and filter change intervals are critical factors in maintaining engine performance and durability.

Cummins Inc. recommends the use of a high-quality 15W-40 multiviscosity heavy-duty engine oil that meets the requirements in Cummins Engineering Specification CES 20071 or CES 20076 (such as Valvoline® Premium Blue® or Premium Blue® 2000). American Petroleum Institute (API) specification CH-4 can be used as an alternative to CES 20071. Oils that meet API specification CG-4 can be used, but at a reduced drain interval according to the Oil Drain Intervals chart listed in Section 2. The oil grades CC, CD, CE, and CF have been obsoleted by API and must not be used.

Shortened drain intervals can be required with monograde oils as determined by close monitoring of the oil condition with scheduled oil sampling. Use of single-grade oils can affect engine oil control.

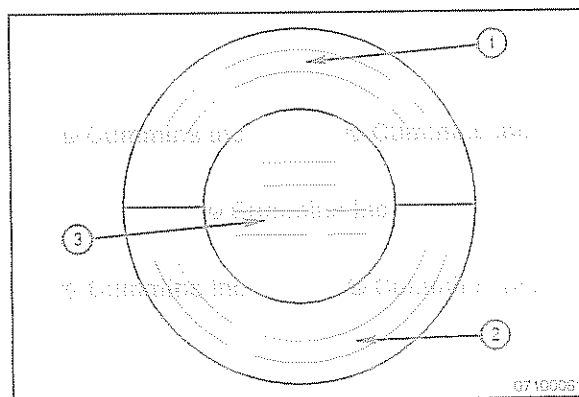
Synthetic engine oils, API category III, are recommended for use in Cummins engines operating in ambient temperature conditions consistently below -25°C [-13°F]. Above this temperature, petroleum-based multigrade lubricants are recommended. Synthetic 0W-30 oils that meet API category III can be used in operations where the ambient temperature never exceeds 0°C [32°F]. Multiviscosity oils rated 0W-30 do not offer the same level of protection against fuel dilution as do higher multigrade oils. Higher cylinder wear can be experienced when using 0W-30 oils in high-load situations.

For further details and an explanation of engine lubricating oils for Cummins engines, refer to Cummins Engine Oil Recommendations, Bulletin 3810340.

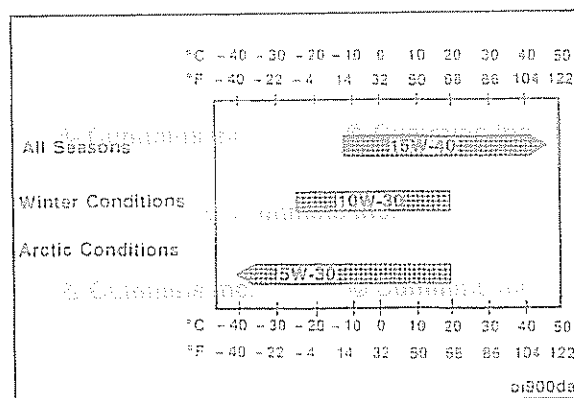
Additional information regarding lubricating oil availability throughout the world is available in the Engine Manufacturing Association (EMA) Lubricating Oils Data Book for Heavy-Duty Automotive and Industrial Engines. The data book can be ordered from Engine Manufacturers Association, Two North LaSalle Street, Chicago, IL 60602; (312) 827-8733, (312) 827-8700, (www.engine-manufacturers.org).

The API service symbols are shown in the accompanying illustration.

1. The upper half of the symbols displays the appropriate oil categories.
2. The lower half contains words to describe oil energy-conserving features.
3. The center section identifies the SAE oil viscosity grade.



Oil viscosity should be chosen according to the typical climate conditions experienced by the user. Use of 15W-40 is recommended for the best engine durability at higher ambient temperature. For temperate or cold conditions, 10W-30 or 5W-30 viscosity can be used for easier starting, improved oil flow, and improved fuel economy.



AfterMarket Oil Additive Usage

Cummins Inc. does not recommend the use of aftermarket oil additives. Current high-quality fully additive engine lubricating oils are very sophisticated, with precise amounts of additives blended into the lubricating oil to meet stringent requirements defined in 1) Cummins Engineering Specification CES 20076 that is similar to API CH-4, in 2) CES 20078 that is similar to API CI-4, and in 3) CES 20081 that is similar to API CJ-4. These furnished oils meet performance characteristics that conform to the lubricant industry standards. Aftermarket lubricating oil additives are not necessary to enhance engine oil performance and in some cases can reduce the finished oil's capability to protect the engine.

New Engine Break-in Oils

Special "break-in" engine lubricating oils are not recommended for new or rebuilt Cummins® engines. Use the same lubricating oil that will be used during normal operation.

Coolant Recommendations and Specifications

Fully Formulated Coolant/Antifreeze

Cummins Inc. recommends using either a 50/50 mixture of good-quality water and fully formulated antifreeze, or fully formulated coolant when filling the cooling system. The fully formulated antifreeze or coolant must meet TMC RP329 or TMC RP330 specifications.

NOTE: Use of products meeting TMC RP329 or RP330 is necessary for 50,000-mile and 150,000-mile service intervals.

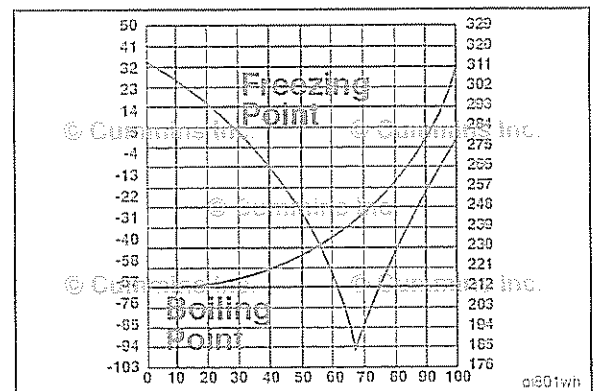
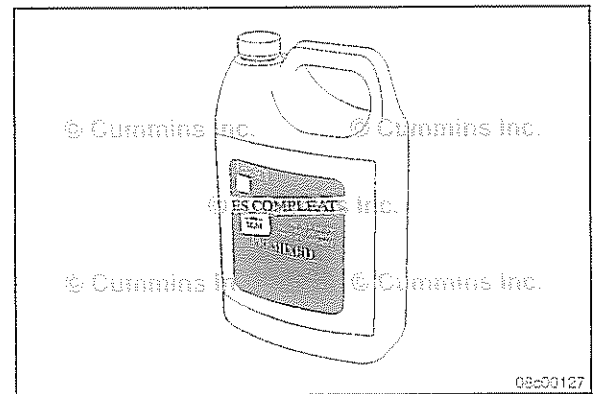
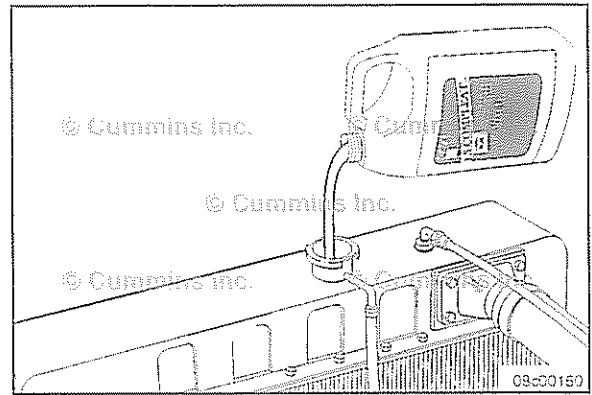
NOTE: Low-silicate antifreeze meeting ASTM D4985 is inadequate for these extended service intervals.

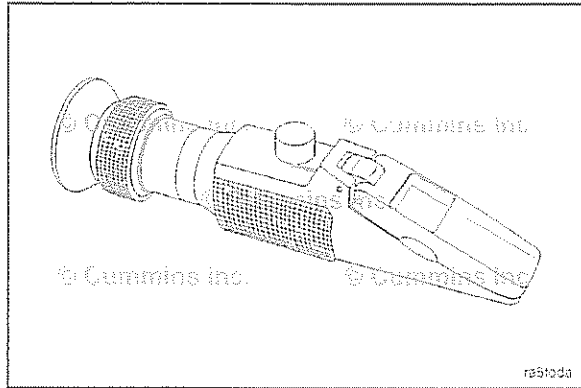
Good-quality water is important for cooling system performance. Excessive levels of calcium and magnesium contribute to scaling problems, and excessive levels of chlorides and sulfates cause cooling system corrosion.

Water Quality	
Calcium Magnesium (Hardness)	Maximum 170 ppm as (CaCO ₃ + MgCO ₃)
Chloride	40 ppm as (Cl)
Sulfate	100 ppm as (SO ₄)

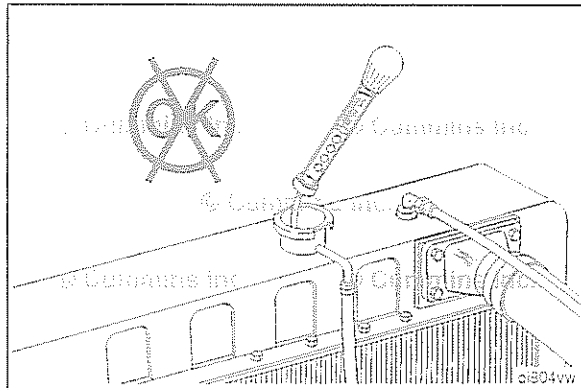
Cummins Inc. recommends using Fleetguard® COMPLETE ES. It is available in glycol forms (ethylene and propylene) and complies with TMC RP329 and RP330 standards.

Fully formulated antifreeze must be mixed with quality water at a 50/50 ratio (40-percent to 60-percent working range). A 50/50 mixture of antifreeze and water gives a -36°C [-34°F] freezing point and a 110°C [228°F] boiling point, which is adequate for locations in North America. The actual lowest freezing point of ethylene glycol antifreeze is 68 percent. Using higher concentrations of antifreeze will raise the freezing point of the solution and increase the possibility of a silica gel problem.

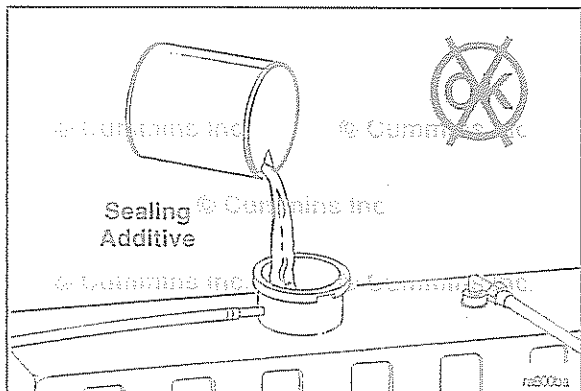




A refractometer must be used to measure the freezing point of the coolant accurately.



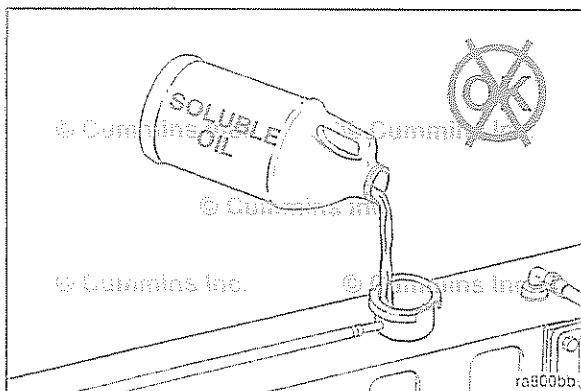
Do not use a floating ball hydrometer. Using a floating ball hydrometer can give an incorrect reading.



Cooling System Sealing Additives

Do not use sealing additives in the cooling systems. The use of sealing additives can:

- Build up in coolant low-flow areas
- Clog coolant filters
- Plug radiator and oil cooler.



Cooling System Soluble Oils

Do not use soluble oils in the cooling system. The use of soluble oils can:

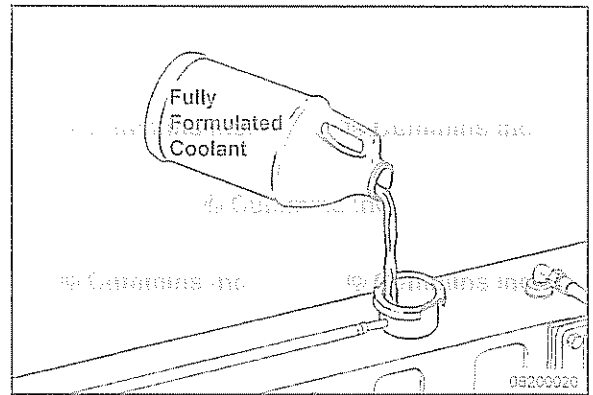
- Allow cylinder liner pitting
- Corrode brass, aluminum, and copper
- Damage heat transfer surfaces
- Damage seals and hoses.

Supplemental Coolant Additive (SCA)



The Signature, ISX, and QSX15 engines use aluminum parts that are in contact with the coolant. Improper coolant, coolant filter selection, and maintenance will likely result in perforation of one of these parts. Insufficient concentration of the coolant additives will result in liner pitting and engine failure.

Fully formulated products contain SCAs and are required to protect the cooling system from scale and fouling, solder corrosion, and general corrosion. The coolant filter is required to protect the cooling system from abrasive materials, debris, and precipitated coolant additives.



Fleetguard® DCA4 Service Filters and Liquid Precharge

Signature, ISX, and QSX15 Series Coolant Filter Options

WF2125 - This filter is designed for use with Fleetguard®'s extended service cooling system, which extends cooling system service to 1 year, 150,000 miles, or 4000 hours, whichever comes first. This filter is used for cooling systems up to 20 gallons. Refer to Coolant Requirements and Maintenance, Bulletin 3666132. Fleetguard® ES coolant must be used for all fill and top-off, which is critical for extended cooling system maintenance intervals.

WF2126 - This filter is designed for extended service intervals up to 50,000 miles when using TMC RP329 or RP330 coolants and is used for cooling systems up to 20 gallons. Refer to Coolant Requirements and Maintenance, Bulletin 3666132.

WF2127 - This filter has been designed for extended service intervals of 50,000 to 150,000 miles. It has no chemical additives and can be used in the following systems:

- Cooling systems above 20 gallons in capacity.
- See maintenance chart below.

When using WF2127 filter, the following volumes of treatment must be added at the designated mileages:

Cooling System Capacity	50,000-Mile Service Interval with RP329/330 Coolant or Treated Water (Fleetcool or DCA4)	150,000-Mile Service Interval with ES Coolant (ES Liquid)
0 to 20 gallons	1 qt (10 units)	1 qt (15 units)
20 to 40 gallons	2 qt (20 units)	2 qt (30 units)
40 to 60 gallons	3 qt (30 units)	3 qt (45 units)
60 to 80 gallons	4 qt (40 units)	4 qt (60 units)
80 to 100 gallons	5 qt (50 units)	5 qt (75 units)

NOTE: Filters must meet Cummins SAM 10,769. Fleetguard® filters meet Cummins SAM 10,769. The standard filter for the Signature engine is Fleetguard® WF2126. For systems larger than 100 gallons, use 1 quart per 15 gallons. Consult vehicle manufacturer for total cooling system capacity.

Testing SCA Concentration Level CC-2602 Test Kit

Precautions and Instructions for Proper Kit Use

Carefully follow the instructions to test the coolant. Take the appropriate action recommended by the kit.

- The coolant sample to be tested must be between 10 and 54 °C [50 and 130°F]. If the sample is too cold or too hot, you will get incorrect results.
- To get the best color match results, compare test strip pads to the color chart in daylight or under cool white fluorescent lighting. If unsure about a specific color match when a test does fall between two colors on the color chart, choose the lower numbered (or lettered) block. It is safer to underestimate your results than to overestimate.
- The test strips do have a limited shelf life and are sensitive to humidity and extreme heat. Proper handling and storage is necessary to protect the life of the strips.
- Keep the cap tightly sealed on the test strip bottle except when removing a strip. Store away from direct sunlight and in an area where the temperature will generally stay below 32°C [90°F].
- Do not use the test strips after the expiration date stamped on the bottle.

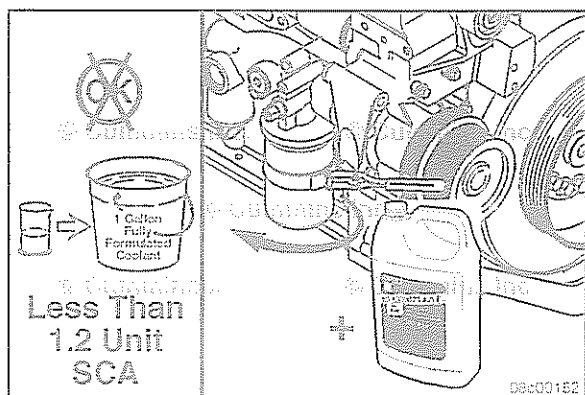
- Discard the kit if the top pad on the unused strips have turned light brown.
- Use one strip at a time and take care not to touch any of the pads on the strip. Doing so will contaminate the pads and skew the test results.
- If the strip container is left uncapped for 24 hours, moisture in the air will render the strips useless, even though no discoloration will be evident.
- Only use the color chart supplied with the kit.
- Following the correct test times is very important. Use a clock or stopwatch.
- Do not utilize the test kit to maintain minimum SCA concentration levels (i.e., 1.5 units).
- When performing service that requires draining the cooling system, take special precautions to collect coolant in a clean non-galvanized container, seal coolant to prevent contamination, and save for reuse.

Coolant Testing

- Probablizer:
 - 3318169S Plug - Installs on the engine for easy coolant sampling
 - 3318168S Cap - Use with Monitor C bottle to sample coolant
 - CC2700 Monitor C™ - Use lab analysis of coolant samples for more detailed analysis.

CC2602 Coolant Test Kit

- Works with any SCA formulation. Call 1-800-22FILTER (800-223-4583) if you have this test kit and the color chart does not show the number of units of SCA gallon of coolant.)



Test Intervals



Do not remove the pressure cap from a hot engine. Wait until coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

Testing is recommended if the operator is not sure of his cooling system condition due to leaks, uncontrolled topping off of the system, or major coolant loss.

If the concentration is below 1.2 units per gallon, replace the filter and precharge with liquid.

Check the SCA concentration level at least every 6 months, and anytime the coolant condition is unknown or corrosion is apparent within the cooling system.

Use Fleetguard® coolant test kit, CC2602, to check the concentration level. Instructions are included with the test kit.

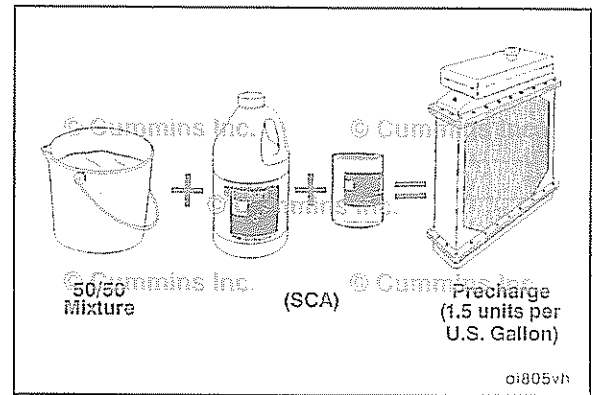
Call the following numbers to get answers to any questions you have about cooling system maintenance:

Cummins	Fleetguard®
1-800-DIESELS	1-800-22FILTER
1-800-343-7357	1-800-223-4583

Coolant Replacement Requirements

Drain and flush the cooling system after 6000 hours or 3 years of service. However, if Fleetguard®'s ES coolant and ES filters are used, check chloride, sulfate, and pH levels according to Coolant Requirements and Maintenance, Bulletin 3666132, to determine whether the coolant must be replaced. Refill with either new fully formulated coolant or ES coolant.

NOTE: Dispose of used coolant/antifreeze in accordance with federal, state, and local laws and regulations.



Drive Belt Tension

Tension Chart

SAE Belt Size	Belt Tension Gauge Part No.		Belt Tension New		Belt Tension Range Used *	
	Click-type	Burroughs	N	lbf	N	lbf
0.380 in	3822524		620	140	270 to 490	60 to 110
0.440 in	3822524		620	140	270 to 490	60 to 110
1/2 in	3822524	ST-1138	620	140	270 to 490	60 to 110
11/16 in	3822524	ST-1138	620	140	270 to 490	60 to 110
3/4 in	3822524	ST-1138	620	140	270 to 490	60 to 110
7/8 in	3822524	ST-1138	620	140	270 to 490	60 to 110
4 rib	3822524	ST-1138	620	140	270 to 490	60 to 110
5 rib	3822524	ST-1138	670	150	270 to 530	60 to 120
6 rib	3822525	ST-1293	710	160	290 to 580	65 to 130
8 rib	3822525	ST-1293	890	200	360 to 710	80 to 160
10 rib	3822525	3823138	1110	250	440 to 890	100 to 200
12 rib	3822525	3823138	1330	300	530 to 1070	120 to 240
12 rib K section	3822525	3823138	1330	300	890 to 1070	200 to 240

NOTE: This chart does not apply to automatic belt tensioners.

* A belt is considered used if it has been in service for ten minutes or longer.

* If used belt tension is less than the minimum value, tighten the belt to the maximum used belt value.

Engine Component Torque Values

General Information

Component	Wrench Size	Torque Value	
		N•m	ft-lb
Oil Pan Drain Plug	1-1/4	88	65
Turbocharger Mounting Nuts	16 mm	68	50
Fan Drive Idler Pulley Shaft Locknut		165 to 190	120 to 140
Injector/Valve Adjusting Screw Locknut	3/4 in.	61	45
Engine Brake Adjusting Screw Locknut		50	40
Fuel Pump Mounting Capscrews	7/16 in	47	35
Fuel Pump Bracket to Cylinder Block		47	35
Fuel Pump Bracket to Fuel Pump Housing	7/16 in	11	95 in-lb
Rocker Lever Cover Capscrews	13 mm	15	130 in-lb
Injector Hold-down Clamps		75	55
Crankcase Breather Cover Bolts	6 mm	.68	6 in-lb
Aftertreatment V-Band Clamps	10 mm	20	180 in-lb
Differential Pressure Sensing Tube Nuts	5/8	55	40
Aftertreatment Mounting Capscrews	10 mm	20	180 in-lb

Capscrew Markings and Torque Values

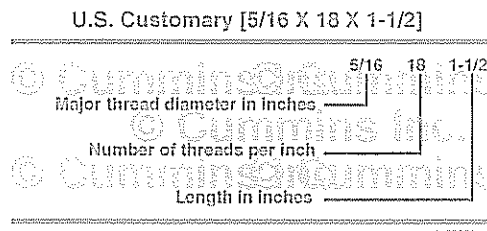
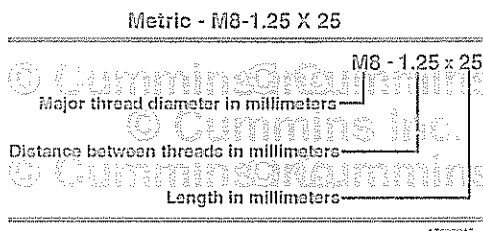
General Information



When replacing capscrews, always use a capscrew of the same measurement and strength as the capscrew being replaced. Using the wrong capscrews can result in engine damage.

Metric capscrews and nuts are identified by the grade number stamped on the head of the capscrew or on the surface of the nuts. U.S. Customary capscrews are identified by radial lines stamped on the head of the capscrew.

The following examples indicate how capscrews are identified:



NOTES:

1. Always use the torque values listed in the following tables when specific torque values are not available.
2. Do not use the torque values in place of those specified in other sections of this manual.
3. The torque values in the table are based on the use of lubricated threads.
4. When the ft-lb value is less than 10, convert the ft-lb value to in-lb to obtain a better torque with an in-lb torque wrench. Example: 6 ft-lb equals 72 in-lb.

Capscrew Markings and Torque Values - Metric

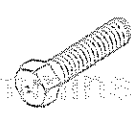
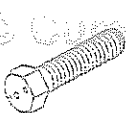
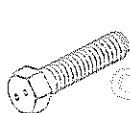
Commercial Steel Class

8.8

10.9

12.9

Capscrew Head Markings



Body Size

Diameter

mm

6

7

8

10

12

14

16

18

20

Torque

Cast Iron

N•m

9

14

23

45

80

125

195

280

400

Aluminum

N•m

7

11

18

30

55

90

140

180

—

Cast Iron

N•m

13

18

33

65

115

180

280

390

550

Torque

Cast Iron

N•m

10

14

25

50

85

133

200

285

400

Aluminum

N•m

7

11

18

30

55

90

140

180

—

Torque

Cast Iron

N•m

14

23

40

70

125

195

290

400

—

Aluminum

N•m

9

18

29

50

95

145

210

290

—

7

11

18

30

55

90

140

180

—

Capscrew Markings and Torque Values - U.S. Customary

SAE Grade Number		5				8			
Capscrew Head Markings									
These are all SAE Grade 5 (3 line)									
									
		Capscrew Torque - Grade 5 Capscrew				Capscrew Torque - Grade 8 Capscrew			
Capscrew Body Size	Cast Iron		Aluminum		Cast Iron		Aluminum		
	N•m	ft-lb	N•m	ft-lb	N•m	ft-lb	N•m	ft-lb	
1/4 - 20	9	7	8	6	15	11	8	6	
1/4 - 28	12	9	9	7	18	13	9	7	
5/16 - 18	20	15	16	12	30	22	16	12	
5/16 - 24	23	17	19	14	33	24	19	14	
3/8 - 16	40	30	25	20	55	40	25	20	
3/8 - 24	40	30	35	25	60	45	35	25	
7/16 - 14	60	45	45	35	90	65	45	35	
7/16 - 20	65	50	55	40	95	70	55	40	
1/2 - 13	95	70	75	55	130	95	75	55	
1/2 - 20	100	75	80	60	150	110	80	60	
9/16 - 12	135	100	110	80	190	140	110	80	
9/16 - 18	150	110	115	85	210	155	115	85	
5/8 - 11	180	135	150	110	255	190	150	110	
5/8 - 18	210	155	160	120	290	215	160	120	
3/4 - 10	325	240	255	190	460	340	255	190	
3/4 - 16	365	270	285	210	515	380	285	210	
7/8 - 9	490	360	380	280	745	550	380	280	
7/8 - 14	530	390	420	310	825	610	420	310	
1 - 8	720	530	570	420	1100	820	570	420	
1 - 14	800	590	650	480	1200	890	650	480	

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Section W - Warranty

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All Engines United States and Canada Industrial (Off-Highway)

Coverage

Products Warranted

This warranty applies to new Engines sold by Cummins and delivered to the first user on or after April 1, 1999, that are used in industrial (off-highway) applications in the United States* and Canada, except for Engines used in marine, generator drive and certain defense applications, for which different warranty coverage is provided.

Base Engine Warranty

This warranty covers any failures of the Engine, under normal use and service, which result from a defect in material or factory workmanship (Warrantable Failures).

Coverage begins with the sale of the Engine by Cummins. Coverage continues for two years or 2,000 hours of operation, whichever occurs first, from the date of delivery of the Engine to the first user, or from the date the unit is first leased, rented or loaned, or when the Engine has been operated for 50 hours, whichever occurs first. If the 2,000 hour limit is exceeded during the first year, Coverage continues until the end of the first year.

Extended Major Components Warranty

The Extended Major Components Warranty covers Warrantable Failures of the Engine cylinder block, camshaft, crankshaft and connecting rods (Covered Parts).

Bushing and bearing failures are not covered.

This Coverage begins with the expiration of the Base Engine Warranty and ends three years or 10,000* hours of operation from the date of delivery of the Engine to the first user, or from the date the unit is first leased, rented or loaned, or from when the Engine has been operated for 50 hours, whichever occurs first.

*3,000 hours for A series engines.

Consumer Products

The warranty on Consumer Products in the United States is a LIMITED warranty. CUMMINS IS NOT RESPONSIBLE FOR INCIDENTAL OR CONSEQUENTIAL DAMAGES. Any implied warranties applicable to Consumer Products in the United States terminate concurrently with the expiration of the express warranties applicable to the product. In the United States, some states do not allow the exclusion of incidental or consequential damages, or limitations on how long an implied warranty lasts, so the limitations or exclusions herein may not apply to you.

These warranties are made to all Owners in the chain of distribution, and Coverage continues to all subsequent Owners until the end of the periods of Coverage.

Cummins' Responsibilities

During The Base Engine Warranty

Cummins will pay for all parts and labor needed to repair the damage to the Engine resulting from a Warrantable Failure.

Cummins will pay for the lubricating oil, antifreeze, filter elements, and other maintenance items that are not reusable due to the Warrantable Failure.

Cummins will pay reasonable costs for mechanics to travel to and from the equipment site, including meals, mileage and lodging, when the repair is performed at the site of the failure.

Cummins will pay reasonable labor costs for Engine removal and reinstallation when necessary to repair a Warrantable Failure.

During The Extended Major Components Warranty

Cummins will pay for the repair or, at its option, replacement of the defective Covered Part and any Covered Part damaged by a Warrantable Failure of the defective Covered part.

Owner's Responsibilities

During The Base Engine Warranty

Owner is responsible for the cost of lubricating oil, antifreeze, filter elements and other maintenance items provided during warranty repairs unless such items are not reusable due to the Warrantable Failure.

During The Extended Major Components Warranty

Owner is responsible for the cost of all labor needed to repair the Engine, including the labor to remove and reinstall the Engine. When Cummins elects to repair a part instead of replacing it, Owner is not responsible for the labor needed to repair the part.

Owner is responsible for the cost of all parts required for the repair except for the defective Covered Part and any Covered Part damaged by a Warrantable Failure of the defective Covered Part.

Owner is responsible for the cost of lubricating oil, antifreeze, filter elements and other maintenance items replaced during repair of a Warrantable Failure.

During The Base Engine and Extended Major Components Warranties

Owner is responsible for the operation and maintenance of the Engine as specified in the applicable Cummins Operation and Maintenance Manual. Owner is also responsible for providing proof that all recommended maintenance has been performed.

Before the expiration of the applicable warranty, Owner must notify a Cummins distributor, authorized dealer or other repair location approved by Cummins of any Warrantable Failure and make the Engine available for repair by such facility. Locations in the United States and Canada are listed in the Cummins Off Highway Authorized Dealer Directory.

Owner is responsible for communication expenses, meals, lodging and similar costs incurred as a result of a Warrantable Failure.

Owner is responsible for non-Engine repairs, "downtime" expenses, cargo damage, fines, all applicable taxes, all business costs and other losses resulting from a Warrantable Failure.

Limitations

Cummins is not responsible for failures or damage resulting from what Cummins determines to be abuse or neglect, including, but not limited to: operation without adequate coolants or lubricants; overfueling; overspeeding; lack of maintenance of lubricating, cooling or intake systems; improper storage, starting, warm-up, run-in or shutdown practices; unauthorized modifications of the Engine. Cummins is also not responsible for failures caused by incorrect oil or fuel or by water, dirt or other contaminants in the fuel or oil.

For power units and fire pumps (package units), this warranty applies to accessories, except for clutches and filters, supplied by Cummins which bear the name of another company.

Except for power units and fire pumps, this warranty does not apply to accessories which bear the name of another company. Such non-warranted accessories include, but are not limited to: alternators, starters, fans¹, air conditioning compressors, clutches, filters, transmissions, torque converters, steering pumps, and non-Cummins fan drives, engine compression brakes and air compressors.

Cummins Compusave units are covered by a separate warranty.

Before a claim for excessive oil consumption will be considered, Owner must submit adequate documentation to show that consumption exceeds Cummins published standards.

Failures of belts and hoses supplied by Cummins are not covered beyond the first 500 hours or one year of operation, whichever occurs first.

Parts used to repair a Warrantable Failure may be new Cummins parts, Cummins-approved rebuilt parts, or repaired parts. Cummins is not responsible for failures resulting from the use of parts not approved by Cummins.

A new Cummins or Cummins-approved rebuilt part used to repair a Warrantable Failure assumes the identity of the part it replaced and is entitled to the remaining coverage hereunder.

CUMMINS DOES NOT COVER WEAR OR WEAROUT OF COVERED PARTS.

CUMMINS IS NOT RESPONSIBLE FOR INCIDENTAL OR CONSEQUENTIAL DAMAGES.

THESE WARRANTIES SET FORTH HEREIN ARE THE SOLE WARRANTIES MADE BY CUMMINS IN REGARD TO THESE ENGINES. CUMMINS MAKES NO OTHER WARRANTIES, EXPRESS OR IMPLIED, OR OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE.

This warranty gives you specific legal rights, and you may also have other rights which vary from state to state.

Emission Warranty

Products Warranted

This emission warranty applies to new Engines marketed by Cummins that are used in the United States¹ in vehicles designed for Industrial off-highway use. This warranty applies to Engines delivered to the ultimate purchaser on or after April 1, 1999 for engines up to 750 horsepower, on or after January 1, 2000 for engines 751 horsepower and over.

Coverage

Cummins warrants to the ultimate purchaser and each subsequent purchaser that the Engine is designed, built and equipped so as to conform at the time of sale by Cummins with all U.S. Federal emission regulations applicable at the time of manufacture and that it is free from defects in workmanship or material which would cause it not to meet these regulations within the longer of the following periods: (A) Five years or 3,000 hours of operation, whichever

occurs first, as measured from the date of delivery of the Engine to the ultimate purchaser, or (B) The Base Engine Warranty.

If the vehicle in which the Engine is installed is registered in the state of California, a separate California Emission Warranty also applies.

Limitations

Failures, other than those resulting from defects in materials, or workmanship, are not covered by this warranty.

Cummins is not responsible for failures or damage resulting from what Cummins determines to be abuse or neglect, including, but not limited to: operation without adequate coolant or lubricants; overfueling; overspeeding; lack of maintenance of lubricating, cooling or intake systems; improper storage, starting, warm-up, run-in or shutdown practices; unauthorized modifications of the Engine. Cummins is also not responsible for failures caused by incorrect fuel or by water, dirt or other contaminants in the fuel.

Cummins is not responsible for non-Engine repairs, "downtime" expenses, cargo damage, fines, all business costs or other losses resulting from a Warrantable Failure.

CUMMINS IS NOT RESPONSIBLE FOR INCIDENTAL OR CONSEQUENTIAL DAMAGES.

* Includes American Samoa, the Commonwealth of Northern Mariana Islands, Guam, Puerto Rico, and the U.S. Virgin Islands.

* * Alternators, starters, and fans ARE covered for the duration of the base engine warranty on A series and B3.3 engines.

All Engines International Industrial (Off-Highway)

Coverage

PRODUCTS WARRANTED

This warranty applies to new Engines sold by Cummins and delivered to the first user on or after April 1, 1999, that are used in industrial (off-highway) applications anywhere in the world where Cummins-approved service is available, except the United States* and Canada. Different warranty coverage is provided for Engines used in marine, generator drive and certain defense applications.

BASE ENGINE WARRANTY

This warranty covers any failures of the Engine, under normal use and service, which result from a defect in material or factory workmanship (Warrantable Failure).

Coverage begins with the sale of the Engine by Cummins. Coverage continues for two years or 2,000 hours of operation, whichever occurs first, from the date of delivery of the Engine to the first user, or from the date the unit is first leased, rented or loaned, or when the Engine has been operated for 50 hours, whichever occurs first. If the 2,000 hour limit is exceeded during the first year, coverage continues until the end of the first year.

EXTENDED MAJOR COMPONENTS WARRANTY

The Extended Major Components Warranty covers Warrantable Failures of the Engine cylinder block, camshaft, crankshaft and connecting rods (Covered Parts).

Bushing and bearing failures are not covered.

This coverage begins with the expiration of the Base Engine Warranty and ends three years or 10,000* hours of operation, from the date of delivery of the Engine to the first user, or from the date the unit is first leased, rented or loaned, or when the Engine has been operated for 50 hours, whichever occurs first.

*3,000 hours for A series engines.

These warranties are made to all Owners in the chain of distribution, and Coverage continues to all subsequent Owners until the end of the periods of Coverage.

Cummins' Responsibilities

DURING THE BASE ENGINE WARRANTY

Cummins will pay for all parts and labor needed to repair the damage to the Engine resulting from a Warrantable Failure.

Cummins will pay for the lubricating oil, antifreeze, filter elements, and other maintenance items that are not reusable due to a Warrantable Failure.

Cummins will pay reasonable costs for mechanics to travel to and from the equipment site, including meals, mileage and lodging, when the repair is performed at the site of the failure.

Cummins will pay reasonable labor costs for Engine removal and reinstallation when necessary to repair a Warrantable Failure.

DURING THE EXTENDED MAJOR COMPONENTS WARRANTY

Cummins will pay for the repair or, at its option, replacement of the defective Covered Part and any Covered Part damaged by a Warrantable Failure of the defective Covered part.

Owner's Responsibilities

DURING THE BASE ENGINE WARRANTY

Owner is responsible for the cost of lubricating oil, antifreeze, filter elements and other maintenance items replaced during warranty repairs unless such items are not reusable due to the Warrantable Failure.

DURING THE EXTENDED MAJOR COMPONENTS WARRANTY

Owner is responsible for the cost of all labor needed to repair the Engine, including the labor to remove and reinstall the Engine. When Cummins elects to repair a part instead of replacing it, Owner is not responsible for the labor needed to repair the part.

Owner is responsible for the cost of all parts required for the repair except for the defective Covered Part and any Covered Part damaged by a Warrantable Failure of the defective Covered Part.

Owner is responsible for the cost of lubricating oil, antifreeze, filter elements and other maintenance items replaced during repair of a Warrantable Failure.

DURING THE BASE ENGINE AND EXTENDED MAJOR COMPONENTS WARRANTIES

Owner is responsible for the operation and maintenance of the Engine as specified in the applicable Cummins Operation and Maintenance Manual. Owner is also responsible for providing proof that all recommended maintenance has been performed.

Before the expiration of the applicable warranty, Owner must notify a Cummins distributor, authorized dealer or other repair location approved by Cummins of any Warrantable Failure and make the product available for repair by such facility. Locations are listed in the Cummins International Sales and Service Directory.

Owner is responsible for communication expenses, meals, lodging and similar costs incurred as a result of a Warrantable Failure.

Owner is responsible for non-Engine repairs, "downtime" expenses, cargo damage, fines, all applicable taxes, all business costs and other losses resulting from a Warrantable Failure.

Limitations

Cummins is not responsible for failures or damage resulting from what Cummins determines to be abuse or neglect, including, but not limited to: operation without adequate coolants or lubricants; overfueling; overspeeding; lack of maintenance of lubricating, cooling or intake systems; improper storage, starting, warm-up, run-in or shutdown practices; unauthorized modifications of the Engine. Cummins is also not responsible for failures caused by incorrect oil or fuel or by water, dirt or other contaminants in the fuel or oil.

For power units and fire pumps (package units) the warranty applies to accessories, except for clutches and filters supplied by Cummins which bear the name of another company.

Non-Cummins starters, alternators, power steering pumps and air compressors supplied by Cummins that are not supplied as part of a package unit are covered for six months from the date of delivery of the Engine to the first user, or from the date the unit is first leased, rented or loaned, or when the Engine has been operated for 50 hours, whichever occurs first. Cummins branded parts noted above supplied by Cummins are covered for the entire Base Warranty period.

Except for the accessories noted previously, Cummins does not warrant accessories which bear the name of another company. Such non-warranted accessories include, but are not limited to: alternators, starters, fans*, air conditioning compressors, clutches, filters, transmissions, torque converters, steering pumps, non-Cummins fan drives, and air cleaners.

Cummins Compusave units are covered by a separate warranty.

Before a claim for excessive oil consumption will be considered, Owner must submit adequate documentation to show that consumption exceeds Cummins published standards.

Failures of belts and hoses supplied by Cummins are not covered beyond the first 500 hours or one year of operation, whichever occurs first.

Parts used to repair a Warrantable Failure may be new Cummins parts, Cummins-approved rebuilt parts, or repaired parts. Cummins is not responsible for failures resulting from the use of parts not approved by Cummins.

A new Cummins or Cummins-approved rebuilt part used to repair a Warrantable Failure assumes the identity of the part it replaced and is entitled to the remaining coverage hereunder.

CUMMINS DOES NOT COVER WEAR OR WEAROUT OF COVERED PARTS.

CUMMINS IS NOT RESPONSIBLE FOR INCIDENTAL OR CONSEQUENTIAL DAMAGES.

THESE WARRANTIES SET FORTH HEREIN ARE THE SOLE WARRANTIES MADE BY CUMMINS IN REGARD TO THESE ENGINES. CUMMINS MAKES NO OTHER WARRANTIES, EXPRESS OR IMPLIED, OR OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE.

In case of consumer sales, in some countries, the Owner has statutory rights which cannot be affected or limited by the terms of this warranty.

Nothing in this warranty excludes or restricts any contractual rights the Owner may have against third parties.

* Alternators, starters, and fans ARE covered for the duration of the base engine warranty on A series and B3.3 engines.

All Engines WorldwideCommercial Marine PropulsionAuxiliary Products

Coverage

Products Warranted Marine

This warranty applies to new Engines sold by Cummins Inc., herein after "Cummins", and used in Marine propulsion and marine auxiliary applications anywhere in the world where Cummins approved service is available and delivered to the first user on or after July 1, 2002. This warranty excludes all engines branded and sold as Cummins Mercruiser Diesel products. The 'Product' consists of a new Cummins Engine, as well as accessories, which are approved and supplied by Cummins and which are either installed by Cummins or a Cummins authorized distributor. These Products have the following designation:

MARINE PROPULSION AND MARINE AUXILIARY (EXCLUDING GENERATOR DRIVE ENGINES)

Intermittent Rating

This power rating is intended for intermittent use in variable load applications where full power is limited to two hours out of every eight hours of operation. Also, reduced power operations must be at or below 200 RPM of the maximum rated RPM. This rating is an ISO3046 Fuel Stop Power Rating and is for applications that operate less than 1500 hours per year.

Medium Continuous Rating

This power rating is intended for continuous use in variable load applications where full power is limited to six hours out of every twelve hours of operation. Also, reduced power operations must be at or below 200 RPM of the maximum rated RPM. This rating is an ISO3046 Fuel Stop Power Rating and is for applications that operate less than 3000 hours per year.

Heavy Duty Rating

This power rating is intended for continuous use in variable load applications where full power is limited to eight hours out of every ten hours of operation. Also, reduced power must be at least 200 RPM below the maximum rated RPM. This rating is an ISO3046 Fuel Stop Power Rating and is for applications that operate less than 5,000 hours per year.

Continuous Rating

This power rating is intended for continuous use in applications requiring uninterrupted service at full power. This rating is an ISO3046 Standard Power Rating.

MARINE GENERATOR DRIVE

Prime Power

Engines with this rating are available for an unlimited number of hours per year in variable load applications. Variable load is not to exceed a 70 percent average of the Prime Power rating during any operating period of 250 hours. Total operating time at 100 percent Prime Power shall not exceed 500 hours per year.

A 10 percent overload capability is available for a period of one hour within a twelve hour period of operation. Total operating time at the 10 percent overload power shall not exceed 25 hours per year. This power rating conforms to ISO 8528 guidelines.

Continuous Power

Engines with this rating are available for supplying utility power at a constant 100 percent load for an unlimited number of hours per year. No overload capability is available for this rating.

Continuous Power ratings differ from Prime Power ratings in that the Continuous Load ratings are significantly reduced from the Prime Power ratings. Continuous Load ratings have no load factor or application restrictions. This power rating conforms to ISO 8528 guidelines.

Base Engine Warranty

This warranty covers any failures of the Product, under normal use and service, which result from a defect in Cummins material or factory workmanship (Warrantable Failure). Coverage begins with the sale of the Engine by Cummins and ends at the time and hours stated in the following table. The Duration commences on either the date of delivery of the Product to the first end-user, or the date the unit is first leased, rented or loaned, or when the Product has been operated for 50 hours, whichever occurs first.

Table with 3 columns: Rating, Years, Hours. Rows include Intermittent (1 year, 1500 hours) and Medium Continuous (1 year, 3000 hours). Header indicates Duration is Whichever Occurs First.

Heavy Duty	1	5000
Continuous	1	Unlimited
Prime Power	1	Unlimited
Continuous Power	1	Unlimited

Extended Major Components Warranty

The Extended Major Components Warranty applies to engines other than B and C Series. It covers Warrantable Failures of the engine cylinder block, camshaft, crankshaft and connecting rods (Covered Parts). Bushing and bearing failures are not covered. This coverage begins with the expiration of the Base Engine Warranty and ends at three years or 10,800 hours of operation, whichever occurs first, after the date of delivery to the first end-user, or the date the unit is first leased, rented, or loaned, or when the Product has been operated for 50 hours, whichever occurs first.

These warranties are made to all Owners in the chain of distribution and Coverage continues to all subsequent Owners until the end of the periods of Coverage.

Cummins Responsibilities During Base Engine Warranty

Cummins will pay for all parts and labor needed to repair the damage to the Product resulting from a Warrantable Failure when performed during normal business hours. All labor costs will be paid in accordance with Cummins published Standard Repair Time guidelines.

When it is necessary for mechanics to make on-site warranty repairs, Cummins will pay reasonable travel expenses, including meals, mileage and lodging, for mechanics to travel to and from the repair dock.

Cummins will pay for the lubricating oil, antifreeze, filter elements, and other maintenance items that are not reusable due to the Warrantable Failure.

Cummins will pay for reasonable labor costs for Engine removal and reinstallation when necessary to repair a Warrantable Failure.

During the Extended Major Components Warranty

Cummins will pay for the repair or, at its option, replacement of the defective Covered Part and any Covered Part damaged by a Warrantable Failure of the defective Covered Part.

Owner Responsibilities

During the Base Engine Warranty

Owner is responsible for the cost of lubricating oil, antifreeze, filter elements, and other maintenance items replaced during warranty repairs unless such items are not reusable due to the Warrantable Failure.

During the Extended Major Components Warranty

Owner is responsible for the cost of all labor needed to repair the Engine, including the labor cost for Engine removal and reinstallation. When Cummins elects to repair a part instead of replacing it, the Owner is not responsible for the labor needed to repair the part.

Owner is responsible for the cost of all parts required for the repair except for the defective Covered Part and any Covered Part damaged by a Warrantable Failure of the defective Covered Part.

Owner is responsible for the cost of lubricating oil, antifreeze, filter elements and other maintenance items replaced during repair of a Warrantable Failure.

During the Base Engine and Extended Major Components Warranties

Owner is responsible for the operation and maintenance of the Product as specified in the applicable Cummins Operation and Maintenance Manual. Owner is also responsible for providing proof that all recommended maintenance has been performed.

Before the expiration of the applicable warranty, Owner must notify a Cummins distributor, authorized dealer, or other repair location approved by Cummins of any Warrantable Failure and make the Engine available for repair by such facility. Locations in the United States and Canada are listed in the Cummins U.S. and Canada Sales and Service Directory; other locations are listed in the Cummins International Sales and Service Directory.

In the event of any Product failure, Owner is responsible for the cost of towing the boat to the repair dock and for all associated docking and harbor charges.

Owner is responsible for communication expenses, meals, lodging, and similar costs incurred as a result of a Warrantable Failure.

Owner is responsible for maintaining the Engine hourmeter in good working order at all times and to ensure that the hourmeter accurately reflects the total hours of operation of the Product.

Owner is responsible for the costs to investigate complaints, unless the problem is caused by a defect in Cummins material or factory workmanship.

Owner is responsible for non-Engine repairs, "downtime" expenses, cargo damage, fines, all applicable taxes, all business costs, and other losses resulting from a Warrantable Failure.

Limitations

Cummins is not responsible for failures or damage resulting from what Cummins determines to be abuse or neglect, including, but not limited to: operation without adequate coolants or lubricants; overfueling; overspeeding; lack of maintenance of cooling, lubricating or intake systems; improper storage, starting, warm-up, run-in or shutdown practices; unauthorized modifications to the Engine. Cummins is also not responsible for failures caused by incorrect oil or fuel or by water, dirt or other contaminants in the fuel or oil.

Cummins is not responsible for failures resulting from:

- 1. Use or application of the Product inconsistent with its rating designation set forth above.
2. Incorrect installation.

Before a claim for excessive oil consumption will be considered, Owner must submit adequate documentation to show that oil consumption exceeds Cummins published standards.

Cummins is not responsible for failures of marine maintenance components supplied by Cummins beyond ninety days after the coverage start date. Marine maintenance components include, but are not limited to: sea water pump impellers; zinc plugs; oil filters; fuel filters; air filters; water filters; fuel/water separator filters; expansion tank pressure caps.

Failure of belts and hoses supplied by Cummins are not covered beyond 90 days after the date of delivery of the Product to the first user, or the date the unit is first leased, rented or loaned, or when the Product has been operated for 50 hours, whichever occurs first.

Except for the accessories noted previously, Cummins does not warrant accessories which bear the name of another company.

Parts used in warranty repairs may be new Cummins parts, Cummins approved rebuilt parts, or repaired parts. Cummins is not responsible for failures resulting from the use of parts not supplied by Cummins.

Parts used to repair a Warrantable Failure may be new Cummins parts, Cummins approved rebuilt parts, or repaired parts. Cummins is not responsible for failures resulting from the use of parts not approved by Cummins.

A new Cummins or Cummins-approved rebuilt part used to replace a Warranted Part assumes the identity of the Warranted Part it replaced and is entitled to the remaining coverage hereunder.

Cummins Inc. reserves the right to interrogate Electronic Control Module (ECM) data for purposes of failure analysis.

CUMMINS DOES NOT COVER WEAR OR WEAROUT OF COVERED PARTS.

CUMMINS IS NOT RESPONSIBLE FOR INCIDENTAL OR CONSEQUENTIAL DAMAGES.

THIS WARRANTY AND THE EMISSION WARRANTY SET FORTH HEREIN ARE THE SOLE WARRANTIES MADE BY CUMMINS IN REGARD TO THESE ENGINES. CUMMINS MAKES NO OTHER WARRANTIES, EXPRESS OR IMPLIED, OR OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE.

In the United States* * and Canada, this warranty gives you specific legal rights, and you may also have other rights which vary from state to state.

Outside the United States* * and Canada, in case of consumer sales, in some countries the Owner has statutory rights which cannot be affected or limited by the terms of this warranty.

Nothing in this warranty excludes or restricts any contractual rights the Owner may have against third parties.

Emission Warranty

Products Warranted

This Emission Warranty applies to new Engines certified to United States EPA 40 CFR 94 sold by Cummins that are installed on vessels flagged or registered in the United States* *.

Coverage

Cummins warrants to the first user and each subsequent purchaser that the Engine is designed, built and equipped so as to conform at the time of sale by Cummins with all United States Federal emission regulations applicable at the time of manufacture and that it is free from defects in material or factory workmanship which would cause it not to meet these regulations within the longer of the following periods. (A) Five years or 5,000 hours of operation, whichever occurs first, as measured from the date of delivery of the engine to the first user, or the date the unit is first leased,

rented or loaned, or when the Engine has been operated for 50 hours, whichever occurs first, or (B) The Base Engine Warranty.

Limitations

An owner may elect to have maintenance, replacement or repair of the emission control parts performed by a facility other than a Cummins distributor, an authorized dealer or a repair location approved by Cummins, and may elect to use parts other than new genuine Cummins or Cummins approved rebuilt parts and assemblies for such maintenance, replacement or repair; however, the cost of such service or parts and subsequent failures resulting from such service or parts will not be covered under this emission control system warranty.

Failures, other than those resulting from defects in materials or factory workmanship, are not covered by this warranty.

Cummins is not responsible for failures or damage resulting from what Cummins determines to be abuse or neglect, including, but not limited to: operation without adequate coolants or lubricants; overfueling; overspeeding, lack of maintenance of lubricating, cooling or intake systems; improper storage, starting, warm-up, run-in or shutdown practices; unauthorized modifications of the Engine. Cummins is also not responsible for failures caused by incorrect oil or fuel or by water, dirt or other contaminants in the fuel or oil.

Cummins is not responsible for non-Engine repairs, "downtime" expenses, cargo damage, fines, all applicable taxes, all business costs or other losses resulting from a Warrantable Failure.

CUMMINS IS NOT RESPONSIBLE FOR INCIDENTAL OR CONSEQUENTIAL DAMAGES.

* * United States includes American Samoa, the Commonwealth of Northern Mariana Islands, Guam, Puerto Rico, and the U.S. Virgin Islands.

Recreational Applications Marine Propulsion Products

Coverage

Products Warranted Cummins MerCruiser Diesel

This warranty applies to new Cummins Engines sold by Cummins MerCruiser Diesel., herein after "CMD" that are used in Marine propulsion applications anywhere in the world where Cummins approved service is available* and delivered to the first user on or after September 1, 1998. The 'Product' consists of a new Cummins Engine, as well as accessories as listed below, which are approved and supplied by CMD and which are either installed by CMD or a CMD authorized distributor. These Products have the following designation:

High Output Rating

This power rating is for use in variable load applications where full power is limited to (1) hour out of every (8) hours of operation. Also, reduced power operations must be at or below 200 RPM of the maximum rated RPM. This rating is an ISO3046 Fuel Stop Power Rating and is for pleasure/non-revenue generating applications that operate less than 300 hours per year (500 hours for QSMII).

Base Engine Warranty

This warranty covers any failures of the Product, under normal use and service, which result from a defect in Cummins material or factory workmanship (Warrantable Failure). Coverage begins with the sale of the Engine by CMD and continues for the Duration stated below. The Duration commences on either the date of delivery of the Product to the first user, or the date the unit is first leased, rented or loaned, or when the Product has been operated for 50 hours, whichever occurs first.

Extended Major Components Warranty

The Extended Major Components Warranty covers Warrantable Failures of the following Engine parts of castings (Covered parts):

- Engine Cylinder Block Casting
- Engine Cylinder Head Casting
- Engine Camshaft Forging
- Engine Crankshaft Forging
- Engine Connecting Rods
- Engine Gear Train Gears
- Crankshaft Gear
- Camshaft Gear
- Camshaft Idler Gear
- Accessory Drive Gear
- Fuel Pump Gear
- Engine Gear Cover & Housing
- Flywheel Housing

Bushing and Bearing failures are not covered.

Extended Major Components Warranty continues beyond the expiration of the Base Engine Warranty and continues for the Duration stated below. The Duration commences on either the date of delivery of the Product to the first user, or the date the unit is first leased, rented or loaned, or when the Product has been operated for 50 hours, whichever occurs first.

Consumer Products

The warranty on Consumer Products in the United States is a limited warranty. CMD IS NOT RESPONSIBLE FOR INCIDENTAL OR CONSEQUENTIAL DAMAGES. Any implied warranties applicable to Consumer Products terminate concurrently with the expiration of the express warranties applicable to the Product. In the United States, some states do not allow the exclusion of incidental or consequential damages, or limitations on how long an implied warranty lasts, so the limitations or exclusions herein may not apply to you.

These warranties are made to all Owners in the chain of distribution and Coverage continues to all subsequent Owners until the end of the periods of Coverage.

Warranty Coverage						
Coverage Category	Coverage Duration*		Repair Charge Paid By Cummins			
	Months	Hours B,C/ QSM	Parts	Labor	Removal & Installation Labor	Travel

Warranty Coverage						
	Coverage Duration*		Repair Charge Paid By Cummins			
Base Engine Warranty**	24	600/1000	Yes	Yes	Yes	Yes - Up to 6 hours
Extended Major Components	72	1800/3000	Yes***	Yes	Yes	No
* Whichever occurs first. ** Other accessories, ie. Marine Gear and Drive Plate, Instruments, Instrument Panel, Gage Senders, Vibration Isolators, and Fuel/Water Separator are covered for a period of twelve (12) months or 300 hours, whichever occurs first (12 months / 500 hours for QSM11). *** Covered Parts as listed above.						

CMD Responsibilities

During Base Engine Warranty

CMD will pay for all parts and labor needed to repair the damage to the product resulting from a Warrantable Failure when performed during normal business hours. All labor costs will be paid in accordance with Cummins published Standard Repair Time guidelines.

When it is necessary for mechanics to make on-site warranty repairs, CMD will pay up to six hours total travel expenses, including meals, mileage and lodging, for mechanics to travel to and from the repair dock.

CMD will pay for the lubricating oil, antifreeze, filter elements, and other maintenance items that are not reusable due to the Warrantable Failure.

CMD will pay for reasonable labor costs for Engine removal and reinstallation when necessary to repair a Warrantable Failure.

During the Extended Major Components Warranty

CMD will pay for the repair or, at its option, replacement of the defective Covered Part and any Covered Part damaged by a Warrantable Failure of the defective Covered Part.

Owner Responsibilities

During the Extended Major Components Warranty

Owner is responsible for the cost of all parts required for the repair except for the defective Covered Part and any Covered Part damaged by a Warrantable Failure of the defective Covered Part.

During the Base Engine and Extended Major Components Warranties

Owner is responsible for the cost of lubricating oil, antifreeze, filter elements, and other maintenance items replaced during warranty repairs unless such items are not reusable due to the Warrantable Failure.

Owner is responsible for the operation and maintenance of the Product as specified in the applicable CMD Operation and Maintenance Manual. Owner is also responsible for providing proof that all recommended maintenance has been performed.

Before the expiration of the applicable warranty, Owner must notify a Cummins distributor, authorized dealer, or other repair location approved by CMD of any Warrantable Failure and make the Product available for repair by such facility.

In the event of any Product failure, Owner is responsible for the cost of towing the boat to the repair dock and for all associated docking and harbor charges.

Owner is responsible for communication expenses, meals, lodging, and similar costs incurred as a result of a Warrantable Failure.

Owner is responsible for maintaining the Engine hourmeter in good working order at all times and to ensure that the hourmeter accurately reflects the total hours of operation of the Product.

Owner is responsible for the costs to investigate complaints, unless the problem is caused by a defect in Cummins material or factory workmanship.

Owner is responsible for non-Engine repairs, "downtime" expenses, cargo damage, fines, all applicable taxes, all business costs, and other losses resulting from a Warrantable Failure.

Limitations

CMD is not responsible for failures or damage resulting from what CMD determines to be abuse or neglect, including, but not limited to: operation without adequate coolants or lubricants; overfueling; overspeeding; lack of maintenance of cooling, lubricating or intake systems; improper storage, starting, warm-up, run-in or shutdown practices;

unauthorized modifications to the Engine. CMD is also not responsible for failures caused by incorrect oil or fuel or by water, dirt or other contaminants in the fuel or oil.

CMD is not responsible for failure resulting from:

1. Use or application of the Product inconsistent with its rating designation set forth above.
2. Incorrect installation.

Before a claim for excessive oil consumption will be considered, Owner must submit adequate documentation to show that oil consumption exceeds CMD published standards.

CMD is not responsible for failures of maintenance components supplied by Cummins beyond 90 days after the coverage duration start date. Maintenance components include, but are not limited to: sea water pump impellers; zinc plugs; oil filters; fuel filters; air filters; water filters; fuel/water separator filters; expansion tank pressure caps.

Failure of belts and hoses supplied by Cummins are not covered beyond 90 days after warranty start date.

Except for the accessories noted previously, Cummins does not warrant accessories which bear the name of another company.

Parts used in warranty repairs may be new Cummins parts, Cummins approved rebuilt parts, or repaired parts. CMD is not responsible for failures resulting from the use of parts not supplied by Cummins.

A new Cummins or Cummins-approved rebuilt part used to replace a Warranted Part assumes the identity of the Warranted Part it replaced and is entitled to the remaining coverage hereunder.

CUMMINS DOES NOT COVER WEAR OR WEAROUT OF COVERED PARTS.

CUMMINS IS NOT RESPONSIBLE FOR INCIDENTAL OR CONSEQUENTIAL DAMAGES.

THESE WARRANTIES SET FORTH HEREIN ARE THE SOLE WARRANTIES MADE BY CUMMINS IN REGARD TO THESE ENGINES. CUMMINS MAKES NO OTHER WARRANTIES, EXPRESS OR IMPLIED, OR OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE.

In the United States ** and Canada, this warranty gives you specific legal rights, and you may also have other rights which vary from state to state.

Outside the United States ** and Canada, in case of consumer sales, in some countries the Owner has statutory rights which cannot be affected or limited by the terms of this warranty.

Nothing in this warranty excludes or restricts any contractual rights the Owner may have against third parties.

* Locations in the United States and Canada are listed in the Cummins United States and Canada Sales and Service Directory; other locations are listed in the Cummins International Sales and Service Directory.

** United States includes American Samoa, the Commonwealth of Northern Mariana Islands, Guam, Puerto Rico, and the U.S. Virgin Islands.

QSB5.9/QSC8.3/QSL9/QSM11 Recreational Applications Marine Propulsion Worldwide

Coverage

Engines Included in this Coverage

Marine Propulsion

QSB5.9

QSC8.3

QSL9

QSM11

Products Warranted

This warranty applies to new Engines sold by Cummins MerCruiser Diesel LLC. Inc., herein after "CMD", that are branded as Cummins MerCruiser Diesel products and used in Marine applications anywhere in the world where CMD approved service is available* and delivered to the first user on or after Jan 1, 2004. This warranty excludes all engines branded and sold as Cummins Marine Diesel products. The 'Product' consists of a new CMD Engine, as well as accessories, which are approved and supplied by CMD and which are either installed by CMD or a CMD authorized distributor. These Products have the following designation:

High Output Rating

Intended for use in variable load applications where full power is limited to one hour out of every eight hours of operation. Reduced power must be at or below 200 RPM of the maximum RPM. This power rating is for pleasure craft applications that operate less than 500 hours per year.

Base Engine Warranty

This warranty covers any failures of the Product, under normal use and service, which result from a defect in CMD material or factory workmanship (Warrantable Failure). Coverage begins with the sale of the Engine by CMD and continues for the Duration stated in the following table. The Duration commences on either the date of delivery of the Product to the first user, or the date the unit is first leased, rented or loaned, or when the Product has been operated for 125 hours, whichever occurs first.

Extended Major Components Warranty

The Extended Major Components Warranty provision is concurrent with the Base Engine Warranty and continues beyond the expiration of the Base Engine Warranty for an additional four years or 2000 hours, whichever occurs first.

The Extended Major Components Warranty covers Warrantable Failures of the following Engine parts or castings (Covered parts):

- Engine Cylinder Block Casting
- Engine Cylinder Head Casting
- Engine Camshaft Forging
- Engine Crankshaft Forging
- Engine Connecting Rod Forging
- Engine Gear Train Gears:
 - Crankshaft Gear
 - Camshaft Gear
 - Camshaft Idler Gear
 - Accessory Drive Gear
 - Fuel Pump Gear
- Engine Gear Cover Casting
- Flywheel Housing Casting

Bushing and bearing failures are NOT covered.

Extended Major Components Warranty continues beyond the expiration of the Base Engine Warranty for the duration stated below. The Duration commences on either the date of delivery of the Product to the first user, or the date the unit is first leased, rented or loaned, or when the Product has been operated for 125 hours, whichever occurs first.

Warranty Coverage						
Coverage Category	Coverage Duration*		Repair Charge Paid By CMD			
	Months	Hours	Parts	Labor	Removal & Installation Labor	Travel
Base Engine Warranty	24	1000	Yes	Yes	Yes	Yes
Extended Major Components	72	3000	Yes	Yes	Yes	Yes
* Whichever occurs first.						

Consumer Products

The warranty on Consumer Products in the United States is a limited warranty. **CMD IS NOT RESPONSIBLE FOR INCIDENTAL OR CONSEQUENTIAL DAMAGES.** Any implied warranties applicable to Consumer Products terminate concurrently with the expiration of the express warranties applicable to the Product. In the United States, some states do not allow the exclusion of incidental or consequential damages, or limitations on how long an implied warranty lasts, so the limitations or exclusions herein may not apply to you.

These warranties are made to all Owners in the chain of distribution and Coverage continues to all subsequent Owners until the end of the periods of Coverage.

CMD Responsibilities

During the Base Engine Warranty

CMD will pay for all parts and labor needed to repair the damage to the Product resulting from a Warrantable Failure when performed during normal business hours. All labor costs will be paid in accordance with Cummins published Standard Repair Time guidelines.

When it is necessary for mechanics to make on-site warranty repairs, CMD will pay reasonable travel expenses, including meals, mileage and lodging, for mechanics to travel to and from the repair dock.

CMD will pay for the lubricating oil, antifreeze, filter elements, and other maintenance items that are not reusable due to the Warrantable Failure.

CMD will pay for reasonable labor costs for Engine removal and reinstallation when necessary to repair a Warrantable Failure.

During the Extended Major Components Warranty

CMD will pay for parts and labor for the repair or, at its option, the replacement of the defective covered part and any covered part damaged by a warrantable failure of the defective covered part.

Owner Responsibilities

During Both the Base Engine and the Extended Major Components Warranties

Owner is responsible for the cost of lubricating oil, antifreeze, filter elements, and other maintenance items replaced during warranty repairs unless such items are not reusable due to the Warrantable Failure.

During the Extended Major Components Warranties

Owner is responsible for the cost of all parts and labor required for the repair except for the defective Covered Part and any Covered Part damaged by a Warrantable Failure of the defective Covered Part.

Owner is responsible for the operation and maintenance of the Product as specified in the applicable Cummins or CMD Operation and Maintenance Manual. Owner is also responsible for providing proof that all recommended maintenance has been performed.

Exceeding the operational parameters of the HO rating will void this warranty. The owner of the boat is ultimately responsible for ensuring the engine is properly operated and maintained. The warranty will be void on any engines that are misapplied, not maintained properly or misused.

Before the expiration of the applicable warranty, Owner must notify a CMD distributor, authorized dealer, or other repair location approved by CMD of any Warrantable Failure and make the Engine available for repair by such facility. Locations in the United States and Canada are listed in the Cummins U.S. and Canada Sales and Service Directory; other locations are listed in the Cummins International Sales and Service Directory.

In the event of any Product failure, Owner is responsible for the cost of towing the boat to the repair dock and for all associated docking and harbor charges.

Owner is responsible for communication expenses, meals, lodging, and similar costs incurred as a result of a Warrantable Failure.

Owner is responsible for maintaining the Engine hourmeter in good working order at all times and to ensure that the hourmeter accurately reflects the total hours of operation of the Product.

Owner is responsible for the costs to investigate complaints, unless the problem is caused by a defect in CMD material or factory workmanship.

Owner is responsible for non-Engine repairs, "downtime" expenses, cargo damage, fines, all applicable taxes, all business costs, and other losses resulting from a Warrantable Failure.

Limitations

CMD is not responsible for failures or damage resulting from what CMD determines to be abuse or neglect, including, but not limited to: operation without adequate coolants or lubricants; overfueling; overspeeding; lack of maintenance of cooling, lubricating or intake systems; improper storage, starting, warm-up, run-in or shutdown practices; unauthorized modifications to the Engine. CMD is also not responsible for failures caused by incorrect oil or fuel or by water, dirt or other contaminants in the fuel or oil.

CMD is not responsible for failure resulting from:

1. Use or application of the Product inconsistent with its rating designation set forth above.
2. Incorrect installation.
3. Engines that do not reach rated RPM because of vessel overloading.

Before a claim for excessive oil consumption will be considered, Owner must submit adequate documentation to show that oil consumption exceeds CMD published standards.

CMD does not warrant accessories that are not supplied by CMD factory.

CMD is not responsible for failures of maintenance components supplied by Cummins beyond 90 days after the coverage duration start date. Maintenance components include, but are not limited to: sea water pump impellers; zinc plugs; oil filters; fuel filters; air filters; water filters; and fuel/water separator filters.

Except for the accessories noted previously, CMD does not warrant accessories which bear the name of another company.

Parts used in warranty repairs may be new CMD parts, CMD-approved rebuilt parts, or repaired parts. CMD is not responsible for failures resulting from the use of parts not supplied by CMD.

A new CMD or CMD-approved rebuilt part used to replace a Warranted Part assumes the identity of the Warranted Part it replaced and is entitled to the remaining coverage hereunder.

CMD DOES NOT COVER WEAR OR WEAROUT OF COVERED PARTS.

CMD IS NOT RESPONSIBLE FOR INCIDENTAL OR CONSEQUENTIAL DAMAGES.

THESE WARRANTIES SET FORTH HEREIN ARE THE SOLE WARRANTIES MADE BY CMD IN REGARD TO THESE ENGINES. CMD MAKES NO OTHER WARRANTIES, EXPRESS OR IMPLIED, OR OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE.

In the United States* * and Canada, this warranty gives you specific legal rights, and you may also have other rights which vary from state to state.

Outside the United States* * and Canada, in case of consumer sales, in some countries the Owner has statutory rights which cannot be affected or limited by the terms of this warranty.

Nothing in this warranty excludes or restricts any contractual rights the Owner may have against third parties.

* Locations in the United States and Canada are listed in the Cummins United States and Canada Sales and Service Directory; other locations are listed in the Cummins International Sales and Service Directory.

* * United States includes American Samoa, the Commonwealth of Northern Mariana Islands, Guam, Puerto Rico, and the U.S. Virgin Islands.

QSB5.9/QSC8.3/QSL9/QSM11 Commercial Applications Marine Propulsion
Worldwide

Coverage

Engines Included in this Coverage

Marine Propulsion

QSB5.9

QSC8.3

QSL9

QSM11

Products Warranted

This warranty applies to new QSB5.9, QSC8.3, QSL9 and QSM11 Engines sold by Cummins MerCruiser Diesel LLC., herein after "CMD", that are branded as Cummins MerCruiser Diesel products and used in Marine propulsion applications anywhere in the world where CMD approved service is available* and delivered to the first user on or after Jan 1, 2004. This warranty excludes all engines branded and sold as Cummins Marine Diesel products. The 'Product' consists of a new CMD Engine, as well as accessories, which are approved and supplied by CMD and which are either installed by CMD or a CMD authorized distributor. These Products have the following designation:

MARINE PROPULSION

Government Service (GS) Rating

Intended for use in variable load applications where full power is limited to one hour out of every eight hours of operation. Also reduced power must be at or below 200 RPM of the maximum rated RPM. This power rating is only for use in National, State or Local government non-revenue producing applications.

Intermittent Rating

This power rating is intended for intermittent use in variable load applications where full power is limited to two hours out of every eight hours of operation. Also, reduced power operations must be at or below 200 RPM of the maximum rated RPM. This rating is an ISO3046 Fuel Stop Power Rating and is for applications that operate less than 1500 hours per year.

Medium Continuous Rating

This power rating is intended for continuous use in variable load applications where full power is limited to six hours out of every twelve hours of operation. Also, reduced power operations must be at or below 200 RPM of the maximum rated RPM. This rating is an ISO3046 Fuel Stop Power Rating and is for applications that operate less than 3000 hours per year.

Heavy Duty Rating

This power rating is intended for continuous use in variable load applications where full power is limited to eight hours out of every ten hours of operation. Also, reduced power must be at least 200 RPM below the maximum rated RPM. This rating is an ISO3046 Fuel Stop Power Rating and is for applications that operate less than 5,000 hours per year.

Continuous Rating

This power rating is intended for continuous use in applications requiring uninterrupted service at full power. This rating is an ISO3046 Standard Power Rating.

Base Engine Warranty

This warranty covers any failures of the Product, under normal use and service, which result from a defect in CMD material or factory workmanship (Warrantable Failure). Coverage begins with the sale of the Engine by CMD and continues for the Duration stated in the following table. The Duration commences on either the date of delivery of the Product to the first end-user, or the date the unit is first leased, rented or loaned, or when the Product has been operated for 50 hours, whichever occurs first.

Table with 3 columns: Rating, QSB, QSC, QSL, and QSM11. It details warranty coverage periods for different engine ratings, noting that coverage ends at the first occurrence of months or hours of usage.

Warranty Coverage Periods				
Rating	QSB, QSC, QSL		QSM11	
	Months	Hours	Months	Hours
GS	24	1000	24	1000
Intermittent	24	3000	24	3000
Medium Continuous Duty	24	5000	24	6000
Heavy Duty	24	5500	24	8000
Continuous	24	6500	24	9000

Cummins MerCruiser Diesel Responsibilities

During Engine Warranty

CMD will pay for all parts and labor needed to repair the damage to the Product resulting from a Warrantable Failure when performed during normal business hours. All labor costs will be paid in accordance with Cummins published Standard Repair Time guidelines.

When it is necessary for mechanics to make on-site warranty repairs CMD will pay reasonable travel expenses, including meals, mileage and lodging, for mechanics to travel to and from the repair dock.

CMD will pay for the lubricating oil, antifreeze, filter elements, and other maintenance items that are not reusable due to the Warrantable Failure.

CMD will pay for reasonable labor costs for Engine removal and reinstallation when necessary to repair a Warrantable Failure.

Owner Responsibilities

During the Engine Warranty

Owner is responsible for the operation and maintenance of the Product as specified in the applicable CMD Operation and Maintenance Manual. Owner is also responsible for providing proof that all recommended maintenance has been performed. This warranty does not cover normal wear and tear of covered parts.

Before the expiration of the applicable warranty, Owner must notify a CMD service provider, distributor, authorized dealer, or other repair location approved by CMD of any Warrantable Failure and make the Engine available for repair by such facility. Locations in the United States and Canada are listed in the Cummins U.S. and Canada Sales and Service Directory; other locations are listed in the CMD International Sales and Service Directory.

Owner is responsible for the cost of lubricating oil, antifreeze, filter elements, and other maintenance items replaced during warranty repairs unless such items are not reusable due to the Warrantable Failure.

Owner is responsible for communication expenses, meals, lodging, and similar costs incurred as a result of a Warrantable Failure.

Owner is responsible for non-Engine repairs, "downtime" expenses, cargo damage, fines, all applicable taxes, all business costs, and other losses resulting from a Warrantable Failure.

In the event of any Product failure, Owner is responsible for the cost of towing the boat to the repair dock and for all associated docking and harbor charges.

Owner is responsible for maintaining the Engine hourmeter in good working order at all times and to ensure that the hourmeter accurately reflects the total hours of operation of the Product.

Owner is responsible for the costs to investigate complaints, unless the problem is caused by a defect in CMD material or factory workmanship.

Limitations

1. Maintenance Component Limitations

CMD will replace certain maintenance components if they fail within 90 days or less after the base coverage starts. Maintenance components include but are not limited to: sea water pump impellers, zinc plugs, oil filters, fuel filters, air filters, water filters, fuel/water separator filters, expansion tank pressure caps, belts, hoses.

2. Other Component Limitations

CMD does not warrant components that are not supplied by CMD factory.

3. CMD supplied alternators and starters limitation

Warranty coverage is limited to 2 years or 2000 hours, whichever expires first.

Consumer Products

The warranty on Consumer Products in the United States is a limited warranty. **CMD IS NOT RESPONSIBLE FOR INCIDENTAL OR CONSEQUENTIAL DAMAGES.** Any implied warranties applicable to Consumer Products terminate concurrently with the expiration of the express warranties applicable to the Product. In the United States, some states do not allow the exclusion of incidental or consequential damages, or limitations on how long an implied warranty lasts, so the above limitations or exclusions may not apply to you.

CMD is not responsible for failures or damage resulting from what CMD determines to be abuse or neglect, including, but not limited to: operation without adequate coolants or lubricants; overfueling; overspeeding; lack of maintenance of cooling, lubricating or intake systems; improper storage, starting, warm-up, run-in or shutdown practices; unauthorized modifications to the engine. CMD is also not responsible for failures caused by incorrect oil or fuel or by water, dirt or other contaminants in the fuel or oil.

CMD is not responsible for failures resulting from:

1. Use or application of the Product inconsistent with its rating designation set forth above.
2. Incorrect installation.

Before a claim for excessive oil consumption will be considered, Owner must submit adequate documentation to show that oil consumption exceeds CMD published standards.

CMD is not responsible for failures of maintenance components supplied by CMD beyond 90 days after the coverage duration start date. Maintenance components include, but are not limited to: sea water pump impellers; zinc plugs; oil filters; fuel filters; air filters; water filters; fuel/water separator filters.

Parts used in warranty repairs may be new CMD parts, CMD-approved rebuilt parts, or repaired parts. CMD is not responsible for failures resulting from the use of parts not supplied by CMD.

A new CMD or CMD-approved rebuilt part used to replace a Warranted Part assumes the identity of the Warranted Part it replaced and is entitled to the remaining coverage hereunder.

CMD DOES NOT COVER WEAR OR WEAROUT OF COVERED PARTS.

CMD IS NOT RESPONSIBLE FOR INCIDENTAL OR CONSEQUENTIAL DAMAGES.

THESE WARRANTIES SET FORTH HEREIN ARE THE SOLE WARRANTIES MADE BY CMD IN REGARD TO THESE ENGINES. CMD MAKES NO OTHER WARRANTIES, EXPRESS OR IMPLIED, OR OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE.

Emission Warranty

Products Warranted

This Emission Warranty applies to new Engines certified to United States EPA 40 CFR 94 sold by CMD that are installed in vessels flagged or registered in the United States**.

Coverage

CMD warrants to the first user and each subsequent purchaser that the Engine is designed, built, and equipped so as to conform at the time of sale by CMD with all U.S. Federal emission regulations applicable at the time of manufacture and that it is free from defects in workmanship or material which would cause it not to meet these regulations within the longer of the following periods: (A) Five years or 5,000 hours of operation, whichever occurs first, The Emissions Warranty starts from the date of delivery of the Engine to the first user, or the date the unit is first leased, rented, or loaned, or when the Engine has been operated for 50 hours, whichever occurs first, or (B) The Base Engine Warranty.

Limitations

The owner may elect to have maintenance, replacement, or repair of the emission control parts performed by a facility other than a CMD distributor, an authorized dealer or a repair location approved by CMD, and may elect to use parts other than new genuine CMD or CMD-approved rebuilt parts and assemblies for such maintenance, replacement or repair; however, the cost of such service or parts and subsequent failures resulting from such service or parts will not be covered under this emission control system warranty.

Failures, except those resulting from a defect in materials, or factory workmanship, are not covered by this WARRANTY.

CMD IS NOT RESPONSIBLE FOR INCIDENTAL OR CONSEQUENTIAL DAMAGES.

In the United States** and Canada, this warranty gives you specific legal rights, and you may also have other rights which vary from state to state.

Outside the United States** and Canada, in case of consumer sales, in some countries the Owner has statutory rights which cannot be affected or limited by the terms of this warranty.

Nothing in this warranty excludes or restricts any contractual rights the Owner may have against third parties.

* Locations in the United States and Canada are listed in the Cummins United States and Canada Sales and Service Directory; other locations are listed in the Cummins International Sales and Service Directory.

* * United States includes American Samoa, the Commonwealth of Northern Mariana Islands, Guam, Puerto Rico, and the U.S. Virgin Islands.

California Emission Control System Warranty, Off-Highway

Products Warranted

This Emission Control System Warranty applies to off-road diesel engines certified with the California Air Resources Board beginning with the year 1996 for engines up to 750 horsepower, beginning with the year 2000 for 751 horsepower and over, marketed by Cummins, and registered in California for use in industrial off-highway applications.

Your Warranty Rights and Obligations

The California Air Resources Board and Cummins Engine Company, Inc., are pleased to explain the emission control system warranty on your engine. In California, new off-road diesel engines must be designed, built and equipped to meet the State's stringent anti-smog standards. Cummins must warrant the emission control system on your engine for the periods of time listed below provided there has been no abuse, neglect or improper maintenance of your engine.

Your emission control system may include parts such as the fuel injection system and the air induction system. Also included may be hoses, belts, connectors and other emission-related assemblies.

Where a warrantable condition exists, Cummins will repair your off-road diesel engine at no cost to you including diagnosis, parts and labor.

Manufacturer's Warranty Coverage

This warranty coverage is provided for 5 years or 3,000 hours of engine operation, whichever first occurs from the date of delivery of the engine to the first user. If any emission-related part on your engine is defective, the part will be repaired or replaced by Cummins.

Coverage

This emission control system warranty applies to the following M11 and N14 emission control parts:

Fuel Pump

AFC Plunger
AFC Spring
AFC/ASA Valve
AFC Setting
Throttle Shaft
No Air Setting
Static Timing

Injectors (STC)

Cup
Calibration
Top Stop
Spring
Spring Retainer
Sleeve
Check Ball Spring
Retainer Clip

Turbocharger

Compressor Wheel
Turbine Wheel
Turbine Oil Seal

Intake Manifold

Charge Air Cooler
Aftercooler

Exhaust Manifold

Oil Control Valve (STC)

Plunger
Spring
Oil Transfer Connection
Assembly

Injectors (CELECT™)

Body
Cup
Needle
Nozzle Spring
Barrel

Electronic Control System

Control Module
Boost Pressure Sensor
Coolant Temperature Sensor

Owner's Warranty Responsibilities

As the off-road diesel engine owner, you are responsible for the performance of the required maintenance listed in your Cummins Operation and Maintenance Manual. Cummins recommends that you retain all receipts covering maintenance on your off-road diesel engine, but Cummins cannot deny warranty solely for the lack of receipts or for your failure to ensure the performance of all scheduled maintenance.

You are responsible for presenting your off-road diesel engine to a Cummins dealer as soon as a problem exists. The warranty repairs should be completed in a reasonable amount of time, not to exceed 30 days.

As the off-road diesel engine owner, you should also be aware that Cummins may deny you warranty coverage if your off-road diesel engine or a part has failed due to abuse, neglect, improper maintenance or unapproved modifications.

Your engine is designed to operate on diesel fuel only. Use of any other fuel may result in your engine no longer operating in compliance with California's emissions requirements.

If you have any questions regarding your warranty rights and responsibilities, you should contact Cummins Customer Assistance Department at 1-800-343-7357 (1-800-DIESELS) or the California Air Resources Board at 9528 Telstar Avenue, El Monte, CA 91731.

Prior to the expiration of the applicable warranty, Owner must give notice of any warranted emission control failure to a Cummins distributor, authorized dealer or other repair location approved by Cummins and deliver the engine to such facility for repair. Repair locations are listed in Cummins United States and Canada Service Directory.

Owner is responsible for incidental costs such as: communication expenses, meals, lodging incurred by Owner or employees of Owner as a result of a warrantable failure.

Owner is responsible for business costs and losses, "downtime" expenses, and cargo damage resulting from a warrantable failure. CUMMINS IS NOT RESPONSIBLE FOR OTHER INCIDENTAL OR CONSEQUENTIAL DAMAGES. INCIDENTAL OR CONSEQUENTIAL DAMAGES INCLUDE BUT ARE NOT LIMITED TO FINES, THEFT, VANDALISM OR COLLISIONS.

Replacement Parts

Cummins recommends that any service parts used for maintenance, repair or replacement of emission control systems be new, genuine Cummins or Cummins approved rebuilt parts and assemblies, and that the engine be serviced by a Cummins distributor, authorized dealer or the repair location approved by Cummins. The owner may elect to have maintenance, replacement or repair of the emission control parts performed by a facility other than a Cummins distributor, an authorized dealer or a repair location approved by Cummins, and may elect to use parts other than new genuine Cummins or Cummins approved rebuilt parts and assemblies for such maintenance, replacement or repair; however, the cost of such service or parts will not be covered under this emission control system warranty.

Cummins Responsibilities

Repairs and service will be performed by any Cummins distributor, authorized dealer or other repair location approved by Cummins using new, genuine Cummins or Cummins approved rebuilt parts and assemblies. Cummins will repair any of the emission control parts found by Cummins to be defective without charge for parts or labor (including diagnosis which results in determination that there has been a failure of a warranted emission control part).

Emergency Repairs

In the case of an emergency where a Cummins distributor, authorized dealer, or other repair location approved by Cummins is not available, repairs may be performed by any available repair location using any replacement parts. Cummins will reimburse the Owner for expenses (including diagnosis), not to exceed the manufacturer's suggested retail price for all warranted parts replaced and labor charges based on the manufacturer's recommended time allowance for the warranty repair and the geographically appropriate hourly labor rate. A part not being available within 30 days or a repair not being complete within 30 days constitutes an emergency. Replaced parts and paid invoices must be presented at a Cummins authorized repair facility as a condition of reimbursement for emergency repairs not performed by a Cummins distributor, authorized dealer, or other repair location approved by Cummins.

Warranty Limitations

Cummins is not responsible for failures resulting from Owner or operator abuse or neglect, such as: operation without adequate coolant, fuel or lubricants; overfueling; overspeeding; lack of maintenance of lubricating, cooling or air intake systems; improper storage, starting, warm-up, run-in or shutdown practices.

The manufacturer warrants to the ultimate purchaser and each subsequent purchaser that the engine is designed, built, and equipped so as to conform with all applicable regulations adopted by the Air Resources Board, and that it is free from defects in materials and workmanship which cause the failure of a warranted part.

Any warranted part which is not scheduled for replacement as required maintenance, or which is scheduled only for regular inspection to the effect of "repair or replace as necessary" is warranted for the warranty period.

Any warranted part which is scheduled for replacement as required maintenance is warranted for the period of time prior to the first scheduled replacement point for that part.

The owner will not be charged for diagnostic labor which leads to the determination that a warranted part is defective, if the diagnostic work is performed at a warranty station.

The manufacturer is liable for damages to other engine components caused by the failure under warranty of any warranted part.

Cummins is not responsible for failures resulting from improper repair or the use of parts which are not genuine Cummins or Cummins approved parts.

These warranties, together with the express commercial warranties and emission warranty are the sole warranties of Cummins. There are no other warranties, express or implied, or of merchantability or fitness for a particular purpose.

[illegible]

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